THE HODGE CONJECTURE FOR GAUSSIAN AND EISENSTEIN ABELIAN SIXFOLDS

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Abstract. We prove the Hodge conjecture for Gaussian and Eisenstein abelian sixfolds, that is, abelian sixfolds whose endomorphism algebra contains the Gaussian integers or the Eisenstein integers. In particular, the Weil classes on Gaussian and Eisenstein abelian sixfolds of Weil type are algebraic for every discriminant of the Hermitian form. We also prove the Hodge conjecture for any nonsimple abelian sixfold, except for the sixfolds isogenous to products of a fourfold and a surface of one explicit type.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Hodge proposed the following conjecture in his talk at the International Congress of Mathematicians in 1950 [Hod52]. We refer to [Del06] for the modern formulation.

Conjecture 1.1 (Hodge)**.** *Let Y be a smooth projective variety and p a non-negative integer. Then the cycle class map*

$$\mathrm{cl} : CH^p(Y) \otimes \mathbb{Q} \to H^{2p}(Y, \mathbb{Q}) \cap H^{p,p}(Y)$$

is surjective.

The elements of $H^{2p}(Y, \mathbb{Q}) \cap H^{p,p}(Y)$ are called *Hodge classes*, and the elements of the image of cl are called *algebraic classes*. For $p = 1$, Conjecture 1.1 holds by the Lefschetz $(1,1)$ theorem (cf. [Moo99, (2.10)]), and Conjecture 1.1 holds for any Y with $\dim Y \leq 3$ (cf. [Mar26, Section 1.1]). The analogue of Conjecture 1.1 with integral coefficients is false. Engel–de Gaay Fortman–Schreieder proved that it fails for very general principally polarized abelian varieties of dimension at least 4 [EFS25]. Thus we only consider the conjecture with rational coefficients.

Date: October 27, 2026.

2020 *Mathematics Subject Classification.* 14C30, 14K20, 14C25, 14H40, 14D07.

Key words and phrases. Hodge conjecture, abelian varieties, Weil classes, Hurwitz spaces, period maps, complex reflection groups.

Abelian varieties are the main testing ground for Conjecture 1.1. For an abelian variety A, the Hodge classes on A are the elements of $H^*(A,\mathbb{Q})$ that are invariant under an algebraic group, the Hodge group of A, acting on the cohomology (cf. [Moo99, (1.11), (1.12)]). Thus the Hodge classes on A can be computed from the Hodge structure on $H^1(A,\mathbb{Q})$. In many cases the Hodge classes on A are spanned by products of divisor classes, hence algebraic. Tankeev proved that the Hodge classes on A are spanned by products of divisor classes when A is simple of prime dimension [Tan82], and Ribet [Rib83] and Murty [Mur84] extended the theorem of Tankeev to criteria in terms of the Hodge group. Write

$$\mathrm{End}^0(A) := \mathrm{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

We say that A is of CM type if $\mathrm{End}^0(A)$ contains a commutative semisimple $\mathbb{Q}$-subalgebra of dimension $2\dim A$. An example of Mumford, presented by Pohlmann [Poh68], is the first abelian variety whose Hodge classes are not all spanned by products of divisor classes. It is a fourfold of CM type.

Weil found Hodge classes of a different kind [Wei77] (cf. [Gor99, 4.9–4.11, 8.2, 8.3]). Let A be an abelian variety with $\dim A = 2n$, and let K be an imaginary quadratic field with a unital embedding $K \hookrightarrow \mathrm{End}^0(A)$. Write σ and $\bar\sigma$ for the complex embeddings of K, and write $H^1(A,\mathbb{C})_\sigma$ and $H^1(A,\mathbb{C})_{\bar\sigma}$ for the subspaces of $H^1(A,\mathbb{C})$ on which K acts through σ and through $\bar\sigma$, respectively. If

$$\dim\left(H^{1,0}(A) \cap H^1(A,\mathbb{C})_\sigma\right) = \dim\left(H^{1,0}(A) \cap H^1(A,\mathbb{C})_{\bar\sigma}\right) = n,$$

then we say that K has multiplicities (n,n) on $H^{1,0}(A)$. In this case, the *Weil plane* of A with respect to K is the two-dimensional subspace $W_K(A) \subset H^{2n}(A,\mathbb{Q})$ with

$$W_K(A) \otimes_{\mathbb{Q}} \mathbb{C} = \wedge^{2n} H^1(A,\mathbb{C})_\sigma \oplus \wedge^{2n} H^1(A,\mathbb{C})_{\bar\sigma}.$$

It consists of Hodge classes, and its elements are called *Weil classes*. For $n \geq 2$ and a very general A with an action of K of multiplicities (n,n), the intersection of the Weil plane with the subspace of $H^{2n}(A,\mathbb{Q})$ spanned by products of n divisor classes is zero (cf. [Mar26, Section 1.1]). Weil suggested looking for counterexamples to Conjecture 1.1 among Weil classes.

We now recall the Hermitian forms attached to polarizations. Consider a polarization of A whose Rosati involution restricts to complex conjugation on K. Such a polarization is called *compatible*. Its Riemann form ψ on $H_1(A,\mathbb{Q})$ can be written as

$$\psi = \mathrm{Tr}_{K/\mathbb{Q}}(\delta h),$$

where h is a K-Hermitian form on $H_1(A,\mathbb{Q})$ and $\delta \in K$ is a fixed non-zero element with $\bar\delta = -\delta$ (cf. [Del82, Lemma 4.6], [Mar25, Section 1.1]). We call h the *Hermitian form* of the polarization. The discriminant of h is the class of $\det h$ in $\mathbb{Q}^*/\mathrm{Nm}_{K/\mathbb{Q}}(K^*)$, where $\det h$ is computed in a K-basis of $H_1(A,\mathbb{Q})$. The discriminant does not depend on the basis or on δ. We say that h is split if $H_1(A,\mathbb{Q})$ contains a K-subspace of K-dimension n on which h is zero, and we say that A is split if A admits a compatible polarization whose Hermitian form is split. When K has multiplicities (n,n) on $H^{1,0}(A)$, h is split if and only if the discriminant of h is $[(-1)^n]$ [Bro26, Proposition 4].

We now list the known results on Conjecture 1.1 for abelian varieties and on the algebraicity of Weil classes.

(1) Moonen–Zarhin determined the Hodge classes on abelian varieties of dimension at most 5 [MZ95, MZ99]. Markman proved Conjecture 1.1 for abelian fourfolds [Mar25, Corollary 1.6.1], and it follows that Conjecture 1.1 holds for abelian varieties of dimension at most 5 [Mar26, Corollary 1.3].

(2) Weil classes on fourfolds with an action of K of multiplicities $(2,2)$ are algebraic in the following cases. Schoen proved that the Weil classes are algebraic for $K = \mathbb{Q}(\sqrt{-3})$ and any discriminant [Sch98, Theorem, p. 329]; the earlier statement [Sch88, Theorem 3.2] overlooked an invariant of the polarization [Sch98, p. 330]. For $K = \mathbb{Q}(i)$ and discriminant

[1], van Geemen proved that the Weil classes are algebraic [vG96]. Markman proved that the Weil classes are algebraic for any K when the discriminant is [1], that is, when the fourfold is split [Mar23]. Other proofs of the result of Markman were given by Floccari–Fu [FF25, Theorems 1.1–1.2], whose results hold for all powers of the fourfold, and by van Geemen–Rapagnetta [vGR26, Theorems 0.1 and 0.3].

(3) Weil classes on sixfolds with an action of K of multiplicities $(3,3)$ are algebraic in the following cases. Schoen proved that the Weil classes are algebraic on the sixfolds of a family with $K=\mathbb{Q}(\sqrt{-3})$ [Sch98, §§11–13], and Koike proved that the Weil classes are algebraic on the sixfolds of a family with $K=\mathbb{Q}(i)$ [Koi04, Corollary 2.1]. The sixfolds in both families are split (cf. [Mar26, Section 1.1]). Markman proved that the Weil classes are algebraic on any split sixfold, for any K [Mar25, Theorem 1.5.1].

(4) Flapan proved Conjecture 1.1 for simple abelian sixfolds of type I or II [Fla17, Corollary 13.5], where the types refer to the Albert classification of $\mathrm{End}^0(A)$ (cf. [Moo99, (1.19), (2.4)]). Abdulali proved Conjecture 1.1 for the sixfolds of type III in certain three-dimensional families [Abd99].

(5) Hazama reduced Conjecture 1.1 for abelian varieties of CM type to Hodge classes of codimension two [Haz03] (cf. [vHS26, Theorem 2.2.1]). Conjecture 1.1 was proved for any abelian variety of CM type in [OAI26a, Theorem 1.1], and another proof was given by van Hoften–Sempliner [vHS26, Theorem I].

(6) Conjecture 1.1 holds for all powers of a sixfold A with an action of an imaginary quadratic field K of multiplicities $(3,3)$ such that A is split [OAI26b, Theorem 1.1].

We now explain where the known methods stop for sixfolds. The theorem of Markman only treats split sixfolds, that is, sixfolds with a compatible polarization whose Hermitian form has discriminant $[-1]$. Its proof in [Mar25, Section 9.3] starts from a Hermitian form on K^{2n} of discriminant $[(-1)^n]$, which is $[-1]$ for $n=3$. Mostaed pointed out that, outside discriminant $[-1]$, there were no known results on Conjecture 1.1 for Weil classes on sixfolds [Mos26, Introduction]. In two cases, one can change the polarization of a sixfold A with an action of K of multiplicities $(3,3)$ so that the discriminant becomes $[-1]$. The first case is when A contains an abelian subvariety of odd dimension that is stable under K (Lemma 8.1). The second case is when A is simple, $\mathrm{End}^0(A)$ is a CM field L of degree 4, and L contains an imaginary quadratic subfield K of multiplicities $(3,3)$ (Lemma 7.3).

However, if A is simple and $\mathrm{End}^0(A)=K$, then any two compatible polarizations differ by a positive rational multiple, so the discriminant is an invariant of A and K. This is Lemma 2.11, which only uses that the Néron–Severi group of A has rank one. Given a K-Hermitian form h of signature (n,n) on K^{2n}, the abelian varieties of dimension $2n$ with an action of K of multiplicities (n,n) and a compatible polarization whose Hermitian form is isometric to h form a family of dimension n^2, and, for $n\geq 2$, a very general member of the family has Néron–Severi group of rank one (cf. [Mar25, Section 1.1], [Mar26, Section 1.1], [vG94, Theorem 5.2]). For example, let $K=\mathbb{Q}(i)$ and

$$h=\mathrm{diag}(1,1,1,-1,-1,-3).$$

Then $\det h=-3$, and the class of -3 in $\mathbb{Q}^*/\mathrm{Nm}_{K/\mathbb{Q}}(K^*)$ is different from the class of -1, because 3 is not a sum of two squares of rational numbers. Thus a very general sixfold A in the family attached to h admits no compatible polarization whose Hermitian form has discriminant $[-1]$, and the theorem of Markman does not cover A.

Similarly, if $\mathrm{End}^0(A)$ is a CM field of degree 6 that contains an imaginary quadratic subfield K of multiplicities $(3,3)$, then the discriminant does not depend on the compatible polarization (Proposition 7.2). If $\mathrm{End}^0(A)\cong(-1,-1)_{\mathbb{Q}}$, the quaternion algebra over $\mathbb{Q}$ with basis $1,i,j,k$, $i^2=j^2=-1$, and $ij=-ji=k$, so that A is of type III, then there exist examples of A such that for any imaginary quadratic subfield $K\subset\mathrm{End}^0(A)$, no compatible polarization has a Hermitian form of discriminant $[-1]$ (Proposition 7.8). The examples in Proposition 7.8 only show that

the method of Markman does not reach all sixfolds of type III. They are covered by our main theorem, since $(-1,-1)_{\mathbb{Q}}$ contains the subfield $\mathbb{Q}(i)$.

A definite quaternion algebra D over $\mathbb{Q}$ contains a subfield isomorphic to $\mathbb{Q}(i)$ (respectively $\mathbb{Q}(\sqrt{-3})$) if and only if no prime at which D is ramified splits in $\mathbb{Q}(i)$ (respectively $\mathbb{Q}(\sqrt{-3})$) [Voi21, Proposition 14.6.7]. For example, let D be the definite quaternion algebra over $\mathbb{Q}$ ramified exactly at 13 and at the real place. Since $13 = 2^2 + 3^2$ and $-3 \equiv 6^2 \pmod{13}$, the prime 13 splits in both $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$, so D contains no subfield isomorphic to $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, and a sixfold A of type III with $\mathrm{End}^0(A) \cong D$ is not covered by Theorem 1.2.

The goal of this paper is to prove Conjecture 1.1 for any abelian sixfold A such that $\mathrm{End}^0(A)$ contains a subfield isomorphic to $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$. We call A a *Gaussian abelian sixfold* if $\mathrm{End}^0(A)$ contains a subfield isomorphic to $\mathbb{Q}(i)$, and an *Eisenstein abelian sixfold* if $\mathrm{End}^0(A)$ contains a subfield isomorphic to $\mathbb{Q}(\sqrt{-3})$.

In this paper, we prove the following theorem.

Theorem 1.2. *Let K be $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, and let A be an abelian variety of dimension 6. Assume that there exists a unital $\mathbb{Q}$-algebra embedding $K \hookrightarrow \mathrm{End}^0(A)$, where*

$$\mathrm{End}^0(A) = \mathrm{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Then for any integer $0 \leq p \leq 6$, the cycle class map

$$\mathrm{cl} : CH^p(A) \otimes \mathbb{Q} \to H^{2p}(A, \mathbb{Q}) \cap H^{p,p}(A)$$

is surjective.

The main new input in the proof of Theorem 1.2 is the following theorem on Weil classes.

Theorem 1.3. *Let K be $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, and let A be an abelian sixfold with a unital embedding $K \hookrightarrow \mathrm{End}^0(A)$ such that the multiplicities of K on $H^{1,0}(A)$ are $(3,3)$. Then any class in $W_K(A)$ is the class of an algebraic cycle of codimension 3 with rational coefficients.*

Nonsimple sixfolds. For nonsimple abelian sixfolds, we prove the Hodge conjecture with the exception of one explicit family of products. For an abelian variety A, a number field L with a unital embedding $L \hookrightarrow \mathrm{End}^0(A)$, and a complex embedding τ of L, we denote by $H^1(A, \mathbb{C})_\tau$ the subspace of $H^1(A, \mathbb{C})$ on which L acts through τ, and we call $\dim\left(H^{1,0}(A) \cap H^1(A, \mathbb{C})_\tau\right)$ the *multiplicity* of L on $H^{1,0}(A)$ at τ.

Theorem 1.4. *Let A be an abelian sixfold which is not simple. Then the cycle class map*

$$\mathrm{cl} : CH^p(A) \otimes \mathbb{Q} \to H^{2p}(A, \mathbb{Q}) \cap H^{p,p}(A)$$

is surjective for any integer $0 \leq p \leq 6$, unless A is isogenous to a product $B \times S$ satisfying the following conditions.

1. *B is a simple abelian fourfold, $F := \mathrm{End}^0(B)$ is a CM field of degree 4, and for some ordering $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$ of the complex embeddings of F, the multiplicities of F on $H^{1,0}(B)$ at $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$ are $(2,1,0,1)$.*
2. *S is an abelian surface which is either simple of CM type or isogenous to a product of two non-isogenous elliptic curves with complex multiplication.*
3. *$U_B \cong T_S$ as rational Hodge structures, where $U_B \subset H^2(B, \mathbb{Q})$ is the Hodge substructure with*

$$U_B \otimes \mathbb{C} = \bigoplus_{\tau} \wedge^2 H^1(B, \mathbb{C})_\tau,$$

the sum running over the four complex embeddings τ of F, and $T_S \subset H^2(S, \mathbb{Q})$ is the orthogonal complement of $NS(S) \otimes \mathbb{Q}$ with respect to the cup product.

For a product $B \times S$ of Theorem 1.4, the Hodge classes in $H^4(B \times S, \mathbb{Q})$ modulo the span of products of two classes of divisors form a four-dimensional space, and the Hodge conjecture holds for $B \times S$ if and only if $H^4(B \times S, \mathbb{Q})$ contains an algebraic class which is a Hodge class but

not a linear combination of products of classes of divisors (Proposition 8.14). Moreover, for any abelian variety A isogenous to a product $B \times S$ of Theorem 1.4, there exists no unital embedding of an imaginary quadratic field into $\mathrm{End}^0(A)$ (Lemma 8.15). Thus no exceptional case occurs in Theorem 1.2. In Section 8, Proposition 8.14 and Lemma 8.15 are formulated for the pairs (B, S) satisfying the three conditions of Theorem 1.4, and we call such pairs *exceptional pairs* (Definition 8.12).

Sketch of the proof. The proof of Theorem 1.2 in Section 9 distinguishes cases according to $\mathrm{End}^0(A)$. If A is simple, the proof distinguishes nine cases, according to the Albert type of $\mathrm{End}^0(A)$ [Moo99, (1.19)], its degree when it is a CM field (Lemma 2.5), and whether $\mathrm{End}^0(A)$ contains an imaginary quadratic subfield of multiplicities $(3, 3)$ on $H^{1,0}(A)$. In four of them, the Hodge classes are spanned by products of divisor classes (the proof of [Fla17, Corollary 13.5] for type II, Proposition 7.1, Theorem 7.4, and Proposition 7.2). In one, A is of CM type, which is covered by [OAI26a, Theorem 1.1]. In one, $\mathrm{End}^0(A)$ is a quartic CM field with an imaginary quadratic subfield of multiplicities $(3, 3)$, and a totally positive rescaling of the polarization changes the discriminant to $[-1]$ (Lemma 7.3), so that Theorem 2.14 can be used.

In the remaining three, $\mathrm{End}^0(A)$ is K, a sextic CM field, or a definite quaternion algebra, and K has multiplicities $(3, 3)$. Then the Hodge classes are spanned by products of divisor classes and by $W_K(A)$, or, for a definite quaternion algebra, by its translates under $\mathrm{End}^0(A)^*$ (Definition-Theorem 7.6), and the discriminant cannot be changed to $[-1]$ in general (Lemma 2.11, Proposition 7.2, and Proposition 7.8). Theorem 1.3 is used exactly in the last three cases. For nonsimple sixfolds which are not of CM type, Theorem 1.4 uses only known cycles.

Weil planes of non-split sixfolds. The determinant cycles of Schoen [Sch88, Theorem 2.0], and their extension to finite groups G acting on curves X (Theorem 3.11), need paired branching (Definition 3.4), that is, the branch points come in pairs with mutually inverse local monodromies. Paired branching yields a G-invariant Lagrangian subspace of $H_1(X, \mathbb{Q})$ (Proposition 3.7), so the Jacobian factors cut out by self-adjoint idempotents are split (Corollary 3.8), and so is any abelian variety K-linearly isogenous to one of them (Lemma 2.10 (4)). Hence a sixfold which is not split is not K-linearly isogenous to any such factor.

We pass to dimension 8 instead. For the Hermitian form h_A of a compatible polarization of A, the form $h_A \perp \mathrm{diag}(1, (\det h_A)^{-1})$ has signature $(4, 4)$ and determinant 1, hence is split by Landherr's theorem (Lemma 6.1). Thus $A \times S$ can be K-linearly isogenous to a split eightfold A_X for an abelian surface S with Hermitian form $\mathrm{diag}(1, (\det h_A)^{-1})$, and contraction with a non-zero class in $W_K(S)$, which is the class of a $\mathbb{Q}$-divisor, maps $W_K(A_X)$ isomorphically onto $W_K(A)$ (Lemma 6.6) (cf. [Sch98, Proposition 10], which passes from Weil classes of sixfolds to Weil classes of fourfolds by a product with an abelian surface). The price is a family of split eightfolds with algebraic Weil planes whose periods reach $A \times S$ up to isogeny, and the period rank of the family has to be at least 15 (see below).

Let a finite group G act on curves X with $X/G \cong \mathbb{P}^1$ and paired branching, with k pairs of branch points, and let V be a faithful absolutely irreducible representation of G of dimension d with character field K. By the Chevalley–Weil formula [Can18, Theorem 3.1], the multiplicity of V in $H^1(X, \mathbb{C})$ is $2\sum_t (d - \dim V^t) - 2d$, where t runs over one branch monodromy in each pair and $V^t \subset V$ is the subspace fixed by t, while the family is $(2k-3)$-dimensional. For an eightfold the multiplicity is 8, that is, $\sum_t (d - \dim V^t) = d + 4$, and each summand is at least 1. Rank 15 needs $k \geq 9$, so $d \geq 5$, and t acts on V as a reflection for all but at most $d - 5$ of the pairs.

For $\mathbb{Q}(\sqrt{-3})$, the Shephard–Todd group G_{34} with $d = 6$ and ten pairs of reflections yields a 17-dimensional family, which we call the *Eisenstein family* (Construction-Theorem 4.4). For $\mathbb{Q}(i)$, the representation of the Weyl group $W(E_8)$ has character field $\mathbb{Q}$, so for $G = \mu_4 W(E_8)$ and $d = 8$ some branch monodromy is not in $W(E_8)$, and it fixes a subspace of dimension at most 4. Since $d + 4 = 12$, the families of $\mu_4 W(E_8)$ are at most 15-dimensional, and eight pairs of reflections and one pair of copies of an involution outside $W(E_8)$ yield a 15-dimensional family,

which we call the *Gaussian family* (Construction-Theorem 4.5). For other K, we do not know such a group. The assumption on K is used only in Construction-Theorems 4.4 and 4.5.

Weil classes on Jacobian factors. Let $M_d(K)$ be the block of $\mathbb{Q}[G]$ attached to A_X, whose irreducible complex representations are V and the complex conjugate of V. The evaluation map at a suitable vector of V embeds the multiplicity space $M(V) \cong \mathrm{Hom}_G(V, H^1(X,\mathbb{C}))$ in $H^1(X,\mathbb{C})$, and the Künneth formula places its top exterior power $\wedge^{2n}M(V)$ in $H^{2n}(X^{2n},\mathbb{C})$, where $2n = \dim A_X$. After complexification, the Weil plane of A_X is spanned by the images of $\wedge^{2n}M(V)$ and of the complex conjugate line under an algebraic correspondence from X^{2n} to A_X given by a Poincaré line bundle and the small diagonal of A_X^{2n}.

Theorem 3.1 treats one cyclic group, and an induction formula with rational coefficients, such as induction from cyclic subgroups, yields only a tensor power of the top exterior power of $M(V)$ after clearing denominators. Integral Brauer induction [Bol90, Corollary (2.12)] writes $[V] = [R_+] - [R_-]$, where R_+ and R_- are sums of representations induced from characters α_i of subgroups $L_i \leq G$. Each term corresponds to the cyclic quotient cover $X/\ker(\alpha_i) \to X/L_i$, on which paired branching becomes the condition of Theorem 3.1, so the top exterior powers of $M(R_+)$ and of $M(R_-^\vee)$ are spanned by algebraic classes. The top exterior power of $M(R_+)$ is the top exterior power of $M(V) \oplus M(R_-)$, and contraction with an algebraic class spanning the top exterior power of $M(R_-^\vee)$, through the cup product pairing $M(R_-) \times M(R_-^\vee) \to \mathbb{C}$, leaves the top exterior power of $M(V)$ (Theorem 3.11).

The proof of [Sch88, Theorem 2.0] blows up the intersection of two branch divisors with mutually inverse exponents and uses [Sch88, Lemma 2.8], whose conclusion Appendix B proves for any cyclic order (Remark B.1). On the fiber of the preimage of the exceptional divisor over a point of the center of the blow-up, a generator of the inertia group of the pair of branch divisors is the value at a root of unity of a family of rotations of the lines covering the exceptional line, hence homotopic to the identity on that fiber, and the Leray spectral sequence over the center shows that the inertia group acts trivially on the cohomology of that preimage, which excludes the primitive characters (Definition-Lemma B.2, Proposition B.5).

The period rank. The boundary points used below are admissible covers over a stable curve of genus 0 whose dual graph is a tree, where each leaf carries both branch points of one pair and the internal vertices carry no branch points. The nodes correspond to the edges of the tree, and their smoothing parameters q_I are those of Definition-Proposition 5.2. The caterpillar boundary points below are those of Construction-Theorems 4.4 (4) and 4.5 (4).

At a boundary point where A_X has totally toric reduction (Definition 4.3), the period map is the sum of a holomorphic function and of the logarithms of the parameters q_I times the logarithmic monodromy forms, which are computed by the Picard–Lefschetz formula [ADF+20, §3.2, pp. 174–175]. The rank is at least the dimension of the span of the logarithmic monodromy forms (Lemma 5.3), which in both families are explicit matrices $Q_I \in \mathrm{Herm}_4(K)$ (Definition-Proposition 5.2). Families of non-split abelian varieties of Weil type have no such boundary points [Bro26, Proposition 4], which is a second reason to work with split eightfolds.

For the Eisenstein family, the seventeen matrices Q_I at one caterpillar boundary point already span a 15-dimensional space (Proposition 5.4), so the rank is at least 15 (Corollary 5.5). For the Gaussian family, the fifteen matrices Q_I span only 13 dimensions, and Q_I belongs to the 13-dimensional span for any set I of leaves, so no trivalent tree does better (Proposition 5.6). Instead, we use the plumbing expansion [HN20, Theorem 3.3, Corollary 4.6] beyond the logarithmic terms. In the plumbing parameters s_1, s_9 of the nodes of the covering curve at the leaves 1 and 9, which are $q_{\{1\}}^2$ and $q_{\{9\}}^2$ up to units, the terms of degree at most 2 add no direction modulo the 13-dimensional span, except the coefficient of $s_1 s_9$, which adds two, so the rank is $13 + 2 = 15$ (Theorem 5.7). This is the only analytic input beyond the Picard–Lefschetz formula.

Reaching a given sixfold. Since A is given, possibly with $\mathrm{End}^0(A)$ larger than K, an argument for a general member of a family does not suffice. We have $\dim \mathbb{D}(h_0) = 16$ for the period domain of the eightfolds with the split form $h_0 = H_K^{\oplus 4}$ on K^8 (Definition-Proposition 2.12), but

we only know that each period image contains a hypersurface Y of an open subset of $\mathbb{D}(h_0)$ (Construction-Theorem 6.3).

With A fixed, let S_s be the abelian surfaces of Construction-Theorem 6.2, with Hermitian form $\operatorname{diag}(1, (\det h_A)^{-1})$ and period s. Then the periods of $A \times S_s$ form a disc, which need not intersect Y. However, the tangent vectors of the disc have rank one in the tangent space $T_P\mathbb{D}(h_0) \cong M_4(\mathbb{C})$ at any point P of the disc, and the orbit of a rank one matrix under the stabilizer $\mathrm{U}(4) \times \mathrm{U}(4)$ of P spans $M_4(\mathbb{C})$, so a real unitary translate of the disc intersects Y transversally (Lemma 6.4). Transversality survives small changes of the translation, and $\mathrm{U}(h_0)(\mathbb{Q})$ is dense in $\mathrm{U}(h_0)(\mathbb{R})$ [PR94, Theorem 7.7], so a rational translation also works, and after multiplication by an integer it induces a K-linear isogeny $A \times S_s \to A_X$ (Proposition 6.5). Thus the surface accounts for the sixteenth dimension. An explicit cycle would require an explicit choice of s and of the curve X.

Nonsimple sixfolds. Rescaling a compatible polarization by $q \in \mathbb{Q}_{>0}$ multiplies $\det h$ by $q^6 = \mathrm{Nm}_{K/\mathbb{Q}}(q^3)$, so the discriminant does not change. Rescaling only a K-stable factor of odd dimension d' multiplies $\det h$ by $q^{d'}$, and $q = -1/\det h$ changes the discriminant to $[-1]$ (Lemma 8.1). Thus Theorem 2.14 proves that $W_K(A)$ is algebraic for a nonsimple sixfold A which contains a K-stable abelian subvariety of odd dimension, or a K-stable abelian surface on which K has multiplicities $(2,0)$ or $(0,2)$, since such a surface contains a K-stable elliptic curve. The remaining nonsimple sixfolds of Weil type are treated with Theorem 2.16 and [MZ99, Theorem (0.1)(iv)] (Theorem 8.2).

The Hodge groups of products are computed with a Goursat-type criterion (Lemma 8.6). Modulo a four-dimensional space of Hodge classes on the products $B \times S$ of Theorem 1.4 (Proposition 8.14), the Hodge classes are spanned by products of pullbacks of Hodge classes on the factors and by Weil classes of products of factors on which an imaginary quadratic field acts, and known cycles reach both kinds. An action of an imaginary quadratic field on $B \times S$ would force $\mathrm{End}^0(B)$ to be biquadratic and S to be isogenous to $E_1 \times E_2$ with non-isomorphic fields $\mathrm{End}^0(E_1)$ and $\mathrm{End}^0(E_2)$, which cannot both receive the acting field (Lemma 8.15).

Structure of the paper. In Section 2, we recall preliminary material and fix notation. In particular, we introduce Weil planes, discriminants of Hermitian forms, and unitary period domains, and recall the theorem of Markman on Weil classes of sixfolds of discriminant $[-1]$. In Section 3, using the determinant cycles of Schoen, we construct algebraic Weil classes on the Jacobian factors of Galois covers of curves whose branch points come in pairs with mutually inverse local monodromies, and we show that such a Jacobian factor has a split Hermitian form when it is cut out by a self-adjoint idempotent. In Section 4, we construct two families of abelian eightfolds with split Hermitian forms and algebraic Weil classes, one for $\mathbb{Q}(\sqrt{-3})$ and one for $\mathbb{Q}(i)$. In Section 5, we show that the Hodge structures of both families vary in at least 15 independent directions.

In Section 6, we prove Theorem 1.3. In Section 7, we study the Hodge classes on simple abelian sixfolds. In Section 8, we study the Hodge classes on nonsimple abelian sixfolds and prove Theorem 1.4. In Section 9, we prove Theorem 1.2.

In Appendix A, we list the explicit monodromy forms of both families and the exact data of the rank computations in Section 5. In Appendix B, we show that for a cyclic cover of a smooth variety, of arbitrary order and with simple normal crossing branch divisor, blowing up the intersection of two branch divisors with mutually inverse local monodromies and normalizing the pullback of the cover induces, by pullback, an isomorphism on the part of the cohomology on which the cyclic group acts by primitive characters.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

This section fixes notation and collects facts on Hodge classes and Hodge groups of abelian varieties, the reduction to codimensions two and three, abelian varieties of Weil type and unitary period domains, and known results on the Hodge conjecture for abelian varieties.

2.1. Hodge classes and Hodge groups.

Definition 2.1. Let A be an abelian variety of dimension d and p an integer with $0 \le p \le d$. We denote by

$$\mathrm{Hdg}^p(A) := H^{2p}(A,\mathbb{Q}) \cap H^{p,p}(A)$$

the space of *Hodge classes* of codimension p, and by $\mathcal{D}^p(A) \subset \mathrm{Hdg}^p(A)$ the subspace spanned by products of p classes of divisors. The *Hodge algebra* of A is the $\mathbb{Q}$-subalgebra $\bigoplus_p \mathrm{Hdg}^p(A)$ of $H^*(A,\mathbb{Q})$, and the *divisor algebra* of A is the subalgebra $\bigoplus_p \mathcal{D}^p(A)$ of the Hodge algebra. A class in $H^{2p}(A,\mathbb{Q})$ is *algebraic* if it belongs to the image of the cycle class map

$$\mathrm{cl} : CH^p(A) \otimes \mathbb{Q} \to H^{2p}(A,\mathbb{Q}).$$

We say that $\mathrm{HC}^p(A)$ holds if any class in $\mathrm{Hdg}^p(A)$ is algebraic. We say that $\mathrm{HC}(A)$ holds if $\mathrm{HC}^p(A)$ holds for any $0 \le p \le d$.

We write $\mathrm{Hdg}^*(A) := \bigoplus_p \mathrm{Hdg}^p(A)$ for the Hodge algebra and $\mathcal{D}^*(A) := \bigoplus_p \mathcal{D}^p(A)$ for the divisor algebra.

Definition 2.2 (cf. [MZ99, (1.2)], [Moo99, (1.11)])**.** Let A be an abelian variety, $V := H^1(A,\mathbb{Q})$, and $\mathbb{U} := \{z \in \mathbb{C}^* \mid |z| = 1\}$. The *Hodge circle* of A is the homomorphism

$$\eta_A : \mathbb{U} \to \mathrm{GL}(V \otimes \mathbb{R})$$

acting by z on $H^{1,0}(A)$ and by $\bar{z}$ on $H^{0,1}(A)$. The *Hodge group* $\mathrm{Hg}(A)$ is the smallest algebraic subgroup of $\mathrm{GL}(V)$ defined over $\mathbb{Q}$ whose real points contain $\eta_A(\mathbb{U})$, and the *Mumford–Tate group* $\mathrm{MT}(A)$ is the algebraic subgroup $\mathbb{G}_m \cdot \mathrm{Hg}(A)$ of $\mathrm{GL}(V)$ generated by the homotheties and $\mathrm{Hg}(A)$. We say that A is of *CM type* if $\mathrm{End}^0(A)$ contains a commutative semisimple $\mathbb{Q}$-subalgebra of dimension $2\dim A$.

In [MZ99, (1.2)] the Hodge group is defined on $H_1(A,\mathbb{Q})$, which is the dual of $H^1(A,\mathbb{Q})$, and passing to the dual does not affect the statements below.

Lemma 2.3 (cf. [MZ99, (1.2), (1.4)], [Moo99, (1.11), (1.13), (2.3)])**.** *Let A and A' be abelian varieties and $V := H^1(A,\mathbb{Q})$.*

(1) $\mathrm{Hg}(A)$ *is connected and reductive, and the commutant of* $\mathrm{Hg}(A)$ *in* $\mathrm{End}_{\mathbb{Q}}(V)$ *is the image of* $\mathrm{End}^0(A)$.

(2) For any p, $\mathrm{Hdg}^p(A)$ *is the subspace of* $\mathrm{Hg}(A)$*-invariants in* $H^{2p}(A,\mathbb{Q}) = \wedge^{2p} V$.

(3) We have

$$\mathrm{Hg}(A \times A') \subset \mathrm{Hg}(A) \times \mathrm{Hg}(A'),$$

and both projections are surjective.

(4) A is of CM type if and only if $\mathrm{Hg}(A)$ *is a torus.*

Proof. (1) and (2) are [MZ99, (1.2), (1.4)]. (3) is [Moo99, (1.13)]. By [Moo99, (2.3), last paragraph], A is of CM type if and only if $\mathrm{MT}(A)$ is a torus. By [Moo99, (1.11)], $\mathrm{MT}(A)$ is the almost direct product of $\mathbb{G}_m$ and $\mathrm{Hg}(A)$, so $\mathrm{MT}(A)$ is a torus if and only if $\mathrm{Hg}(A)$ is a torus. This implies (4). □

Definition 2.4 (cf. [Moo99, (1.22)])**.** Let L be a CM field with maximal totally real subfield L^+. We denote by

$$\mathrm{U}_L := \ker\left(\mathrm{Nm}_{L/L^+} : \mathrm{Res}_{L/\mathbb{Q}}\,\mathbb{G}_m \to \mathrm{Res}_{L^+/\mathbb{Q}}\,\mathbb{G}_m\right)$$

the *norm-one torus* of L.

Let V be an L-vector space with $\dim_L V < +\infty$ and ψ a non-degenerate alternating $\mathbb{Q}$-bilinear form on V with $\psi(ax, y) = \psi(x, \bar{a}y)$ for any $a \in L$ and $x, y \in V$. We denote by $\mathrm{U}_L(V, \psi)$ the algebraic group over $\mathbb{Q}$ of L-linear automorphisms of V preserving ψ, by

$$\det_L : \mathrm{U}_L(V, \psi) \to \mathrm{Res}_{L/\mathbb{Q}} \mathbb{G}_m$$

the L-determinant, which takes values in U_L, and by $\mathrm{SU}_L(V, \psi)$ the kernel of $\det_L$.

Lemma 2.5. *Let A be a simple abelian sixfold such that $\mathrm{End}^0(A)$ is of Albert type IV. Then $\mathrm{End}^0(A)$ is a CM field of degree 2, 4, 6, or 12 over $\mathbb{Q}$.*

Proof. Let $D := \mathrm{End}^0(A)$. Since A is simple, D is a division algebra, so $H_1(A, \mathbb{Q})$ is a free D-module, and $[D : \mathbb{Q}]$ divides $\dim_{\mathbb{Q}} H_1(A, \mathbb{Q}) = 12$. By Albert's classification [Moo99, (1.19)], the center F of D is a CM field. Write $[F : \mathbb{Q}] = 2e_0$ and $[D : F] = d_0^2$, where e_0 and d_0 are positive integers. Then $2e_0 d_0^2$ divides 12 (cf. [Moo99, (2.4)]), so d_0^2 divides 6. Thus $d_0 = 1$, $D = F$ is a CM field, and $[D : \mathbb{Q}] = 2e_0 \in \{2, 4, 6, 12\}$. □

2.2. **Reduction to codimensions two and three.**

Lemma 2.6. *Let $f : A \to A'$ be an isogeny between abelian varieties. Then for any integer $0 \leq p \leq \dim A$, $\mathrm{HC}^p(A)$ holds if and only if $\mathrm{HC}^p(A')$ holds. If $\dim A = 6$, then $\mathrm{HC}(A)$ holds if and only if $\mathrm{HC}^2(A)$ and $\mathrm{HC}^3(A)$ hold.*

Proof. Since f is an isogeny, $f_* H_1(A, \mathbb{Z})$ is a sublattice of finite index in $H_1(A', \mathbb{Z})$, hence $f^* : H^1(A', \mathbb{Q}) \to H^1(A, \mathbb{Q})$ is an isomorphism. Since cup product induces an isomorphism of Hodge structures $\wedge^* H^1(B, \mathbb{Q}) \cong H^*(B, \mathbb{Q})$ for any abelian variety B [Moo99, (2.1)], $f^* : H^*(A', \mathbb{Q}) \to H^*(A, \mathbb{Q})$ is an isomorphism of Hodge structures. Thus $f^* \mathrm{Hdg}^p(A') = \mathrm{Hdg}^p(A)$ for any p. By the projection formula,

$$f_* f^* \alpha = (\deg f)\alpha$$

for any $\alpha \in H^*(A', \mathbb{Q})$. Moreover, cl commutes with f^* and f_* [Ful98, Corollary 19.2].

Assume that $\mathrm{HC}^p(A)$ holds and let $\alpha \in \mathrm{Hdg}^p(A')$. Then $f^*\alpha \in \mathrm{Hdg}^p(A)$, so $f^*\alpha = \mathrm{cl}(z)$ for some $z \in CH^p(A) \otimes \mathbb{Q}$, and

$$\alpha = \frac{1}{\deg f} f_* f^* \alpha = \mathrm{cl}\left(\frac{1}{\deg f} f_* z\right).$$

Conversely, assume that $\mathrm{HC}^p(A')$ holds and let $\beta \in \mathrm{Hdg}^p(A)$. Then $\beta = f^*\alpha$ for some $\alpha \in \mathrm{Hdg}^p(A')$, and $\alpha = \mathrm{cl}(z')$ for some $z' \in CH^p(A') \otimes \mathbb{Q}$, so $\beta = \mathrm{cl}(f^* z')$.

Now assume that $\dim A = 6$. If $\mathrm{HC}(A)$ holds, then $\mathrm{HC}^2(A)$ and $\mathrm{HC}^3(A)$ hold by Definition 2.1. Assume that $\mathrm{HC}^2(A)$ and $\mathrm{HC}^3(A)$ hold. The space $\mathrm{Hdg}^0(A)$ is spanned by $\mathrm{cl}(A)$, and $\mathrm{Hdg}^6(A)$ is spanned by the class of a point. $\mathrm{HC}^1(A)$ holds by the Lefschetz $(1,1)$ theorem (cf. [Moo99, (2.10)]). Let H be an ample divisor on A and $\Theta := \mathrm{cl}(H)$. For $q \in \{1, 2\}$, the hard Lefschetz theorem (cf. [Del82, Example 2.1(c)]) implies that

$$\Theta^{6-2q} \cup - : H^{2q}(A, \mathbb{Q}) \to H^{12-2q}(A, \mathbb{Q})$$

is an isomorphism. Since $\Theta^{6-2q} \cup -$ maps $H^{a,b}(A)$ into $H^{a+6-2q, b+6-2q}(A)$ for any a, b, it maps $\mathrm{Hdg}^q(A)$ onto $\mathrm{Hdg}^{6-q}(A)$. Thus any $\alpha \in \mathrm{Hdg}^{6-q}(A)$ is of the form $\Theta^{6-2q} \cup \mathrm{cl}(z)$ for some $z \in CH^q(A) \otimes \mathbb{Q}$, and

$$\alpha = \mathrm{cl}(H^{6-2q} \cdot z)$$

by [Ful98, Corollary 19.2]. Hence $\mathrm{HC}^5(A)$ and $\mathrm{HC}^4(A)$ hold, and $\mathrm{HC}(A)$ holds. □

2.3. **Weil type and Hermitian forms.** In this subsection, we introduce the Weil plane of an abelian variety with an action of an imaginary quadratic field, and the discriminant of the Hermitian form of a compatible polarization.

Definition 2.7. Let A be an abelian variety of dimension d, L a number field, and $\iota : L \hookrightarrow \mathrm{End}^0(A)$ a unital embedding. For a complex embedding $\tau : L \hookrightarrow \mathbb{C}$, we denote by $H^1(A,\mathbb{C})_\tau$ the subspace of $H^1(A,\mathbb{C})$ on which $\iota(a)^*$ acts by $\tau(a)$ for any $a \in L$, and we call $\dim H^{1,0}(A) \cap H^1(A,\mathbb{C})_\tau$ the *multiplicity* of L on $H^{1,0}(A)$ at τ. If $L = K$ is an imaginary quadratic field with complex embeddings $\sigma, \bar{\sigma}$, the *multiplicities* of K on $H^{1,0}(A)$ are the pair (r,s), where r and s are the multiplicities at σ and at $\bar{\sigma}$, respectively. We say that (A,ι) is of *Weil type* if $L = K$ is an imaginary quadratic field, $d = 2n$ is even, and the multiplicities of K on $H^{1,0}(A)$ are (n,n). In this case, the *Weil plane* $W_K(A) \subset H^{2n}(A,\mathbb{Q})$ is the two-dimensional subspace such that

$$W_K(A) \otimes \mathbb{C} = \wedge^{2n} H^1(A,\mathbb{C})_\sigma \oplus \wedge^{2n} H^1(A,\mathbb{C})_{\bar{\sigma}}.$$

Definition 2.8. Let A be an abelian variety, K an imaginary quadratic field, and $\iota : K \hookrightarrow \mathrm{End}^0(A)$ a unital embedding. An abelian subvariety $A' \subset A$ is *K-stable* if $H_1(A',\mathbb{Q}) \subset H_1(A,\mathbb{Q})$ is stable under $\iota(K)$. In this case, we denote by $\iota|_{A'} : K \hookrightarrow \mathrm{End}^0(A')$ the restricted embedding, and for K-stable $A', A'' \subset A$ we let K act on $A' \times A''$ through $\iota|_{A'} \times \iota|_{A''}$. Let Y be an abelian variety with a unital embedding $\iota_Y : K \hookrightarrow \mathrm{End}^0(Y)$. A homomorphism $f : Y \to A$ is *K-linear* if $f \circ \iota_Y(k) = \iota(k) \circ f$ in $\mathrm{Hom}(Y,A) \otimes_{\mathbb{Z}} \mathbb{Q}$ for any $k \in K$.

Definition 2.9 (cf. [Mar25, Section 1.1])**.** Let A be an abelian variety of dimension d, K an imaginary quadratic field, and $\iota : K \hookrightarrow \mathrm{End}^0(A)$ a unital embedding, through which K acts on $H_1(A,\mathbb{Q})$. A *compatible polarization* of (A,ι) is a class $\Theta \in NS(A) \otimes \mathbb{Q}$ of an ample $\mathbb{Q}$-divisor whose Riemann form ψ on $H_1(A,\mathbb{Q})$ satisfies $\psi(kx,y) = \psi(x,\bar{k}y)$ for any $k \in K$ and $x, y \in H_1(A,\mathbb{Q})$. When ι is the inclusion of a subfield $K \subset \mathrm{End}^0(A)$, we write (A,K) for (A,ι). Fix $\delta \in K$ with $\bar{\delta} = -\delta$ and $\mathrm{Im}\,\sigma(\delta) > 0$, where σ is the complex embedding of K in Definition 2.7. The *Hermitian form* of a compatible polarization Θ with Riemann form ψ is the unique K-Hermitian form h on $H_1(A,\mathbb{Q})$, conjugate-linear in the first variable, such that $\psi(x,y) = \mathrm{Tr}_{K/\mathbb{Q}}(\delta h(x,y))$ for any $x, y \in H_1(A,\mathbb{Q})$ (it exists by [Del82, Lemma 4.6]). The *discriminant* $\mathrm{disc}(h)$ is the class of $\det h$ in $\mathbb{Q}^*/\mathrm{Nm}_{K/\mathbb{Q}}(K^*)$, where $\det h$ is computed in a K-basis of $H_1(A,\mathbb{Q})$. We say that h is *split* if $H_1(A,\mathbb{Q})$ contains a K-subspace of K-dimension $d/2$ on which h vanishes.

We say that (A,ι) is *split* if it admits a compatible polarization whose Hermitian form is split. The discriminant is the invariant of a polarization that enters Markman's theorem (Theorem 2.14), and Lemmas 8.1 and 7.3 change it in two cases.

Lemma 2.10. *Let A be an abelian variety of dimension $2n$, K an imaginary quadratic field, and $\iota : K \hookrightarrow \mathrm{End}^0(A)$ a unital embedding.*

(1) There exists a compatible polarization of (A,ι).
(2) If the multiplicities of K on $H^{1,0}(A)$ are (r,s), then the Hermitian form of any compatible polarization has signature (r,s) or (s,r). If (A,ι) is of Weil type, then $W_K(A) \subset \mathrm{Hdg}^n(A)$.
(3) Let Θ be a compatible polarization and $A' \subset A$ a K-stable abelian subvariety. Then there exists a K-stable abelian subvariety $A'' \subset A$ such that the addition map $f : A' \times A'' \to A$ is a K-linear isogeny and

$$f^*\Theta = \mathrm{pr}_1^*(\Theta|_{A'}) + \mathrm{pr}_2^*(\Theta|_{A''}).$$

(4) Let A' be an abelian variety with a unital embedding $\iota' : K \hookrightarrow \mathrm{End}^0(A')$, and $f : A' \to A$ a K-linear isogeny. Then f^* maps $W_K(A)$ isomorphically onto $W_K(A')$ when (A,ι) is of Weil type, and a class $w \in H^*(A,\mathbb{Q})$ is algebraic if and only if f^*w is algebraic. If Θ is a compatible polarization of (A,ι), then $f^*\Theta$ is a compatible polarization of (A',ι'), and f_* is an isometry between the Hermitian forms of $f^*\Theta$ and Θ.

(5) *If the Hermitian form h of a compatible polarization has signature (n,n), then h is split if and only if* $\mathrm{disc}(h) = [(-1)^n]$.
(6) $\mathrm{disc}(h)$ *does not change if δ is replaced in Definition 2.9 by any $\delta' \in K$ with $\bar{\delta}' = -\delta' \neq 0$, and for any compatible polarization Θ and any $q \in \mathbb{Q}_{>0}$, the Hermitian form of $q\Theta$ has discriminant equal to the discriminant of the Hermitian form of Θ.*

Proof. (1) Write $K = \mathbb{Q}(j)$, where $j^2 = -c$ for some $c \in \mathbb{Q}_{>0}$, and write j for $\iota(j)$. Let $\Theta_0 \in NS(A) \otimes \mathbb{Q}$ be an ample class with Riemann form ψ_0, and let $\Theta := \Theta_0 + c^{-1} j^* \Theta_0$. For a positive integer m with $mj \in \mathrm{End}(A)$, the endomorphism mj is an isogeny, so $j^*\Theta_0 = m^{-2}(mj)^*\Theta_0$ is ample, and hence Θ is ample. The Riemann form of Θ is

$$\psi(x,y) = \psi_0(x,y) + c^{-1}\psi_0(jx, jy),$$

and $j^2 = -c$ implies that $\psi(jx, jy) = c\psi(x,y)$. Replacing y with jy, we get $\psi(jx, y) = -\psi(x, jy)$. Thus $\psi(kx, y) = \psi(x, \bar{k}y)$ for any $k = a + bj$ with $a, b \in \mathbb{Q}$.

(2) The Hermitian form h exists and is unique by [Del82, Lemma 4.6]. We identify $H_1(A, \mathbb{R})$ with $H_1(A, \mathbb{Q}) \otimes_{K,\sigma} \mathbb{C}$ through σ, and we extend h to a $\mathbb{C}$-Hermitian form on $H_1(A, \mathbb{Q}) \otimes_{K,\sigma} \mathbb{C}$, still denoted by h, so that $\psi(x,y) = 2\,\mathrm{Re}(\sigma(\delta)h(x,y))$. The complex structure J of $H_1(A, \mathbb{R})$ commutes with K, so J is $\mathbb{C}$-linear on $H_1(A, \mathbb{Q}) \otimes_{K,\sigma} \mathbb{C}$. Under the pairing between H_1 and H^1, the eigenspaces of J for i and $-i$ are dual to $H^{1,0}(A) \cap H^1(A, \mathbb{C})_\sigma$ and $H^{0,1}(A) \cap H^1(A, \mathbb{C})_\sigma$, respectively. Since complex conjugation maps $H^{1,0}(A) \cap H^1(A, \mathbb{C})_{\bar{\sigma}}$ onto $H^{0,1}(A) \cap H^1(A, \mathbb{C})_\sigma$, the eigenspaces for i and $-i$ are of dimensions r and s. For x in the eigenspace for $\pm i$,

$$\psi(x, Jx) = 2\,\mathrm{Re}\left(\pm i\sigma(\delta)h(x,x)\right) = \mp 2\,\mathrm{Im}\,\sigma(\delta)h(x,x).$$

Since $\psi(x, Jx) > 0$ for any $x \neq 0$ (cf. [Moo99, (1.15), (2.2)]), h is definite on both eigenspaces, with opposite signs. Hence h has signature (r,s) or (s,r). If (A, ι) is of Weil type, then $W_K(A)$ consists of Hodge classes by [Del82, Proposition 4.4], that is, $W_K(A) \subset \mathrm{Hdg}^n(A)$.

(3) Let ψ be the Riemann form of Θ, $U := H_1(A', \mathbb{Q})$, and $U^\perp \subset H_1(A, \mathbb{Q})$ the ψ-orthogonal complement of U. The restriction of ψ to U is the Riemann form of the ample class $\Theta|_{A'}$, hence non-degenerate. Thus $H_1(A, \mathbb{Q}) = U \oplus U^\perp$, and $U^\perp$ is a sub-Hodge structure (cf. [Moo99, (1.15)]). By [Moo99, (2.2)], there exists an abelian subvariety $A'' \subset A$ with $H_1(A'', \mathbb{Q}) = U^\perp$. For $v \in U^\perp$, $x \in U$, and $k \in K$, we have $\psi(kv, x) = \psi(v, \bar{k}x) = 0$, so A'' is K-stable. The addition map f induces the K-linear isomorphism $U \oplus U^\perp \to H_1(A, \mathbb{Q})$, so f is a K-linear isogeny. Since U and $U^\perp$ are ψ-orthogonal, the Riemann form of $f^*\Theta$ is

$$\psi(x' + x'', y' + y'') = \psi(x', y') + \psi(x'', y'')$$

for $x', y' \in U$ and $x'', y'' \in U^\perp$, which is the Riemann form of

$$\mathrm{pr}_1^*(\Theta|_{A'}) + \mathrm{pr}_2^*(\Theta|_{A''}).$$

Hence

$$f^*\Theta = \mathrm{pr}_1^*(\Theta|_{A'}) + \mathrm{pr}_2^*(\Theta|_{A''}).$$

(4) The map $f^* : H^1(A, \mathbb{Q}) \to H^1(A', \mathbb{Q})$ is a K-linear isomorphism of Hodge structures. Thus f^* maps $H^1(A, \mathbb{C})_\tau$ onto $H^1(A', \mathbb{C})_\tau$ for $\tau \in \{\sigma, \bar{\sigma}\}$, preserves the multiplicities, and maps $\wedge^{2n} H^1(A, \mathbb{C})_\tau$ onto $\wedge^{2n} H^1(A', \mathbb{C})_\tau$. Hence f^* maps $W_K(A)$ isomorphically onto $W_K(A')$. The pullback of an algebraic class is algebraic. Conversely, if f^*w is algebraic, then $w = (\deg f)^{-1} f_* f^* w$ is algebraic. The class $f^*\Theta$ is ample since f is finite, and the Riemann form $\psi'(x,y) = \psi(f_*x, f_*y)$ of $f^*\Theta$ satisfies

$$\psi'(kx, y) = \psi(kf_*x, f_*y) = \psi(f_*x, \bar{k}f_*y) = \psi'(x, \bar{k}y)$$

since f_* is K-linear. The form $(x,y) \to h(f_*x, f_*y)$ is K-Hermitian with trace form ψ', so it is the Hermitian form of $f^*\Theta$ by uniqueness. Since $f_* : H_1(A', \mathbb{Q}) \to H_1(A, \mathbb{Q})$ is a K-linear isomorphism, f_* is an isometry.

(5) This is [Bro26, Proposition 4]. The discriminant in [Bro26] is also the class in $\mathbb{Q}^*/\,\mathrm{Nm}_{K/\mathbb{Q}}(K^*)$ of the determinant of the Hermitian matrix in a K-basis.

(6) Any $\delta' \in K$ with $\bar{\delta}' = -\delta' \neq 0$ is of the form $\delta' = t\delta$ for some $t \in \mathbb{Q}^*$. Replacing δ with $t\delta$ replaces h with $t^{-1}h$ and multiplies $\det h$ by $t^{-2n} = \mathrm{Nm}_{K/\mathbb{Q}}(t^{-n})$. Replacing Θ with $q\Theta$ replaces ψ with $q\psi$ and h with qh, and multiplies $\det h$ by $q^{2n} = \mathrm{Nm}_{K/\mathbb{Q}}(q^n)$. In both cases, $\mathrm{disc}(h)$ does not change. □

Lemma 2.11. *Let A be an abelian variety of even dimension, K an imaginary quadratic field, and $\iota : K \hookrightarrow \mathrm{End}^0(A)$ a unital embedding. Assume that $NS(A) \otimes \mathbb{Q}$ is spanned by one class. Then $\mathrm{disc}(h) = \mathrm{disc}(h')$ for the Hermitian forms h and h' of any two compatible polarizations of (A, ι). In particular, $NS(A) \otimes \mathbb{Q}$ is spanned by one class, and hence the conclusion holds, when A is simple and $\mathrm{End}^0(A) = K$.*

Proof. Let Θ_1 and Θ_2 be compatible polarizations of (A, ι). Since $NS(A) \otimes \mathbb{Q}$ is spanned by one class, $\Theta_2 = q\Theta_1$ for some $q \in \mathbb{Q}$, and $q > 0$ since Θ_1 and Θ_2 are ample. Hence $\mathrm{disc}(h) = \mathrm{disc}(h')$ by Lemma 2.10 (6).

Assume that A is simple and $\mathrm{End}^0(A) = K$. Let

$$\lambda_1, \lambda_2 \in \mathrm{Hom}(A, A^\vee) \otimes_{\mathbb{Z}} \mathbb{Q}$$

be the polarizations induced by two ample classes in $NS(A) \otimes \mathbb{Q}$. Then $\lambda_1^{-1}\lambda_2 \in \mathrm{End}^0(A) = K$ is symmetric for the Rosati involution of λ_1. The Rosati involution is a positive involution of K, hence complex conjugation, so $\lambda_1^{-1}\lambda_2 \in \mathbb{Q}$. Thus any two ample classes in $NS(A) \otimes \mathbb{Q}$ are proportional. Since the ample cone is open in $NS(A) \otimes \mathbb{R}$, any element of $NS(A) \otimes \mathbb{Q}$ is a difference of two ample classes in $NS(A) \otimes \mathbb{Q}$. Hence $NS(A) \otimes \mathbb{Q}$ is spanned by one class. □

In Definition-Proposition 2.12, K is an imaginary quadratic field with the complex embedding σ of Definition 2.7, δ is the element fixed in Definition 2.9, and the case of K^2 is used for the abelian surfaces of Section 6 and the case of K^8 for the abelian eightfolds of Sections 5 and 6.

Definition-Proposition 2.12. Let (V, h) be either K^2 with a K-Hermitian form h of signature $(1, 1)$, or K^8 with a K-Hermitian form h isometric to $h_0 := H_K^{\oplus 4}$, where H_K is the Hermitian matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Write $\dim_K V = 2n$, so that h has signature (n, n), and let $\mathrm{U}(h)$ be the unitary group of h, an algebraic group over $\mathbb{Q}$ with $\mathrm{U}(h)(\mathbb{R}) \cong \mathrm{U}(n, n)$. Identify $V \otimes_{\mathbb{Q}} \mathbb{R}$ with $V \otimes_{K,\sigma} \mathbb{C}$ through the isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{C}$ given by σ. Let $\mathbb{D}(h)$ be the set of n-dimensional complex subspaces $P \subset V \otimes_{K,\sigma} \mathbb{C}$ on which h is positive definite, and let $P^\perp$ be the h-orthogonal complement of P. Then $\mathbb{D}(h)$ is a connected complex manifold of dimension n^2, the group $\mathrm{U}(h)(\mathbb{R})$ acts transitively on $\mathbb{D}(h)$, and the tangent space of $\mathbb{D}(h)$ at P is $\mathrm{Hom}_{\mathbb{C}}(P, P^\perp)$.

Moreover, let $\Lambda \subset V$ be a lattice stable under the ring of integers of K such that $\psi := \mathrm{Tr}_{K/\mathbb{Q}}(\delta h)$ is integral on Λ, and let $\epsilon := -1$ if $\mathrm{Im}\,\sigma(\delta) > 0$ and $\epsilon := 1$ otherwise. For $P \in \mathbb{D}(h)$, let J_P be the complex structure on $V \otimes_{\mathbb{Q}} \mathbb{R}$ acting by ϵi on P and by $-\epsilon i$ on $P^\perp$. Then $(V \otimes_{\mathbb{Q}} \mathbb{R}, J_P)/\Lambda$ is an abelian variety with a polarization of Riemann form ψ, normalized by $\psi(v, J_P v) > 0$ for $v \neq 0$, and a K-action of multiplicities (n, n). Conversely, let A be an abelian variety with a unital embedding $K \hookrightarrow \mathrm{End}^0(A)$, a compatible polarization, and a K-linear isometry ϕ from the Hermitian form of the polarization to (V, h). Then exactly one of the eigenspaces of the complex structure of $H_1(A, \mathbb{R})$ for the eigenvalues i and $-i$, transported by ϕ to $V \otimes_{K,\sigma} \mathbb{C}$, belongs to $\mathbb{D}(h)$. We call the eigenspace belonging to $\mathbb{D}(h)$ the *period* of A with respect to ϕ.

Proof. We extend h to a $\mathbb{C}$-Hermitian form on $V \otimes_{K,\sigma} \mathbb{C}$, still denoted by h, so that $\psi(x, y) = 2\,\mathrm{Re}(\sigma(\delta)h(x, y))$. Then $\mathrm{U}(h)(\mathbb{R})$ is the group of $\mathbb{C}$-linear automorphisms of $V \otimes_{K,\sigma} \mathbb{C}$ preserving h.

Let $P_0 \in \mathbb{D}(h)$. Since h has signature (n, n), h is negative definite on $P_0^\perp$ and $V \otimes_{K,\sigma} \mathbb{C} = P_0 \oplus P_0^\perp$. Choose bases of P_0 and $P_0^\perp$ in which h is the identity matrix and minus the identity matrix, respectively. Any $P \in \mathbb{D}(h)$ satisfies $P \cap P_0^\perp = 0$, so P is the graph of a linear map $Z : P_0 \to P_0^\perp$, and the graph is in $\mathbb{D}(h)$ if and only if

$$h(x + Zx, x + Zx) = x^*(I - Z^*Z)x > 0 \quad \text{for any } x \in P_0 \setminus \{0\},$$

that is, $I - Z^*Z$ is positive definite. The graphs of linear maps $P_0 \to P_0^\perp$ form the standard affine chart of the Grassmannian of n-dimensional subspaces at P_0, with tangent space $\mathrm{Hom}_{\mathbb{C}}(P_0, P_0^\perp)$ at P_0. In the chart, $\mathbb{D}(h)$ is the set of $n \times n$ complex matrices Z with $I - Z^*Z$ positive definite, which is a convex open subset of $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$. Hence $\mathbb{D}(h)$ is a connected complex manifold of dimension n^2, and, since $P_0 \in \mathbb{D}(h)$ is arbitrary, the tangent space at any $P \in \mathbb{D}(h)$ is $\mathrm{Hom}_{\mathbb{C}}(P, P^\perp)$. For $P, P' \in \mathbb{D}(h)$, choose bases of P, $P^\perp$, P', and $P'^\perp$ in which h is the identity matrix on P and P' and minus the identity matrix on $P^\perp$ and $P'^\perp$. The $\mathbb{C}$-linear map sending the bases of P and $P^\perp$ to the bases of P' and $P'^\perp$ preserves h and maps P onto P'. Thus $\mathrm{U}(h)(\mathbb{R})$ acts transitively on $\mathbb{D}(h)$.

Let $P \in \mathbb{D}(h)$. The complex structure J_P is $\mathbb{C}$-linear on $V \otimes_{K,\sigma} \mathbb{C}$, so J_P commutes with K. Since P and $P^\perp$ are h-orthogonal, $h(J_P x, J_P y) = h(x, y)$, and hence $\psi(J_P x, J_P y) = \psi(x, y)$. For $v = v_+ + v_-$ with $v_+ \in P$ and $v_- \in P^\perp$,

$$\psi(v, J_P v) = 2\,\mathrm{Re}\,\big(\epsilon i \sigma(\delta)(h(v_+, v_+) - h(v_-, v_-))\big) = -2\epsilon\,\mathrm{Im}\,\sigma(\delta)\big(h(v_+, v_+) - h(v_-, v_-)\big),$$

which is positive for $v \neq 0$ since $-\epsilon\,\mathrm{Im}\,\sigma(\delta) > 0$. By the Riemann conditions (cf. [Moo99, (1.15), (2.2)]), $A_P := (V \otimes_{\mathbb{Q}} \mathbb{R}, J_P)/\Lambda$ is an abelian variety with a polarization of Riemann form ψ. The ring of integers of K preserves Λ and commutes with J_P, so K acts on A_P. The eigenspaces of J_P for i and $-i$ in $V \otimes_{K,\sigma} \mathbb{C}$ are P and $P^\perp$ in some order, both of dimension n, and they are dual to $H^{1,0}(A_P) \cap H^1(A_P, \mathbb{C})_\sigma$ and $H^{0,1}(A_P) \cap H^1(A_P, \mathbb{C})_\sigma$, respectively. Since complex conjugation maps $H^{1,0}(A_P) \cap H^1(A_P, \mathbb{C})_{\bar\sigma}$ onto $H^{0,1}(A_P) \cap H^1(A_P, \mathbb{C})_\sigma$, the multiplicities of K on $H^{1,0}(A_P)$ are (n, n).

Conversely, let J be the complex structure of $H_1(A, \mathbb{R})$ transported by ϕ to $V \otimes_{K,\sigma} \mathbb{C}$, and let E_+ and E_- be the eigenspaces of J for i and $-i$. Since ϕ is a K-linear isometry, ϕ transports the Riemann form of the polarization to ψ, and J is $\mathbb{C}$-linear. For $x \in E_\pm$,

$$\psi(x, Jx) = 2\,\mathrm{Re}\,\big(\pm i\sigma(\delta)h(x, x)\big) = \mp 2\,\mathrm{Im}\,\sigma(\delta)h(x, x),$$

and $\psi(x, Jx) > 0$ for $x \neq 0$. Thus h is definite on E_+ and on E_-, with opposite signs. Since h has signature (n, n) and $E_+ \oplus E_- = V \otimes_{K,\sigma} \mathbb{C}$, we get $\dim E_+ = \dim E_- = n$. Hence exactly one of E_+ and E_- belongs to $\mathbb{D}(h)$, namely E_+ if $\epsilon = 1$ and E_- if $\epsilon = -1$. In particular, for the fixed δ, the period P of A is the eigenspace on which h is positive definite, and $J = J_P$. □

Lemma 2.13. *Let (V, h), A, and ϕ be as in Definition-Proposition 2.12. Then $\epsilon = -1$, and the period of A with respect to ϕ is the eigenspace E_- of the complex structure of $H_1(A, \mathbb{R})$ for the eigenvalue $-i$, transported by ϕ to $V \otimes_{K,\sigma} \mathbb{C}$. If $V \otimes_{K,\sigma} \mathbb{C}$ is identified through ϕ with the subspace of $H_1(A, \mathbb{C})$ on which K acts through σ, then E_- is the annihilator in $V \otimes_{K,\sigma} \mathbb{C}$ of $H^{1,0}(A) \cap H^1(A, \mathbb{C})_\sigma$ under the pairing between $H_1(A, \mathbb{C})$ and $H^1(A, \mathbb{C})$. For a family of abelian varieties over a complex manifold with a K-action, compatible polarizations, and locally constant K-linear isometries ϕ to (V, h), the period varies holomorphically.*

Proof. Since $\mathrm{Im}\,\sigma(\delta) > 0$ (Definition 2.9), we have $\epsilon = -1$. Let J be the complex structure of $H_1(A, \mathbb{R})$, transported by ϕ to $V \otimes_{K,\sigma} \mathbb{C}$, and let E_+ be the eigenspace of J for i. Since J commutes with K, J is $\mathbb{C}$-linear, and for $x \in E_\pm$,

$$\psi(x, Jx) = 2\,\mathrm{Re}\,\big(\pm i\sigma(\delta)h(x, x)\big) = \mp 2\,\mathrm{Im}\,\sigma(\delta)h(x, x),$$

where $\psi = \mathrm{Tr}_{K/\mathbb{Q}}(\delta h)$ and h is extended to a $\mathbb{C}$-Hermitian form on $V \otimes_{K,\sigma} \mathbb{C}$. Since $\psi(x, Jx) > 0$ for $x \neq 0$ (cf. [Moo99, (1.15), (2.2)]), h is positive definite on E_- and negative definite on E_+. As h has signature (n, n), we get $\dim E_- = \dim E_+ = n$, and E_- is the eigenspace which belongs to $\mathbb{D}(h)$, that is, the period of A with respect to ϕ.

Let $H_\sigma \subset H_1(A, \mathbb{C})$ be the subspace on which K acts through σ, identified with $V \otimes_{K,\sigma} \mathbb{C}$ through ϕ. Forms of type $(1, 0)$ vanish on the eigenspace of the complex structure for $-i$ in $H_1(A, \mathbb{C})$, so $H^{1,0}(A) \cap H^1(A, \mathbb{C})_\sigma$ vanishes on E_-. By Lemma 2.10 (2), the multiplicities of K on $H^{1,0}(A)$ are (n, n), that is, $\dim(H^{1,0}(A) \cap H^1(A, \mathbb{C})_\sigma) = n$. Since $\dim H_\sigma = 2n$ and H_σ pairs

perfectly with $H^1(A,\mathbb{C})_\sigma$, the annihilator of $H^{1,0}(A)\cap H^1(A,\mathbb{C})_\sigma$ in H_σ is n-dimensional and contains E_-. Hence E_- is the annihilator.

Since K preserves $H^{1,0}(A)$, the space $H^{1,0}(A)\cap H^1(A,\mathbb{C})_\sigma$ is the image of $H^{1,0}(A)$ under the projection of $H^1(A,\mathbb{C})$ onto $H^1(A,\mathbb{C})_\sigma$ along $H^1(A,\mathbb{C})_{\bar\sigma}$. In a family as in the statement, $H^{1,0}$ varies holomorphically, and the projection, the pairing, and the isometries ϕ are locally constant. Hence the image of $H^{1,0}$, which is n-dimensional at any point, varies holomorphically, and so does the annihilator of the image. □

2.4. **Markman's theorem and abelian varieties of small dimension.** In this subsection, we recall Markman's theorem on the Weil planes of abelian sixfolds of Weil type, and the Hodge conjecture for abelian varieties of dimension at most five.

Theorem 2.14 ([Mar25, Theorem 1.5.1])**.** *Let A be an abelian sixfold, K an imaginary quadratic field, and $\iota : K \hookrightarrow \mathrm{End}^0(A)$ a unital embedding such that (A,ι) is of Weil type. Assume that (A,ι) admits a compatible polarization whose Hermitian form h satisfies $\mathrm{disc}(h) = [-1]$. Then any class in $W_K(A)$ is algebraic.*

Definitions 2.7 and 2.9 are reformulations of the definitions of [Mar25, Section 1.1]. In [Mar25, Section 1.1], a polarization is compatible with ι if $\iota(k)$ multiplies the polarization class by $\mathrm{Nm}_{K/\mathbb{Q}}(k)$ for any non-zero element k of K. In terms of the Riemann form, this is condition (2) of Lemma 2.15 below, which is equivalent to the condition in Definition 2.9 by that lemma. The discriminant in [Mar25, Section 1.1] is the class of the determinant of the Hermitian form, as in Definition 2.9, and it does not depend on the normalizations of δ and of the polarization (Lemma 2.10 (6)).

Lemma 2.15. *Let K be an imaginary quadratic field, V a K-vector space of finite dimension, and ψ a $\mathbb{Q}$-bilinear form on V. Then the following are equivalent.*

(1) $\psi(kx,y) = \psi(x,\bar k y)$ for any $k\in K$ and $x,y\in V$.
(2) $\psi(kx,ky) = \mathrm{Nm}_{K/\mathbb{Q}}(k)\psi(x,y)$ for any $k\in K^*$ and $x,y\in V$.

Proof. If (1) holds, then for any $k\in K^*$ and $x,y\in V$,

$$\psi(kx,ky) = \psi(x,\bar k k y) = \mathrm{Nm}_{K/\mathbb{Q}}(k)\psi(x,y).$$

Assume that (2) holds, and let $k\in K$ and $x,y\in V$. For all but at most one $t\in\mathbb{Q}$, we have $1+tk\neq 0$, and applying (2) to $1+tk$, we get

$$\psi(x,y) + t\big(\psi(kx,y)+\psi(x,ky)\big) + t^2\psi(kx,ky) = \big(1 + t\,\mathrm{Tr}_{K/\mathbb{Q}}(k) + t^2\,\mathrm{Nm}_{K/\mathbb{Q}}(k)\big)\psi(x,y).$$

Both sides are polynomials in t of degree at most two whose values coincide for infinitely many $t\in\mathbb{Q}$, so their coefficients of t coincide, that is,

$$\psi(kx,y)+\psi(x,ky) = \mathrm{Tr}_{K/\mathbb{Q}}(k)\psi(x,y).$$

Since $\bar k = \mathrm{Tr}_{K/\mathbb{Q}}(k) - k$, we get

$$\psi(x,\bar k y) = \mathrm{Tr}_{K/\mathbb{Q}}(k)\psi(x,y) - \psi(x,ky) = \psi(kx,y).$$

□

Theorem 2.16 (cf. [Mar26, Corollary 1.3])**.** *Let A be an abelian variety with $\dim A\le 5$. Then $\mathrm{HC}(A)$ holds.*

For abelian varieties of dimension at most three, Theorem 2.16 is classical (cf. [Mar26, Section 1.1]). For abelian fourfolds, Theorem 2.16 is [Mar25, Corollary 1.6.1]. A simple abelian fivefold has prime dimension, so the Hodge algebra of a simple abelian fivefold coincides with the divisor algebra [Tan82] (cf. [MZ99, Theorem (2.7)]). For a non-simple abelian fivefold A, [MZ99, Theorem (0.2)] reduces $\mathrm{HC}(A)$ to the algebraicity of the classes in the Weil planes of abelian fourfolds of Weil type, which follows from the case of abelian fourfolds.

3. Weil classes on Jacobian factors of covers with paired branching

In this section, we prove that the Weil plane of the Jacobian factor A_X of Set-up 3.5 is algebraic when the action of G on X has paired branching (Theorem 3.11). The determinant cycles of Schoen were extended to generalized Prym varieties of abelian covers in [PZ25, Theorem 1.2]. The Weil classes on abelian eightfolds used in Section 6 are algebraic by Theorem 3.11.

3.1. Determinant cycles of Schoen.

In this subsection, we recall the theorem of Schoen on algebraic cycles on self-products of a curve with a cyclic group of automorphisms, and we define paired branching for actions of arbitrary finite groups on curves.

Theorem 3.1 ([Sch88, Theorem 2.0]). *Let C be a smooth projective connected curve, $\ell \geq 2$ an integer, and $\mathbb{Z}/\ell \hookrightarrow \mathrm{Aut}(C)$ an injective homomorphism. For a branch point b of the quotient map $C \to C/(\mathbb{Z}/\ell)$ and a point $x \in C$ over b, let $a_b \in \mathbb{Z}/\ell$ be the element of the stabilizer of x that acts on the tangent space T_xC by $\exp(2\pi i/s)$, where s is the order of the stabilizer. The element a_b does not depend on the choice of x. Let $V \subset H^1(C,\mathbb{Q})$ be the largest $\mathbb{Z}/\ell$-submodule such that $V \otimes \mathbb{C}$ is a sum of eigenspaces for primitive characters of $\mathbb{Z}/\ell$, and for a primitive character χ, let $V_\chi \subset V \otimes \mathbb{C}$ be the χ-eigenspace. Assume that $\dim_{\mathbb{Q}(\mu_\ell)} V = 2k$ is even and that the tuple $(a_b)_b$, with b running over the branch points, can be ordered as $(a_1, -a_1, \ldots, a_r, -a_r)$. Then the subspace of $H^{2k}(C^{2k},\mathbb{Q})$ whose complexification is*

$$\bigoplus_\chi \wedge^{2k} V_\chi \subset H^{2k}(C^{2k},\mathbb{C})$$

is spanned by classes of algebraic cycles, where χ runs over the primitive characters of $\mathbb{Z}/\ell$ and $\wedge^{2k} V_\chi$ is embedded in $H^{2k}(C^{2k},\mathbb{C})$ by the Künneth formula.

In Theorem 3.1, μ_ℓ is the group of ℓ-th roots of unity. Since only primitive characters occur in $V \otimes \mathbb{C}$, the action of $\mathbb{Q}[\mathbb{Z}/\ell]$ on V factors through $\mathbb{Q}(\mu_\ell)$, so V is a $\mathbb{Q}(\mu_\ell)$-vector space.

Remark 3.2. In Theorem 3.1, the tuple $(a_b)_b$ differs from the $2r$-tuple of [Sch88, §1, p. 11] only by the diagonal action of $(\mathbb{Z}/\ell)^*$, so the condition on $(a_b)_b$ in Theorem 3.1 is the condition that the $2r$-tuple of Schoen is simple in the sense of [Sch88, §1, p. 11]. Schoen writes the function field of C as generated over the function field of $C/(\mathbb{Z}/\ell)$ by an element y with $y^\ell = f$, and the entry of his tuple at b is the class modulo ℓ of $a := \mathrm{mult}_b \,\mathrm{div}(f)$. Let γ be the generator of $\mathbb{Z}/\ell$ with $\gamma^* y = \exp(2\pi i/\ell) y$, let u be a local coordinate at b, and let w be a local coordinate at x. Then $y^\ell = u^a \cdot (\text{unit})$ and $\mathrm{mult}_x \,\mathrm{div}(u) = s$, hence

$$y = w^{as/\ell} \cdot (\text{unit}).$$

The stabilizer of x is generated by $\gamma^{\ell/s}$, and comparing the leading terms in $(\gamma^{\ell/s})^* y = \exp(2\pi i/s) y$ shows that γ^a acts on T_xC by $\exp(2\pi i/s)$. Hence a_b is the image of a under the isomorphism $\mathbb{Z}/\ell \to \langle\gamma\rangle$ sending 1 to γ, and replacing γ with another generator multiplies the tuple by an element of $(\mathbb{Z}/\ell)^*$.

Schoen works over an algebraically closed field, with étale cohomology with coefficients in $\mathbb{Q}_q$ for a prime number q. Over $\mathbb{C}$, the comparison isomorphism between étale and Betti cohomology (cf. [Har77, Appendix C, 3.7]) commutes with automorphisms of C^{2k} and is compatible with the cycle class maps (cf. [Del82, Example 2.1(a)]). Hence the $\mathbb{Q}$-span of classes of algebraic cycles has equal dimension in Betti cohomology and in étale cohomology, and Theorem 3.1 is a special case of [Sch88, Theorem 2.0].

Remark 3.3. The proof of [Sch88, Theorem 2.0] uses [Sch88, Lemma 2.8], whose conclusion Appendix B proves for any $\ell \geq 2$ (Remark B.1), so Theorem 3.1, as used in this paper, follows from [Sch88] and Appendix B.

Definition 3.4. Let G be a finite group acting faithfully on a smooth projective connected curve X. For a point $x \in X$ with non-trivial stabilizer G_x, let $t_x \in G_x$ be the generator acting

on the tangent space T_xX by $\exp(2\pi i/|G_x|)$. We say that the action has *paired branching* if the multiset of conjugacy classes $[t_x]$, with one point x for each G-orbit with non-trivial stabilizer, can be partitioned into pairs of the form $\{[t],[t^{-1}]\}$.

Since $t_{gx} = gt_xg^{-1}$ for any $g \in G$ and $x \in X$, the multiset in Definition 3.4 does not depend on the choice of the points x. Paired branching is the condition on $(a_b)_b$ in Theorem 3.1 for actions of G. In **Step 1** of the proof of Theorem 3.11, we prove that it descends to the cyclic covers $X/N \to X/L$, where $L \subset G$ is a subgroup and $N \subset L$ is a normal subgroup with L/N cyclic.

3.2. **Invariant Lagrangians and split Hermitian forms.** In this subsection, we prove that paired branching implies the existence of a G-invariant Lagrangian subspace of $H_1(X,\mathbb{Q})$ (Proposition 3.7), and we deduce that self-adjoint idempotents of $\mathbb{Q}[G]$ cut out split Hermitian forms (Corollary 3.8).

Set-up 3.5. Let G, X, ϵ, K, e, m be as follows.

(1) G is a finite group acting faithfully on a smooth projective connected curve X.
(2) $\epsilon \in \mathbb{Q}[G]$ is a central primitive idempotent such that $\epsilon\mathbb{Q}[G] \cong M_d(K)$ for a positive integer d and an imaginary quadratic field K.
(3) $e \in \epsilon\mathbb{Q}[G]$ is a primitive idempotent.
(4) m is a positive integer with $me \in \mathbb{Z}[G]$.
(5) $eH_1(X,\mathbb{Q}) \neq 0$.

Then $\mathbb{Q}[G]$ acts on $J(X)$ through $\mathrm{End}^0(J(X))$ and on $H_1(X,\mathbb{Q})$ by push-forward, $g \in G$ acting by g_*, and we denote by $*$ the involution of $\mathbb{Q}[G]$ with $g^* = g^{-1}$ for $g \in G$. Let

$$A_X := \mathrm{im}(me : J(X) \to J(X)),$$

with the polarization induced from the principal polarization of $J(X)$. We identify $H_1(A_X,\mathbb{Q})$ with

$$eH_1(X,\mathbb{Q}) \subset H_1(J(X),\mathbb{Q}) = H_1(X,\mathbb{Q}).$$

We identify K with the center of $\epsilon\mathbb{Q}[G]$. The action of K on A_X through $\epsilon\mathbb{Q}[G]$ defines a unital embedding $K \hookrightarrow \mathrm{End}^0(A_X)$.

Remark 3.6. In Set-up 3.5, e is a primitive idempotent of $\epsilon\mathbb{Q}[G]$ rather than the central idempotent ϵ for the following reason. Let $e_1 = e, e_2, \ldots, e_d$ be orthogonal primitive idempotents of $\epsilon\mathbb{Q}[G] \cong M_d(K)$ with $e_1 + \cdots + e_d = \epsilon$. Since the e_i are pairwise conjugate by units of $\epsilon\mathbb{Q}[G]$, the decomposition

$$\epsilon H_1(X,\mathbb{Q}) = \bigoplus_{i=1}^{d} e_i H_1(X,\mathbb{Q})$$

is a direct sum of d Hodge substructures isomorphic to $eH_1(X,\mathbb{Q})$, and the abelian subvariety of $J(X)$ cut out by ϵ is isogenous to A_X^d. Since any non-zero idempotent $e' \in \epsilon\mathbb{Q}[G]$ is conjugate to $e_1 + \cdots + e_r$ for some $r \geq 1$, the abelian subvariety of $J(X)$ cut out by e' is isogenous to A_X^r, so Theorem 3.11 treats the Weil plane $W_K(A_X)$. Set-up 3.5 does not require $e^* = e$. For an action with paired branching, Theorem 3.11 implies that $W_K(A_X)$ is algebraic for any primitive idempotent e, and the condition $e^* = e$ is used only to prove that the Hermitian form of the induced polarization is split (Corollary 3.8). If $e^* = e$, then the decomposition

$$H_1(X,\mathbb{Q}) = eH_1(X,\mathbb{Q}) \oplus (1-e)H_1(X,\mathbb{Q})$$

is orthogonal for the intersection form.

Proposition 3.7. *Let G be a finite group acting faithfully on a smooth projective connected curve X with paired branching. Then $H_1(X,\mathbb{Q})$ contains a G-invariant Lagrangian subspace for the intersection form.*

Proof. We replace X by a surface X' with a free G-action through an equivariant cobordism, and we reduce an invariant Lagrangian of $H_1(X', \mathbb{Q})$ back to X.

Step 1. In this step, we construct an equivariant 1-handle for one pair of orbits.

Let O_+ and O_- be the G-orbits of one pair $\{[t], [t^{-1}]\}$ of Definition 3.4. Since $t_{gx} = gt_x g^{-1}$ for any $g \in G$ and $x \in X$, there exist points $x_+ \in O_+$ and $x_- \in O_-$ with $t_{x_-} = t_{x_+}^{-1}$. Let $t := t_{x_+}$, $I := \langle t \rangle = G_{x_+} = G_{x_-}$, $r := |I| \geq 2$, and $\zeta := \exp(2\pi i/r)$. Since I is finite, there exist local coordinates z_+ at x_+ and z_- at x_- such that t acts by $z_+ \to \zeta z_+$ and by $z_- \to \zeta^{-1} z_-$. The points $gx_\pm$ with $gI \in G/I$ are pairwise distinct. Possibly replacing $z_\pm$ with $c^{-1} z_\pm$ for a small real number $c > 0$, we may assume that the closed discs $D_\pm := \{|z_\pm| \leq 1\}$ are defined, and that the discs gD_+ and gD_-, with $gI \in G/I$ and with $\{O_+, O_-\}$ running over the pairs of Definition 3.4, are pairwise disjoint.

Let $D^2 := \{z \in \mathbb{C} \mid |z| \leq 1\}$, let I act on $D^2 \times [0,1]$ by $t \cdot (z,u) = (\zeta z, u)$, and let

$$E := G \times_I (D^2 \times [0,1])$$

be the quotient of $G \times D^2 \times [0,1]$ by $(gt, z, u) \sim (g, \zeta z, u)$, with G acting by left multiplication. We write $[g, z, u]$ for the class of (g, z, u). For $z \in D^2$, let $p_\pm(z) \in D_\pm$ be the point with $z_\pm(p_\pm(z)) = z$, and define

$$\phi : G \times_I (D^2 \times \{0,1\}) \to X, \quad \phi([g,z,0]) = g \cdot p_+(z), \quad \phi([g,z,1]) = g \cdot p_-(\bar{z}).$$

The map ϕ is well defined, since

$$gt \cdot p_+(z) = g \cdot p_+(\zeta z) \quad \text{and} \quad gt \cdot p_-(\bar{z}) = g \cdot p_-(\zeta^{-1}\bar{z}) = g \cdot p_-(\overline{\zeta z}).$$

Moreover, ϕ is a G-equivariant embedding with image $\bigcup_{g \in G} g(D_+ \sqcup D_-)$.

Step 2. In this step, we construct a compact oriented 3-manifold M with an orientation-preserving G-action and with $\partial M = (-X) \sqcup X'$, where G acts freely on X'.

Let G act on $X \times [0,1]$ trivially on $[0,1]$, and give $X \times [0,1]$ the product orientation, so that $\partial(X \times [0,1]) = (-X \times \{0\}) \sqcup X \times \{1\}$. Let M be the 3-manifold obtained from $X \times [0,1]$ by attaching, for any pair $\{O_+, O_-\}$, the G-manifold E of **Step 1** along ϕ to $X \times \{1\}$, and by rounding the corners G-equivariantly. Give $D^2 \times [0,1]$ the product orientation. Then the discs $D^2 \times \{0\}$ and $D^2 \times \{1\}$ carry the boundary orientations of $-D^2$ and D^2, while ϕ preserves the orientation of D^2 on $D^2 \times \{0\}$ and reverses it on $D^2 \times \{1\}$. Thus ϕ reverses the boundary orientations at both ends of E, the orientations of $X \times [0,1]$ and of the copies of $D^2 \times [0,1]$ extend to an orientation of M, and G preserves the orientation of M.

The boundary of M is $(-X) \sqcup X'$, where X' is obtained from X by removing the interiors of the discs $gD_\pm$ of all the pairs and gluing in the annuli $G \times_I (\partial D^2 \times [0,1])$. Removing finitely many disjoint open discs from X leaves a connected surface, so X' is connected. Any point of X with non-trivial stabilizer belongs to an orbit O_+ or O_- of some pair, hence it is not in X'. If $h \in G$ fixes a point $[g, z, u]$ of an annulus, then $hg = gt^k$ for some integer k and $\zeta^k z = z$. Since $|z| = 1$, we have $t^k = 1$, hence $h = 1$. Thus G acts freely on X', and G preserves the orientation of X'.

Step 3. In this step, we construct a G-invariant Lagrangian subspace Λ of $H_1(\partial M, \mathbb{Q})$.

We have

$$H_1(\partial M, \mathbb{Q}) = H_1(X, \mathbb{Q}) \oplus H_1(X', \mathbb{Q}),$$

and the intersection form of ∂M is the negative of the intersection form of X on the first summand and the intersection form of X' on the second summand. Let $j : \partial M \to M$ be the inclusion and

$$\Lambda := \mathrm{Ker}\left(j_* : H_1(\partial M, \mathbb{Q}) \to H_1(M, \mathbb{Q})\right).$$

By the exact sequence of the pair $(M, \partial M)$, Λ is the image of the boundary map

$$\partial : H_2(M, \partial M; \mathbb{Q}) \to H_1(\partial M, \mathbb{Q}).$$

Under Poincaré–Lefschetz duality with rational coefficients, ∂ corresponds to the restriction $H^1(M,\mathbb{Q}) \to H^1(\partial M,\mathbb{Q})$, which is the dual of j_*, followed by Poincaré duality on ∂M. Hence $\Lambda = \Lambda^\perp$. Since j is G-equivariant, Λ is G-invariant.

Step 4. In this step, we conclude the proof.

By **Step 2**, G acts freely on the connected closed oriented surface X' by orientation-preserving homeomorphisms. By [BM24, Theorem 1], there exists a G-invariant Lagrangian subspace $L' \subset H_1(X',\mathbb{Q})$. Let

$$W := 0 \oplus L' \subset H_1(\partial M,\mathbb{Q}).$$

Then W is isotropic, $W^\perp = H_1(X,\mathbb{Q}) \oplus L'$, and $W^\perp/W = H_1(X,\mathbb{Q})$ with the negative of the intersection form of X. Let $L_X \subset H_1(X,\mathbb{Q})$ be the isotropic reduction of Λ along W, that is, the image of $\Lambda \cap W^\perp$ in $W^\perp/W$. Since Λ and W are G-invariant, L_X is G-invariant. Since Λ is isotropic, L_X is isotropic.

It remains to compute $\dim L_X$. Let $2\beta := \dim H_1(\partial M,\mathbb{Q})$, $\kappa := \dim W$, and $a := \dim(\Lambda \cap W)$. The map $W \to \mathrm{Hom}(\Lambda,\mathbb{Q})$ induced by the intersection form has kernel $W \cap \Lambda^\perp = W \cap \Lambda$, hence rank $\kappa - a$. The transpose map $\Lambda \to \mathrm{Hom}(W,\mathbb{Q})$ has kernel $\Lambda \cap W^\perp$ and rank $\kappa - a$. Hence

$$\dim L_X = \dim(\Lambda \cap W^\perp) - \dim(\Lambda \cap W) = (\beta - \kappa + a) - a = \beta - \kappa = \tfrac{1}{2}\dim(W^\perp/W).$$

Thus L_X is a G-invariant Lagrangian subspace of $H_1(X,\mathbb{Q})$. □

Corollary 3.8. *Let G be a finite group acting faithfully on a smooth projective connected curve X with paired branching, and let $*$ be the involution of $\mathbb{Q}[G]$ of Set-up 3.5. Let $e \in \mathbb{Q}[G]$ be an idempotent with $e^* = e$ such that $e\mathbb{Q}[G]e$ is an imaginary quadratic field K on which $*$ restricts to complex conjugation. Then the K-Hermitian form on $eH_1(X,\mathbb{Q})$ induced by the intersection form is split. In particular, $\dim_K eH_1(X,\mathbb{Q}) = 2n$ is even, and the discriminant of the form is $[(-1)^n]$.*

Proof. Let $H := H_1(X,\mathbb{Q})$ and let ψ be the intersection form on H. Since G preserves ψ, we have $\psi(ax,y) = \psi(x,a^*y)$ for any $a \in \mathbb{Q}[G]$ and $x,y \in H$. Since $e^* = e$, we have $\psi(ex,(1-e)y) = \psi(x,e(1-e)y) = 0$, so $H = eH \oplus (1-e)H$ is an orthogonal decomposition and ψ is non-degenerate on eH. The field $K = e\mathbb{Q}[G]e$ acts on eH with unit e, and $\psi(kx,y) = \psi(x,\bar{k}y)$ for any $k \in K$ and $x,y \in eH$. Let h be the K-Hermitian form on eH, conjugate-linear in the first variable, such that $\psi(x,y) = \mathrm{Tr}_{K/\mathbb{Q}}(\delta h(x,y))$ for any $x,y \in eH$, where $\delta \in K$ with $\bar{\delta} = -\delta \neq 0$ is the element fixed in Definition 2.9. Since ψ is non-degenerate on eH, so is h.

Let $L_X \subset H$ be the G-invariant Lagrangian subspace of Proposition 3.7, and let $L' := eL_X$. Since L_X is a $\mathbb{Q}[G]$-submodule, $L' = L_X \cap eH$. For $w \in eH$ and $x \in L_X$, we have $\psi(w,ex) = \psi(ew,x) = \psi(w,x)$. Hence the orthogonal of L' in eH is

$$eH \cap L_X^\perp = eH \cap L_X = L',$$

and L' is a Lagrangian subspace of eH. Moreover, L' is K-stable, since $eae \cdot ex = e(aex) \in eL_X$ for any $a \in \mathbb{Q}[G]$ and $x \in L_X$.

Let $x,y \in L'$. For any $k \in K$, we have $ky \in L'$, hence

$$\mathrm{Tr}_{K/\mathbb{Q}}(\delta k h(x,y)) = \mathrm{Tr}_{K/\mathbb{Q}}(\delta h(x,ky)) = \psi(x,ky) = 0.$$

Since $\delta \neq 0$ and the trace form of $K/\mathbb{Q}$ is non-degenerate, $h(x,y) = 0$. Thus h vanishes on the K-subspace L', and $\dim_K L' = \frac{1}{2}\dim_K eH$. In particular, $\dim_K eH = 2n$ for a non-negative integer n, and h is split.

Let $u_1,\dots,u_n$ be a K-basis of L'. Since h is non-degenerate, there exist $v_1,\dots,v_n \in eH$ with $(h(u_i,v_j))_{i,j} = I_n$. Replacing v_j with $v_j - \frac{1}{2}\sum_{l=1}^n h(v_l,v_j)u_l$ for any j, we may moreover assume that $h(v_i,v_j) = 0$ for any i,j. Then the Gram matrix of h in $u_1,\dots,u_n,v_1,\dots,v_n$ is

$$\begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix},$$

which is invertible. Hence $u_1, \dots, u_n, v_1, \dots, v_n$ is a K-basis of eH, and $\det h = (-1)^n$. Since the discriminant does not depend on the choice of δ (Lemma 2.10 (6)), the discriminant of h is $[(-1)^n]$. □

3.3. **Weil planes of Jacobian factors.** In this subsection, we prove Theorem 3.11.

Notation 3.9. Let $F \subset \mathbb{C}$ be a subfield, Y a smooth projective variety, not necessarily connected, and $M \subset H^1(Y, F)$ a subspace with $\dim_F M = a$. We view $\wedge^a M$ as the line of antisymmetric tensors in $M^{\otimes a}$, and we embed $H^1(Y, F)^{\otimes a}$ in $H^a(Y^a, F)$ by the Künneth formula

$$x_1 \otimes \cdots \otimes x_a \to \mathrm{pr}_1^* x_1 \cup \cdots \cup \mathrm{pr}_a^* x_a.$$

For $a = 0$, Y^0 is a point and $\wedge^0 M = F$. We say that a class in $H^{2p}(Y, F) = H^{2p}(Y, \mathbb{Q}) \otimes_{\mathbb{Q}} F$ is *algebraic* if it is an F-linear combination of classes of algebraic cycles of codimension p on Y.

Lemma 3.10. *Let C be a smooth projective connected curve of genus $g \geq 1$. Then the line $\wedge^{2g} H^1(C, \mathbb{Q}) \subset H^{2g}(C^{2g}, \mathbb{Q})$, embedded by the Künneth formula, is spanned by the class of an algebraic cycle.*

Proof. Fix a point $d_0 \in C$, and let $\pi_1 := \Delta_C - d_0 \times C - C \times d_0$, where $\Delta_C \subset C \times C$ is the diagonal. The Künneth components of $[\Delta_C]$ in $H^2(C, \mathbb{Q}) \otimes H^0(C, \mathbb{Q})$ and in $H^0(C, \mathbb{Q}) \otimes H^2(C, \mathbb{Q})$ are $[d_0 \times C]$ and $[C \times d_0]$, so $[\pi_1]$ is the Künneth component $\kappa \in H^1(C, \mathbb{Q})^{\otimes 2}$ of $[\Delta_C]$. As a correspondence, Δ_C acts on $H^1(C, \mathbb{Q})$ as the identity, and $d_0 \times C$ and $C \times d_0$ act on $H^1(C, \mathbb{Q})$ as zero. Thus κ induces the identity of $H^1(C, \mathbb{Q})$ through the cup product pairing on the first factor, and κ has rank $2g$ as a tensor. The involution of $C \times C$ exchanging the factors fixes π_1, and by the graded commutativity of the cup product it acts on $H^1(C, \mathbb{Q})^{\otimes 2}$ as -1 times the permutation of the tensor factors. Hence κ is antisymmetric, and there exists a basis $x_1, y_1, \dots, x_g, y_g$ of $H^1(C, \mathbb{Q})$ with

$$\kappa = \sum_{i=1}^{g} (x_i \otimes y_i - y_i \otimes x_i).$$

For $1 \leq j \leq g$, let $p_j : C^{2g} \to C^2$ be the projection to the factors $2j - 1$ and $2j$. The class of the cycle $p_1^* \pi_1 \cdots p_g^* \pi_1$ is the image of $\kappa^{\otimes g}$ under the Künneth formula. The symmetric group S_{2g} acts on C^{2g} by permuting the factors, and by the graded commutativity of the cup product, τ^* acts on the image of $H^1(C, \mathbb{Q})^{\otimes 2g}$ as $\mathrm{sgn}(\tau)$ times the permutation of the tensor factors for any $\tau \in S_{2g}$. Thus $((2g)!)^{-1} \sum_{\tau \in S_{2g}} \tau^*$ acts there as the antisymmetrizer, and it maps algebraic classes to algebraic classes. The antisymmetrization of $\kappa^{\otimes g}$ is $2^g g!$ times the antisymmetrization of $x_1 \otimes y_1 \otimes \cdots \otimes x_g \otimes y_g$, which spans $\wedge^{2g} H^1(C, \mathbb{Q})$. □

Theorem 3.1 treats one cyclic automorphism, while G may be non-commutative. The block $\epsilon\mathbb{Q}[G]$ has two irreducible complex representations, V and the complex conjugate $\bar{V}$ of V, and the complexified Weil plane is spanned by the images of the top exterior powers $\det \mathrm{Hom}_G(V, H^1(X, \mathbb{C}))$ and $\det \mathrm{Hom}_G(\bar{V}, H^1(X, \mathbb{C}))$ under an algebraic correspondence. To reach $\det \mathrm{Hom}_G(V, H^1(X, \mathbb{C}))$, we need an induction formula with integer coefficients [Bol90], since clearing the denominators of a formula with rational coefficients only yields a tensor power of $\det \mathrm{Hom}_G(V, H^1(X, \mathbb{C}))$.

Theorem 3.11. *In Set-up 3.5, assume that the action of G on X has paired branching and that $A_X \neq 0$. Then $\dim A_X = 2n$ is even, the multiplicities of K on $H^{1,0}(A_X)$ are (n, n), and any class in $W_K(A_X)$ is algebraic. If moreover $e^* = e$, then the polarization of A_X induced from $J(X)$ is a compatible polarization of (A_X, K) with split Hermitian form.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, for a subgroup $L \leq G$ and a homomorphism $\alpha : L \to F^*$, where $F := \mathbb{Q}(\exp(2\pi i/|G|)) \subset \mathbb{C}$, we let $U_\alpha \subset H^1(X/\ker(\alpha), F)$ be the subspace on which $L/\ker(\alpha)$

acts by push-forward through α, and we prove that $\dim_F U_\alpha$ is even and that the top exterior power of U_α is spanned by an algebraic class.

Algebraic correspondences act on cohomology compatibly with the cycle class map (cf. [Ful98, Chapter 16]). In particular, pullbacks and push-forwards along morphisms, exterior products, and intersection products with classes of algebraic cycles map algebraic classes to algebraic classes. For $x \in X$ with $G_x \neq 1$, the group G_x acts faithfully on T_xX, so G_x is cyclic and the generator t_x of Definition 3.4 is unique.

Let H be a finite group acting faithfully with paired branching on a smooth projective connected curve Y, and let R be a non-trivial irreducible complex representation of H. For $y \in Y$ with $H_y \neq 1$ and $1 \leq k < |H_y|$, let $\mu_k(R, t_y)$ be the multiplicity of $\exp(2\pi ik/|H_y|)$ as an eigenvalue of t_y on R. By the Chevalley–Weil formula [Can18, Theorem 3.1],

$$\dim \operatorname{Hom}_H(R, H^{1,0}(Y)) = (g(Y/H) - 1) \dim R + \sum_y \sum_{k=1}^{|H_y|-1} \mu_k(R, t_y) \left(1 - \frac{k}{|H_y|}\right),$$

where y runs over one point in each H-orbit with non-trivial stabilizer. For the trivial representation the formula has an additional summand 1, and we use it only for non-trivial R. If the orbits of y and y' form one of the pairs of Definition 3.4, with $t_{y'}$ conjugate to t_y^{-1}, then $r := |H_y| = |H_{y'}|$ and $\mu_k(R, t_{y'}) = \mu_{r-k}(R, t_y)$. Thus the contributions of y and y' add up to $\sum_{k=1}^{r-1} \mu_k(R, t_y) = \dim R - \dim R^{t_y}$, where $R^{t_y} \subset R$ is the subspace fixed by t_y, and

$$\dim \operatorname{Hom}_H(R, H^{1,0}(Y)) = (g(Y/H) - 1) \dim R + \sum_{\{y,y'\}} \left(\dim R - \dim R^{t_y}\right), \tag{3.1}$$

the sum running over the pairs. The right-hand side of (3.1) does not change when R is replaced by the dual $R^\vee$, which is isomorphic to the complex conjugate $\bar{R}$. Since $H^{0,1}(Y) = \overline{H^{1,0}(Y)}$, we have $\dim \operatorname{Hom}_H(R, H^{0,1}(Y)) = \dim \operatorname{Hom}_H(\bar{R}, H^{1,0}(Y))$, so $\dim \operatorname{Hom}_H(R, H^1(Y, \mathbb{C}))$ is twice the right-hand side of (3.1).

Let $L \leq G$ be a subgroup, $\alpha : L \to F^*$ a homomorphism, $N := \ker(\alpha)$, and $\ell := |\alpha(L)|$. An element of L acting trivially on X/N maps a point x with $G_x = 1$ into Nx, so it belongs to N, and L/N acts faithfully on the connected curve X/N. Let $x \in X$ with $G_x \neq 1$, let $g \in G$, and let $q \geq 1$ be the smallest integer with $gt_x^q g^{-1} \in L$. Then $L_{gx} = L \cap gG_xg^{-1}$ is generated by $t_{gx}^q = gt_x^q g^{-1}$, which acts on $T_{gx}X$ by $\exp(2\pi i/|L_{gx}|)$. Let y be the image of gx in X/N. The stabilizer $(L/N)_y$ is the image of L_{gx}, and in a local coordinate z at gx on which L_{gx} acts linearly, the quotient map $X \to X/N$ is $z \to z^{|N_{gx}|}$. Hence the image of t_{gx}^q in L/N is the generator t_y for the action of L/N on X/N when $(L/N)_y \neq 1$, and it is trivial otherwise. The L-orbits in Gx are the orbits Lgx, with g running over representatives of the double cosets of L and G_x in G. Let the orbit of $x' \in X$ be paired with the orbit of x as in Definition 3.4. Since $t_{hx'} = ht_{x'}h^{-1}$ for $h \in G$, possibly replacing x' with a point of Gx', we may assume that $t_{x'} = t_x^{-1}$. Then $G_{x'} = G_x$, the integer q for (x', g) equals the integer q for (x, g), and $t_{gx'}^q = (t_{gx}^q)^{-1}$. Thus the images in X/N of gx and gx' have generators in L/N inverse to each other, or stabilizers in L/N that are both trivial. Hence the action of L/N on X/N has paired branching. If $\ell \geq 2$, we identify L/N with $\mathbb{Z}/\ell$ through α and $\exp(2\pi ik/\ell) \to k$, and then the tuple $(a_b)_b$ of Theorem 3.1 for $X/N \to X/L$ can be ordered as $(a_1, -a_1, a_2, -a_2, \dots)$.

We embed $\wedge^{\dim U_\alpha} U_\alpha$ in $H^{\dim U_\alpha}((X/N)^{\dim U_\alpha}, F)$ as in Notation 3.9.

- If $U_\alpha = 0$, the class of the point spans $\wedge^0 U_\alpha = F$.
- If $U_\alpha \neq 0$ and $\ell = 1$, then $N = L$, $U_\alpha = H^1(X/L, F)$, and $g(X/L) \geq 1$. Lemma 3.10 implies the claim.
- If $U_\alpha \neq 0$ and $\ell \geq 2$, then $U_\alpha \otimes_F \mathbb{C}$ is the eigenspace in $H^1(X/N, \mathbb{C})$ of the faithful character of L/N induced by α, and $U_\alpha \otimes_F \mathbb{C}$ is one of the spaces V_χ of Theorem 3.1. For a faithful character χ of L/N, we have $\chi(t_y) \neq 1$ for any $y \in X/N$ with $(L/N)_y \neq 1$, so by (3.1) the multiplicity of χ in $H^1(X/N, \mathbb{C})$ is $2(g(X/L) - 1)$ plus twice the number

of pairs for the action of L/N on X/N. In particular, it does not depend on χ, and it is equal to $\dim_F U_\alpha =: 2k$. Let $W \subset H^{2k}((X/N)^{2k}, \mathbb{Q})$ be the subspace of Theorem 3.1 for the action of L/N on X/N, which is spanned by algebraic classes. Let

$$\Pi_\alpha := \ell^{-2k} \sum_{(u_1,\dots,u_{2k})\in(L/N)^{2k}} \alpha(u_1)^{-1}\cdots\alpha(u_{2k})^{-1}(u_1\times\cdots\times u_{2k})_*,$$

acting on $H^*((X/N)^{2k}, F)$. Then Π_α maps algebraic classes to algebraic classes, and on the image of $H^1(X/N, F)^{\otimes 2k}$ it acts as the $2k$-th tensor power of the projector of $H^1(X/N, F)$ onto U_α. Hence Π_α kills $\wedge^{2k}V_\chi$ for $V_\chi \neq U_\alpha \otimes_F \mathbb{C}$ and maps $W \otimes_{\mathbb{Q}} F$ onto $\wedge^{2k}U_\alpha$.

Step 2. In this step, we prove that $\det M(R)$, defined below, is spanned by an algebraic class when R is a direct sum of induced modules $\mathrm{Ind}_L^G \alpha$, and we write the class of V, defined below, in terms of such modules.

We let G act on $H^1(X, F)$ by push-forward. Let R be an $F[G]$-module with a basis $r_1, \dots, r_s$, and let $r_1^\vee, \dots, r_s^\vee$ be the dual basis of $R^\vee$. Let

$$M(R) := \left(R^\vee \otimes_F H^1(X, F)\right)^G \cong \mathrm{Hom}_{F[G]}(R, H^1(X, F)),$$

where G acts diagonally. Let $X_s := X\times\{1, \dots, s\}$. We identify $R^\vee\otimes_F H^1(X, F)$ with $H^1(X_s, F) = H^1(X, F)^{\oplus s}$, so that $r_j^\vee \otimes \xi$ is ξ on $X\times\{j\}$, and we let $\det M(R) := \wedge^a M(R) \subset H^a(X_s^a, F)$, where $a := \dim_F M(R)$. A change of basis of R changes the coordinates by an F-linear combination of the identifications between the components of X_s, and the corresponding combination of identifications between the components of X_s^a maps algebraic classes to algebraic classes and acts on the image of $H^1(X_s, F)^{\otimes a}$ as the a-th tensor power of the coordinate change. Hence the property that $\det M(R)$ is spanned by an algebraic class does not depend on the basis.

Let $L \leq G$ be a subgroup, $\alpha : L \to F^*$ a homomorphism, $N := \ker(\alpha)$, and $\pi : X \to X/N$ the quotient map. Let $\mathrm{Ind}_L^G \alpha := F[G] \otimes_{F[L]} F_\alpha$, where F_α is F with L acting through α, with the basis $r_j := g_j \otimes 1$, where $g_1 = 1, g_2, \dots, g_s$ are representatives of G/L. Frobenius reciprocity is the isomorphism

$$\mathrm{Hom}_{F[G]}(\mathrm{Ind}_L^G \alpha, H^1(X, F)) \cong \mathrm{Hom}_{F[L]}(F_\alpha, \mathrm{Res}_L^G H^1(X, F)), \quad f \to f(r_1)$$

(cf. [Ful97, Section 7.3, p. 93]), whose inverse maps ξ to the homomorphism $r_j \to g_{j*}\xi$. The right-hand side is the α-eigenspace $H^1(X, F)^{(L,\alpha)}$ of L. Thus $M(\mathrm{Ind}_L^G \alpha) \subset H^1(X_s, F)$ is the image of $H^1(X, F)^{(L,\alpha)}$ under $\xi \to (g_{j*}\xi)_j$, which is the pullback along $\gamma : X_s \to X$, $(x, j) \to g_j^{-1}x$. On cohomology, $\pi_*\pi^* = |N|$ and $\pi^*\pi_* = \sum_{h\in N} h^*$. Since $\alpha|_N = 1$, any $\xi \in H^1(X, F)^{(L,\alpha)}$ is N-invariant, so $\xi = \pi^*(|N|^{-1}\pi_*\xi)$. As π^* is injective and intertwines the push-forward actions of L/N and L, the pullback π^* maps U_α of **Step 1** isomorphically onto $H^1(X, F)^{(L,\alpha)}$. Hence $(\pi\circ\gamma)^*$ maps U_α isomorphically onto $M(\mathrm{Ind}_L^G \alpha)$, and the pullback along $(\pi\circ\gamma)^{\times a} : X_s^a \to (X/N)^a$ maps $\wedge^a U_\alpha$ onto $\det M(\mathrm{Ind}_L^G \alpha)$. By **Step 1**, $\det M(\mathrm{Ind}_L^G \alpha)$ is spanned by an algebraic class, and $\dim M(\mathrm{Ind}_L^G \alpha) = \dim U_\alpha$ is even.

Let $R = R_1 \oplus \cdots \oplus R_p$ with the union of bases of the R_i, so that X_s is the disjoint union of the X_{s_i} and $M(R) = \bigoplus_i M(R_i)$. Let $a_i := \dim M(R_i)$, $a := \sum_i a_i$, and assume that

$$\det M(R_i) \subset H^{a_i}(X_{s_i}^{a_i}, F) \subset H^{a_i}(X_s^{a_i}, F)$$

(extension by zero, $X_{s_i}^{a_i}$ being a union of connected components of $X_s^{a_i}$) is spanned by an algebraic class z_i for any i. The symmetric group S_a acts on X_s^a by permuting the factors, and by the graded commutativity of the cup product, τ^* acts on the image of $H^1(X_s, F)^{\otimes a}$ as $\mathrm{sgn}(\tau)$ times the permutation of the tensor factors for any $\tau \in S_a$. Thus $(a!)^{-1}\sum_{\tau\in S_a}\tau^*$ acts there as the antisymmetrizer, and it maps $z_1 \times \cdots \times z_p$ to an algebraic class spanning $\det M(R)$.

Let $\sigma, \bar\sigma$ be the complex embeddings of K. The field F contains the $|G|$-th roots of unity, so F is a splitting field for G by Brauer's theorem (cf. [Pas21, Introduction]). Let V be an $F[G]$-module such that $V \otimes_F \mathbb{C}$ is irreducible, ϵ acts on V as the identity, and K acts on $V \otimes_F \mathbb{C}$ through

σ, and let $\bar{V}$ be the analogous module for $\bar{\sigma}$. Then $\dim_F V = \dim_F \bar{V} = d$, and $\sigma(K) \subset F$. By [Bol90, Corollary (2.12)], there exist subgroups $L_1, \dots, L_p \leq G$, homomorphisms $\alpha_i : L_i \to \mathbb{C}^*$, and non-zero integers c_i such that

$$[V] = \sum_{i=1}^{p} c_i \left[\mathrm{Ind}_{L_i}^G \alpha_i\right]$$

in the group of virtual characters of G. The values of α_i are $|G|$-th roots of unity, so α_i takes values in F^*. Let

$$R_+ := \bigoplus_{c_i>0} \left(\mathrm{Ind}_{L_i}^G \alpha_i\right)^{\oplus c_i}, \qquad R_- := \bigoplus_{c_i<0} \left(\mathrm{Ind}_{L_i}^G \alpha_i\right)^{\oplus(-c_i)}.$$

Since $F[G]$-modules are determined up to isomorphism by their characters, there exists an isomorphism of $F[G]$-modules

$$T : R_+ \to V \oplus R_-.$$

Since

$$(\mathrm{Ind}_L^G \alpha)^\vee \cong \mathrm{Ind}_L^G(\alpha^{-1}),$$

the module $R_-^\vee$ is also a direct sum of induced modules. Hence $\det M(R_+)$ and $\det M(R_-^\vee)$ are spanned by algebraic classes for any bases of R_+ and $R_-^\vee$, and $\dim M(R_+)$ and $\dim M(R_-)$ are even.

Step 3. In this step, we prove that $\det M(V)$ and $\det M(\bar{V})$ are spanned by algebraic classes of codimension n, where $n := \dim \mathrm{Hom}_G(V \otimes_F \mathbb{C}, H^{1,0}(X))$.

The representation $V \otimes_F \mathbb{C}$ is non-trivial, since the simple factor of $\mathbb{Q}[G]$ belonging to the trivial representation is $\mathbb{Q}$. By (3.1) for $H = G$ and $Y = X$, we have $\dim \mathrm{Hom}_G(\bar{V} \otimes_F \mathbb{C}, H^{1,0}(X)) = n$ and $\dim_F M(V) = 2n$. Let $a_+ := \dim M(R_+)$ and $a_- := \dim M(R_-)$. As T induces an isomorphism $M(V \oplus R_-) \cong M(R_+)$, we have $a_+ - a_- = 2n$.

Fix a basis of V and a basis of R_-, and let $r'_1, \dots, r'_{d+s}$ be their union, where $s := \dim R_-$. We equip R_+ with the basis $T^{-1}(r'_1), \dots, T^{-1}(r'_{d+s})$ and $R_-^\vee$ with the dual basis of the basis of R_-. Let $Y := X_{d+s}$, the disjoint union of X_d and X_s. Since $f \to f \circ T^{-1}$ maps $M(R_+)$ onto $M(V \oplus R_-)$, we have $M(R_+) = M(V) \oplus M(R_-)$ as subspaces of $H^1(Y, F) = H^1(X_d, F) \oplus H^1(X_s, F)$, and $M(R_-)$ and $M(R_-^\vee)$ are subspaces of $H^1(X_s, F)$. By **Step 2**, there exist algebraic classes $z_+ \in H^{a_+}(Y^{a_+}, F)$ spanning $\wedge^{a_+}(M(V) \oplus M(R_-))$ and z_- spanning $\det M(R_-^\vee)$.

For $\xi, \eta \in H^1(X_s, F)$, let $\langle \xi, \eta \rangle := \int_{X_s} \xi \cup \eta$. We identify $H^1(X_s, F)$ with $R_-^\vee \otimes_F H^1(X, F)$ through the basis of R_- as in **Step 2**, and with $R_- \otimes_F H^1(X, F)$ through the dual basis of $R_-^\vee$. Then $\langle , \rangle$ is the tensor product of the evaluation map $R_-^\vee \otimes R_- \to F$ and the cup product pairing on $H^1(X, F)$, and $\rho := |G|^{-1} \sum_{g \in G} g$ acts on both spaces. Hence $\langle , \rangle$ is a perfect pairing between $R_-^\vee \otimes_F H^1(X, F)$ and $R_- \otimes_F H^1(X, F)$ invariant under G. The invariance of $\langle , \rangle$ implies $\langle \rho\xi, \eta \rangle = \langle \xi, \rho\eta \rangle$, so $\langle , \rangle$ restricts to a perfect pairing

$$M(R_-) \times M(R_-^\vee) \to F. \tag{3.2}$$

In particular, $\dim M(R_-^\vee) = a_-$ and $z_- \in H^{a_-}(X_s^{a_-}, F)$.

Let $\Delta_- \subset Y \times X_s$ be the diagonal of X_s, where $X_s \subset Y$. For $1 \leq j \leq a_-$, let $D_j \subset Y^{a_+} \times X_s^{a_-}$ be the pullback of Δ_- along the projection to the $(2n + j)$-th factor of Y^{a_+} and the j-th factor of $X_s^{a_-}$, and let $\varpi : Y^{a_+} \times X_s^{a_-} \to Y^{2n}$ be the projection to the first $2n$ factors. The map

$$w \to \varpi_* \left(w \cup [D_1] \cup \cdots \cup [D_{a_-}]\right)$$

maps algebraic classes to algebraic classes and lowers the codimension by a_-. On the image of $H^1(Y, F)^{\otimes a_+} \otimes H^1(X_s, F)^{\otimes a_-}$, it maps

$$x_1 \otimes \cdots \otimes x_{a_+} \otimes y_1 \otimes \cdots \otimes y_{a_-}$$

to

$$\pm\prod_{j=1}^{a_-}\langle x_{2n+j}|_{X_s}, y_j\rangle\cdot x_1\otimes\cdots\otimes x_{2n},$$

with a sign depending only on n and a_-. Let $\xi_1,\dots,\xi_{2n}$ be a basis of $M(V)$, $\eta_1,\dots,\eta_{a_-}$ a basis of $M(R_-)$, and $\eta'_1,\dots,\eta'_{a_-}$ a basis of $M(R_-^\vee)$. Up to non-zero factors, z_+ is the antisymmetrization of

$$\xi_1\otimes\cdots\otimes\xi_{2n}\otimes\eta_1\otimes\cdots\otimes\eta_{a_-}$$

and z_- is the antisymmetrization of $\eta'_1\otimes\cdots\otimes\eta'_{a_-}$. Since $\xi_i|_{X_s}=0$, only the terms of z_+ with $\eta_1,\dots,\eta_{a_-}$ in the last a_- positions contribute, and the image of $z_+\times z_-$ is a non-zero multiple of

$$\det\left(\langle\eta_k,\eta'_l\rangle\right)_{1\le k,l\le a_-}$$

times the antisymmetrization of $\xi_1\otimes\cdots\otimes\xi_{2n}$. The determinant is non-zero since (3.2) is perfect. Restricting to $X_d^{2n}\subset Y^{2n}$, we obtain an algebraic class spanning $\det M(V)\subset H^{2n}(X_d^{2n},F)$. The class has codimension $a_+/2+a_-/2-a_-=n$. The argument for $\bar V$ in place of V proves the claim for $\det M(\bar V)$.

Step 4. In this step, we prove the statements on A_X.

We first show that $*$ restricts to complex conjugation on K. The involution $*$ permutes the central primitive idempotents of $\mathbb{Q}[G]$, and ϵ^* acts as the identity on the dual of $V\otimes_F\mathbb{C}$, which is isomorphic to $\bar V\otimes_F\mathbb{C}$. Hence $\epsilon^*=\epsilon$, and $*$ restricts to an automorphism of K whose square is id_K. For $b=\sum_g\beta_g g\in\mathbb{Q}[G]$, the trace of the left multiplication by bb^* on $\mathbb{Q}[G]$ is $|G|\sum_g\beta_g^2$. For $k\in K$, the left multiplication by k on $\mathbb{Q}[G]$ vanishes on $(1-\epsilon)\mathbb{Q}[G]$ and has trace $d^2\,\mathrm{Tr}_{K/\mathbb{Q}}(k)$ on $\epsilon\mathbb{Q}[G]\cong M_d(K)$. Thus $\mathrm{Tr}_{K/\mathbb{Q}}(kk^*)>0$ for any $k\in K\setminus\{0\}$. As $\mathrm{Tr}_{K/\mathbb{Q}}(\delta^2)=2\delta^2<0$, where $\delta\in K$ with $\bar\delta=-\delta\neq0$ is the element fixed in Definition 2.9, the restriction of $*$ to K is not id_K, so $k^*=\bar k$ for any $k\in K$.

We use the following facts (cf. [BL04, Ch. 11]). Fix $x_0\in X$ and let $\mathrm{AJ}: X\to J(X)$, $x\to\mathcal{O}_X(x-x_0)$, be the Abel–Jacobi map.

(1) The push-forward
$$\mathrm{AJ}_*: H_1(X,\mathbb{Q})\to H_1(J(X),\mathbb{Q})$$
is an isomorphism, and $H_1(A_X,\mathbb{Q})\subset H_1(J(X),\mathbb{Q})$ is identified with $eH_1(X,\mathbb{Q})$ through AJ_*.

(2) The first Chern class $c_1(\mathcal{P})\in CH^1(X\times J(X))$ of a Poincaré line bundle $\mathcal{P}$ induces, as a correspondence from X to $J(X)$, the inverse of
$$\mathrm{AJ}^*: H^1(J(X),\mathbb{Q})\to H^1(X,\mathbb{Q})$$
up to sign.

(3) The Riemann form of the principal polarization of $J(X)$ is the intersection form on $H_1(X,\mathbb{Z})$.

Let

$$\Lambda:=c_1(\mathcal{P})|_{X\times A_X}\in CH^1(X\times A_X),$$

and let $\iota: K\to\mathrm{End}^0(A_X)$ be the unital embedding of Set-up 3.5. We view $H^1(X,\mathbb{Q})$ as $\mathrm{Hom}(H_1(X,\mathbb{Q}),\mathbb{Q})$ and $H^1(A_X,\mathbb{Q})$ as $\mathrm{Hom}(eH_1(X,\mathbb{Q}),\mathbb{Q})$. By (1) and (2), $\Lambda_*\xi=\pm\xi|_{eH_1(X,\mathbb{Q})}$ for any $\xi\in H^1(X,\mathbb{Q})$. The push-forward action of $b\in\mathbb{Q}[G]$ on $H^1(X,\mathbb{Q})$ is $(b_*\xi)(y)=\xi(b^*y)$. Hence Λ_* vanishes on $(1-e^*)_*H^1(X,\mathbb{Q})$ and maps $e^*_*H^1(X,\mathbb{Q})$ isomorphically onto $H^1(A_X,\mathbb{Q})$. Moreover, for $k\in K$,

$$\iota(k)^*\Lambda_*\xi=\pm\xi(k\cdot)|_{eH_1(X,\mathbb{Q})}=\Lambda_*(\bar k_*\xi).$$

For $\tau\in\{\sigma,\bar\sigma\}$, let $H^1(A_X,F)_\tau\subset H^1(A_X,F)$ be the subspace on which $\iota(k)^*$ acts by $\tau(k)$ for any $k\in K$, so that $H^1(A_X,F)=H^1(A_X,F)_\sigma\oplus H^1(A_X,F)_{\bar\sigma}$. Let $H^1(X,F)_V$ and $H^1(X,F)_{\bar V}$ be the images of the evaluation maps $V\otimes_F M(V)\to H^1(X,F)$ and $\bar V\otimes_F M(\bar V)\to H^1(X,F)$.

Since $\epsilon^* = \epsilon$, the element e^* is a primitive idempotent of $\epsilon\mathbb{Q}[G] \cong M_d(K)$. Thus e^* acts on V and on $\bar{V}$ as idempotents of rank one, and

$$e^*_* H^1(X,F) = e^*_* H^1(X,F)_V \oplus e^*_* H^1(X,F)_{\bar{V}}.$$

Let $v \in V$ span e^*V. As v generates V, the map $M(V) \to e^*_* H^1(X,F)_V$, $h \to h(v)$, is an isomorphism. On $e^*_* H^1(X,F)_V$, the element $\bar{k}_*$ acts by $\sigma(\bar{k}) = \bar{\sigma}(k)$, so Λ_* maps $e^*_* H^1(X,F)_V$ isomorphically onto $H^1(A_X,F)_{\bar{\sigma}}$, and similarly $e^*_* H^1(X,F)_{\bar{V}}$ onto $H^1(A_X,F)_\sigma$. Let $H^1(X,\mathbb{C})_V := H^1(X,F)_V \otimes_F \mathbb{C}$. As Λ_* is a morphism of Hodge structures, it maps $e^*_* H^1(X,\mathbb{C})_V \cap H^{1,0}(X)$ onto $H^{1,0}(A_X) \cap H^1(A_X,\mathbb{C})_{\bar{\sigma}}$, and

$$\dim\left(e^*_* H^1(X,\mathbb{C})_V \cap H^{1,0}(X)\right) = \dim \mathrm{Hom}_G(V \otimes_F \mathbb{C}, H^{1,0}(X)) = n.$$

By **Step 3**, $\dim \mathrm{Hom}_G(\bar{V} \otimes_F \mathbb{C}, H^{1,0}(X)) = n$ as well. Thus the multiplicities of K on $H^{1,0}(A_X)$ are (n,n), and $\dim A_X = 2n$ is the multiplicity of $V \otimes_F \mathbb{C}$ in $H^1(X,\mathbb{C})$. Thus (A_X,ι) is of Weil type.

We apply **Step 3** with a basis $v_1 = v, v_2, \ldots, v_d$ of V. The pullback along $X = X \times \{1\} \subset X_d$ maps $h \in M(V)$ to $h(v)$, so the pullback along $(X \times \{1\})^{2n} \subset X_d^{2n}$ maps the class of **Step 3** to an algebraic class w_V spanning $\wedge^{2n} e^*_* H^1(X,F)_V \subset H^{2n}(X^{2n},F)$. The correspondence $\Lambda^{\times 2n} \in CH^{2n}(X^{2n} \times A_X^{2n})$ acts on the image of $H^1(X,F)^{\otimes 2n}$ as $\pm\Lambda_*^{\otimes 2n}$, so it maps w_V to an algebraic class spanning

$$\wedge^{2n} H^1(A_X,F)_{\bar{\sigma}} \subset H^{2n}(A_X^{2n},F).$$

The pullback along the small diagonal $A_X \to A_X^{2n}$ maps $x_1 \otimes \cdots \otimes x_{2n}$ to $x_1 \cup \cdots \cup x_{2n}$. By the graded commutativity of the cup product, it maps the antisymmetrization of $x_1 \otimes \cdots \otimes x_{2n}$ for a basis $x_1, \ldots, x_{2n}$ of $H^1(A_X,F)_{\bar{\sigma}}$ to $x_1 \cup \cdots \cup x_{2n}$, which spans $\wedge^{2n} H^1(A_X,F)_{\bar{\sigma}} \subset H^{2n}(A_X,F)$ since $H^*(A_X,F) = \wedge^* H^1(A_X,F)$. The argument for $\bar{V}$ proves that $\wedge^{2n} H^1(A_X,F)_\sigma \subset H^{2n}(A_X,F)$ is spanned by an algebraic class as well.

Let $Z \subset H^{2n}(A_X,\mathbb{Q})$ be the $\mathbb{Q}$-span of the classes of algebraic cycles of codimension n. The algebraic classes in $H^{2n}(A_X,F)$ form $Z \otimes_{\mathbb{Q}} F$. Both $W_K(A_X) \otimes_{\mathbb{Q}} F$ and

$$\wedge^{2n} H^1(A_X,F)_\sigma \oplus \wedge^{2n} H^1(A_X,F)_{\bar{\sigma}}$$

are F-subspaces of $H^{2n}(A_X,F)$ whose complexifications are equal by Definition 2.7, so they are equal. Hence $W_K(A_X) \otimes_{\mathbb{Q}} F \subset Z \otimes_{\mathbb{Q}} F$, so $((W_K(A_X) + Z)/Z) \otimes_{\mathbb{Q}} F = 0$ and $W_K(A_X) \subset Z$.

Assume that $e^* = e$. By (3), the Riemann form ψ of the polarization of A_X induced from $J(X)$ is the restriction of the intersection form to $eH_1(X,\mathbb{Q})$. The intersection form is invariant under G, so $\psi(kx,y) = \psi(x,k^*y) = \psi(x,\bar{k}y)$ for any $k \in K$ and $x,y \in eH_1(X,\mathbb{Q})$. Thus the induced polarization is a compatible polarization of (A_X,K). The map $K \to e\mathbb{Q}[G]e$, $k \to ke$, is an isomorphism, since e is a primitive idempotent of $M_d(K)$, and $(ke)^* = ek^* = \bar{k}e$. Thus $e\mathbb{Q}[G]e$ is an imaginary quadratic field on which $*$ restricts to complex conjugation, and ke acts on $eH_1(X,\mathbb{Q})$ as k. By Corollary 3.8, the Hermitian form of the induced polarization is split. □

4. Two families of split eightfolds

The goal of this section is to construct, for $K = \mathbb{Q}(\omega)$, where $\omega := \exp(2\pi i/3)$, so that $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$, and for $K = \mathbb{Q}(i)$, a family of abelian eightfolds with split Hermitian forms such that the Weil plane of any member is algebraic (Construction-Theorems 4.4 and 4.5). Section 6 uses (1)–(3) for any member of both families, and Section 5 computes the rank of the period map at the boundary points of totally toric reduction in (4).

4.1. The Eisenstein family.

In this subsection, we construct the family for $K = \mathbb{Q}(\omega)$ from the Shephard–Todd group G_{34}.

Definition 4.1 (cf. [Völ96, Ch. 10])**.** Let G be a finite group, and let $(g_1, \ldots, g_k)$ be a tuple of elements of G which generates G and satisfies $g_1 \cdots g_k = 1$. For distinct points $p_1, \ldots, p_k \in \mathbb{P}^1$, a point $p_0 \in \mathbb{P}^1 \setminus \{p_1, \ldots, p_k\}$, and loops $\gamma_1, \ldots, \gamma_k$ at p_0 around $p_1, \ldots, p_k$ which generate $\pi_1(\mathbb{P}^1 \setminus \{p_1, \ldots, p_k\}, p_0)$ with the single relation $\gamma_1 \cdots \gamma_k = 1$, the *branch tuple* of a connected

G-cover $X \to \mathbb{P}^1$ branched at $p_1, \dots, p_k$ with respect to $\gamma_1, \dots, \gamma_k$ is the tuple of monodromies of $\gamma_1, \dots, \gamma_k$, which is determined by the cover up to simultaneous conjugation in G. A *Hurwitz component* is a connected component of the space of connected G-covers of $\mathbb{P}^1$ with ordered distinct branch points whose branch tuple with respect to some such loops is $(g_1, \dots, g_k)$.

Lemma 4.2. *Let G and $(g_1, \dots, g_k)$ be as in Definition 4.1.*

(1) For any distinct points $p_1, \dots, p_k \in \mathbb{P}^1$, there exists a connected G-cover $X \to \mathbb{P}^1$ branched at $p_1, \dots, p_k$ whose branch tuple with respect to some loops as in Definition 4.1 is $(g_1, \dots, g_k)$.

(2) Admissible G-covers over stable curves of genus 0 with k marked points form a smooth stack which is proper and quasi-finite over $\overline{M}_{0,k}$, and the G-covers of $\mathbb{P}^1$ with k ordered branch points are the admissible G-covers over smooth curves.

Proof. Item (1) is Riemann's existence theorem [Ful69, Section 1], and item (2) is [ACV03, Theorem 3.0.2, Corollary 3.0.5, Theorem 4.3.2]: by [ACV03, Theorem 4.3.2] admissible G-covers form the stack $\mathcal{B}^{\mathrm{bal}}_{0,k}(G)$ of balanced twisted G-covers, which is open and closed in the smooth stack $\mathcal{B}_{0,k}(G)$ [ACV03, §2.2, Theorem 3.0.2] (it is the locus $u = 0$). □

Definition 4.3 (cf. [ADF+20, §3.2])**.** In Set-up 3.5, let a G-cover $X \to \mathbb{P}^1$ degenerate to an admissible G-cover $X_0 \to C_0$ over a stable curve C_0 of genus 0, and let Γ be the dual graph of X_0, with the induced action of G. The *toric rank* of A_X at $X_0 \to C_0$ is $\dim_{\mathbb{Q}} eH_1(\Gamma, \mathbb{Q})$, and A_X has *totally toric reduction* at $X_0 \to C_0$ if the toric rank of A_X at $X_0 \to C_0$ equals $\dim A_X$.

The number of pairs of reflections is determined by the dimension. If the branch tuple of a G_{34}-cover $X \to \mathbb{P}^1$ consists of k pairs of reflections, then by the Chevalley–Weil formula [Can18, Theorem 3.1], the representation $\mathbb{C}^6$ of G_{34} occurs in $H^1(X, \mathbb{C})$ with multiplicity $2k - 12$, and A_X in Construction-Theorem 4.4 (3) is an abelian eightfold exactly when $2k - 12 = 8$, that is, $k = 10$. For a matrix M with entries in K, we write $M^* := \bar{M}^t$. For a unitary matrix $g \in G$, $g^* = g^{-1}$, so on G this is the involution $*$ of Set-up 3.5. In Construction-Theorems 4.4 and 4.5, $\epsilon\mathbb{Q}[G]$ is identified with $M_6(K)$, resp. $M_8(K)$, through the action of G on K^6, resp. K^8. A trivalent caterpillar is a trivalent tree whose internal vertices form a path, the inertia of a node of an admissible G-cover is the stabilizer of a point of the covering curve over the node, and the parameter of a node is the smoothing parameter q of the local equation $xy = q$ of the node.

Construction-Theorem 4.4. Let K, $r_1, \dots, r_{10}$, $t_1, \dots, t_{10}$, and G be as follows.

(i) $K = \mathbb{Q}(\omega)$, and K^6 carries the Hermitian form x^*y and the standard basis $b_1, \dots, b_6$.

(ii) The vectors $r_1, \dots, r_{10} \in K^6$ are

$$r_1 = b_2 - b_3,\ r_2 = b_3 - b_4,\ r_3 = b_1 - b_2,\ r_4 = b_1 - \omega^2 b_2,\ r_5 = (1,1,1,1,1,1),\ r_6 = b_4 - b_5,$$

$$r_7 = (1,1,1,1,\omega,\omega^2),\ r_8 = (1,\omega,1,\omega,1,\omega),\ r_9 = (1,\omega,\omega^2,1,\omega^2,\omega),\ r_{10} = (1,\omega^2,\omega^2,1,\omega^2,1).$$

(iii) $t_j := I - 2r_j r_j^*/(r_j^* r_j)$ for $1 \le j \le 10$.

(iv) $G := \langle t_1, \dots, t_6 \rangle$.

Then the following hold.

(1)
- G is the Shephard–Todd group G_{34}, of order $108 \cdot 9!$, and $t_7, \dots, t_{10} \in G$.
- The representation K^6 of G is absolutely irreducible with character field K, and the central idempotent $\epsilon \in \mathbb{Q}[G]$ of K^6 satisfies $\epsilon\mathbb{Q}[G] \cong M_6(K)$.
- The element $e \in \mathbb{Q}[G]$ which is zero in the other simple factors and corresponds to the matrix unit E_{11} in $M_6(K)$ is a primitive idempotent with $e^* = e$.

(2)
- The tuple $(t_1, t_1, t_2, t_2, \dots, t_{10}, t_{10})$ generates G and has product 1.
- For the Hurwitz component $\mathcal{H}$ of connected G-covers $X \to \mathbb{P}^1$ with branch tuple $(t_1, t_1, \dots, t_{10}, t_{10})$ and ordered distinct branch points, taken modulo the action of $\mathrm{PGL}_2(\mathbb{C})$ on $\mathbb{P}^1$, we have $\dim \mathcal{H} = 17$.
- Any X in $\mathcal{H}$ has genus $1 + 4|G|$.

(3) For any X in $\mathcal{H}$ and any positive integer m with $me \in \mathbb{Z}[G]$, $A_X := \mathrm{im}(me : J(X) \to J(X))$ is an abelian eightfold, the multiplicities of K on $H^{1,0}(A_X)$ are $(4,4)$, the Hermitian form of the polarization induced from $J(X)$ is isometric to $h_0 = H_K^{\oplus 4}$, and any class in $W_K(A_X)$ is algebraic.

(4)
- The closure of $\mathcal{H}$ in the stack of admissible G-covers over $\overline{M}_{0,20}$ contains an admissible G-cover over a stable curve of genus 0 whose dual graph is a trivalent caterpillar with ten leaves and eight internal vertices, where the leaf i carries the branch points of the pair (t_i, t_i), and whose seventeen nodes separate the leaf sets $\{i\}$ for $1 \le i \le 10$ and $\{1, \dots, j\}$ for $2 \le j \le 8$.
- The admissible G-cover over the stable curve with caterpillar dual graph has trivial node inertias, and all normalization components of the covering curve are rational.
- The seventeen node parameters can be smoothed independently in $\mathcal{H}$.
- A_X has totally toric reduction of toric rank 8 at the boundary point.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove (1).

Write $v = \sum_{k=1}^6 x_k b_k$ for $v \in K^6$. For any j, t_j is the unitary reflection of order 2 with $t_j r_j = -r_j$ which fixes the hyperplane $r_j^* v = 0$ pointwise, and a unitary reflection of order 2 is determined by the hyperplane which it fixes. For $j = 1, \dots, 6$, the equations $r_j^* v = 0$ are

$$x_2 - x_3 = 0, \quad x_3 - x_4 = 0, \quad x_1 - x_2 = 0,$$
$$x_1 - \omega x_2 = 0, \quad x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 0, \quad x_4 - x_5 = 0.$$

They are the hyperplanes of the six reflections which generate the group $[3\ 1; 2]^3$ in [ST54, §10, pp. 298–299]. By [ST54, §10, p. 294], $[3\ 1; 2]^3 = [2\ 1; 3]^3$, under which name it is no. 34 of [ST54, Table VII]. Thus $G = G_{34}$, and $|G| = 108 \cdot 9!$ [ST54, Table VII, no. 34].

By [AGMV08, p. 189], the reflection hyperplanes of G_{34} are the hyperplanes $x_j - \omega^c x_k = 0$ with $j \ne k$ and $c \in \mathbb{Z}$, and the hyperplanes

$$\sum_{k=1}^6 \omega^{c_k} x_k = 0 \quad \text{with } c_1, \dots, c_6 \in \mathbb{Z} \text{ and } \sum_{k=1}^6 c_k \equiv 0 \pmod 3.$$

The six hyperplanes above belong to the list of [AGMV08, p. 189]. Hence G is contained in the reflection group of [AGMV08, p. 189]. Since both groups have order $108 \cdot 9!$, they are equal, and the reflection hyperplanes of G are the hyperplanes of [AGMV08, p. 189]. For $j = 7, 8, 9, 10$, the coordinates of r_j are $\omega^{c_1}, \dots, \omega^{c_6}$ with $\sum_k c_k = 3, 3, 6, 6$ respectively, and the equation $r_j^* v = 0$ has coefficients $\omega^{-c_1}, \dots, \omega^{-c_6}$, whose exponents also have sum divisible by 3. Thus $t_7, \dots, t_{10} \in G$.

The group G_{34} is an irreducible reflection group in U_6 [ST54, Table VII, no. 34], so the representation K^6 is absolutely irreducible. By [ST54, §10, pp. 298–299], the characteristic polynomial of $t_5 t_4 t_3 t_1 t_2 t_6$ is

$$z^6 - \omega z^5 + \omega^2 z^4 - z^3 + \omega z^2 - \omega^2 z + 1.$$

All matrices in G have entries in K, and the trace of $t_5 t_4 t_3 t_1 t_2 t_6$ is $\omega \notin \mathbb{Q}$. Hence the character field of K^6 is K (cf. [MM10, Section 1 and Table 1]).

The algebra $\epsilon\mathbb{Q}[G]$ is simple, and the center of $\epsilon\mathbb{Q}[G]$ is the character field K. Since K^6 realizes the representation over K, the action of G on K^6 induces a K-algebra homomorphism

$$\epsilon\mathbb{Q}[G] \to \mathrm{End}_K(K^6) = M_6(K),$$

which is injective since $\epsilon\mathbb{Q}[G]$ is simple. By absolute irreducibility, the K-span of G in $M_6(K)$ is $M_6(K)$. Thus both sides have K-dimension 36, and $\epsilon\mathbb{Q}[G] \cong M_6(K)$.

Any $g \in G$ is unitary, so the homomorphism $\mathbb{Q}[G] \to M_6(K)$ sends x^* to the conjugate transpose of the image of x. In particular, the kernel of $\mathbb{Q}[G] \to M_6(K)$ is stable under $*$, and $\epsilon^* = \epsilon$. Since E_{11} is a primitive idempotent of $M_6(K)$ and $E_{11}^* = E_{11}$, e is a primitive idempotent, and e^* is an element of $\epsilon\mathbb{Q}[G]$ which corresponds to E_{11}. Thus $e^* = e$.

Step 2. In this step, we prove (2).

Any t_j is an involution, so $(t_1, t_1, \dots, t_{10}, t_{10})$ has product 1. It generates G since $t_1, \dots, t_6$ generate G. By Lemma 4.2, $\mathcal{H}$ is non-empty and finite over the space of twenty ordered distinct points of $\mathbb{P}^1$ modulo $\mathrm{PGL}_2(\mathbb{C})$. Thus $\dim \mathcal{H} = 20 - 3 = 17$.

Let X be in $\mathcal{H}$. The cover $X \to \mathbb{P}^1$ has degree $|G|$, and over any of the twenty branch points it has $|G|/2$ points, each with ramification index 2. By the Hurwitz formula [Har77, Ch. IV, Corollary 2.4],

$$2g(X) - 2 = -2|G| + 20 \cdot \frac{|G|}{2} = 8|G|,$$

so $g(X) = 1 + 4|G|$.

Step 3. In this step, we prove (3).

Let X be in $\mathcal{H}$. Let $V = \mathbb{C}^6$, on which G acts through the matrices t_j, and let $\bar{V}$ be the complex conjugate representation. Since V is not the trivial representation and every local monodromy is an involution, the Chevalley–Weil formula [Can18, Theorem 3.1] for $X \to \mathbb{P}^1$ says that V occurs in $H^{1,0}(X)$ with multiplicity $-\dim V + \frac{1}{2}\sum_x \dim \ker(t_x + 1)$, the sum running over the branch points, where t_x is the local monodromy at x. Any t_j has a one-dimensional (-1)-eigenspace on V, so

$$-\dim V + \frac{1}{2}\sum_x \dim \ker(t_x + 1) = -6 + 20 \cdot \frac{1}{2} = 4.$$

The Chevalley–Weil formula for $\bar{V}$ also yields 4. Since $H^{0,1}(X) = \overline{H^{1,0}(X)}$, both V and $\bar{V}$ occur in $H^1(X, \mathbb{C})$ with multiplicity 8, and hence also in $H_1(X, \mathbb{C})$. The character field K is not real, so $V \not\cong \bar{V}$. By (1), e acts on V and on $\bar{V}$ as an idempotent of rank 1 and annihilates any other irreducible representation of G. Thus $\dim_{\mathbb{C}} eH_1(X, \mathbb{C}) = 16$, and $\dim A_X = 8$.

The stabilizer of a point of X over a branch point with local monodromy t_j is conjugate to $\langle t_j \rangle$, and $t_j = t_j^{-1}$. Thus the classes $[t_x]$ of Definition 3.4 are $[t_1], [t_1], \dots, [t_{10}], [t_{10}]$, and the action of G on X has paired branching. By (1), X, ϵ, e, and m satisfy the conditions of Set-up 3.5, and $e^* = e$. By Theorem 3.11, the multiplicities of K on $H^{1,0}(A_X)$ are $(4, 4)$, any class in $W_K(A_X)$ is algebraic, and the polarization of A_X induced from $J(X)$ is a compatible polarization of (A_X, K) whose Hermitian form h is split.

It remains to show that h is isometric to h_0. By Definition 2.9, $H_1(A_X, \mathbb{Q})$ contains a K-subspace L with $\dim_K L = 4$ and $h|_L = 0$. Let $x_1, \dots, x_4$ be a K-basis of L. Since h is non-degenerate and $\dim_K H_1(A_X, \mathbb{Q}) = 8$, there exist $y_1, \dots, y_4 \in H_1(A_X, \mathbb{Q})$ such that $h(x_j, y_k)$ is 1 if $j = k$ and 0 otherwise. Replacing y_k with $y_k - \frac{1}{2}\sum_{l=1}^4 h(y_l, y_k)x_l$, we may also assume that $h(y_j, y_k) = 0$ for any j, k. Then $x_1, y_1, \dots, x_4, y_4$ is a K-basis of $H_1(A_X, \mathbb{Q})$ in which the Gram matrix of h is $h_0 = H_K^{\oplus 4}$.

Step 4. In this step, we prove (4).

Let C_0 be a stable curve of genus 0 with twenty marked points [Knu83, Sections 1–2] whose components are $L_1, \dots, L_{10}$ and $M_1, \dots, M_8$, all isomorphic to $\mathbb{P}^1$, such that $M_{k-1} \cap M_k \neq \emptyset$ for $2 \leq k \leq 8$, both L_1 and L_2 intersect M_1, $L_{k+1} \cap M_k \neq \emptyset$ for $2 \leq k \leq 7$, both L_9 and L_{10} intersect M_8, and L_i carries the marked points of the pair (t_i, t_i). The dual graph of C_0 is a trivalent caterpillar with ten leaves and eight internal vertices. The node $L_i \cap M_k$ separates $\{i\}$, and for $2 \leq j \leq 8$, the node $M_{j-1} \cap M_j$ separates $\{1, \dots, j\}$.

We construct an admissible G-cover $X_0 \to C_0$ as follows. The local monodromies at the marked points of L_i have product $t_i^2 = 1$, so the cover is unramified over the node of L_i. Over M_k, we take $|G|$ copies of $\mathbb{P}^1$ mapping isomorphically to M_k, labelled by G. Over L_i, we take, for any coset $g\langle t_i \rangle \in G/\langle t_i \rangle$, a copy P_g of the double cover $\mathbb{P}^1 \to L_i$ branched at the marked points of L_i, and we label the points of P_g over the node of L_i by g and gt_i. We glue the points with equal labels over any node of C_0, and we let G act on the labels by left multiplication. Then $X_0 \to C_0$ is an admissible G-cover with local monodromy t_i at the marked points of L_i, all node inertias are trivial, and all normalization components of X_0 are rational. The curve X_0 is

connected since $t_1, \dots, t_{10}$ generate G. It has $8|G| + 10 \cdot |G|/2 = 13|G|$ components and $17|G|$ nodes, so the arithmetic genus of X_0 is $17|G| - 13|G| + 1 = 1 + 4|G|$.

Since all node inertias are trivial, $X_0 \to C_0$ is étale over a neighborhood of any node of C_0, and the plumbing $xy = q$ of a node of C_0 lifts to the plumbing $xy = q$ at any node of X_0 over it. Thus the seventeen node parameters can be smoothed independently and G-equivariantly (Lemma 4.2 (2)). After adding a level structure which kills automorphisms, the Hurwitz space of G-covers carries a universal family of curves [RW06, Section 4], and the smoothings of $X_0 \to C_0$ are fibers of the universal family. Since the planar embedding of the caterpillar above has leaf order $(1, \dots, 10)$, the smooth fibers of the smoothing are G-covers of $\mathbb{P}^1$ with branch tuple $(t_1, t_1, \dots, t_{10}, t_{10})$, and they belong to $\mathcal{H}$. Thus $X_0 \to C_0$ is in the closure of $\mathcal{H}$.

Let Γ be the dual graph of X_0, and let $C_1(\Gamma, \mathbb{Q})$ and $C_0(\Gamma, \mathbb{Q})$ be the $\mathbb{Q}[G]$-modules of 1-chains and 0-chains of Γ, where we orient the edges of Γ over a node of C_0 compatibly with an orientation of that node. Since Γ is connected, we have the equality of virtual representations

$$[H_1(\Gamma, \mathbb{Q})] = [C_1(\Gamma, \mathbb{Q})] - [C_0(\Gamma, \mathbb{Q})] + [\mathbb{Q}].$$

The edges of Γ over a node of C_0 form a free G-orbit, the vertices over M_k form a free G-orbit, and the vertices over L_i form the G-set $G/\langle t_i \rangle$. Thus

$$C_1(\Gamma, \mathbb{Q}) \cong \mathbb{Q}[G]^{\oplus 17}, \qquad C_0(\Gamma, \mathbb{Q}) \cong \mathbb{Q}[G]^{\oplus 8} \oplus \bigoplus_{i=1}^{10} \mathbb{Q}[G/\langle t_i \rangle].$$

The representation V occurs with multiplicity 6 in $\mathbb{C}[G]$ and with multiplicity $\dim V^{t_i} = 5$ in $\mathbb{C}[G/\langle t_i \rangle]$ by Frobenius reciprocity, and it does not occur in $\mathbb{C}$. Hence V occurs in $H_1(\Gamma, \mathbb{C})$ with multiplicity

$$17 \cdot 6 - 8 \cdot 6 - 10 \cdot 5 = 4,$$

and $\bar{V}$ also occurs with multiplicity 4. As in **Step 3**, e acts on V and on $\bar{V}$ as an idempotent of rank 1 and annihilates any other irreducible representation. Thus

$$\dim_{\mathbb{Q}} eH_1(\Gamma, \mathbb{Q}) = 8 = \dim A_X,$$

and A_X has totally toric reduction of toric rank 8 at $X_0 \to C_0$ (Definition 4.3). □

The reflections $t_1, \dots, t_6$ in Construction-Theorem 4.4 generate G. Apart from $t_7, \dots, t_{10} \in G$ in (1), the proof of (1)–(4) of Construction-Theorem 4.4 uses only that $t_7, \dots, t_{10}$ are reflections in G, and the vectors $r_7, \dots, r_{10}$ themselves are used only in the computation of a span in Proposition 5.4.

4.2. **The Gaussian family.** In this subsection, we construct the family for $K = \mathbb{Q}(i)$ from the Weyl group $W(E_8)$. The representation of $W(E_8)$ on $\mathbb{Q}^8$ is real, and we add $t_9 = i\Omega$ to make the character field equal to $\mathbb{Q}(i)$.

Construction-Theorem 4.5. Let K, $a_1, \dots, a_8$, $t_1, \dots, t_9$, Ω, and G be as follows.

(i) $K = \mathbb{Q}(i)$, and K^8 carries the Hermitian form x^*y and the standard basis $b_1, \dots, b_8$.

(ii) The vectors $a_1, \dots, a_8 \in \mathbb{Q}^8 \subset K^8$ are

$$a_1 = \tfrac{1}{2}(b_1 + b_8 - b_2 - b_3 - b_4 - b_5 - b_6 - b_7),\ a_2 = b_1 + b_2,\ a_3 = -b_1 + b_2,\ a_j = -b_{j-2} + b_{j-1}\ (4 \le j \le 8).$$

(iii) $t_j := I - a_j a_j^t$ for $1 \le j \le 8$, $\Omega \in M_8(\mathbb{Z})$ is the block-diagonal matrix with four diagonal blocks $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and $t_9 := i\Omega$.

(iv) $G := \langle t_1, \dots, t_9 \rangle$.

Then the following hold.

(1)
- $\langle t_1, \dots, t_8 \rangle$ is the Weyl group $W(E_8)$, $\Omega \in W(E_8)$, and $G = \mu_4 W(E_8)$.
- The representation K^8 of G is absolutely irreducible with character field K, and the central idempotent $\epsilon \in \mathbb{Q}[G]$ of K^8 satisfies $\epsilon\mathbb{Q}[G] \cong M_8(K)$.

- The element $e \in \mathbb{Q}[G]$ which is zero in the other simple factors and corresponds to E_{11} in $M_8(K)$ is a primitive idempotent with $e^* = e$.

(2)
- The tuple $(t_1, t_1, \dots, t_9, t_9)$ generates G and has product 1.
- For the Hurwitz component $\mathcal{H}$ of connected G-covers $X \to \mathbb{P}^1$ with branch tuple $(t_1, t_1, \dots, t_9, t_9)$ and ordered distinct branch points, taken modulo the action of $\mathrm{PGL}_2(\mathbb{C})$ on $\mathbb{P}^1$, we have $\dim \mathcal{H} = 15$.
- Any X in $\mathcal{H}$ has genus $1 + 7|G|/2$.

(3) For any X in $\mathcal{H}$ and any positive integer m with $me \in \mathbb{Z}[G]$, $A_X := \mathrm{im}(me : J(X) \to J(X))$ is an abelian eightfold, the multiplicities of K on $H^{1,0}(A_X)$ are $(4, 4)$, the Hermitian form of the polarization induced from $J(X)$ is isometric to $h_0 = H_K^{\oplus 4}$, and any class in $W_K(A_X)$ is algebraic.

(4)
- For any trivalent tree with nine labeled leaves, the leaf j carrying the branch points of the pair (t_j, t_j), the closure of $\mathcal{H}$ in the stack of admissible G-covers over $\overline{M}_{0,18}$ contains an admissible G-cover over the corresponding stable curve of genus 0, with trivial node inertias and fifteen node parameters which can be smoothed independently in $\mathcal{H}$.
- For the caterpillar with leaf order $(1, 3, 4, 2, 5, 6, 7, 8, 9)$, whose fifteen nodes separate the nine singleton leaf sets and, for $2 \le k \le 7$, the set of the first k leaves in the order $(1, 3, 4, 2, 5, 6, 7, 8, 9)$, all normalization components of the covering curve are rational, and A_X has totally toric reduction of toric rank 8 at the boundary point.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove (1).

The vectors $a_1, \dots, a_8$ are the simple roots of the root system E_8 in $\mathbb{R}^8$ in the numbering of [Bou02, Ch. VI, Plate VII], and the roots of E_8 are the vectors $\pm b_j \pm b_k$ with $j < k$ and the vectors $\frac{1}{2}\sum_{k=1}^8 \pm b_k$ with an even number of minus signs [Bou02, Ch. VI, Plate VII]. Since $a_j^t a_j = 2$, t_j is the reflection $x \to x - (a_j^t x)a_j$ in a_j for $1 \le j \le 8$. Thus $\langle t_1, \dots, t_8 \rangle = W(E_8)$ [Bou02, Ch. VI, §1.5], and $W(E_8)$ contains the reflection in any root, since by definition $W(E_8)$ is the group generated by the reflections in the roots.

For $j \ne k$, the reflection in $b_j - b_k$ exchanges the coordinates j and k, and the product of the reflections in $b_j - b_k$ and in $b_j + b_k$ changes the signs of the coordinates j and k. Hence $W(E_8)$ contains any signed permutation matrix with an even number of entries -1. Any diagonal block of Ω is a transposition followed by one change of sign, so Ω is a signed permutation matrix with four entries -1, and $\Omega \in W(E_8)$. Thus $iI = t_9\Omega^{-1} \in G$, and G is generated by $W(E_8)$ and iI. Since iI is central of order 4, $G = \mu_4 W(E_8)$.

Let $P_j := (I - t_j)/2 = a_j a_j^t/2$ for $1 \le j \le 8$, which is the image of an element of $\mathbb{Q}[W(E_8)]$. If j and k are adjacent in the Dynkin diagram of E_8, then $a_j^t a_k = -1$. Hence, for any path $j = j_0, j_1, \dots, j_l = k$ in the Dynkin diagram,

$$P_{j_0} P_{j_1} \cdots P_{j_l} = \frac{(-1)^l}{2^{l+1}} a_j a_k^t.$$

The Dynkin diagram is connected, and $a_1, \dots, a_8$ is a basis of $\mathbb{Q}^8$, so the 64 matrices $a_j a_k^t$ form a $\mathbb{Q}$-basis of $M_8(\mathbb{Q})$. Thus the image of $\mathbb{Q}[W(E_8)]$ in $M_8(\mathbb{Q})$ is $M_8(\mathbb{Q})$, and the representation K^8 of G is absolutely irreducible. The image of $\mathbb{Q}[G]$ in $M_8(K)$ contains $M_8(\mathbb{Q})$ and iI, so it is $M_8(K)$. The trace of iI is $8i$, so the character field of K^8 is K. Since $M_8(K)$ is simple, the surjection $\mathbb{Q}[G] \to M_8(K)$ restricts to an isomorphism $\epsilon\mathbb{Q}[G] \cong M_8(K)$.

Any $g \in G$ is unitary, so the surjection $\mathbb{Q}[G] \to M_8(K)$ sends x^* to the conjugate transpose of the image of x. In particular, the kernel of $\mathbb{Q}[G] \to M_8(K)$ is stable under $*$, and $\epsilon^* = \epsilon$. Since E_{11} is a primitive idempotent of $M_8(K)$ and $E_{11}^* = E_{11}$, e is a primitive idempotent, and e^* is an element of $\epsilon\mathbb{Q}[G]$ which corresponds to E_{11}. Thus $e^* = e$.

Step 2. In this step, we prove (2).

For $1 \le j \le 8$, t_j is a reflection, so $t_j^2 = I$. Since $\Omega^2 = -I$, $t_9^2 = -\Omega^2 = I$. Thus $(t_1, t_1, \dots, t_9, t_9)$ has product 1, and it generates G. By Lemma 4.2, $\mathcal{H}$ is non-empty and finite over the space of eighteen ordered distinct points of $\mathbb{P}^1$ modulo $\mathrm{PGL}_2(\mathbb{C})$. Thus $\dim \mathcal{H} = 18 - 3 = 15$.

Let X be in $\mathcal{H}$. Over any of the eighteen branch points, the cover $X \to \mathbb{P}^1$ of degree $|G|$ has $|G|/2$ points, each with ramification index 2. By the Hurwitz formula [Har77, Ch. IV, Corollary 2.4],

$$2g(X) - 2 = -2|G| + 18 \cdot \frac{|G|}{2} = 7|G|,$$

so $g(X) = 1 + 7|G|/2$.

Step 3. In this step, we prove (3).

Let X be in $\mathcal{H}$. Let $V = \mathbb{C}^8$, on which G acts through the matrices t_j, and let $\bar{V}$ be the complex conjugate representation. For $1 \le j \le 8$, the (-1)-eigenspace of t_j on V is $\mathbb{C}a_j$. The matrix t_9 is Hermitian and unitary with trace 0, and the (-1)-eigenspace of t_9 on V is spanned by the four vectors $b_{2k-1} - ib_{2k}$ for $1 \le k \le 4$. Since V is not the trivial representation, the Chevalley–Weil formula of **Step 3** of the proof of Construction-Theorem 4.4 shows that V occurs in $H^{1,0}(X)$ with multiplicity

$$-8 + 16 \cdot \frac{1}{2} + 2 \cdot \left(4 \cdot \frac{1}{2}\right) = 4.$$

The Chevalley–Weil formula for $\bar{V}$ also yields 4. Since $H^{0,1}(X) = \overline{H^{1,0}(X)}$, both V and $\bar{V}$ occur in $H_1(X, \mathbb{C})$ with multiplicity 8, and $V \ncong \bar{V}$ since the character field K is not real. By (1), e acts on V and on $\bar{V}$ as an idempotent of rank 1 and annihilates any other irreducible representation of G. Thus $\dim_{\mathbb{C}} eH_1(X, \mathbb{C}) = 16$, and $\dim A_X = 8$.

The stabilizer of a point of X over a branch point with local monodromy t_j is conjugate to $\langle t_j \rangle$, and $t_j = t_j^{-1}$. Thus the classes $[t_x]$ of Definition 3.4 are $[t_1], [t_1], \dots, [t_9], [t_9]$, and the action of G on X has paired branching. By (1), X, ϵ, e, and m satisfy the conditions of Set-up 3.5, and $e^* = e$. By Theorem 3.11, the multiplicities of K on $H^{1,0}(A_X)$ are $(4, 4)$, any class in $W_K(A_X)$ is algebraic, and the polarization of A_X induced from $J(X)$ is a compatible polarization of (A_X, K) whose Hermitian form h is split.

It remains to show that h is isometric to h_0. By Definition 2.9, $H_1(A_X, \mathbb{Q})$ contains a K-subspace L with $\dim_K L = 4$ and $h|_L = 0$. Let $x_1, \dots, x_4$ be a K-basis of L. Since h is non-degenerate and $\dim_K H_1(A_X, \mathbb{Q}) = 8$, there exist $y_1, \dots, y_4 \in H_1(A_X, \mathbb{Q})$ such that $h(x_j, y_k)$ is 1 if $j = k$ and 0 otherwise. Replacing y_k with $y_k - \frac{1}{2}\sum_{l=1}^4 h(y_l, y_k)x_l$, we may also assume that $h(y_j, y_k) = 0$ for any j, k. Then $x_1, y_1, \dots, x_4, y_4$ is a K-basis of $H_1(A_X, \mathbb{Q})$ in which the Gram matrix of h is $h_0 = H_K^{\oplus 4}$.

Step 4. In this step, we prove (4).

Let $\mathcal{T}$ be a trivalent tree with nine labeled leaves, and let C_0 be a stable curve of genus 0 with eighteen marked points [Knu83, Sections 1–2] whose dual graph is $\mathcal{T}$, in which the component of the leaf j carries the marked points of the pair (t_j, t_j). The tree $\mathcal{T}$ has seven internal vertices and fifteen edges. We construct an admissible G-cover $X_0 \to C_0$ as follows. Let L_j be the component of C_0 of the leaf j. The local monodromies at the marked points of L_j have product $t_j^2 = 1$, so the cover is unramified over the node of L_j. Over an internal component, we take $|G|$ copies of $\mathbb{P}^1$ mapping isomorphically to it, labelled by G. Over L_j, we take, for any coset $g\langle t_j \rangle \in G/\langle t_j \rangle$, a copy P_g of the double cover $\mathbb{P}^1 \to L_j$ branched at the marked points of L_j, and we label the points of P_g over the node of L_j by g and gt_j. We glue the points with equal labels over any node of C_0, and we let G act on the labels by left multiplication. Then $X_0 \to C_0$ is an admissible G-cover with local monodromy t_j at the marked points of L_j, all node inertias are trivial, and all normalization components of X_0 are rational. The curve X_0 is connected since $t_1, \dots, t_9$ generate G. It has $7|G| + 9 \cdot |G|/2 = 23|G|/2$ components and $15|G|$ nodes, so the arithmetic genus of X_0 is $15|G| - 23|G|/2 + 1 = 1 + 7|G|/2$.

Since all node inertias are trivial, $X_0 \to C_0$ is étale over a neighborhood of any node of C_0, and the plumbing $xy = q$ of a node of C_0 lifts to the plumbing $xy = q$ at any node of X_0 over it. Thus the fifteen node parameters can be smoothed independently and G-equivariantly (Lemma 4.2 (2)). After adding a level structure which kills automorphisms, the Hurwitz space of G-covers carries a universal family of curves [RW06, Section 4], and the smoothings of $X_0 \to C_0$ are fibers of the universal family. For a planar embedding of $\mathcal{T}$ with leaf order $(j_1, \dots, j_9)$, the smooth fibers of the smoothing are G-covers of $\mathbb{P}^1$ whose tuple of monodromies with respect to loops as in Definition 4.1 around the branch points, taken in the leaf order, is $(t_{j_1}, t_{j_1}, \dots, t_{j_9}, t_{j_9})$.

We show that such a G-cover belongs to $\mathcal{H}$. The braid group acts on branch tuples by Hurwitz moves, and the G-covers obtained by moving the branch points along a braid belong to one connected component [Völ96, Ch. 10]. The Hurwitz move at the position k sends the entries (g, g') at the positions k and $k+1$ to $(gg'g^{-1}, g)$. For two involutions $s, s' \in G$, the Hurwitz moves at the positions $2, 1, 3, 2$ send

$$(s, s, s', s') \to (s, ss's, s, s') \to (s', s, s, s') \to (s', s, ss's, s) \to (s', s', s, s).$$

Thus any two adjacent pairs of $(t_{j_1}, t_{j_1}, \dots, t_{j_9}, t_{j_9})$ can be exchanged, and the branch points of any pair keep their order. Moreover, for an involution $s \in G$, the Hurwitz move at the first position of a pair (s, s) exchanges the branch points of the pair and does not change the tuple. Since any permutation of the nine pairs is a product of exchanges of adjacent pairs, Hurwitz moves of both kinds transform the tuple $(t_{j_1}, t_{j_1}, \dots, t_{j_9}, t_{j_9})$ with respect to loops around the branch points in the leaf order into the tuple $(t_1, t_1, \dots, t_9, t_9)$ with respect to loops as in Definition 4.1. Hence the smooth fibers of the smoothing belong to $\mathcal{H}$, and $X_0 \to C_0$ is in the closure of $\mathcal{H}$.

Now let $\mathcal{T}$ be the caterpillar with leaf order $(1, 3, 4, 2, 5, 6, 7, 8, 9)$. All normalization components of X_0 are rational by construction. Let Γ be the dual graph of X_0, and let $C_1(\Gamma, \mathbb{Q})$ and $C_0(\Gamma, \mathbb{Q})$ be the $\mathbb{Q}[G]$-modules of 1-chains and 0-chains of Γ, where we orient the edges of Γ over a node of C_0 compatibly with an orientation of that node. Since Γ is connected, we have the equality of virtual representations

$$[H_1(\Gamma, \mathbb{Q})] = [C_1(\Gamma, \mathbb{Q})] - [C_0(\Gamma, \mathbb{Q})] + [\mathbb{Q}].$$

The edges of Γ over a node of C_0 form a free G-orbit, the vertices over an internal component form a free G-orbit, and the vertices over L_j form the G-set $G/\langle t_j \rangle$. Thus

$$C_1(\Gamma, \mathbb{Q}) \cong \mathbb{Q}[G]^{\oplus 15}, \qquad C_0(\Gamma, \mathbb{Q}) \cong \mathbb{Q}[G]^{\oplus 7} \oplus \bigoplus_{j=1}^{9} \mathbb{Q}[G/\langle t_j \rangle].$$

The representation V occurs with multiplicity 8 in $\mathbb{C}[G]$ and with multiplicity $\dim V^{t_j}$ in $\mathbb{C}[G/\langle t_j \rangle]$ by Frobenius reciprocity, which is 7 for $1 \le j \le 8$ and 4 for $j = 9$, and it does not occur in $\mathbb{C}$. Hence V occurs in $H_1(\Gamma, \mathbb{C})$ with multiplicity

$$15 \cdot 8 - 7 \cdot 8 - (8 \cdot 7 + 4) = 4,$$

and $\bar{V}$ also occurs with multiplicity 4. As in **Step 3**, e acts on V and on $\bar{V}$ as an idempotent of rank 1 and annihilates any other irreducible representation. Thus

$$\dim_{\mathbb{Q}} eH_1(\Gamma, \mathbb{Q}) = 8 = \dim A_X,$$

and A_X has totally toric reduction of toric rank 8 at $X_0 \to C_0$ (Definition 4.3). □

5. Period rank of the Eisenstein and Gaussian families

In this section we show that the local period maps of the Eisenstein and Gaussian families of Section 4 have rank at least 15, which is used in Section 6. We obtain the lower bounds from the period expansion at a caterpillar boundary point, starting with the logarithmic monodromy forms.

Definition-Lemma 5.1. Let $(K, \mathcal{H})$ be $(\mathbb{Q}(\omega), \mathcal{H})$ of Construction-Theorem 4.4 or $(\mathbb{Q}(i), \mathcal{H})$ of Construction-Theorem 4.5, and let $\mathcal{U} \subset \mathcal{H}$ be a simply connected open subset. Choose K-linear isometries ϕ_X from the Hermitian forms of A_X to (K^8, h_0), for $X \in \mathcal{U}$, which are locally constant with respect to the local system $\{H_1(A_X, \mathbb{Q})\}_{X\in\mathcal{U}}$. The *local period map* on $\mathcal{U}$ sends $X \in \mathcal{U}$ to the period of A_X with respect to ϕ_X (Definition-Proposition 2.12). Two local period maps on $\mathcal{U}$ differ by the action of an element of $\mathrm{U}(h_0)(\mathbb{Q})$. In particular, their differentials have equal rank at any point of $\mathcal{U}$.

Proof. The field K acts on $H_1(A_X, \mathbb{Q})$ through the center of $\epsilon\mathbb{Q}[G]$ (Set-up 3.5), and the Riemann form of A_X is induced from $J(X)$, so the Hermitian form of A_X is locally constant on $\mathcal{U}$. By Construction-Theorem 4.4 (3), resp. Construction-Theorem 4.5 (3), there exists a K-linear isometry at one point of $\mathcal{U}$, and since $\mathcal{U}$ is simply connected, it extends to a locally constant family $(\phi_X)_{X\in\mathcal{U}}$ of K-linear isometries. If $(\phi'_X)_{X\in\mathcal{U}}$ is another such family, then $\phi'_X \circ \phi_X^{-1}$ is a K-linear isometry of (K^8, h_0) which is locally constant in X, hence equal to a constant $\theta \in \mathrm{U}(h_0)(\mathbb{Q})$. Thus the period of A_X with respect to ϕ'_X is the image under θ of the period with respect to ϕ_X. Since θ acts on $\mathbb{D}(h_0)$ by a biholomorphism, the differentials have equal rank. □

5.1. **Monodromy forms at boundary points.**

Definition-Proposition 5.2. Let $(K, G, e, \mathcal{H})$ be either $(\mathbb{Q}(\omega), G, e, \mathcal{H})$ of Construction-Theorem 4.4, with the caterpillar of (4) there, or $(\mathbb{Q}(i), G, e, \mathcal{H})$ of Construction-Theorem 4.5, with the caterpillar of (4) there. For an edge of the caterpillar with node parameter q_I, where I is the set of leaves cut off by the edge, let T_I be the local monodromy around $q_I = 0$ on $H_1(A_X, \mathbb{Q})$, let $\nu_I := T_I - 1$, and let ψ be the Riemann form of A_X. The bilinear form $\psi(x, \nu_I y)$ on $H_1(A_X, \mathbb{Q})$ factors through the quotient $eH_1(\Gamma, \mathbb{Q})$, where Γ is the dual graph of the covering nodal curve, and we call the induced form B_I on $eH_1(\Gamma, \mathbb{Q})$ the *logarithmic monodromy form* at q_I. In the first case let $R := [r_1, \ldots, r_{10}]$, let R_0 be the matrix formed by the first 6 columns of R, and let $c := |G|/6$. In the second case let $u_k := b_{2k-1} - ib_{2k}$ for $1 \le k \le 4$, let $R := [a_1, a_3, a_4, a_2, a_5, a_6, a_7, a_8, u_1, u_2, u_3, u_4]$, let R_0 be the matrix formed by the first 8 columns of R, let $c := |G|/8$, and number the leaves of the caterpillar, and of any trivalent tree with nine leaves, so that the leaf m carries the branch points of the pair $(t_{\pi(m)}, t_{\pi(m)})$, where $(\pi(1), \ldots, \pi(9)) = (1, 3, 4, 2, 5, 6, 7, 8, 9)$. Write $R = [R_0, R_1]$, and let

$$N := \begin{pmatrix} -R_0^{-1}R_1 \\ I_4 \end{pmatrix},$$

so that $\ker R = NK^4$. For a set I of leaves, let P_I be the diagonal projector onto the columns of R attached to the leaves in I, where in the second case the ninth leaf is attached to the four columns $u_1, \ldots, u_4$, and let

$$Q_I := (RP_IN)^*(RP_IN) \in \mathrm{Herm}_4(K),$$

where $\mathrm{Herm}_4(K)$ is the 16-dimensional $\mathbb{Q}$-vector space of 4×4 Hermitian matrices over K. Then there exists an isomorphism of $\mathbb{Q}$-vector spaces between $eH_1(\Gamma, \mathbb{Q})$ and $\ker R$ under which

$$B_I(Nv, Nw) = \mathrm{Tr}_{K/\mathbb{Q}}(cv^*Q_Iw)$$

for any $v, w \in K^4$. In the second case, for the boundary point of any trivalent tree $\mathcal{T}$ of Construction-Theorem 4.5 (4), with Γ the dual graph of the covering nodal curve over $\mathcal{T}$, there also exists an isomorphism of $\mathbb{Q}$-vector spaces between $eH_1(\Gamma, \mathbb{Q})$ and $\ker R$ under which the logarithmic monodromy form B_I at the node parameter of the edge cutting off any set I of leaves satisfies $B_I(Nv, Nw) = \mathrm{Tr}_{K/\mathbb{Q}}(cv^*Q_Iw)$ for any $v, w \in K^4$.

Proof. The proof is long, so we divide it into several steps.

Let $n := 6$ in the first case and $n := 8$ in the second case, and let $V := K^n$ with the action of G by matrices. The matrix R_0 is invertible. In the first case, with $r_1, \ldots, r_6$ as in Construction-Theorem 4.4 (ii), coordinates 6, 5, 4, and 3 show that a K-linear relation among $r_1, \ldots, r_6$ has

zero coefficients at r_5, r_6, r_2, and r_1, and then $1 \neq \omega^2$ shows that the coefficients at r_3 and r_4 vanish. In the second case, a_1 is the only column of R_0 with a non-zero eighth coordinate, and $a_2, \dots, a_8$ are linearly independent in the span of $b_1, \dots, b_7$, because by the coordinates of a_j in Construction-Theorem 4.5 (ii), each of $a_2, \dots, a_8$ belongs to $\mathrm{Span}_K\{b_1, \dots, b_7\}$, and $b_1 = \frac{1}{2}(a_2 - a_3)$, $b_2 = \frac{1}{2}(a_2 + a_3)$, and $b_{j-1} = a_j + b_{j-2}$ for $4 \leq j \leq 8$ belong to $\mathrm{Span}_K\{a_2, \dots, a_8\}$.

Let $\mathcal{T}$ be the caterpillar in the first case, and the caterpillar or any trivalent tree of Construction-Theorem 4.5 (4) in the second case. For a leaf m of $\mathcal{T}$, let $\eta_m := t_m$ in the first case and $\eta_m := t_{\pi(m)}$ in the second case, and let $E_m := \ker(\eta_m + 1) \subset V$. The columns of R attached to the leaf m form a K-basis of E_m, because t_j is the identity on $r_j^{\perp}$, resp. $a_j^{\perp}$, and sends r_j to $-r_j$, resp. a_j to $-a_j$, for any reflection t_j. For $t_9 = i\Omega$, we have $\Omega u_k = iu_k$, so $t_9 u_k = -u_k$ for $1 \leq k \leq 4$, and $\dim_K E_9 = 4$ because $\Omega^2 = -I$ and Ω is real.

Step 1. In this step, we describe Γ.

All node inertias are trivial by Construction-Theorem 4.4 (4), resp. Construction-Theorem 4.5 (4). Thus the cover is unramified over any internal component of the base curve, which has no branch points, so the cover consists of $|G|$ copies of $\mathbb{P}^1$ over it. Over the component of the leaf m, which carries two branch points with inertia $\langle \eta_m \rangle$ and one node, the cover consists of $|G|/2$ rational double covers of $\mathbb{P}^1$ branched over the branch points of the leaf m, indexed by $G/\langle \eta_m \rangle$.

Hence, after labelling the sheets over the internal components by G, so that G acts by left multiplication and the nodes over the internal edges of $\mathcal{T}$ join sheets with equal labels, the following hold: over any edge of $\mathcal{T}$ there are exactly $|G|$ nodes, labelled by G, and the component $g\langle \eta_m \rangle$ over the leaf m is attached to the sheets g and $g\eta_m$ by the nodes with labels g and $g\eta_m$ over the edge of the leaf m. The description of Γ in this step holds for the caterpillar of Construction-Theorem 4.4 (4) and for any trivalent tree of Construction-Theorem 4.5 (4).

Step 2. In this step, we compute two sums over G.

The generators t_j of G are unitary matrices, so $g^* = g^{-1}$ for $g \in G$. We show that for any $y, z \in V$,

$$\sum_{g \in G} \overline{(b_1^* g y)}(b_1^* g z) = c y^* z \tag{5.1}$$

and

$$\sum_{g \in G} (b_1^* g y)(b_1^* g z) = 0. \tag{5.2}$$

For any $Y \in M_n(K)$, the matrix $\sum_{g \in G} g^* Y g$ commutes with G, hence is a scalar matrix because V is absolutely irreducible (Construction-Theorem 4.4 (1), resp. Construction-Theorem 4.5 (1)). Taking traces, we get

$$\sum_{g \in G} g^* Y g = \frac{|G|}{n} \operatorname{tr}(Y) I.$$

Taking $Y = b_1 b_1^*$ implies (5.1). In the second case, $iI \in G$, and replacing g by $(iI)g$ multiplies the left side of (5.2) by -1, so it is zero. In the first case, view V as a complex representation through the complex embedding σ of K of Definition 2.7. The left side of (5.2) is a G-invariant bilinear form on V. If it is non-zero, then it induces a non-zero G-equivariant map $V \to V^\vee$, which is an isomorphism by Schur's lemma. The character of $V^\vee$ is the complex conjugate of the character of V, while the character of V is not real-valued because the character field is K (Construction-Theorem 4.4 (1)). Explicitly, $\operatorname{tr}(t_5 t_4 t_3 t_1 t_2 t_6) = \omega \notin \mathbb{R}$. Thus (5.2) holds.

For $y \in V$, let $\varphi_y : G \to \mathbb{Q}$ be the function $\varphi_y(g) := \mathrm{Tr}_{K/\mathbb{Q}}(b_1^* g y)$. Expanding the traces and using (5.1) and (5.2), we get

$$\sum_{g \in G} \varphi_y(g) \varphi_z(g) = \mathrm{Tr}_{K/\mathbb{Q}}(c y^* z) \tag{5.3}$$

for any $y, z \in V$. In particular, $\varphi_y = 0$ implies $2cy^*y = 0$, so $y \to \varphi_y$ is injective.

Step 3. In this step, we construct the isomorphism between $\ker R$ and $eH_1(\Gamma, \mathbb{Q})$.

Orient any edge of $\mathcal{T}$ toward the set I of leaves it cuts off, and orient the nodes over it accordingly. For a cycle $z \in H_1(\Gamma, \mathbb{Q})$ and a leaf m, let $f_m(g) \in \mathbb{Q}$ be the coefficient of z at the node with label g over the edge of the leaf m. By **Step 1**, the Kirchhoff condition at the component $g\langle\eta_m\rangle$ is $f_m(g\eta_m) = -f_m(g)$. For a fixed $g \in G$, the sheets and the nodes with label g over the internal part of $\mathcal{T}$ form a tree, so the coefficients of z there are determined by the leaf flows. Namely, the coefficient of z at the node with label g over the edge cutting off I is

$$\kappa_{I,g}(z) = \sum_{m \in I} f_m(g), \tag{5.4}$$

and the remaining Kirchhoff condition is $\sum_m f_m(g) = 0$. Conversely, any family $(f_m)_m$ satisfying both conditions is the family of leaf flows of a cycle, namely of the chain with coefficients $f_m(g)$ over the edges of the leaves and $\kappa_{I,g}$ of (5.4) over the internal edges. Thus, with

$$F_m := \{f : G \to \mathbb{Q} \mid f(g\eta_m) = -f(g) \text{ for any } g \in G\},$$

on which G acts by $(hf)(g) = f(h^{-1}g)$, the group $H_1(\Gamma, \mathbb{Q})$ is G-equivariantly isomorphic to the kernel of the summation map $\bigoplus_m F_m \to \mathbb{Q}^G$. The action of G on F_m is the push-forward action on Γ, and

$$F_m \cong \operatorname{Ind}_{\langle\eta_m\rangle}^G(\text{sign}).$$

Since the summation map is G-equivariant, $eH_1(\Gamma, \mathbb{Q})$ is the kernel of the summation map $\bigoplus_m eF_m \to e\mathbb{Q}^G$.

For $x = \sum_h x_h h \in \mathbb{Q}[G]$ and $y \in V$, we have $(x\varphi_y)(g) = \operatorname{Tr}_{K/\mathbb{Q}}((xb_1)^* gy)$, where x acts on V through the matrix $\sum_h x_h h$. Since e acts on V as E_{11}, we get $e\varphi_y = \varphi_y$. If $y \in E_m$, then $\varphi_y(g\eta_m) = \varphi_{\eta_m y}(g) = -\varphi_y(g)$. Thus $y \to \varphi_y$ is an injective $\mathbb{Q}$-linear map $E_m \to eF_m$. We show that it is an isomorphism. The algebra $\epsilon\mathbb{Q}[G] \cong M_n(K)$ has the unique simple module V, so $\epsilon F_m \cong V^{\oplus \mu_m}$ for an integer $\mu_m \geq 0$, and $\dim_{\mathbb{Q}} eF_m = 2\mu_m$. Since $V \otimes_{\mathbb{Q}} \mathbb{C}$ is the sum of $V \otimes_{K,\sigma} \mathbb{C}$ and its complex conjugate, μ_m is the multiplicity of $V \otimes_{K,\sigma} \mathbb{C}$ in

$$\operatorname{Ind}_{\langle\eta_m\rangle}^G(\text{sign}) \otimes \mathbb{C},$$

which equals $\dim_K E_m$ by Frobenius reciprocity (cf. [Ful97, Section 7.3, p. 93]). Thus $\mu_m = 1$, except for the ninth leaf in the second case, where $\mu_9 = 4$, and in any case $\dim_{\mathbb{Q}} eF_m = \dim_{\mathbb{Q}} E_m$.

Let $\gamma \in K^{10}$ in the first case and $\gamma \in K^{12}$ in the second case. Since the columns of R attached to the leaf m form a K-basis of E_m, the map $\gamma \to (RP_{\{m\}}\gamma)_m$ is an isomorphism onto $\bigoplus_m E_m$, and $\sum_m RP_{\{m\}}\gamma = R\gamma$. Let $\Phi(\gamma)$ be the family of leaf flows $f_m := \varphi_{RP_{\{m\}}\gamma}$. The sum of f_m over all leaves m is $\varphi_{R\gamma}$, which vanishes if and only if $R\gamma = 0$ by **Step 2**. Thus Φ is an isomorphism of $\mathbb{Q}$-vector spaces from $\ker R$ to $eH_1(\Gamma, \mathbb{Q})$, and by (5.4),

$$\kappa_{I,g}(\Phi(\gamma)) = \operatorname{Tr}_{K/\mathbb{Q}}(b_1^* gRP_I\gamma). \tag{5.5}$$

Reversing the orientation of an edge replaces I by the complement I^c and $\kappa_{I,g}$ by $-\kappa_{I,g}$, which is compatible with (5.5) because $RP_{I^c}\gamma = -RP_I\gamma$ for $\gamma \in \ker R$.

Step 4. In this step, we compute B_I by the Picard–Lefschetz formula.

Let $\widetilde{T}_I$ be the local monodromy around $q_I = 0$ on $H_1(X, \mathbb{Q})$ and let ψ_J be the Riemann form of $J(X)$. The specialization map

$$\operatorname{sp} : H_1(X, \mathbb{Q}) \to H_1(\Gamma, \mathbb{Q})$$

is surjective (cf. [ADF+20, §3.2]). Since all node inertias are trivial, the cover is étale near the $|G|$ nodes over the edge cutting off I, so all of them have the local equation $xy = q_I$ in the family, and by the Picard–Lefschetz formula in the form of [ADF+20, §3.2, pp. 174–175], their terms add. Thus

$$\psi_J(x, (\widetilde{T}_I - 1)y) = \sum_{g \in G} \kappa_{I,g}(\operatorname{sp} x)\kappa_{I,g}(\operatorname{sp} y)$$

for any $x, y \in H_1(X, \mathbb{Q})$.

The group G acts on the family fiberwise, so $\widetilde{T}_I$ commutes with G, hence with e, and sp is G-equivariant. The Riemann form ψ of A_X is the restriction of ψ_J to $H_1(A_X, \mathbb{Q}) = eH_1(X, \mathbb{Q})$, and T_I is the restriction of $\widetilde{T}_I$ (cf. [ADF+20, §3.6]). Thus, for $x, y \in eH_1(X, \mathbb{Q})$, the form $\psi(x, \nu_I y)$ depends only on $\operatorname{sp} x$ and $\operatorname{sp} y$, which range over $\operatorname{sp}(eH_1(X, \mathbb{Q})) = eH_1(\Gamma, \mathbb{Q})$, and

$$B_I(z, z') = \sum_{g \in G} \kappa_{I,g}(z)\kappa_{I,g}(z')$$

for $z, z' \in eH_1(\Gamma, \mathbb{Q})$. We use the sign convention of [ADF+20, §3.2], under which B_I is positive semi-definite. The opposite convention changes all B_I by a common sign, which does not change any span dimension or rank below. Let $v, w \in K^4$, and let $y := RP_INv$ and $y' := RP_INw$. By (5.5) and (5.3),

$$B_I(\Phi(Nv), \Phi(Nw)) = \sum_{g \in G} \varphi_y(g)\varphi_{y'}(g) = \operatorname{Tr}_{K/\mathbb{Q}}(cy^*y') = \operatorname{Tr}_{K/\mathbb{Q}}(cv^*Q_Iw),$$

where the last equality holds because $P_I^* = P_I$. Since **Steps 1–4** hold for the caterpillar of Construction-Theorem 4.4 (4) and for any trivalent tree of Construction-Theorem 4.5 (4), this finishes the proof. □

Lemma 5.3. *Let k be a positive integer,*

$$\Delta^k := \{q \in \mathbb{C}^k \mid |q_j| < 1 \text{ for } 1 \le j \le k\},$$

$(\Delta^*)^k := \{q \in \Delta^k \mid q_1 \cdots q_k \neq 0\}$, *and W a complex vector space with $\dim W < +\infty$. Let ϖ be a holomorphic map from the universal cover of $(\Delta^*)^k$ to W of the form $\varpi(q) = \sum_{j=1}^k M_j \log q_j + \phi(q)$, where $M_j \in W$ and $\phi : \Delta^k \to W$ is holomorphic. If $M_1, \dots, M_k$ span a subspace of dimension s, then the differential of ϖ has rank at least s on a non-empty open subset of the universal cover of $(\Delta^*)^k$.*

Proof. The universal cover is locally biholomorphic to $(\Delta^*)^k$ through q, and in the coordinates q,

$$q_j \frac{\partial \varpi}{\partial q_j} = M_j + q_j \frac{\partial \phi}{\partial q_j}.$$

Choose indices $j_1, \dots, j_s$ such that $M_{j_1}, \dots, M_{j_s}$ are linearly independent, and linear forms $\lambda_1, \dots, \lambda_s$ on W such that $\det(\lambda_a(M_{j_b}))_{a,b} \neq 0$. The function

$$D(q) := \det\Big(\lambda_a(M_{j_b}) + q_{j_b}\lambda_a\Big(\frac{\partial \phi}{\partial q_{j_b}}\Big)\Big)_{a,b}$$

is holomorphic on Δ^k and $D(0) \neq 0$. Hence $D \neq 0$ on a neighbourhood of 0, which contains points of $(\Delta^*)^k$. At any point of the universal cover over a point $q \in (\Delta^*)^k$ with $D(q) \neq 0$, the vectors $q_{j_b}\partial\varpi/\partial q_{j_b}$ for $1 \le b \le s$ are linearly independent, and so are the vectors $\partial\varpi/\partial q_{j_b}$ because $q_{j_b} \neq 0$. Such points form a non-empty open subset of the universal cover, and the differential of ϖ has rank at least s there. □

5.2. **Logarithmic rank of the Eisenstein and Gaussian families.** We order the coordinates of $Q \in \operatorname{Herm}_4(K)$ as

$$(d_1, d_2, d_3, d_4, x_{12}, y_{12}, x_{13}, y_{13}, x_{14}, y_{14}, x_{23}, y_{23}, x_{24}, y_{24}, x_{34}, y_{34}),$$

where $d_1, \dots, d_4 \in \mathbb{Q}$ are the diagonal entries of Q and, for $j < k$, $x_{jk}, y_{jk} \in \mathbb{Q}$ are given by writing the (j,k)-entry of Q as $x_{jk} + y_{jk}\omega$ if $K = \mathbb{Q}(\omega)$ and as $x_{jk} + iy_{jk}$ if $K = \mathbb{Q}(i)$. We write n_j for the j-th row of N.

Proposition 5.4. *In the case $K = \mathbb{Q}(\omega)$ of Definition-Proposition 5.2, write $Q_i := Q_{\{i\}}$ for $1 \le i \le 10$ and $Q_{\le j} := Q_{\{1,\dots,j\}}$ for $2 \le j \le 8$. Then $Q_1, \dots, Q_{10}, Q_{\le 2}, \dots, Q_{\le 8}$ span a 15-dimensional $\mathbb{Q}$-subspace of the 16-dimensional $\mathbb{Q}$-vector space $\mathrm{Herm}_4(K)$. The annihilator of*

$$\mathrm{Span}_{\mathbb{Q}}\{Q_1, \dots, Q_{10}, Q_{\le 2}, \dots, Q_{\le 8}\}$$

in the dual space is spanned by

$$\ell(Q) = -3x_{12} + y_{12} - x_{14} - 2y_{14} - 4x_{24} + 4y_{24},$$

and the $\mathbb{Q}$-linear relations among $Q_1, \dots, Q_{10}, Q_{\le 2}, \dots, Q_{\le 8}$ are spanned by

$$2Q_1 + 2Q_2 + Q_4 - Q_5 + Q_6 - 2Q_7 - 4Q_{\le 2} - 2Q_{\le 4} + 2Q_{\le 5} = 0 \quad \text{and} \quad Q_7 - Q_{\le 6} + Q_{\le 7} = 0.$$

Proof. The matrix N is listed in Table 1, and R_0 times the first six rows of N equals $-[r_7, r_8, r_9, r_{10}]$. For $1 \le i \le 10$, we have $RP_{\{i\}}N = r_i n_i$, hence

$$Q_i = (r_i^* r_i) n_i^* n_i,$$

and for $2 \le j \le 8$,

$$Q_{\le j} = \Big(\sum_{i \le j} r_i n_i\Big)^* \Big(\sum_{i \le j} r_i n_i\Big).$$

Since $RN = 0$, we also have $Q_{\le j} = Q_{\{j+1,\dots,10\}}$, for example $Q_{\le 8} = (r_9 n_9 + r_{10} n_{10})^* (r_9 n_9 + r_{10} n_{10})$. We compute $Q_1, \dots, Q_{10}, Q_{\le 2}, \dots, Q_{\le 8}$ by exact arithmetic in $K = \mathbb{Q}(\omega)$, with

$$(a + b\omega)(a' + b'\omega) = (aa' - bb') + (ab' + a'b - bb')\omega \quad \text{and} \quad \overline{a + b\omega} = (a - b) - b\omega$$

for $a, b, a', b' \in \mathbb{Q}$. Their coordinate vectors form the 17×16 matrix of Tables 2 and 3.

Substituting the rows of Tables 2 and 3 into ℓ shows that ℓ vanishes on all seventeen matrices, so they span a subspace of dimension at most 15. The 15×15 minor on the rows of

$$Q_1, \dots, Q_{10}, Q_{\le 2}, Q_{\le 3}, Q_{\le 4}, Q_{\le 6}, Q_{\le 8}$$

and on the columns other than y_{24} is the non-zero number (A.2), which we compute by Gauss elimination with the pivots listed in Section A.1. Hence the span is 15-dimensional, and the annihilator is the line spanned by $\ell \ne 0$. The space of relations is 2-dimensional because $17 - 15 = 2$. The relations in the proposition follow by adding the corresponding rows, and they are linearly independent because only the second relation involves $Q_{\le 7}$ and the first relation is non-trivial. □

Corollary 5.5. *Let $K = \mathbb{Q}(\omega)$ and let $\mathcal{H}$ be as in Construction-Theorem 4.4. Then the local period map of the family $\{A_X\}_{X \in \mathcal{H}}$ to $\mathbb{D}(h_0)$ has rank at least 15 on a non-empty open subset of $\mathcal{H}$.*

Proof. Let $(q_I)_I$ be the seventeen node parameters at the boundary point of Construction-Theorem 4.4 (4). Since $\dim \mathcal{H} = 17$ and the seventeen node parameters can be smoothed independently in $\mathcal{H}$, the functions $(q_I)_I$ are local coordinates of the closure of $\mathcal{H}$ at the boundary point (Lemma 4.2 (2)), possibly after passing to a finite map from Δ^{17} to the closure of $\mathcal{H}$ which sends $(\Delta^*)^{17}$ into $\mathcal{H}$. A finite map is a local biholomorphism at a general point, so it does not change the rank of the local period map there. By Definition-Lemma 5.1, it suffices to show that some local period map on the universal cover of $(\Delta^*)^{17}$ has rank at least 15 on a non-empty open subset.

Let

$$\mathrm{sp} : H_1(X, \mathbb{Q}) \to H_1(\Gamma, \mathbb{Q})$$

be the specialization map, and let ψ_J be the Riemann form of $J(X)$. The map sp is surjective and G-equivariant, and $\ker(\mathrm{sp})$ is spanned by the vanishing cycles, which are isotropic for ψ_J (cf. [ADF+20, §3.2]).

Forgetting the K-action, the map $P \to J_P$ of Definition-Proposition 2.12 is a holomorphic embedding of $\mathbb{D}(h_0)$ into the Siegel space of the complex structures J on $H_1(A_X, \mathbb{R})$ with $\psi(Jx, Jy) = \psi(x, y)$ and $\psi(x, Jx) > 0$ for $x \ne 0$, since $\mathrm{Im}\, \sigma(\delta) > 0$ in Definition 2.9, so that the

$(-i)$-eigenspace of J_P in $H_1(A_X,\mathbb{C})$ is $P\oplus\overline{P^\perp}$ (Lemma 2.13), which varies holomorphically with P, since $P^\perp$ varies antiholomorphically with P, h_0 being conjugate-linear in the first variable. Thus the local period map and the composite of the local period map with the embedding $P\to J_P$ have equal rank at any point, and in a rational symplectic basis of $(H_1(A_X,\mathbb{Q}),\psi)$ the composite is the Siegel period matrix $Z\in\mathrm{Sym}_8(\mathbb{C})$ of A_X, viewed as a function on the universal cover of $(\Delta^*)^{17}$. The subspace $e\ker(\mathrm{sp})$ of $eH_1(X,\mathbb{Q})$ is isotropic for ψ, and

$$\dim_{\mathbb{Q}} e\ker(\mathrm{sp}) = 16-\dim_{\mathbb{Q}} eH_1(\Gamma,\mathbb{Q}) = 8$$

because A_X has totally toric reduction of toric rank 8 at the boundary point (Construction-Theorem 4.4 (4)), so $e\ker(\mathrm{sp})$ is Lagrangian. Choose a rational symplectic basis $\alpha_1,\dots,\alpha_8,\beta_1,\dots,\beta_8$ of $(eH_1(X,\mathbb{Q}),\psi)$ such that $\alpha_1,\dots,\alpha_8$ span $e\ker(\mathrm{sp})$. Since $e^*=e$, the decomposition

$$H_1(X,\mathbb{Q}) = eH_1(X,\mathbb{Q})\oplus(1-e)H_1(X,\mathbb{Q})$$

is orthogonal for ψ_J and preserved by the family, and

$$\ker(\mathrm{sp}) = e\ker(\mathrm{sp})\oplus(1-e)\ker(\mathrm{sp})$$

because $\ker(\mathrm{sp})$ is G-stable. Thus, in a symplectic basis of $H_1(X,\mathbb{Q})$ adapted to the decomposition and to $\ker(\mathrm{sp})$, the period matrix of $J(X)$ is block diagonal, and the block of $eH_1(X,\mathbb{Q})$ is the Siegel period matrix Z of A_X with respect to $\alpha_1,\dots,\alpha_8,\beta_1,\dots,\beta_8$. By the expansion of [ADF+20, §3.2, pp. 174–175], restricted to the e-part as in [ADF+20, §3.6],

$$Z=\frac{1}{2\pi i}\sum_I M_I\log q_I+\phi(q),$$

where ϕ is holomorphic on Δ^{17} and $M_I\in\mathrm{Sym}_8(\mathbb{Q})$ is the matrix of B_I with respect to the basis $\mathrm{sp}\,\beta_1,\dots,\mathrm{sp}\,\beta_8$ of $eH_1(\Gamma,\mathbb{Q})$. A change of rational symplectic basis preserving the vanishing subspace keeps the form of the expansion.

The $\mathbb{Q}$-linear map from $\mathrm{Herm}_4(K)$ to the symmetric $\mathbb{Q}$-bilinear forms on K^4 sending Q to $(v,w)\to\mathrm{Tr}_{K/\mathbb{Q}}(cv^*Qw)$ is injective, because if $\mathrm{Tr}_{K/\mathbb{Q}}(cv^*Qw)=0$ for any $v,w\in K^4$, then replacing w by aw for $a\in K$ shows $v^*Qw=0$, so $Q=0$. By Definition-Proposition 5.2 and Proposition 5.4, the matrices M_I span a 15-dimensional $\mathbb{Q}$-subspace of $\mathrm{Sym}_8(\mathbb{Q})$. Rational matrices which are linearly independent over $\mathbb{Q}$ are linearly independent over $\mathbb{C}$, so the matrices $M_I/(2\pi i)$ span a 15-dimensional subspace of $\mathrm{Sym}_8(\mathbb{C})$. By Lemma 5.3 with $W=\mathrm{Sym}_8(\mathbb{C})$, the differential of Z has rank at least 15 on a non-empty open subset of the universal cover of $(\Delta^*)^{17}$, and Corollary 5.5 follows. □

For the Gaussian family the logarithmic monodromy forms span only a 13-dimensional space, and no other trivalent tree improves the bound 13, so in Subsection 5.3 we use the regular term of the period expansion.

Proposition 5.6. *In the case $K=\mathbb{Q}(i)$ of Definition-Proposition 5.2, write $L_j:=Q_{\{j\}}$ for $1\le j\le 9$ and $Q_{\le k}:=Q_{\{1,\dots,k\}}$ for $2\le k\le 7$. Then $L_1,\dots,L_9,Q_{\le 2},\dots,Q_{\le 7}$ span a 13-dimensional $\mathbb{Q}$-subspace $\mathcal{S}\subset\mathrm{Herm}_4(K)$. The annihilator of $\mathcal{S}$ is spanned by $x_{12}-x_{13}$, $y_{12}-y_{13}$, and $5x_{12}-y_{12}-x_{14}+y_{14}$, and the $\mathbb{Q}$-linear relations among $L_1,\dots,L_9,Q_{\le 2},\dots,Q_{\le 7}$ are spanned by $L_1-2L_8=0$ and $L_6-2L_7-4L_9-2Q_{\le 6}+6Q_{\le 7}=0$. Moreover, $Q_I\in\mathcal{S}$ for any set I of the nine leaves. In particular, for any trivalent tree whose leaves carry the branch pairs of $t_1,\dots,t_9$, the logarithmic monodromy forms of the corresponding boundary point span a $\mathbb{Q}$-subspace of dimension at most 13.*

Proof. The matrix N is listed in Table 5, and R_0 times the first eight rows of N equals $-[u_1,u_2,u_3,u_4]$. For $1\le m\le 8$, the m-th column of R is $a_{\pi(m)}$, so $RP_{\{m\}}N=a_{\pi(m)}n_m$ and $L_m=2n_m^*n_m$. Since $u_1,\dots,u_4$ are orthogonal with $u_k^*u_k=2$, we have $L_9=2I_4$. We compute $L_1,\dots,L_9,Q_{\le 2},\dots,Q_{\le 7}$ by exact arithmetic in $K=\mathbb{Q}(i)$, and their coordinate vectors form the 15×16 matrix of Tables 6 and 7.

Substituting the rows of Tables 6 and 7 shows that the three linear forms in the proposition vanish on all fifteen matrices. They are linearly independent, because x_{13}, y_{13}, and y_{14} occur only in the first, the second, and the third form, respectively. Thus $\dim_{\mathbb{Q}} \mathcal{S} \leq 13$. The 13×13 minor on the rows of $L_1, \ldots, L_7, L_9, Q_{\leq 2}, \ldots, Q_{\leq 6}$ and on the columns

$$d_1, d_2, d_3, d_4, x_{12}, y_{12}, x_{14}, x_{23}, y_{23}, x_{24}, y_{24}, x_{34}, y_{34}$$

is the non-zero number (A.3), which we compute by Gauss elimination with the pivots listed in Section A.2. Hence $\dim_{\mathbb{Q}} \mathcal{S} = 13$, and the three forms span the annihilator. The space of relations is 2-dimensional because $15 - 13 = 2$. The relations in the proposition follow by adding the corresponding rows, and they are linearly independent because only the first relation involves L_1 and the second relation is non-trivial.

The leaves are numbered by π so that, for $2 \leq m \leq 8$, the simple root $a_{\pi(m)}$ of the leaf m is adjacent in the Dynkin diagram of E_8 to exactly one of $a_{\pi(1)}, \ldots, a_{\pi(m-1)}$, and hence the forms $Q_{\{1,\ldots,m\}}$ of the caterpillar determine all cross terms one by one. We show that $Q_I \in \mathcal{S}$ for any set I of leaves. Recall from Construction-Theorem 4.5 that $a_1 = \frac{1}{2}(b_1 - b_2 - \cdots - b_7 + b_8)$, $a_2 = b_1 + b_2$, $a_3 = -b_1 + b_2$, and $a_j = -b_{j-2} + b_{j-1}$ for $4 \leq j \leq 8$. Since a_1 has eight coordinates $\pm\frac{1}{2}$ and each of $a_2, \ldots, a_8$ has two coordinates ± 1 and the others 0, we have $a_j^t a_j = 2$. Next, $a_1^t a_2 = \frac{1}{2}(1-1) = 0$, $a_1^t a_3 = \frac{1}{2}(-1-1) = -1$, and $a_1^t a_k = \frac{1}{2}(1-1) = 0$ for $4 \leq k \leq 8$, because the coordinates $2, \ldots, 7$ of a_1 are all $-\frac{1}{2}$. Moreover $a_2^t a_3 = -1 + 1 = 0$ and $a_2^t a_4 = a_3^t a_4 = -1$. For $j \in \{2, 3\}$ and $5 \leq k \leq 8$, and for $4 \leq j < k \leq 8$ with $k \neq j+1$, the supports of a_j and a_k are disjoint, so $a_j^t a_k = 0$, while $a_j^t a_{j+1} = -1$ for $4 \leq j \leq 7$, the only common coordinate being $j - 1$. Hence, for $j \neq k$, $a_j^t a_k = -1$ if $\{j, k\}$ is one of $\{1, 3\}$, $\{2, 4\}$, $\{3, 4\}$, $\{4, 5\}$, $\{5, 6\}$, $\{6, 7\}$, and $\{7, 8\}$, and $a_j^t a_k = 0$ otherwise. Thus, in the numbering of the leaves, $a_{\pi(m)}^t a_{\pi(m')} = -1$ if $\{m, m'\}$ belongs to

$$\mathcal{E} := \{\{1,2\},\{2,3\},\{3,4\},\{3,5\},\{5,6\},\{6,7\},\{7,8\}\},$$

and $a_{\pi(m)}^t a_{\pi(m')} = 0$ for any other pair $m \neq m'$. Let $C_{mm'} := -(n_m^* n_{m'} + n_{m'}^* n_m)$ for $\{m, m'\} \in \mathcal{E}$. For $I \subset \{1, \ldots, 8\}$, expanding

$$Q_I = \Big(\sum_{m \in I} a_{\pi(m)} n_m\Big)^* \Big(\sum_{m \in I} a_{\pi(m)} n_m\Big)$$

yields

$$Q_I = \sum_{m \in I} L_m + \sum_{\{m,m'\} \in \mathcal{E},\ m,m' \in I} C_{mm'}. \tag{5.6}$$

If $9 \in I$, then $RP_{I^c}N = -RP_I N$ because $RN = 0$, so $Q_I = Q_{I^c}$ with $I^c \subset \{1, \ldots, 8\}$. Hence it suffices to show that $C_{mm'} \in \mathcal{S}$ for $\{m, m'\} \in \mathcal{E}$. The graph with vertices $1, \ldots, 8$ and edges $\mathcal{E}$ is a tree, and any $2 \leq m \leq 8$ is adjacent to exactly one $p(m) < m$, namely $p(2) = 1$, $p(3) = 2$, $p(4) = 3$, $p(5) = 3$, $p(6) = 5$, $p(7) = 6$, and $p(8) = 7$. By (5.6),

$$C_{p(m)m} = Q_{\leq m} - Q_{\leq m-1} - L_m \quad \text{for } 2 \leq m \leq 8,$$

where $Q_{\leq 1} := L_1$ and $Q_{\leq 8} := Q_{\{1,\ldots,8\}}$, which equals $Q_{\{9\}} = L_9$. All terms on the right belong to $\mathcal{S}$, so $C_{mm'} \in \mathcal{S}$ for $\{m, m'\} \in \mathcal{E}$, and $Q_I \in \mathcal{S}$ for any I.

Finally, let $\mathcal{T}$ be a trivalent tree whose leaves carry the branch pairs of $t_1, \ldots, t_9$. By Construction-Theorem 4.5 (4), the closure of $\mathcal{H}$ contains the corresponding boundary point, and by Definition-Proposition 5.2, the logarithmic monodromy form at the edge cutting off I corresponds to $(v, w) \to \mathrm{Tr}_{K/\mathbb{Q}}(cv^* Q_I w)$ under an isomorphism between $eH_1(\Gamma, \mathbb{Q})$ and $\ker R$. The map

$$Q \to ((v, w) \to \mathrm{Tr}_{K/\mathbb{Q}}(cv^* Q w))$$

is $\mathbb{Q}$-linear and injective, since $\mathrm{Tr}_{K/\mathbb{Q}}(cv^* Q(aw)) = 0$ for any $a \in K$ implies $v^* Q w = 0$. Thus the logarithmic monodromy forms of $\mathcal{T}$ span a $\mathbb{Q}$-subspace of dimension at most $\dim_{\mathbb{Q}} \mathcal{S} = 13$. □

5.3. **Rank of the Gaussian family.** In this subsection, we show that the coefficient of $s_1 s_9$ in the plumbing expansion adds two directions to the span $\mathcal{S}$ of Proposition 5.6, where s_1 and s_9 are the plumbing parameters of the nodes at the leaves 1 and 9.

Theorem 5.7. *Let $K = \mathbb{Q}(i)$ and let $\mathcal{H}$ be as in Construction-Theorem 4.5. Then the local period map of the family $\{A_X\}_{X\in\mathcal{H}}$ to $\mathbb{D}(h_0)$ has rank 15 on a non-empty open subset of $\mathcal{H}$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix coordinates on $\mathcal{H}$ near a boundary point and on $\mathbb{D}(h_0)$, and we compute the logarithmic coefficients of the period map.

We number the leaves as in Definition-Proposition 5.2. Thus the leaf j carries the branch points of the pair $(t_{\pi(j)}, t_{\pi(j)})$, where $(\pi(1),\dots,\pi(9)) = (1,3,4,2,5,6,7,8,9)$, and the j-th column of R is $a_{\pi(j)}$ for $1 \le j \le 8$. Let n_j be the j-th row of N for $1 \le j \le 8$ (Table 5), and let $V_j := RP_{\{j\}}N$ for $1 \le j \le 9$, so that $V_j = a_{\pi(j)} n_j$ for $j \le 8$ and $V_9 = [u_1,\dots,u_4]$. For $Q \in \mathrm{Herm}_4(K)$ we write $d_j := Q_{jj}$ and $Q_{jk} = x_{jk} + iy_{jk}$ for $j < k$, and we extend d_j, x_{jk}, and y_{jk} $\mathbb{C}$-linearly to 16 coordinates on $\mathrm{Herm}_4(K) \otimes_{\mathbb{Q}} \mathbb{C}$. Since $RN = 0$, $a_j^t a_j = 2$, and $\Omega u_k = iu_k$, we have

$$\sum_{j=1}^{9} V_j = RN = 0, \qquad L_j = V_j^* V_j, \qquad t_{\pi(j)} V_j = -V_j \quad (1 \le j \le 9).$$

We claim that for any 8×4 matrices M, M', any $v, w \in K^4$, and the constant $c = |G|/8$ of Definition-Proposition 5.2,

$$\sum_{g\in G} \mathrm{Tr}_{K/\mathbb{Q}}(b_1^* g M v)\, \mathrm{Tr}_{K/\mathbb{Q}}(b_1^* g M' w) = \mathrm{Tr}_{K/\mathbb{Q}}(c v^* M^* M' w). \tag{5.7}$$

Expanding $\mathrm{Tr}_{K/\mathbb{Q}}(y) = y + \bar{y}$, where the bar conjugates only the coefficients in K, (5.7) follows from (5.1) and (5.2) with $y = Mv$ and $z = M'w$, extended $\mathbb{C}$-bilinearly to holomorphic entries.

By Construction-Theorem 4.5 (4), the fifteen node parameters at the caterpillar point can be smoothed independently. Smoothing the six internal nodes, we obtain, for any configuration of nine distinct points $z_1,\dots,z_9 \in \mathbb{P}^1$ with $z_1 = 0$, $z_8 = 1$, and $z_9 = +\infty$, an admissible G-cover over the stable curve formed by a core $\mathbb{P}^1$ and nine tails, where the tail of the leaf j is attached at z_j and carries the branch points of $(t_{\pi(j)}, t_{\pi(j)})$. Since the node inertias are trivial, the cover of the core is the union of the sheets $\{g\} \times \mathbb{P}^1$ for $g \in G$. The closure of the set of such admissible covers contains the caterpillar point, and near any such cover the leaf-node parameters $q_j := q_{\{j\}}$ for $1 \le j \le 9$ and the core coordinates $z_2,\dots,z_7$ are $9 + 6 = 15$ coordinates on the closure of $\mathcal{H}$. We work near the cover with

$$(z_1,\dots,z_8) = z^0 := (0,2,3,4,5,6,7,1).$$

Over the tail of the leaf j the cover consists of $|G|/2$ smooth rational curves, each a double cover of the tail branched at both branch points and attached to the sheets g and $gt_{\pi(j)}$ at z_j. Forgetting the branch points and contracting the rational curves over the tails, we obtain a nodal curve X_0' whose components are the sheets, with $|G|/2$ nodes of index j for each j, the node joining (g, z_j) to $(gt_{\pi(j)}, z_j)$. Let Γ' be the dual graph of X_0'. Since the node inertias are trivial, Γ' is the result of contracting in Γ, for each $g \in G$, the path formed by the sheets g over the internal components of the caterpillar, and then of contracting the vertices of valence 2 given by the rational curves over the tails. Hence $H_1(\Gamma', \mathbb{Q}) = H_1(\Gamma, \mathbb{Q})$.

On the sheet g we use the branch coordinates $\theta_j := z - z_j$ at z_j for $j \le 8$ and $\theta_9 := 1/z$ at $z_9 = +\infty$, and we write θ_j' for the coordinate θ_j on the sheet $gt_{\pi(j)}$. We smooth X_0' by the plumbing, or sewing, $\theta_j\theta_j' = s_j$ at all $|G|/2$ nodes of index j, with one parameter s_j for each j [HN20, Definition 2.2]. The group G permutes the nodes of index j and preserves the branch coordinates, so the plumbing family is G-equivariant.

The stabilizer $\langle gt_{\pi(j)}g^{-1}\rangle$ of the node $\{g, gt_{\pi(j)}\}$ exchanges the branches θ_j and θ_j', acts on the plumbing annulus by $\theta_j \to s_j/\theta_j$, and has exactly two fixed points there, $\theta_j^2 = s_j$. After adjoining

square roots of the s_j and marking the fixed points as branch points, we obtain a smoothing of the tail nodes. Each rational curve over the tail of the leaf j is attached to the sheets at two nodes over the leaf node, and in the local model over the leaf node both nodes have the plumbing parameter q_j. Contracting the rational curve multiplies the plumbing parameters of both nodes up to an analytic function without zeros, so s_j/q_j^2 is an analytic function without zeros. Hence $(q,z)\to(s,z)$ is a finite map, and it is a local isomorphism at the points with $q_1\cdots q_9\neq 0$, since the Jacobian matrix $(\partial s_j/\partial q_k)$ is $\operatorname{diag}(q_1,\dots,q_9)$ times a matrix which is invertible near $q=0$. In particular, the points with $s_1\cdots s_9\neq 0$ are points of $\mathcal{H}$, and the rank of the local period map at such points may be computed in the coordinates (s,z).

Below, matrices have entries in K tensored with holomorphic functions of s and z, and M^* denotes the transpose of M with the coefficients in K conjugated. The operation $*$ does not act on s and z. Thus all identities below are holomorphic identities, with values in $\operatorname{Herm}_4(K)\otimes_{\mathbb{Q}}\mathbb{C}$ when the matrices are 4×4.

A *regular differential* on X_0' is a differential on the normalization of X_0' with at most simple poles at the branches of the nodes and with opposite residues at both branches of any node. Let

$$\Xi_0:=\sum_{j=1}^{8}V_j\frac{dz}{z-z_j},$$

an 8×4 matrix of differentials on $\mathbb{P}^1$ with residue V_j at z_j for $j\le 8$ and residue $-\sum_{j\le 8}V_j=V_9$ at $+\infty$. For $x\in K^4$, let $\Xi_0[x]$ be the differential on X_0' equal to $\operatorname{Tr}_{K/\mathbb{Q}}(b_1^*g\Xi_0 x)$ on the sheet g. Since $t_{\pi(j)}V_j=-V_j$, the residues of $\Xi_0[x]$ at (g,z_j) and at $(gt_{\pi(j)},z_j)$ are opposite, so $\Xi_0[x]$ is regular. A regular differential on X_0' is determined by the residues at the node branches, which form a 1-cycle of Γ', and the differentials $\Xi_0[x]$ for $x\in K^4$ are the regular differentials in the e-part defined over $\mathbb{Q}$. The map sending Nx to the residue cycle of $\Xi_0[x]$ is the isomorphism between $\ker R$ and $eH_1(\Gamma',\mathbb{Q})=eH_1(\Gamma,\mathbb{Q})$ of Definition-Proposition 5.2 (cf. [ADF+20, §3.2]).

For X near the boundary point, the e-parts of the vanishing cycles of the nodes span a K-subspace $\mathcal{W}\subset H_1(A_X,\mathbb{Q})$ with $\dim_K\mathcal{W}=4$, which is isotropic for the Riemann form ψ of A_X and hence for the Hermitian form of the polarization, and $H_1(A_X,\mathbb{Q})/\mathcal{W}\cong eH_1(\Gamma,\mathbb{Q})$. For any $P\in\mathbb{D}(h_0)$ we have $P\cap(\mathcal{W}\otimes_{K,\sigma}\mathbb{C})=0$, since h_0 is positive definite on P and zero on $\mathcal{W}\otimes_{K,\sigma}\mathbb{C}$. Thus P is the graph of a linear map from the span of the lifts of the cycles of Γ to $\mathcal{W}\otimes_{K,\sigma}\mathbb{C}$, and the graphs define a holomorphic chart τ of $\mathbb{D}(h_0)$ with values in $\operatorname{Herm}_4(K)\otimes_{\mathbb{Q}}\mathbb{C}$, the Siegel coordinate with respect to $\mathcal{W}$. On the curves near the boundary point, τ is, up to left multiplication by a constant invertible matrix that depends on the basis $\chi_1,\dots,\chi_4$ below and does not change the rank of the local period map, the matrix of B-periods of the A-normalized differentials in the e-part, where the A-cycles are the vanishing cycles of the nodes, as in the A-normalization of [HN20, Theorem 3.3], and the B-cycles are lifts of the cycles Nv of Γ for $v\in K^4$. Since $\operatorname{Im}\sigma(\delta)>0$ (Definition 2.9), P is the annihilator in $H_1(A_X,\mathbb{Q})\otimes_{K,\sigma}\mathbb{C}$ of $H^{1,0}(A_X)\cap H^1(A_X,\mathbb{C})_\sigma$ (Lemma 2.13), so P is spanned by the cycles $b-\sum_{k=1}^4(\int_b\varphi_k)\chi_k$ and is the graph of $b\to-\sum_{k=1}^4(\int_b\varphi_k)\chi_k$, where b is a lift of a cycle of Γ, $\chi_1,\dots,\chi_4$ is a basis of $\mathcal{W}\otimes_{K,\sigma}\mathbb{C}$ formed by e-parts of vanishing cycles, and $\varphi_1,\dots,\varphi_4$ are the A-normalized differentials in $H^{1,0}(A_X)\cap H^1(A_X,\mathbb{C})_\sigma$, with $\int_{\chi_l}\varphi_k$ equal to 1 for $k=l$ and to 0 otherwise. We normalize τ by requiring that $\operatorname{Tr}_{K/\mathbb{Q}}(cv^*2\pi i\tau w)$ be the B-period along the lift of Nv of the differential whose periods along the vanishing cycles are $2\pi i$ times the residues of $\Xi_0[w]$ (cf. [ADF+20, §3.2, §3.6]).

By the Picard–Lefschetz formula [ADF+20, §3.2, pp. 174–175], the value of τ on a pair of cycles is the sum of a holomorphic function of (s,z) and of $(2\pi i)^{-1}$ times the sum over the nodes of the logarithm of the node parameter times the product of the coordinates of both cycles at the node. At the node joining (g,z_j) and $(gt_{\pi(j)},z_j)$, the coordinate of the residue cycle of $\Xi_0[v]$ is $\operatorname{Tr}_{K/\mathbb{Q}}(b_1^*gV_jv)$, and replacing g with $gt_{\pi(j)}$ changes it only by a sign. Since each of the $|G|/2$ nodes of index j corresponds to two elements of G, (5.7) implies that the coefficient of $\log s_j$ in

$\mathrm{Tr}_{K/\mathbb{Q}}(cv^*\tau w)$ is

$$\frac{1}{2\pi i}\cdot\frac{1}{2}\sum_{g\in G}\mathrm{Tr}_{K/\mathbb{Q}}(b_1^*gV_jv)\,\mathrm{Tr}_{K/\mathbb{Q}}(b_1^*gV_jw)=\frac{1}{4\pi i}\,\mathrm{Tr}_{K/\mathbb{Q}}(cv^*L_jw).$$

We set $Z:=4\pi i\tau$, so that the coefficient of $\log s_j$ in Z is L_j. In Definition-Proposition 5.2 the leaf node with parameter q_j has $|G|$ nodes over it, and the coefficient of $\log q_j$ in $2\pi i\tau$ is L_j. Since $\log s_j=2\log q_j+\log(s_j/q_j^2)$ and the $|G|$ nodes over the leaf node become the $|G|/2$ nodes of index j, the coefficient of $\log s_j$ in $2\pi i\tau$ is $L_j/2$, in accordance with $Z=2\cdot 2\pi i\tau$.

Let $\xi_j:=s_j\partial/\partial s_j$ for $1\le j\le 9$, and let D_Z be the 16×15 matrix whose columns are

$$\xi_1Z,\dots,\xi_9Z,\partial Z/\partial z_2,\dots,\partial Z/\partial z_7,$$

written in the coordinates d_j, x_{jk}, y_{jk}. On a simply connected open subset of $\{s_1\cdots s_9\neq 0\}$, the local period map composed with τ is a branch of τ as a function of (s,z), and the rank of the local period map does not depend on the choices (Definition-Lemma 5.1). Since τ is a chart and multiplying the column ξ_jZ by $1/s_j$ does not change the rank, the rank of the local period map at a point with $s_1\cdots s_9\neq 0$ is $\operatorname{rank} D_Z$.

Step 2. In this step, we expand Z in the plumbing coordinates.

By [HN20, Theorem 3.3], for s small the A-normalized differentials of the plumbing family are given by the convergent solution of the jump problem [HN20, (3.3)–(3.11)]. The hypotheses of [HN20, Theorem 3.3] hold, since X_0' is a stable nodal curve whose components are rational with nine distinct node branches each. By uniqueness of the normalized solution, the A-normalized differential whose periods along the vanishing cycles are $2\pi i$ times the residues of $\Xi_0[x]$ equals $\mathrm{Tr}_{K/\mathbb{Q}}(b_1^*g\Xi x)$ on the sheet g, where $\Xi=\Xi(s,z)$ is an 8×4 matrix of differentials, holomorphic on $\mathbb{P}^1$ minus small discs around $z_1,\dots,z_9$, with $\Xi|_{s=0}=\Xi_0$. More precisely, we fix r with $0<r<1/8$. For z near z^0 the discs $|\theta_j|<r$ for $1\le j\le 9$ are pairwise disjoint, for $|s_j|<r^2$ the plumbing annulus of index j on each sheet is $|s_j|/r<|\theta_j|<r$, and Ξ is holomorphic on the complement U_s in $\mathbb{P}^1$ of the discs $|\theta_j|\le|s_j|/r$. Let $\beta_{j,\ell}=\beta_{j,\ell}(s,z)$ be the 8×4 matrix coefficient of $\theta_j^\ell d\theta_j$ in the regular part of Ξ at z_j. Since the sheet $gt_{\pi(j)}$ carries $t_{\pi(j)}\Xi$ relative to the sheet g and $\theta_j'=s_j/\theta_j$, on a plumbing annulus of index j we have

$$\Xi=V_j\frac{d\theta_j}{\theta_j}+\sum_{\ell\ge 0}\beta_{j,\ell}\theta_j^\ell d\theta_j-\sum_{\ell\ge 0}s_j^{\ell+1}t_{\pi(j)}\beta_{j,\ell}\theta_j^{-\ell-2}d\theta_j. \tag{5.8}$$

This follows from

$$\theta_j'^{\ell}d\theta_j'=-s_j^{\ell+1}\theta_j^{-\ell-2}d\theta_j$$

and from $t_{\pi(j)}V_jd\theta_j'/\theta_j'=V_jd\theta_j/\theta_j$, which holds since $t_{\pi(j)}V_j=-V_j$.

Let $\kappa_{j,\ell}:=dz/(z-z_j)^{\ell+2}$ for $j\le 8$ and $\kappa_{9,\ell}:=-z^\ell dz$. Then $\kappa_{j,\ell}=\theta_j^{-\ell-2}d\theta_j$, and $\kappa_{j,\ell}$ has no pole other than z_j. For $1\le k\le 9$ let

$$\Xi_k^-:=-\sum_{\ell\ge 0}s_k^{\ell+1}t_{\pi(k)}\beta_{k,\ell}\kappa_{k,\ell},$$

the principal part of Ξ on the annulus at z_k without the residue term. The series of negative powers of θ_k in (5.8) converges for $|\theta_k|>|s_k|/r$, so Ξ_k^- is holomorphic on $\mathbb{P}^1$ minus the disc $|\theta_k|\le|s_k|/r$. Let $\Xi^+:=\Xi-\Xi_0-\sum_{k=1}^9\Xi_k^-$, a holomorphic differential on U_s. On the disc $|\theta_j|<r$ the differentials $\Xi_0-V_jd\theta_j/\theta_j$ and Ξ_k^- for $k\neq j$ are holomorphic, and by (5.8) the Laurent expansion of $\Xi-V_jd\theta_j/\theta_j-\Xi_j^-$ on the annulus at z_j has no negative powers of θ_j. Hence Ξ^+ has no principal part in any of the nine discs, and it extends to a holomorphic differential on $\mathbb{P}^1$. Since $\mathbb{P}^1$ has no non-zero holomorphic differentials, $\Xi^+=0$, that is,

$$\Xi=\Xi_0-\sum_{j=1}^9\sum_{\ell\ge 0}s_j^{\ell+1}t_{\pi(j)}\beta_{j,\ell}\kappa_{j,\ell}, \tag{5.9}$$

and the regular Taylor coefficients of the right-hand side at the points z_j determine the $\beta_{j,\ell}$ recursively as power series in s.

We claim that

$$\xi_j Z = L_j - 2\sum_{\ell\geq 0} s_j^{\ell+1}\beta_{j,\ell}^* t_{\pi(j)}\beta_{j,\ell} \qquad (1\leq j\leq 9). \tag{5.10}$$

The differential of the period map is the cup product with the Kodaira–Spencer class. Differentiating $\theta_j\theta_j' = s_j$ with respect to $\log s_j$ at fixed θ_j', we see that the Kodaira–Spencer class of ξ_j is represented on the plumbing annuli of index j by the vector field $\theta_j\partial/\partial\theta_j$. The pairing of the Kodaira–Spencer class of ξ_j with the product of two differentials is the sum over the annuli of index j of the coefficient of $\theta_j^{-2}(d\theta_j)^2$ in the product, up to a constant which does not depend on j. By (5.8), in the product of the expansions of the differentials attached to v and to w, the coefficient of $\theta_j^{-2}(d\theta_j)^2$ is the sum of the coefficients of $\theta_j^{-2}(d\theta_j)^2$ in the product of the residue terms and in the products of $\beta_{j,\ell}\theta_j^\ell d\theta_j$ with

$$-s_j^{\ell+1}t_{\pi(j)}\beta_{j,\ell}\theta_j^{-\ell-2}d\theta_j$$

in both orders, since no other pair of exponents has sum -2. Since $t_{\pi(j)}^* = t_{\pi(j)}$, summing over the $|G|/2$ annuli of index j and applying (5.7) as in **Step 1**, we obtain the right-hand side of (5.10) up to the constant, and the constant is 1 since the leading coefficient is L_j by **Step 1**. In our normalization, the terms of degree 1 in s of the terms with $\ell = 0$ are, including the sign, twice the linear terms of [HN20, Corollary 4.6, (4.11)], whose period matrix corresponds to $2\pi i\tau = Z/2$, and they have the sign of the linear terms of [Yam80, Corollary 6].

Integrating (5.10) and the expansion of τ in **Step 1**, we obtain

$$Z = \sum_{j=1}^{9} L_j\log s_j + F_0(z) + F_1(s,z) + F_2(s,z) + \cdots, \tag{5.11}$$

where F_k is homogeneous of degree k in s and the series converges on a polydisc around $(0,z^0)$. Additive constants and branches of the logarithms do not change D_Z. In particular, D_Z is a holomorphic function of (s,z) on the polydisc.

For $1\leq j<k\leq 8$ let $C_{jk} := V_j^*V_k + V_k^*V_j$, and let $\mathcal{E}$ be the set of pairs $\{j,k\}$, with $j<k$, such that $a_{\pi(j)}^t a_{\pi(k)} = -1$. By Construction-Theorem 4.5,

$$\mathcal{E} = \{\{1,2\},\{2,3\},\{3,4\},\{3,5\},\{5,6\},\{6,7\},\{7,8\}\},$$

and $a_{\pi(j)}^t a_{\pi(k)} = 0$ for the other pairs $j<k$. Hence $C_{jk} = -(n_j^*n_k + n_k^*n_j)$ for $\{j,k\}\in\mathcal{E}$ and $C_{jk}=0$ otherwise. We claim that, up to an additive constant,

$$F_0 = 2\sum_{\{j,k\}\in\mathcal{E}} C_{jk}\log(z_j - z_k). \tag{5.12}$$

A cycle of Γ' passing through the sheet g enters and leaves the sheet at node branches (g,z_j) and (g,z_k), and the part on the sheet g of the B-period along the lift of the cycle is the finite part of the integral from z_j to z_k, the logarithms of the branch coordinates being absorbed in the terms $L_j\log s_j$. We use the primitive $\sum_{l\leq 8} V_l\log(z-z_l)$ of Ξ_0. The finite part of the primitive at z_k is $\sum_{l\neq k, l\leq 8} V_l\log(z_k-z_l)$ for $k\leq 8$, and the finite part at $z_9=+\infty$ is constant. An additive constant in the primitive contributes $\sum_j V_j = 0$ after weighting with the residues. Weighting the finite parts of $\mathrm{Tr}_{K/\mathbb{Q}}(b_1^* g\Xi_0 v)$ at (g,z_k) by the residues $\mathrm{Tr}_{K/\mathbb{Q}}(b_1^* g V_k w)$ and summing over k and over the $|G|$ sheets, by (5.7) we obtain that the term of $\mathrm{Tr}_{K/\mathbb{Q}}(cv^*(2\pi i\tau)w)$ corresponding to F_0 in (5.11) is, up to an additive constant,

$$\mathrm{Tr}_{K/\mathbb{Q}}\Big(cv^*\sum_{k\neq l} V_l^*V_k\log(z_k-z_l)w\Big) = \mathrm{Tr}_{K/\mathbb{Q}}\Big(cv^*\sum_{\{j,k\}\in\mathcal{E}} C_{jk}\log(z_j-z_k)w\Big) + \text{constant}.$$

Since $Z = 2\cdot 2\pi i\tau$ by **Step 1**, (5.12) follows.

Step 3. In this step, we show that the constant term $\mathcal{A}_0 := D_Z|_{s=0}$ has rank 13, and we determine $\ker \mathcal{A}_0$.

By (5.10) and (5.11),

$$\mathcal{A}_0 = \Big[L_1, \dots, L_9, \frac{\partial F_0}{\partial z_2}, \dots, \frac{\partial F_0}{\partial z_7}\Big].$$

Since $C_{jk} = Q_{\{j,k\}} - L_j - L_k$ and $Q_I \in \mathcal{S}$ for any set I of leaves (Proposition 5.6), all columns of $\mathcal{A}_0$ belong to $\mathcal{S} \otimes_{\mathbb{Q}} \mathbb{C}$. Let

$$\rho(Q) := (d_1, d_2, d_3, d_4, x_{12}, y_{12}, x_{14}, x_{23}, y_{23}, x_{24}, y_{24}, x_{34}, y_{34}),$$
$$\Pi(Q) := (x_{12} - x_{13},\ y_{12} - y_{13},\ 5x_{12} - y_{12} - x_{14} + y_{14})$$

for $Q \in \mathrm{Herm}_4(K) \otimes_{\mathbb{Q}} \mathbb{C}$. By Proposition 5.6, $\ker \Pi = \mathcal{S} \otimes_{\mathbb{Q}} \mathbb{C}$.

We have

$$\sum_{\{j,k\}\in\mathcal{E}} C_{jk} = L_9 - \sum_{j=1}^{8} L_j, \qquad 2C_{67} - 4C_{78} = 4L_8 - L_6, \qquad L_1 = 2L_8. \tag{5.13}$$

The first relation follows by expanding $(\sum_{j\le 8} V_j)^*(\sum_{j\le 8} V_j) = V_9^* V_9 = L_9$. The second follows from $C_{jk} = -(n_j^* n_k + n_k^* n_j)$, from $L_j = 2n_j^* n_j$ for $j \le 8$, and from $n_7 = \frac{1-i}{2} n_6 + (1+i) n_8$ (Table 5). The third is the first relation of Proposition 5.6.

Let $\lambda := 2(1 - z_7) - (z_7 - z_6)$, and let

$$\zeta := \lambda \sum_{k=2}^{6} z_k \frac{\partial}{\partial z_k} + (\lambda z_7 + z_7 - z_6) \frac{\partial}{\partial z_7}, \qquad k_1 := \xi_1 - 2\xi_8, \qquad k_2 := \zeta + 2\lambda \sum_{j=1}^{8} \xi_j - 2\lambda\xi_9 + \xi_6 - 4\xi_8.$$

Since $z_1 = 0$ and $z_8 = 1$ are fixed, $\zeta \log(z_j - z_k)$ equals λ for the five pairs $\{j,k\} \in \mathcal{E} \setminus \{\{6,7\},\{7,8\}\}$, equals $\lambda + 1$ for $\{j,k\} = \{6,7\}$, and equals $\lambda - 2$ for $\{j,k\} = \{7,8\}$. The last value follows from

$$\lambda z_7 + z_7 - z_6 = (\lambda - 2)(z_7 - 1),$$

which is equivalent to the definition of λ. Hence (5.12) and (5.13) imply

$$\zeta F_0 = 2\lambda \sum_{\{j,k\}\in\mathcal{E}} C_{jk} + 2C_{67} - 4C_{78} = 2\lambda\Big(L_9 - \sum_{j=1}^{8} L_j\Big) + 4L_8 - L_6,$$

so $\mathcal{A}_0 k_2 = 0$, and $\mathcal{A}_0 k_1 = L_1 - 2L_8 = 0$. Here $\mathcal{A}_0$ is applied to the coefficient columns of k_1 and k_2 at each point, and no derivative of the coefficients of k_1 and k_2 is taken.

Let

$$\Sigma := (\xi_1, \dots, \xi_7, \xi_9, \partial/\partial z_2, \dots, \partial/\partial z_6),$$

and for a matrix M with 15 columns let M_Σ be the matrix of the 13 columns of M in Σ. At z^0,

$$\frac{\partial F_0}{\partial z_2} = C_{12} - 2C_{23}, \quad \frac{\partial F_0}{\partial z_3} = 2C_{23} - 2C_{34} - C_{35}, \quad \frac{\partial F_0}{\partial z_4} = 2C_{34},$$
$$\frac{\partial F_0}{\partial z_5} = C_{35} - 2C_{56}, \quad \frac{\partial F_0}{\partial z_6} = 2C_{56} - 2C_{67}.$$

Since $Q_{\le k} - Q_{\le k-1} = L_k + \sum_{j<k} C_{jk}$, modulo the span of $L_1, \dots, L_7, L_9$, which contains L_8, we have

$$C_{12} \equiv Q_{\le 2}, \quad C_{23} \equiv Q_{\le 3} - Q_{\le 2}, \quad C_{34} \equiv Q_{\le 4} - Q_{\le 3},$$
$$C_{35} \equiv Q_{\le 5} - Q_{\le 4}, \quad C_{56} \equiv Q_{\le 6} - Q_{\le 5},$$
$$C_{67} \equiv Q_{\le 7} - Q_{\le 6} \equiv -\tfrac{2}{3} Q_{\le 6},$$

where $Q_{\le 7} \equiv Q_{\le 6}/3$ follows from the second relation of Proposition 5.6. Let Λ be the 5×5 matrix whose k-th column is the coefficient vector of $\partial F_0/\partial z_{k+1}$ at z^0 with respect to $Q_{\le 2}, \dots, Q_{\le 6}$ modulo the span of $L_1, \dots, L_7, L_9$. Then $\det \Lambda = 40$ (see (A.4)). By (A.3), the thirteen matrices $L_1, \dots, L_7, L_9, Q_{\le 2}, \dots, Q_{\le 6}$ are linearly independent, so they form a basis of $\mathcal{S}$, and ρ restricts

to an isomorphism from $\mathcal{S}\otimes_{\mathbb{Q}}\mathbb{C}$ onto $\mathbb{C}^{13}$. In the basis $(L_1,\dots,L_7,L_9,Q_{\le 2},\dots,Q_{\le 6})$, the matrix of $(\mathcal{A}_0)_\Sigma$ at z^0 is block upper triangular with diagonal blocks I_8 and Λ. Hence, by (A.3), at z^0

$$\det\rho(\mathcal{A}_0)_\Sigma = 40\cdot(-6144) = -245760. \tag{5.14}$$

Thus $\operatorname{rank}\mathcal{A}_0 = 13$ and $\mathcal{A}_0$ has image $\mathcal{S}\otimes_{\mathbb{Q}}\mathbb{C}$ near z^0. At z^0 we have $\lambda=-13$, the vector k_1 has ξ_8-coefficient -2 and $\partial/\partial z_7$-coefficient 0, and k_2 has $\partial/\partial z_7$-coefficient -90. Therefore Σ together with k_1,k_2 is a basis of the tangent space in the coordinates (s,z), and $\ker\mathcal{A}_0$ is spanned by k_1 and k_2 at z^0. Let Ψ_0 be the 15×2 matrix of the coefficients of k_1 and k_2 at z^0 with respect to

$$\xi_1,\dots,\xi_9,\partial/\partial z_2,\dots,\partial/\partial z_7.$$

Step 4. In this step, we compute the terms of Z of degree at most 2 in s_1,s_9 on the locus $s_j=0$ for $j\neq 1,9$.

Since D_Z is holomorphic in (s,z) by **Step 2**, the Taylor coefficients of D_Z, and of any polynomial in the entries of D_Z, at monomials in s_1 and s_9 alone are the Taylor coefficients of their restrictions to the locus $s_j=0$ for $j\neq 1,9$. In particular, all fifteen columns of D_Z are computed before the restriction, and by (5.10) we have $\xi_j Z = L_j$ on the locus for $j\neq 1,9$.

Let

$$\alpha_1 := -\sum_{k=2}^{8}\frac{V_k}{z_k},\qquad \alpha_9 := -\sum_{k=2}^{8} z_k V_k,\qquad \gamma_1 := -\sum_{k=2}^{8}\frac{V_k}{z_k^2},\qquad \gamma_9 := -\sum_{k=2}^{8} z_k^2 V_k.$$

Since $\pi(1)=1$ and $\pi(9)=9$, we have $t_{\pi(1)}=t_1$ and $t_{\pi(9)}=t_9$. Expanding Ξ_0 in $\theta_1=z$ at $z_1=0$ and in $\theta_9=1/z$ at $+\infty$, we see that $\beta_{j,0}=\alpha_j$ and $\beta_{j,1}=\gamma_j$ at $s=0$ for $j=1,9$, where the terms with $k=1$ in α_9 and γ_9 vanish since $z_1=0$. The differential $\kappa_{1,0}=dz/z^2$ has no regular part at z_1, and $\kappa_{9,0}=-dz=\theta_9^{-2}d\theta_9$ has no regular part at $+\infty$. Hence by (5.9), on the locus, $\beta_{1,0}$ has no term linear in s_1, and $\beta_{9,0}$ has no term linear in s_9. The regular part of $\kappa_{9,0}=-dz$ at z_1 is $-d\theta_1$, so (5.9) implies that the coefficient of s_9 in $\beta_{1,0}$ is $t_9\alpha_9$.

For a monomial m in s, let $[m]$ denote the coefficient of m. Since ξ_j multiplies a monomial $s_1^{m_1}\cdots s_9^{m_9}$ by m_j, and F_1 is linear in s, (5.10) implies, on the locus,

$$F_1 = s_1\Phi_1 + s_9\Phi_9,\qquad \Phi_1 := -2\alpha_1^* t_1\alpha_1,\qquad \Phi_9 := -2\alpha_9^* t_9\alpha_9. \tag{5.15}$$

Since ξ_1 multiplies s_1^2 by 2, and $\beta_{1,0}$ has no term linear in s_1, the term with $\ell=1$ of (5.10) for $j=1$, and likewise for $j=9$, implies

$$[s_1^2]F_2 = -\gamma_1^* t_1\gamma_1,\qquad [s_9^2]F_2 = -\gamma_9^* t_9\gamma_9. \tag{5.16}$$

Finally, ξ_1 multiplies s_1s_9 by 1, and substituting $[s_9]\beta_{1,0}=t_9\alpha_9$ into the term with $\ell=0$ of (5.10) for $j=1$, we obtain

$$\mathcal{M} := [s_1s_9]F_2 = -2\big(\alpha_1^* t_1 t_9\alpha_9 + \alpha_9^* t_9 t_1\alpha_1\big). \tag{5.17}$$

Since $\mathcal{M}$ is the coefficient of s_1s_9 in Z, the equation (5.10) for $j=9$ yields $\mathcal{M}$ again, and it suffices to use (5.10) for $j=1$.

Step 5. In this step, we show that Π kills the pure terms of degree at most 2 on the locus, and we compute $\Pi\mathcal{M}$.

Let $w_2,\dots,w_8$ be holomorphic functions of z, on which $*$ acts trivially, and let $\mathcal{J} := \sum_{j=2}^{8} w_j V_j$. Then

$$\mathcal{J}^*\mathcal{J} = \sum_{j=2}^{8} w_j^2 L_j + \sum_{2\le j<k\le 8} w_j w_k C_{jk} \in \mathcal{S}\otimes_{\mathbb{Q}}\mathbb{C}.$$

Moreover,

$$\begin{aligned}(\mathcal{J}^* t_9\mathcal{J})_{12} = (\mathcal{J}^* t_9\mathcal{J})_{13} &= (w_2w_3 - w_2w_4) + i(w_3w_4 - w_2w_4),\\ (\mathcal{J}^* t_9\mathcal{J})_{14} &= (3w_2w_3 - 6w_2w_4 + 2w_3w_4) + i(-2w_2w_3 - 2w_2w_4 + 3w_3w_4).\end{aligned} \tag{5.18}$$

Among the first coordinates of $n_2,\dots,n_8$ only those of n_2 and n_4 are non-zero (Table 5), and $a_{\pi(2)}=a_3=-b_1+b_2$ and $a_{\pi(4)}=a_2=b_1+b_2$, so the first column of $\mathcal{J}$ belongs to Kb_1+Kb_2. Since t_9 preserves Kb_1+Kb_2 and acts on it by $\begin{pmatrix}0&-i\\ i&0\end{pmatrix}$, the first row of $\mathcal{J}^*t_9\mathcal{J}$ depends only on the first two rows of $\mathcal{J}$. Since $a_{\pi(3)}=a_4=-b_2+b_3$ and $a_{\pi(j)}$ has zero first two coordinates for $j\ge 5$, the first two rows of $\mathcal{J}$ are $-w_2n_2+w_4n_4$ and $w_2n_2-w_3n_3+w_4n_4$, and inserting the rows n_2,n_3,n_4 of Table 5 implies (5.18). Substituting (5.18) in the three components of Π, we obtain $\Pi(\mathcal{J}^*t_9\mathcal{J})=0$. Since $\Pi(\mathcal{J}^*t_9\mathcal{J})$ is quadratic in $(w_2,\dots,w_8)$ and vanishes identically, polarization implies

$$\Pi(\mathcal{J}^*t_9\mathcal{J}'+\mathcal{J}'^*t_9\mathcal{J})=0 \tag{5.19}$$

for any $\mathcal{J}=\sum_{j=2}^8 w_jV_j$ and $\mathcal{J}'=\sum_{j=2}^8 w'_jV_j$, where w_j and w'_j are holomorphic functions of z.

Since $a_1^ta_{\pi(j)}$ is -1 for $j=2$ and 0 for $3\le j\le 8$, we have $a_1^t\alpha_1=n_2/z_2$ and $a_1^t\gamma_1=n_2/z_2^2$. Since $t_1=I-a_1a_1^t$ and $n_2^*n_2=L_2/2$, we obtain

$$\Phi_1=-2\alpha_1^*\alpha_1+\frac{2}{z_2^2}n_2^*n_2\in\mathcal{S}\otimes_{\mathbb{Q}}\mathbb{C},\qquad [s_1^2]F_2=-\gamma_1^*\gamma_1+\frac{1}{z_2^4}n_2^*n_2\in\mathcal{S}\otimes_{\mathbb{Q}}\mathbb{C}.$$

Since α_9 and γ_9 are of the form $\mathcal{J}$, the identity $\Pi(\mathcal{J}^*t_9\mathcal{J})=0$ implies $\Pi\Phi_9=\Pi[s_9^2]F_2=0$. Hence

$$\Pi\Phi_1=\Pi\Phi_9=\Pi[s_1^2]F_2=\Pi[s_9^2]F_2=0 \tag{5.20}$$

as functions of $z_2,\dots,z_7$. In particular, Π also kills the derivatives of Φ_1, Φ_9, $[s_1^2]F_2$, and $[s_9^2]F_2$ with respect to $z_2,\dots,z_7$.

In (5.17) we write $t_1=I-a_1a_1^t$. Since α_1 and α_9 are of the form $\mathcal{J}$, the terms with I are killed by Π by (5.19). Since $t_9^*=t_9$ and $a_1^t\alpha_1=n_2/z_2$, we obtain

$$\Pi\mathcal{M}=\frac{2}{z_2}\Pi\big(n_2^*\mathcal{M}_1+\mathcal{M}_1^*n_2\big),\qquad \mathcal{M}_1:=a_1^tt_9\alpha_9=i(z_4n_4-z_5n_5+z_6n_6-z_7n_7), \tag{5.21}$$

where the second equality follows from $t_9=i\Omega$ and from the values $(0,0,-1,1,-1,1,0)$ of $a_1^t\Omega a_{\pi(j)}$ for $j=2,\dots,8$, computed directly from the coordinates of $a_1,\dots,a_8$ and Ω in Construction-Theorem 4.5. Let $Y_k:=i(n_2^*n_k-n_k^*n_2)\in\mathrm{Herm}_4(K)$ for $4\le k\le 7$. Then

$$n_2^*\mathcal{M}_1+\mathcal{M}_1^*n_2=z_4Y_4-z_5Y_5+z_6Y_6-z_7Y_7,$$

and the values of ΠY_k in Table 8 imply

$$f(z):=\Pi\mathcal{M}=\frac{1}{z_2}\big(-z_5+2z_6-z_7,\ -z_5+z_7,\ -2z_5+6z_6-4z_7\big). \tag{5.22}$$

Step 6. In this step, we eliminate the 13 columns in Σ, and we show that the remaining 3×2 matrix has a non-zero 2×2 minor.

By (5.14), $\rho(D_Z)_\Sigma$ is invertible near $(0,z^0)$. Let

$$\mathcal{R}:=\Pi\Big(D_Z\Psi_0-(D_Z)_\Sigma\big(\rho(D_Z)_\Sigma\big)^{-1}\rho D_Z\Psi_0\Big),$$

a 3×2 matrix of holomorphic functions of (s,z) near $(0,z^0)$. Since $\ker\Pi=\mathcal{S}\otimes_{\mathbb{Q}}\mathbb{C}$ and ρ restricts to an isomorphism on $\mathcal{S}\otimes_{\mathbb{Q}}\mathbb{C}$, the components of ρ and Π form 16 linearly independent coordinates on $\mathrm{Herm}_4(K)\otimes_{\mathbb{Q}}\mathbb{C}$. Since Σ together with the columns of Ψ_0 is a basis by **Step 3**, block elimination implies

$$\operatorname{rank}D_Z=13+\operatorname{rank}\mathcal{R}$$

near $(0,z^0)$.

We compute $\mathcal{R}$ on the locus $s_j=0$ for $j\ne 1,9$ at $z=z^0$, modulo terms of degree at least 3 in (s_1,s_9). By **Step 3**, $\Pi\mathcal{A}_0=0$. By (5.15), (5.20), and $\xi_jZ=L_j$ for $j\ne 1,9$, the terms of degree 1 of $\Pi(D_Z)_\Sigma$ vanish, so $\Pi(D_Z)_\Sigma=O((s_1,s_9)^2)$. Since $\mathcal{A}_0\Psi_0=0$ at z^0, we have $D_Z\Psi_0=O(s_1,s_9)$. Hence

$$\Pi(D_Z)_\Sigma\big(\rho(D_Z)_\Sigma\big)^{-1}\rho D_Z\Psi_0=O((s_1,s_9)^3),$$

and $\mathcal{R} \equiv \Pi D_Z \Psi_0$ modulo terms of degree at least 3. By (5.15) and (5.20), $\Pi D_Z \Psi_0$ has no terms of degree 1, and the coefficients of s_1^2 and s_9^2 in $\Pi D_Z \Psi_0$ are zero. On the locus, $[s_1 s_9](\xi_j Z)$ is $\mathcal{M}$ for $j = 1, 9$ and 0 for $j \neq 1, 9$. Hence the coefficient of $s_1 s_9$ in the first column of $\mathcal{R}$ is $f(z^0)$, since $k_1 = \xi_1 - 2\xi_8$. In the second column, the terms ξ_j of k_2 contribute $(2\lambda - 2\lambda) f(z^0) = 0$, and ζ contributes $(\zeta f)(z^0)$. Since f is homogeneous of degree 0 in (z_2, z_5, z_6, z_7) and does not depend on z_3, z_4, we have

$$\zeta f = (z_7 - z_6) \frac{\partial f}{\partial z_7} = \frac{z_7 - z_6}{z_2}(-1, 1, -4). \tag{5.23}$$

At z^0 we have $z_7 - z_6 = 1$, $z_2 = 2$, $z_5 = 5$, $z_6 = 6$, and $z_7 = 7$, so (5.22) and (5.23) imply $f(z^0) = (0, 1, -1)$ and $(\zeta f)(z^0) = (-1/2, 1/2, -2)$. Hence, on the locus at $z = z^0$,

$$\mathcal{R} = s_1 s_9 \begin{pmatrix} 0 & -1/2 \\ 1 & 1/2 \\ -1 & -2 \end{pmatrix} + O((s_1, s_9)^3). \tag{5.24}$$

Since $\mathcal{R}$ has no terms of degree 1 and the coefficients of s_1^2 and s_9^2 in $\mathcal{R}$ vanish, the minor of $\mathcal{R}$ formed by the first two rows is

$$\tfrac{1}{2} s_1^2 s_9^2 + (\text{terms of degree at least 5}). \tag{5.25}$$

Let μ be the 15×15 minor of D_Z formed by the rows ρ and the first two components of Π, with respect to the basis Σ, Ψ_0 of the columns. Then $\mu = \det \rho(D_Z)_\Sigma \cdot \det \mathcal{R}_{\{1,2\}}$, where $\mathcal{R}_{\{1,2\}}$ is formed by the first two rows of $\mathcal{R}$. By (5.14) and (5.25), the coefficient of $s_1^2 s_9^2$ in μ at $z = z^0$ is $-245760 \cdot \frac{1}{2} = -122880$. Thus μ is a non-zero holomorphic germ in $(s_1, \ldots, s_9, z_2, \ldots, z_7)$ at $(0, z^0)$. A non-zero holomorphic germ does not vanish identically on the dense open subset $\{s_1 \cdots s_9 \neq 0\}$, so $\mu \neq 0$ on a non-empty open subset of $\{s_1 \cdots s_9 \neq 0\}$. There $\operatorname{rank} D_Z = 15$, and by **Step 1** the local period map has rank at least 15 on the corresponding non-empty open subset of $\mathcal{H}$. Since $\dim \mathcal{H} = 15$, the rank is 15. □

6. Weil classes on sixfolds of Weil type

In this section, we prove Theorem 1.3. For the sixfold A of Theorem 1.3, we construct an abelian surface S and a K-linear isogeny from $A \times S$ onto one of the eightfolds A_X of Section 4, and we contract $W_K(A_X)$ with the class of a $\mathbb{Q}$-divisor on S.

6.1. Completing the Hermitian form.

In this subsection, we construct, for a sixfold A of Weil type with a compatible polarization, a family of polarized abelian surfaces S_s such that the Hermitian form of $A \times S_s$ is isometric to h_0.

The Hermitian forms of the Jacobian factors of Section 3 are split (Corollary 3.8) and a K-linear isogeny preserves the Hermitian form up to isometry (Lemma 2.10 (4)), whereas the Hermitian form of a compatible polarization of A need not be split.

Lemma 6.1. *Let K be an imaginary quadratic field and h_A a non-degenerate K-Hermitian form on K^6 of signature $(3, 3)$. Then $\det h_A < 0$, and the orthogonal sum $h_A \perp h_S$, where $h_S := \operatorname{diag}(1, (\det h_A)^{-1})$ on K^2, is isometric over K to $h_0 = H_K^{\oplus 4}$.*

Proof. The determinant of a Hermitian matrix is fixed by complex conjugation, so $\det h_A \in \mathbb{Q}$. Under σ, the matrix of h_A becomes a complex Hermitian matrix with three positive and three negative eigenvalues, so $\det h_A < 0$. Thus h_S has signature $(1, 1)$, the form $h_A \perp h_S$ has signature $(4, 4)$, and

$$\det(h_A \perp h_S) = \det h_A \cdot (\det h_A)^{-1} = 1.$$

Since $\det H_K = -1$, we have $\det h_0 = 1$, and h_0 has signature $(4, 4)$. Thus $h_A \perp h_S$ and h_0 are non-degenerate forms of K-dimension 8 and signature $(4, 4)$, and the class of the determinant of each of them in $\mathbb{Q}^* / \operatorname{Nm}_{K/\mathbb{Q}}(K^*)$ is [1]. By Landherr's theorem (cf. [Bay82, Definition 3.4, Lemma 3.5]), $h_A \perp h_S$ is isometric to h_0 over K. □

Construction-Theorem 6.2. Let $K, A, \iota, \Theta, h_A, h_S, a$ be as follows.

(i) K is an imaginary quadratic field, and A is an abelian sixfold with a unital embedding $\iota : K \hookrightarrow \mathrm{End}^0(A)$ such that (A, ι) is of Weil type.
(ii) Θ is a compatible polarization of (A, ι) with Hermitian form h_A on $H_1(A, \mathbb{Q})$, and $\det h_A$ is computed in a K-basis of $H_1(A, \mathbb{Q})$.
(iii) $h_S := \mathrm{diag}(1, (\det h_A)^{-1})$ on $V_S := K^2$.
(iv) a is a K-linear isometry

$$a : (H_1(A, \mathbb{Q}) \oplus V_S, h_A \perp h_S) \to (K^8, h_0),$$

which exists by Lemma 6.1.

Then the following hold.

(1) There exists a lattice $\Lambda_S \subset V_S$ stable under the ring of integers of K such that, for any $s \in \mathbb{D}(h_S)$, there exists a polarized abelian surface S_s with $H_1(S_s, \mathbb{Z}) = \Lambda_S$, a K-action of multiplicities $(1, 1)$, Hermitian form h_S, and period s with respect to id_{V_S}.
(2) The isometry a induces a holomorphic embedding $c : \mathbb{D}(h_S) \to \mathbb{D}(h_0)$ such that, for any $s \in \mathbb{D}(h_S)$, $c(s)$ is the period with respect to a of $A \times S_s$ with the diagonal K-action and the polarization

$$\mathrm{pr}_A^* \Theta + \mathrm{pr}_{S_s}^* \Theta_{S_s},$$

where Θ_{S_s} is the polarization of S_s.
(3) For any $s \in \mathbb{D}(h_S)$, any non-zero tangent vector of the image of c at $P := c(s)$ has rank one as an element of $\mathrm{Hom}_{\mathbb{C}}(P, P^{\perp})$.

Proof. By Lemma 2.10 (2), h_A has signature $(3, 3)$, so h_S has signature $(1, 1)$ by Lemma 6.1, and $\mathbb{D}(h_S)$ is defined.

(1) Let $\psi_S := \mathrm{Tr}_{K/\mathbb{Q}}(\delta h_S)$, where $\delta \in K$ with $\bar{\delta} = -\delta \neq 0$ is the element fixed in Definition 2.9, and let $\Lambda' \subset V_S$ be the direct sum of two copies of the ring of integers of K. Since Λ' is finitely generated and ψ_S takes rational values, there exists a positive integer ℓ such that ψ_S is integral on $\Lambda_S := \ell\Lambda'$, and Λ_S is stable under the ring of integers of K. For $s \in \mathbb{D}(h_S)$, by Definition-Proposition 2.12,

$$S_s := (V_S \otimes_{\mathbb{Q}} \mathbb{R}, J_s)/\Lambda_S$$

is an abelian surface with a polarization Θ_{S_s} of Riemann form ψ_S and a K-action of multiplicities $(1, 1)$, and $H_1(S_s, \mathbb{Z}) = \Lambda_S$. Since $\psi_S(kx, y) = \psi_S(x, \bar{k}y)$ for any $k \in K$, the polarization Θ_{S_s} is compatible, and the Hermitian form of Θ_{S_s} is h_S by the uniqueness in Definition 2.9. The complex structure J_s acts by ϵi on s and by $-\epsilon i$ on $s^{\perp}$, where ϵ is as in Definition-Proposition 2.12. Since h_S is positive definite on s and negative definite on $s^{\perp}$, the period of S_s with respect to id_{V_S} is s.

(2) The Riemann form of

$$\mathrm{pr}_A^* \Theta + \mathrm{pr}_{S_s}^* \Theta_{S_s}$$

on $H_1(A \times S_s, \mathbb{Q}) = H_1(A, \mathbb{Q}) \oplus V_S$ is the orthogonal sum of the Riemann forms of Θ and Θ_{S_s}. Hence

$$\mathrm{pr}_A^* \Theta + \mathrm{pr}_{S_s}^* \Theta_{S_s}$$

is a compatible polarization of $A \times S_s$ with Hermitian form $h_A \perp h_S$, and the period of $A \times S_s$ with respect to a is defined by Definition-Proposition 2.12. We denote the $\mathbb{C}$-linear extension of a to $(H_1(A, \mathbb{Q}) \oplus V_S) \otimes_{K,\sigma} \mathbb{C}$ again by a. Let $P_A \subset H_1(A, \mathbb{Q}) \otimes_{K,\sigma} \mathbb{C}$ be the eigenspace of the complex structure of $H_1(A, \mathbb{R})$ for the eigenvalue ϵi, which does not depend on s.

Let

$$E \subset (H_1(A, \mathbb{Q}) \oplus V_S) \otimes_{K,\sigma} \mathbb{C}$$

be the eigenspace of the complex structure of $H_1(A \times S_s, \mathbb{R})$ with $a(E)$ equal to the period of $A \times S_s$ with respect to a. Then $E = E_A \oplus E_S$, where E_A and E_S are the eigenspaces of the complex structures of $H_1(A, \mathbb{R})$ and $H_1(S_s, \mathbb{R})$ for the eigenvalue of E. Since $a(E) \in \mathbb{D}(h_0)$, the

form h_S is positive definite on E_S. Thus $E_S = s$, the eigenvalue of E is ϵi, and $E_A = P_A$. In particular, h_A is positive definite on P_A, and $\dim_{\mathbb{C}} P_A = 4 - 1 = 3$. Let

$$c(s) := a(P_A \oplus s)$$

for $s \in \mathbb{D}(h_S)$. Then $c(s)$ is the period of $A \times S_s$ with respect to a. Since a is an isometry, $c(s)^\perp = a(P_A^\perp \oplus s^\perp)$, where $P_A^\perp$ and $s^\perp$ are the orthogonal complements with respect to h_A and h_S.

(3) Let $s \in \mathbb{D}(h_S)$ and $P := c(s)$. For $\xi \in \mathrm{Hom}_{\mathbb{C}}(s, s^\perp)$, the sum of P_A and the graph of ξ is the graph of

$$0 \oplus \xi \in \mathrm{Hom}_{\mathbb{C}}(P_A \oplus s, P_A^\perp \oplus s^\perp).$$

Hence, in the chart of $\mathbb{D}(h_0)$ at P given by graphs of elements of $\mathrm{Hom}_{\mathbb{C}}(P, P^\perp)$ (Definition-Proposition 2.12), the map c sends the graph of ξ to the graph of

$$a \circ (0 \oplus \xi) \circ a^{-1} \in \mathrm{Hom}_{\mathbb{C}}(P, P^\perp).$$

The map $\xi \to a \circ (0 \oplus \xi) \circ a^{-1}$ is $\mathbb{C}$-linear and injective, and

$$\mathrm{rank}(a \circ (0 \oplus \xi) \circ a^{-1}) = \mathrm{rank}\, \xi \leq \dim_{\mathbb{C}} s = 1.$$

Thus c is a holomorphic immersion, and any non-zero tangent vector of the image of c at P has rank one. Moreover, c is injective, and the inverse of c on the image of c is the continuous map $P \to a^{-1}(P) \cap (V_S \otimes_{K,\sigma} \mathbb{C})$. Thus c is a holomorphic embedding. □

6.2. **A transversal intersection.** In this subsection, we find $s \in \mathbb{D}(h_S)$, $X \in \mathcal{H}$, and a K-linear isogeny $f : A \times S_s \to A_X$.

The period images of the families of Construction-Theorems 4.4 and 4.5 contain 15-dimensional pieces, which are hypersurfaces in the 16-dimensional domain $\mathbb{D}(h_0)$, while for a fixed A only the surface S_s moves, along the one-dimensional image of c.

Construction-Theorem 6.3. Let $(K, \mathcal{H})$ be $(\mathbb{Q}(\omega), \mathcal{H})$ of Construction-Theorem 4.4 or $(\mathbb{Q}(i), \mathcal{H})$ of Construction-Theorem 4.5. Then there exist a point $X_0 \in \mathcal{H}$, a simply connected open neighborhood $\mathcal{N}$ of X_0, K-linear isometries ϕ_X from the Hermitian forms of A_X to (K^8, h_0) for $X \in \mathcal{N}$ which are locally constant with respect to the local system $\{H_1(A_X, \mathbb{Q})\}_{X \in \mathcal{N}}$, and a 15-dimensional complex submanifold $\mathcal{N}' \subset \mathcal{N}$ through X_0, such that the local period map $p : \mathcal{N} \to \mathbb{D}(h_0)$ defined by the isometries ϕ_X restricts to an isomorphism from $\mathcal{N}'$ onto a smooth complex hypersurface Y of an open subset of $\mathbb{D}(h_0)$.

Proof. By Corollary 5.5 (for $K = \mathbb{Q}(\omega)$) or Theorem 5.7 (for $K = \mathbb{Q}(i)$), there exists a non-empty open subset $\mathcal{U} \subset \mathcal{H}$ on which the local period map has rank at least 15, and the rank of the local period map does not depend on the choices by Definition-Lemma 5.1. By Lemma 4.2 (2), $\mathcal{H}$ is smooth. Let $X_0 \in \mathcal{U}$, let $\mathcal{N} \subset \mathcal{U}$ be a coordinate ball centered at X_0, and choose isometries ϕ_X as in Definition-Lemma 5.1. Let $p : \mathcal{N} \to \mathbb{D}(h_0)$ be the local period map defined by the isometries ϕ_X. Then p is holomorphic by Lemma 2.13, since $H^{1,0}(A_X)$, the restriction of $H^{1,0}(J(X)) = H^{1,0}(X)$ to A_X, varies holomorphically with X.

Since $\mathrm{rank}\, dp_{X_0} \geq 15$, there exists a 15-dimensional complex linear subspace $T \subset T_{X_0}\mathcal{N}$ such that $dp_{X_0}|_T$ is injective. Let $\mathcal{N}_1' \subset \mathcal{N}$ be the intersection of $\mathcal{N}$ with the affine subspace through X_0 tangent to T in the coordinates of $\mathcal{N}$. Since $\dim \mathbb{D}(h_0) = 16$ (Definition-Proposition 2.12), there exist holomorphic coordinates $\zeta_1, \ldots, \zeta_{16}$ on an open neighborhood W of $p(X_0)$ such that the differential of

$$q := (\zeta_1, \ldots, \zeta_{15}) \circ p|_{\mathcal{N}_1'}$$

at X_0 is invertible. By the holomorphic inverse function theorem, there exists an open neighborhood $\mathcal{N}' \subset \mathcal{N}_1'$ of X_0 such that $p(\mathcal{N}') \subset W$ and q maps $\mathcal{N}'$ biholomorphically onto an open subset $\mathcal{B} \subset \mathbb{C}^{15}$. Then

$$Y := p(\mathcal{N}') = \{\zeta_{16} = \zeta_{16} \circ p \circ q^{-1}(\zeta_1, \ldots, \zeta_{15})\} \cap \{(\zeta_1, \ldots, \zeta_{15}) \in \mathcal{B}\}$$

is the graph of a holomorphic function on $\mathcal{B}$, hence a smooth complex hypersurface of the open subset $W \cap \{(\zeta_1, \dots, \zeta_{15}) \in \mathcal{B}\}$ of $\mathbb{D}(h_0)$. The map $p|_{\mathcal{N}'} : \mathcal{N}' \to Y$ is an isomorphism with inverse $q^{-1} \circ (\zeta_1, \dots, \zeta_{15})$. □

Lemma 6.4. *Let K be an imaginary quadratic field, h_S a K-Hermitian form of signature $(1,1)$ on K^2, Y a smooth complex hypersurface of an open subset of $\mathbb{D}(h_0)$, $y \in Y$, and $c : \mathbb{D}(h_S) \to \mathbb{D}(h_0)$ a holomorphic embedding such that for any $s \in \mathbb{D}(h_S)$, any non-zero tangent vector of the image of c at $P := c(s)$ has rank one as an element of $\mathrm{Hom}_{\mathbb{C}}(P, P^{\perp})$. Then for any $s_0 \in \mathbb{D}(h_S)$, there exist $u_0 \in \mathrm{U}(h_0)(\mathbb{R})$ with $u_0 c(s_0) = y$, an open neighborhood $\mathcal{O}$ of u_0 in $\mathrm{U}(h_0)(\mathbb{R})$, and a continuous map $\lambda : \mathcal{O} \to \mathbb{D}(h_S)$ such that $uc(\lambda(u)) \in Y$ for any $u \in \mathcal{O}$.*

Proof. Let $s_0 \in \mathbb{D}(h_S)$ and $P_0 := c(s_0)$. By Definition-Proposition 2.12, there exists $u_1 \in \mathrm{U}(h_0)(\mathbb{R})$ with $u_1 P_0 = y$. For $u \in \mathrm{U}(h_0)(\mathbb{R})$, $P \in \mathbb{D}(h_0)$, and $\xi \in \mathrm{Hom}_{\mathbb{C}}(P, P^{\perp})$, the element u maps the graph of ξ onto the graph of $u\xi u^{-1} \in \mathrm{Hom}_{\mathbb{C}}(uP, uP^{\perp})$, and $uP^{\perp} = (uP)^{\perp}$. Hence the differential of u at P is $\xi \to u\xi u^{-1}$, which preserves the rank. Since $\dim \mathbb{D}(h_S) = 1$, the tangent line of the image of c at P_0 is spanned by an element $\mu_0 \in \mathrm{Hom}_{\mathbb{C}}(P_0, P_0^{\perp})$ of rank one, and $\mu_1 := u_1 \mu_0 u_1^{-1} \in \mathrm{Hom}_{\mathbb{C}}(y, y^{\perp})$ has rank one.

Choose orthonormal bases of y with respect to h_0 and of $y^{\perp}$ with respect to $-h_0$, which identify $\mathrm{Hom}_{\mathbb{C}}(y, y^{\perp})$ with $M_4(\mathbb{C})$. For $g_1, g_2 \in \mathrm{U}(4)$, the automorphism $g_1 \oplus g_2$ of $y \oplus y^{\perp}$ is an element of $\mathrm{U}(h_0)(\mathbb{R})$ fixing y, and acts on $T_y \mathbb{D}(h_0) = M_4(\mathbb{C})$ by $\xi \to g_2 \xi g_1^{-1}$. Write $\mu_1 = vw^*$ with $v, w \in \mathbb{C}^4 \setminus \{0\}$. Since $\mathrm{U}(4)$ acts transitively on the vectors of a given norm in $\mathbb{C}^4$, the orbit of μ_1 contains $|v||w|E_{jk}$ for any $1 \le j, k \le 4$, where E_{jk} are the matrix units, and the matrices E_{jk} span $M_4(\mathbb{C})$. Since $T_y Y$ is a complex hyperplane of $T_y \mathbb{D}(h_0)$, there exist $g_1, g_2 \in \mathrm{U}(4)$ such that $g_2 \mu_1 g_1^{-1} \notin T_y Y$. Let $u_0 := (g_1 \oplus g_2) u_1$. Then $u_0 c(s_0) = y$, and the differential of $u_0 \circ c$ at s_0 maps a non-zero tangent vector to $g_2 \mu_1 g_1^{-1} \notin T_y Y$.

Choose an open neighborhood W of y and a holomorphic function $F : W \to \mathbb{C}$ such that $Y \cap W = F^{-1}(0)$ and $dF_y \neq 0$, so that $\ker dF_y = T_y Y$. Choose a holomorphic coordinate t on an open neighborhood of s_0 in $\mathbb{D}(h_S)$ with $t(s_0) = 0$, and write $c(t)$ for the value of c at the point with coordinate t. The action map

$$\mathrm{U}(h_0)(\mathbb{R}) \times \mathbb{D}(h_0) \to \mathbb{D}(h_0)$$

is real analytic, so

$$\chi(u, t) := F(uc(t))$$

is a smooth function on an open neighborhood of $(u_0, 0)$ in $\mathrm{U}(h_0)(\mathbb{R}) \times \mathbb{C}$, and χ is holomorphic in t. We have

$$\chi(u_0, 0) = F(y) = 0, \quad \frac{\partial \chi}{\partial t}(u_0, 0) = dF_y\Big(d(u_0 \circ c)\Big(\frac{\partial}{\partial t}\Big)\Big) \neq 0.$$

Since χ is holomorphic in t, the derivative of χ at $(u_0, 0)$ with respect to $(\mathrm{Re}\, t, \mathrm{Im}\, t)$ is the multiplication by a non-zero complex number, which is an invertible $\mathbb{R}$-linear endomorphism of $\mathbb{C}$. By the real implicit function theorem, there exist an open neighborhood $\mathcal{O}$ of u_0 in $\mathrm{U}(h_0)(\mathbb{R})$ and a continuous map $\lambda : \mathcal{O} \to \mathbb{D}(h_S)$ with values in the coordinate neighborhood of t such that $uc(\lambda(u)) \in W$ and $\chi(u, t(\lambda(u))) = 0$ for any $u \in \mathcal{O}$. Thus $uc(\lambda(u)) \in F^{-1}(0) = Y \cap W$ for any $u \in \mathcal{O}$. □

Proposition 6.5. *Let $(K, \mathcal{H})$ and Y be as in Construction-Theorem 6.3, and let A, Θ, S_s, and c be as in Construction-Theorem 6.2 for this K. Then there exist $s \in \mathbb{D}(h_S)$, $X \in \mathcal{H}$, a K-linear isogeny $f : A \times S_s \to A_X$, and a positive integer N, such that*

$$f^* \Theta_X = N^2(\mathrm{pr}_A^* \Theta + \mathrm{pr}_{S_s}^* \Theta_{S_s}),$$

where Θ_X and Θ_{S_s} are the polarizations of A_X and S_s.

Proof. Let $\mathcal{N}'$, ϕ_X, and p be as in Construction-Theorem 6.3, and choose $y \in Y$ and $s_0 \in \mathbb{D}(h_S)$. By Lemma 2.10 (2) and Lemma 6.1, h_S has signature $(1,1)$. By Construction-Theorem 6.2 (3), c satisfies the hypothesis of Lemma 6.4. By Lemma 6.4, there exist $u_0 \in \mathrm{U}(h_0)(\mathbb{R})$ with $u_0 c(s_0) = y$,

an open neighborhood $\mathcal{O}$ of u_0 in $\mathrm{U}(h_0)(\mathbb{R})$, and a continuous map $\lambda : \mathcal{O} \to \mathbb{D}(h_S)$ such that $uc(\lambda(u)) \in Y$ for any $u \in \mathcal{O}$.

The algebraic group $\mathrm{U}(h_0)$ over $\mathbb{Q}$ is connected, since

$$\mathrm{U}(h_0) \times_{\mathbb{Q}} \mathbb{C} \cong \mathrm{GL}_{8,\mathbb{C}}\,.$$

By real approximation [PR94, Theorem 7.7], $\mathrm{U}(h_0)(\mathbb{Q})$ is dense in $\mathrm{U}(h_0)(\mathbb{R})$. Choose $u \in \mathrm{U}(h_0)(\mathbb{Q}) \cap \mathcal{O}$, and let $s := \lambda(u)$. Then $uc(s) \in Y$, so there exists $X \in \mathcal{N}'$ such that $p(X) = uc(s)$, that is, $uc(s)$ is the period of A_X with respect to ϕ_X.

By Construction-Theorem 6.2 (2), $c(s)$ is the period of $A \times S_s$ with respect to a. Let

$$g := \phi_X^{-1} \circ u \circ a : H_1(A \times S_s, \mathbb{Q}) \to H_1(A_X, \mathbb{Q}).$$

Then g is a K-linear isometry from the Hermitian form of

$$\mathrm{pr}_A^*\,\Theta + \mathrm{pr}_{S_s}^*\,\Theta_{S_s}$$

to the Hermitian form of Θ_X, so g preserves the Riemann forms. Let

$$E \subset H_1(A \times S_s, \mathbb{Q}) \otimes_{K,\sigma} \mathbb{C}$$

and $E_X \subset H_1(A_X, \mathbb{Q}) \otimes_{K,\sigma} \mathbb{C}$ be the eigenspaces with $a(E) = c(s)$ and $\phi_X(E_X) = p(X)$. Then $g(E) = E_X$ and $g(E^\perp) = E_X^\perp$. The complex structure J of $H_1(A \times S_s, \mathbb{R})$ commutes with K and preserves the Riemann form, hence preserves the Hermitian form, so the eigenspaces of J for i and $-i$ are E and $E^\perp$ in some order. Similarly, the eigenspaces of the complex structure J_X of $H_1(A_X, \mathbb{R})$ are E_X and $E_X^\perp$. Thus $gJg^{-1} = \pm J_X$, and the sign is $+$, since g preserves the Riemann forms and the Riemann forms ψ of

$$\mathrm{pr}_A^*\,\Theta + \mathrm{pr}_{S_s}^*\,\Theta_{S_s}$$

and ψ_X of Θ_X satisfy $\psi(v, Jv) > 0$ and $\psi_X(v, J_X v) > 0$ for $v \neq 0$. Hence g is a K-linear isomorphism of rational Hodge structures.

Since $g(H_1(A \times S_s, \mathbb{Z}))$ is a finitely generated subgroup of $H_1(A_X, \mathbb{Q})$, there exists a positive integer N such that $Ng(H_1(A \times S_s, \mathbb{Z})) \subset H_1(A_X, \mathbb{Z})$. The map $Ng \otimes \mathbb{R}$ is $\mathbb{C}$-linear with respect to J and J_X, so it induces a holomorphic homomorphism $f : A \times S_s \to A_X$ with $f_* = Ng$ on $H_1(\cdot, \mathbb{Q})$. Since Ng is bijective on rational homology, f has finite kernel and is surjective, so f is an isogeny (cf. [Moo99, (2.2)]). Since f_* commutes with the actions of K on rational homology, f is K-linear. The Riemann form of $f^*\Theta_X$ sends (x, y) to $\psi_X(Ngx, Ngy) = N^2\psi(x, y)$. Since a class in $NS(A \times S_s) \otimes \mathbb{Q}$ is determined by the Riemann form,

$$f^*\Theta_X = N^2(\mathrm{pr}_A^*\,\Theta + \mathrm{pr}_{S_s}^*\,\Theta_{S_s}).$$

□

6.3. **Contraction with a divisor class.** In this subsection, we contract $W_K(A_X)$ with a class in $W_K(S_s)$, and we prove Theorem 1.3.

Lemma 6.6. *Let K be an imaginary quadratic field, let A be an abelian sixfold and S an abelian surface with K-actions of multiplicities $(3,3)$ and $(1,1)$, respectively, and let $f : A \times S \to A'$ be a K-linear isogeny onto an abelian eightfold A' with a unital embedding $K \hookrightarrow \mathrm{End}^0(A')$, where $A \times S$ carries the diagonal K-action. Then any $\beta \in W_K(S)$ is the class of a $\mathbb{Q}$-divisor, and for any non-zero $\beta \in W_K(S)$ the linear map sending $\alpha \in W_K(A')$ to*

$$(\mathrm{pr}_A)_*(f^*\alpha \cdot \mathrm{pr}_S^*\,\beta)$$

is an isomorphism from $W_K(A')$ onto $W_K(A)$.

Proof. Since $f^* : H^1(A', \mathbb{Q}) \to H^1(A \times S, \mathbb{Q})$ is an isomorphism of Hodge structures commuting with the actions of K, and hence the multiplicities of K on $H^{1,0}(A')$ are $(4,4)$, the plane $W_K(A')$ is defined. For $\tau \in \{\sigma, \bar{\sigma}\}$, f^* maps $H^1(A', \mathbb{C})_\tau$ onto

$$H^1(A \times S, \mathbb{C})_\tau = \mathrm{pr}_A^*\,H^1(A, \mathbb{C})_\tau \oplus \mathrm{pr}_S^*\,H^1(S, \mathbb{C})_\tau,$$

and $\dim_{\mathbb{C}} H^1(A,\mathbb{C})_\tau = 6$, $\dim_{\mathbb{C}} H^1(S,\mathbb{C})_\tau = 2$. Since f^* is a ring homomorphism, by the Künneth formula,

$$f^* \wedge^8 H^1(A',\mathbb{C})_\tau = \wedge^6 H^1(A,\mathbb{C})_\tau \otimes \wedge^2 H^1(S,\mathbb{C})_\tau \subset H^6(A,\mathbb{C}) \otimes H^2(S,\mathbb{C}) \subset H^8(A\times S,\mathbb{C}),$$

where $\gamma \otimes b$ corresponds to $\mathrm{pr}_A^*\,\gamma \cdot \mathrm{pr}_S^*\, b$.

The pair (S,K) is of Weil type, so $W_K(S) \subset \mathrm{Hdg}^1(S)$ by Lemma 2.10 (2). By the Lefschetz $(1,1)$ theorem (cf. [Moo99, (2.10)]), any $\beta \in W_K(S)$ is the class of a $\mathbb{Q}$-divisor.

Let $\beta \in W_K(S)$ be non-zero, and write $\beta = \beta_\sigma + \beta_{\bar\sigma}$ with $\beta_\tau \in \wedge^2 H^1(S,\mathbb{C})_\tau$. Complex conjugation maps $H^1(S,\mathbb{C})_\sigma$ onto $H^1(S,\mathbb{C})_{\bar\sigma}$ and fixes β, so $\beta_{\bar\sigma} = \overline{\beta_\sigma}$, and both β_σ and $\beta_{\bar\sigma}$ are non-zero. Since

$$H^1(S,\mathbb{C}) = H^1(S,\mathbb{C})_\sigma \oplus H^1(S,\mathbb{C})_{\bar\sigma},$$

the cup product

$$\wedge^2 H^1(S,\mathbb{C})_\sigma \otimes \wedge^2 H^1(S,\mathbb{C})_{\bar\sigma} \to \wedge^4 H^1(S,\mathbb{C}) = H^4(S,\mathbb{C})$$

is an isomorphism, and

$$\wedge^2 H^1(S,\mathbb{C})_\tau \cdot \wedge^2 H^1(S,\mathbb{C})_\tau \subset \wedge^4 H^1(S,\mathbb{C})_\tau = 0.$$

Let $\gamma \in \wedge^6 H^1(A,\mathbb{C})_\sigma$ and $b \in \wedge^2 H^1(S,\mathbb{C})_\sigma$. By the projection formula,

$$(\mathrm{pr}_A)_*\big(\mathrm{pr}_A^*\,\gamma \cdot \mathrm{pr}_S^*\, b \cdot \mathrm{pr}_S^*\,\beta\big) = \Big(\int_S b\cdot\beta_{\bar\sigma}\Big)\gamma,$$

and $\int_S b\cdot\beta_{\bar\sigma} \neq 0$ if $b \neq 0$. The case of $\bar\sigma$ is symmetric. Hence the $\mathbb{C}$-linear extension of the map

$$\alpha \to (\mathrm{pr}_A)_*(f^*\alpha \cdot \mathrm{pr}_S^*\,\beta)$$

maps $\wedge^8 H^1(A',\mathbb{C})_\tau$ isomorphically onto $\wedge^6 H^1(A,\mathbb{C})_\tau$ for $\tau \in \{\sigma,\bar\sigma\}$, and thus maps $W_K(A')\otimes\mathbb{C}$ isomorphically onto $W_K(A)\otimes\mathbb{C}$. Since f^*, pr_S^*, $(\mathrm{pr}_A)_*$, and the cup product are defined over $\mathbb{Q}$, the map sends $W_K(A')$ into $H^6(A,\mathbb{Q}) \cap (W_K(A)\otimes\mathbb{C}) = W_K(A)$. Since $\dim_{\mathbb{Q}} W_K(A') = \dim_{\mathbb{Q}} W_K(A) = 2$, the map is an isomorphism from $W_K(A')$ onto $W_K(A)$. □

Proof of Theorem 1.3. Let $\iota : K \hookrightarrow \mathrm{End}^0(A)$ be the given embedding. Since $\dim A = 6$ and the multiplicities of K on $H^{1,0}(A)$ are $(3,3)$, the pair (A,ι) is of Weil type. By Lemma 2.10 (1), there exists a compatible polarization Θ of (A,ι). Let S_s and c be as in Construction-Theorem 6.2. If $K = \mathbb{Q}(i)$, let $\mathcal{H}$ be as in Construction-Theorem 4.5, and if $K = \mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\omega)$, let $\mathcal{H}$ be as in Construction-Theorem 4.4. Let Y be as in Construction-Theorem 6.3. By Proposition 6.5, there exist $s \in \mathbb{D}(h_S)$, $X \in \mathcal{H}$, and a K-linear isogeny $f : A \times S_s \to A_X$. By Construction-Theorem 4.5 (3) or Construction-Theorem 4.4 (3), any class in $W_K(A_X)$ is algebraic.

Since $\dim_{\mathbb{Q}} W_K(S_s) = 2$, there exists a non-zero $\beta \in W_K(S_s)$, and by Lemma 6.6, $\beta = \mathrm{cl}(D)$ for a $\mathbb{Q}$-divisor D on S_s. Let $w \in W_K(A)$. By Lemma 6.6, there exists $\alpha \in W_K(A_X)$ such that

$$w = (\mathrm{pr}_A)_*\big(f^*\alpha \cdot \mathrm{pr}_{S_s}^*\,\beta\big).$$

Let C be an algebraic cycle of codimension 4 on A_X with rational coefficients such that $\mathrm{cl}(C) = \alpha$. Since the cycle class map commutes with pullback, proper pushforward, and products [Ful98, Corollary 19.2],

$$w = \mathrm{cl}\,\big((\mathrm{pr}_A)_*(f^*C \cdot \mathrm{pr}_{S_s}^*\, D)\big),$$

and

$$(\mathrm{pr}_A)_*(f^*C \cdot \mathrm{pr}_{S_s}^*\, D)$$

is an algebraic cycle of codimension $4+1-2 = 3$ on A with rational coefficients. □

7. Hodge classes on simple sixfolds

In this section, according to the type of $\mathrm{End}^0(A)$, we prove the results on the Hodge classes of simple abelian sixfolds that Section 9 uses in the proof of Theorem 1.2. We treat the simple sixfolds A such that $\mathrm{End}^0(A)$ is an imaginary quadratic field, a quartic or sextic CM field, or a definite quaternion algebra. For each of them, we determine whether the Hodge classes are spanned by products of divisor classes. We also determine whether the discriminant of the Hermitian form of a compatible polarization depends on the polarization, which shows where Theorem 2.14 does not suffice.

7.1. Imaginary quadratic endomorphism field.

In this subsection, we compute the Hodge group and the Hodge classes of a simple abelian sixfold whose endomorphism algebra is an imaginary quadratic field.

Proposition 7.1. *Let A be a simple abelian sixfold with $\mathrm{End}^0(A) = K$ an imaginary quadratic field, let Θ be an ample class, let $V := H^1(A, \mathbb{Q})$, viewed as a 6-dimensional K-vector space, let ψ be the alternating form on V dual to the Riemann form of a polarization of A, and let $(r, 6-r)$ be the multiplicities of K on $H^{1,0}(A)$. Then $\psi(ax, y) = \psi(x, \bar{a}y)$ for any $a \in K$ and $x, y \in V$, $1 \le r \le 5$,*

$$\mathrm{Hg}(A) = \mathrm{SU}_K(V, \psi) \quad \text{if } r = 3, \quad \text{and} \quad \mathrm{Hg}(A) = \mathrm{U}_K(V, \psi) \quad \text{if } r \neq 3.$$

If $r \neq 3$, then $\mathrm{Hdg}^p(A) = \mathbb{Q}\Theta^p$ for any $0 \le p \le 6$. If $r = 3$, then $\mathrm{Hdg}^p(A) = \mathbb{Q}\Theta^p$ for $p \neq 3$,

$$\mathrm{Hdg}^3(A) = \mathbb{Q}\Theta^3 \oplus W_K(A),$$

and $\Theta \cdot W_K(A) = 0$.

Proof of Proposition 7.1. The Rosati involution of the polarization is a positive involution of $\mathrm{End}^0(A) = K$, so it is complex conjugation [Moo99, (1.19)]. Hence $\psi(ax, y) = \psi(x, \bar{a}y)$ for any $a \in K$ and $x, y \in V$. The rest of the proof is long, so we divide it into several steps.

Step 1. In this step, we show that $\mathrm{Hg}(A) \subset \mathrm{U}_K(V, \psi)$, that $1 \le r \le 5$, and that $\mathrm{Hg}(A)$ is a semisimple subgroup of $\mathrm{SU}_K(V, \psi)$ if $r = 3$.

By Lemma 2.3 (1), $\mathrm{Hg}(A)$ is connected and reductive, and the commutant of $\mathrm{Hg}(A)$ in $\mathrm{End}_{\mathbb{Q}}(V)$ is K. The Riemann form of a polarization of A, viewed as an element of $\wedge^2 V = H^2(A, \mathbb{Q})$, is the first Chern class of an ample line bundle. Thus the Riemann form is a Hodge class, and it is fixed by $\mathrm{Hg}(A)$ (Lemma 2.3 (2)). Hence $\mathrm{Hg}(A)$ preserves ψ. Since $\mathrm{Hg}(A)$ also commutes with K, we have $\mathrm{Hg}(A) \subset \mathrm{U}_K(V, \psi)$.

Let $V_\sigma := H^1(A, \mathbb{C})_\sigma$ and $V_{\bar\sigma} := H^1(A, \mathbb{C})_{\bar\sigma}$. Since $\dim_K V = 6$, we have

$$V \otimes \mathbb{C} = V_\sigma \oplus V_{\bar\sigma}, \qquad \dim V_\sigma = \dim V_{\bar\sigma} = 6.$$

For any $x, y \in V_\sigma$ and $a \in K$, we have

$$\sigma(a)\psi(x, y) = \psi(ax, y) = \psi(x, \bar{a}y) = \overline{\sigma(a)}\psi(x, y).$$

Taking $a \in K \setminus \mathbb{Q}$, we get $\psi(x, y) = 0$. Thus V_σ is isotropic, and so is $V_{\bar\sigma}$. Since ψ is non-degenerate, ψ induces a perfect pairing between V_σ and $V_{\bar\sigma}$. Any $g \in \mathrm{U}_K(V, \psi)(\mathbb{C})$ preserves V_σ and $V_{\bar\sigma}$, and $g|_{V_{\bar\sigma}}$ is determined by $g|_{V_\sigma}$ through the pairing induced by ψ. Hence restriction to V_σ induces isomorphisms

$$\mathrm{U}_K(V, \psi)_{\mathbb{C}} \xrightarrow{\cong} \mathrm{GL}(V_\sigma), \qquad \mathrm{SU}_K(V, \psi)_{\mathbb{C}} \xrightarrow{\cong} \mathrm{SL}(V_\sigma),$$

under which the σ-component of $\det_K$ is the determinant on V_σ, and $V_{\bar\sigma} \cong V_\sigma^\vee$ as representations of $\mathrm{U}_K(V, \psi)_{\mathbb{C}}$. In particular, $\mathrm{Hg}(A)_{\mathbb{C}}$ acts faithfully on V_σ.

The commutant of $\mathrm{Hg}(A)_{\mathbb{C}}$ in $\mathrm{End}(V \otimes \mathbb{C})$ is $K \otimes \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$, which acts on V_σ and on $V_{\bar\sigma}$ by scalars. Thus

$$\mathrm{End}_{\mathrm{Hg}(A)_{\mathbb{C}}}(V_\sigma) = \mathbb{C} \qquad \text{and} \qquad \mathrm{Hom}_{\mathrm{Hg}(A)_{\mathbb{C}}}(V_\sigma, V_\sigma^\vee) = 0.$$

Since $\mathrm{Hg}(A)_{\mathbb{C}}$ is reductive, V_σ is a semisimple representation, and hence V_σ is irreducible. The center of $\mathrm{Hg}(A)_{\mathbb{C}}$ acts on V_σ by scalars, and $\mathrm{Hg}(A)_{\mathbb{C}}$ is generated by the center and the derived group $\mathrm{Hg}(A)^{\mathrm{der}}_{\mathbb{C}}$ [Bor91, Proposition 14.2]. Thus V_σ is an irreducible representation of $\mathrm{Hg}(A)^{\mathrm{der}}_{\mathbb{C}}$. Since $\dim V_\sigma > 1$, $\mathrm{Hg}(A)^{\mathrm{der}}_{\mathbb{C}}$ is non-trivial.

Let $\mathfrak{g}$ be the Lie algebra of $\mathrm{Hg}(A)_{\mathbb{C}}$, and let $\mathfrak{g}^{\mathrm{der}} := [\mathfrak{g}, \mathfrak{g}]$ be the Lie algebra of $\mathrm{Hg}(A)^{\mathrm{der}}_{\mathbb{C}}$. Let $\xi \in \mathrm{End}(V \otimes \mathbb{C})$ act on $H^{1,0}(A)$ by 1 and on $H^{0,1}(A)$ by -1. Then $\eta_A(e^{i\theta}) = \exp(i\theta\xi)$ for any $\theta \in \mathbb{R}$. Since $\eta_A(\mathbb{U}) \subset \mathrm{Hg}(A)(\mathbb{R})$ (Definition 2.2), we have $\xi \in \mathfrak{g}$. The action of K preserves the Hodge decomposition, so

$$V_\sigma = \big(V_\sigma \cap H^{1,0}(A)\big) \oplus \big(V_\sigma \cap H^{0,1}(A)\big),$$

and complex conjugation maps $V_{\bar\sigma} \cap H^{1,0}(A)$ onto $V_\sigma \cap H^{0,1}(A)$. By Definition 2.7, ξ acts on V_σ with eigenvalue 1 of multiplicity r and eigenvalue -1 of multiplicity $6 - r$. In particular,

$$\det\big(\eta_A(z)|_{V_\sigma}\big) = z^{2r-6} \quad \text{for any } z \in \mathbb{U}. \tag{7.1}$$

Suppose that $r \in \{0, 6\}$. Then $\eta_A(z)$ acts on V_σ by a scalar λ with $|\lambda| = 1$ and on $V_{\bar\sigma}$ by $\bar\lambda$, that is, by multiplication by the element $a \in K \otimes \mathbb{R}$ with $\sigma(a) = \lambda$, which satisfies $a\bar a = 1$. Thus $\eta_A(\mathbb{U}) \subset \mathrm{U}_K(\mathbb{R})$, where U_K acts on V through the action of K. By the minimality of $\mathrm{Hg}(A)$ (Definition 2.2), $\mathrm{Hg}(A)$ is contained in the image of U_K. Hence $\mathrm{End}_K(V)$ commutes with $\mathrm{Hg}(A)$, and

$$\dim_{\mathbb{Q}} \mathrm{End}_K(V) = 72 > 2 = \dim_{\mathbb{Q}} K,$$

which contradicts Lemma 2.3 (1). Thus $1 \le r \le 5$.

Suppose that $r = 3$. By (7.1), the determinant of $\eta_A(z)$ on V_σ is 1 for any $z \in \mathbb{U}$, so

$$\eta_A(\mathbb{U}) \subset \mathrm{SU}_K(V, \psi)(\mathbb{R}).$$

By the minimality of $\mathrm{Hg}(A)$, $\mathrm{Hg}(A) \subset \mathrm{SU}_K(V, \psi)$. Thus the center of $\mathrm{Hg}(A)_{\mathbb{C}}$ acts faithfully on V_σ by scalars λ with $\lambda^6 = 1$, so the center is finite. Since $\mathrm{Hg}(A)$ is reductive, $\mathrm{Hg}(A)$ is semisimple.

Step 2. In this step, we show that $\mathfrak{g}^{\mathrm{der}} = \mathfrak{sl}(V_\sigma)$ and deduce the formulas for $\mathrm{Hg}(A)$.

The results [Moo99, Theorem (3.11), Corollary (3.13)] are formulated for $\mathrm{MT}(A)$. Since $\mathrm{MT}(A) = \mathbb{G}_m \cdot \mathrm{Hg}(A)$, the derived groups of $\mathrm{MT}(A)$ and $\mathrm{Hg}(A)$ coincide (cf. [Moo99, (1.11)]). Let

$$\mathfrak{g}^{\mathrm{der}} = \mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_m$$

be the decomposition into simple ideals. By [Moo99, Theorem (3.11), Corollary (3.13)], the irreducible representation V_σ of $\mathfrak{g}^{\mathrm{der}}$ is isomorphic to $\rho_1 \otimes \cdots \otimes \rho_m$, where each ρ_j is an irreducible representation of $\mathfrak{s}_j$ which is either trivial or minuscule in the sense of [Moo99, Example (3.6)]. Since $\mathfrak{g}$ acts faithfully on V_σ, each ρ_j is non-trivial, and $\prod_j \dim \rho_j = 6$ with $m \ge 1$. By the list of minuscule representations [Moo99, Example (3.6), Table 3], up to replacing ρ_j with the dual representation, we are in one of the following cases.

(i) $m = 1$, $\mathfrak{s}_1 \cong \mathfrak{sl}_6$, and ρ_1 is the standard representation $\mathbb{C}^6$.
(ii) $m = 1$, $\mathfrak{s}_1 \cong \mathfrak{sl}_4$, and $\rho_1 = \wedge^2\mathbb{C}^4$.
(iii) $m = 1$, $\mathfrak{s}_1 \cong \mathfrak{sp}_6$, and ρ_1 is the standard representation $\mathbb{C}^6$.
(iv) $m = 2$, $\mathfrak{s}_1 \cong \mathfrak{sl}_2$, $\mathfrak{s}_2 \cong \mathfrak{sl}_3$, and $\rho_1 \otimes \rho_2 = \mathbb{C}^2 \otimes \mathbb{C}^3$ is the tensor product of the standard representations.

No other case occurs, since the minuscule representations of type A_l are $\wedge^j\mathbb{C}^{l+1}$ with $1 \le j \le l$, and $\binom{l+1}{j} = 6$ only for $(l+1, j) \in \{(6,1), (6,5), (4,2)\}$. The spin representation of type B_l and the half-spin representations of type D_l have dimensions powers of 2, the standard representation of type C_l is of dimension $2l$, the standard representation of type D_l with $l \ge 4$ is of dimension $2l \ge 8$, and the minuscule representations of types E_6 and E_7 are of dimensions 27 and 56. If $m = 2$, then $\{\dim \rho_1, \dim \rho_2\} = \{2, 3\}$, which only occurs for the standard representations of $\mathfrak{sl}_2$ and of $\mathfrak{sl}_3$, and if $m \ge 3$, then $\prod_j \dim \rho_j \ge 8$.

We exclude case (iv). The center of $\mathfrak{g}$ acts on V_σ by scalars, so we may write $\xi = cI + \xi_1 \otimes 1 + 1 \otimes \xi_2$ on $V_\sigma = \mathbb{C}^2 \otimes \mathbb{C}^3$, where $c \in \mathbb{C}$, $\xi_1 \in \mathfrak{s}_1$, and $\xi_2 \in \mathfrak{s}_2$. Let a_1, a_2 be the eigenvalues of ξ_1 on $\mathbb{C}^2$

and b_1, b_2, b_3 the eigenvalues of ξ_2 on $\mathbb{C}^3$. Triangularizing ξ_1 and ξ_2, we see that the eigenvalues of ξ on V_σ are $c + a_i + b_j$ for $1 \le i \le 2$ and $1 \le j \le 3$, so

$$c + a_i + b_j \in \{1, -1\} \quad \text{for any } i, j.$$

Hence $a_1 - a_2$ and $b_j - b_k$ are real for any j, k. Since $a_1 + a_2 = 0$ and $b_1 + b_2 + b_3 = 0$, all a_i, all b_j, and c are real. If $a_1 \neq a_2$ and b_1, b_2, b_3 are not all equal, let $a < a'$ be the minimum and the maximum of $\{a_1, a_2\}$ and $b < b'$ the minimum and the maximum of $\{b_1, b_2, b_3\}$. Then $a + b < a' + b < a' + b'$ are three distinct values of $a_i + b_j$, while $a_i + b_j \in \{1 - c, -1 - c\}$, a contradiction. Thus $a_1 = a_2 = 0$ or $b_1 = b_2 = b_3 = 0$. Let $\mathfrak{s} := \mathfrak{s}_1$ in the first case and $\mathfrak{s} := \mathfrak{s}_2$ in the second case. Since ξ is diagonalizable on $V \otimes \mathbb{C}$, $\mathrm{ad}(\xi)$ is diagonalizable on $\mathrm{End}(V \otimes \mathbb{C})$, and hence on the invariant subspace $\mathfrak{s}$. On $\mathfrak{s}$, $\mathrm{ad}(\xi)$ is equal to $\mathrm{ad}(\xi_1)$ in the first case and to $\mathrm{ad}(\xi_2)$ in the second case, so any eigenvalue of $\mathrm{ad}(\xi)$ on $\mathfrak{s}$ is of the form $a_i - a_k = 0$ or $b_j - b_k = 0$. Thus

$$[\xi, \mathfrak{s}] = 0.$$

For any $\tau \in \mathrm{Aut}(\mathbb{C}/\mathbb{Q})$, the map $1 \otimes \tau$ on

$$\mathfrak{g} = \mathrm{Lie}\,\mathrm{Hg}(A) \otimes_{\mathbb{Q}} \mathbb{C}$$

is a τ-semilinear automorphism of Lie algebras, so it permutes the simple ideals of $\mathfrak{g}^{\mathrm{der}}$ and preserves their dimensions. Since $\dim \mathfrak{s}_1 = 3 \neq 8 = \dim \mathfrak{s}_2$, $\mathfrak{s}$ is stable under $\mathrm{Aut}(\mathbb{C}/\mathbb{Q})$, so $\mathfrak{s} = \mathfrak{s}_0 \otimes \mathbb{C}$ for a $\mathbb{Q}$-subspace $\mathfrak{s}_0 \subset \mathrm{Lie}\,\mathrm{Hg}(A)$ (cf. [Bor91, Chapter AG, §14]). The adjoint action on $\mathfrak{s}_0$ is a homomorphism

$$\mathrm{Hg}(A) \to \mathrm{GL}(\mathfrak{s}_0)$$

defined over $\mathbb{Q}$, and we let $N \subset \mathrm{Hg}(A)$ be the kernel. Since $[\xi, \mathfrak{s}] = 0$, the operator

$$\mathrm{Ad}(\eta_A(e^{i\theta})) = \exp(i\theta\,\mathrm{ad}(\xi))$$

is the identity on $\mathfrak{s}$ for any $\theta \in \mathbb{R}$, so $\eta_A(\mathbb{U}) \subset N(\mathbb{R})$. By the minimality of $\mathrm{Hg}(A)$, $N = \mathrm{Hg}(A)$. Thus $[\mathfrak{g}, \mathfrak{s}] = 0$, which is impossible since $[\mathfrak{s}, \mathfrak{s}] = \mathfrak{s} \neq 0$. Hence case (iv) does not occur.

We exclude cases (ii) and (iii). Write $\xi = cI + \xi'$ on V_σ, where $c \in \mathbb{C}$ and $\xi' \in \mathfrak{g}^{\mathrm{der}}$. Since the trace of ξ' on V_σ is 0, we have $6c = r - (6 - r)$, and the eigenvalues of ξ' on V_σ are $1 - c$ with multiplicity r and $-1 - c$ with multiplicity $6 - r$. Both eigenvalues occur since $1 \le r \le 5$, and they are distinct. In cases (ii) and (iii), $V_\sigma \cong V_\sigma^\vee$ as representations of $\mathfrak{g}^{\mathrm{der}}$, since $\wedge^2\mathbb{C}^4$ is self-dual through the pairing

$$\wedge^2\mathbb{C}^4 \times \wedge^2\mathbb{C}^4 \to \wedge^4\mathbb{C}^4 \cong \mathbb{C},$$

and the standard representation of $\mathfrak{sp}_6$ is self-dual through the symplectic form. Thus the eigenvalues of ξ' and of $-\xi'$ on V_σ coincide with multiplicities. Thus $\{1-c, -1-c\} = \{c-1, c+1\}$, and comparing the sums of the elements, we get $c = 0$. Hence the eigenvalues 1 and -1 of ξ' have equal multiplicities, that is, $r = 6 - r$, and so $r = 3$. By **Step 1**, $\mathrm{Hg}(A)$ is semisimple, so $\mathfrak{g} = \mathfrak{g}^{\mathrm{der}}$. Since $\mathrm{Hg}(A)_{\mathbb{C}}$ is connected, an isomorphism $V_\sigma \cong V_\sigma^\vee$ of representations of $\mathfrak{g}$ is $\mathrm{Hg}(A)_{\mathbb{C}}$-equivariant, which contradicts

$$\mathrm{Hom}_{\mathrm{Hg}(A)_{\mathbb{C}}}(V_\sigma, V_\sigma^\vee) = 0.$$

Hence cases (ii) and (iii) do not occur.

Thus we are in case (i), and

$$\mathfrak{g}^{\mathrm{der}} \subset \mathfrak{sl}(V_\sigma) \quad \text{with} \quad \dim \mathfrak{g}^{\mathrm{der}} = 35 = \dim \mathfrak{sl}(V_\sigma).$$

Hence $\mathfrak{g}^{\mathrm{der}} = \mathfrak{sl}(V_\sigma)$, and

$$\mathrm{Hg}(A)^{\mathrm{der}}_{\mathbb{C}} = \mathrm{SL}(V_\sigma)$$

since connected algebraic subgroups of $\mathrm{GL}(V_\sigma)$ are determined by their Lie algebras.

If $r = 3$, then by **Step 1**,

$$\mathrm{SL}(V_\sigma) \subset \mathrm{Hg}(A)_{\mathbb{C}} \subset \mathrm{SU}_K(V, \psi)_{\mathbb{C}} \cong \mathrm{SL}(V_\sigma).$$

Thus

$$\mathrm{Hg}(A)_{\mathbb{C}} = \mathrm{SU}_K(V, \psi)_{\mathbb{C}},$$

and hence $\mathrm{Hg}(A) = \mathrm{SU}_K(V,\psi)$ as algebraic subgroups of $\mathrm{GL}(V)$ defined over $\mathbb{Q}$.

If $r \neq 3$, then by (7.1), the determinant $\mathrm{Hg}(A)_{\mathbb{C}} \to \mathbb{G}_m$ on V_σ is non-constant. Since $\mathrm{Hg}(A)_{\mathbb{C}}$ is connected, the determinant is surjective. For any $g \in \mathrm{GL}(V_\sigma)$, choose $h \in \mathrm{Hg}(A)(\mathbb{C})$ with $\det h = \det g$. Then

$$h^{-1}g \in \mathrm{SL}(V_\sigma) \subset \mathrm{Hg}(A)_{\mathbb{C}},$$

so $g \in \mathrm{Hg}(A)(\mathbb{C})$. Thus

$$\mathrm{Hg}(A)_{\mathbb{C}} = \mathrm{GL}(V_\sigma) \cong \mathrm{U}_K(V,\psi)_{\mathbb{C}},$$

and hence $\mathrm{Hg}(A) = \mathrm{U}_K(V,\psi)$.

Step 3. In this step, we compute $\mathrm{Hdg}^p(A)$ for $0 \leq p \leq 6$.

By Lemma 2.3 (2), $\mathrm{Hdg}^p(A)$ is the subspace of $\mathrm{Hg}(A)$-invariants in $H^{2p}(A,\mathbb{Q}) = \wedge^{2p}V$, and taking invariants commutes with extension of scalars to $\mathbb{C}$. The decomposition $V \otimes \mathbb{C} = V_\sigma \oplus V_{\bar\sigma}$ induces

$$\wedge^{2p}(V\otimes\mathbb{C}) = \bigoplus_{\substack{a+b=2p\\ 0\leq a,b\leq 6}} \wedge^a V_\sigma \otimes \wedge^b V_{\bar\sigma},$$

and the cup product maps the product of the (a,b) summand and the (a',b') summand into the $(a+a',b+b')$ summand. Since $V_{\bar\sigma} \cong V_\sigma^\vee$, the $\mathrm{Hg}(A)_{\mathbb{C}}$-invariants in the (a,b) summand are

$$\left(\wedge^a V_\sigma \otimes (\wedge^b V_\sigma)^\vee\right)^{\mathrm{Hg}(A)_{\mathbb{C}}} = \mathrm{Hom}_{\mathrm{Hg}(A)_{\mathbb{C}}}\left(\wedge^b V_\sigma, \wedge^a V_\sigma\right).$$

By **Step 2**, $\mathrm{Hg}(A)_{\mathbb{C}}$ acts on V_σ as $\mathrm{GL}(V_\sigma)$ if $r \neq 3$ and as $\mathrm{SL}(V_\sigma)$ if $r = 3$.

Suppose that $r \neq 3$. For $0 \leq a \leq 6$, $\wedge^a V_\sigma$ is an irreducible representation of $\mathrm{GL}(V_\sigma)$ [Ful97, Section 8.2], and a scalar λ acts on $\wedge^a V_\sigma$ by λ^a. Hence $\wedge^a V_\sigma \ncong \wedge^b V_\sigma$ for $a \neq b$, and by Schur's lemma, the invariants in the (a,b) summand are 0 if $a \neq b$ and one-dimensional if $a = b$. Thus $\dim \mathrm{Hdg}^p(A) = 1$ for any $0 \leq p \leq 6$, and

$$\mathrm{Hdg}^p(A) \otimes \mathbb{C} \subset \wedge^p V_\sigma \otimes \wedge^p V_{\bar\sigma}.$$

Suppose that $r = 3$. For $1 \leq a \leq 5$, $\wedge^a V_\sigma$ is a non-trivial irreducible representation of $\mathrm{SL}(V_\sigma)$, and $\wedge^a V_\sigma \ncong \wedge^b V_\sigma$ for $1 \leq a < b \leq 5$ [Ful97, Section 8.2], while $\wedge^0 V_\sigma$ and $\wedge^6 V_\sigma$ are trivial. By Schur's lemma, the invariants in the (a,b) summand are one-dimensional if $a = b$ or $\{a,b\} = \{0,6\}$, and 0 otherwise. Thus $\dim \mathrm{Hdg}^p(A) = 1$ and

$$\mathrm{Hdg}^p(A) \otimes \mathbb{C} \subset \wedge^p V_\sigma \otimes \wedge^p V_{\bar\sigma} \quad \text{for } p \neq 3,$$

and $\mathrm{Hdg}^3(A) \otimes \mathbb{C}$ is the direct sum of a line in $\wedge^3 V_\sigma \otimes \wedge^3 V_{\bar\sigma}$ and

$$\wedge^6 V_\sigma \oplus \wedge^6 V_{\bar\sigma} = W_K(A) \otimes \mathbb{C}$$

(Definition 2.7, since (A,K) is of Weil type). In particular, $W_K(A) \subset \mathrm{Hdg}^3(A)$.

In both cases, $\Theta \in \mathrm{Hdg}^1(A)$, so $\Theta \otimes 1 \in V_\sigma \otimes V_{\bar\sigma}$, and hence

$$\Theta^p \otimes 1 \in \wedge^p V_\sigma \otimes \wedge^p V_{\bar\sigma} \quad \text{for any } 0 \leq p \leq 6.$$

Since Θ is ample, Θ^6 is a positive multiple of the class of a point, so $\Theta^p \neq 0$ for any $0 \leq p \leq 6$. Hence Θ^p spans the invariant line in $\wedge^p V_\sigma \otimes \wedge^p V_{\bar\sigma}$. If $r \neq 3$, then $\mathrm{Hdg}^p(A) = \mathbb{Q}\Theta^p$ for any $0 \leq p \leq 6$. If $r = 3$, then $\mathrm{Hdg}^p(A) = \mathbb{Q}\Theta^p$ for $p \neq 3$, and

$$\mathrm{Hdg}^3(A) \otimes \mathbb{C} = \mathbb{C}\Theta^3 \oplus \left(W_K(A) \otimes \mathbb{C}\right),$$

so $\mathrm{Hdg}^3(A) = \mathbb{Q}\Theta^3 \oplus W_K(A)$. Finally, the product of $\Theta \otimes 1 \in V_\sigma \otimes V_{\bar\sigma}$ with $\wedge^6 V_\sigma$ belongs to $\wedge^7 V_\sigma \otimes V_{\bar\sigma} = 0$, and the product with $\wedge^6 V_{\bar\sigma}$ belongs to $V_\sigma \otimes \wedge^7 V_{\bar\sigma} = 0$. Thus $\Theta \cdot W_K(A) = 0$. □

7.2. **Sextic CM endomorphism field.** In this subsection, we compute the Hodge group and the Hodge classes of a simple abelian sixfold whose endomorphism algebra is a CM field of degree 6.

Proposition 7.2. *Let A be a simple abelian sixfold with $\mathrm{End}^0(A) = L$ a CM field of degree 6, let $F := L^+$, $V := H^1(A, \mathbb{Q})$, ψ a polarization of V, and*

$$\det_L : \mathrm{U}_L(V, \psi) \to \mathrm{U}_L$$

the L-determinant. Then

$$\mathrm{Hg}(A)^{\mathrm{der}} = \mathrm{U}_L(V, \psi)^{\mathrm{der}}, \quad \mathrm{Hg}(A)^{\mathrm{der}}_{\mathbb{C}} \cong \mathrm{SL}^3_{2,\mathbb{C}},$$

and $\dim \mathrm{Hdg}^1(A) = 3$, $\dim \mathrm{Hdg}^2(A) = 6$, and $\dim \mathcal{D}^3(A) = 7$. Moreover, exactly one of the following holds.

1. *L contains no imaginary quadratic subfield whose multiplicities on $H^{1,0}(A)$ are $(3, 3)$. Then $\mathrm{Hg}(A) = \mathrm{U}_L(V, \psi)$, the identity component Z° of the center of $\mathrm{Hg}(A)$ satisfies $\dim Z^\circ = 3$, and $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p.*
2. *L contains an imaginary quadratic subfield K whose multiplicities on $H^{1,0}(A)$ are $(3, 3)$. Then K is unique, $L = KF$, the three embeddings of L above a fixed embedding of K have multiplicities 0, 1, and 2 in some order,*

   $$\mathrm{Hg}(A) = \{g \in \mathrm{U}_L(V, \psi) \mid \mathrm{Nm}_{L/K}(\det_L g) = 1\},$$

   the identity component Z° of the center of $\mathrm{Hg}(A)$ satisfies $\dim Z^\circ = 2$, $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for $p \neq 3$, $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_K(A)$, and $\mathcal{D}^1(A) \cdot W_K(A) = 0$. Moreover, $\mathrm{disc}(h) = \mathrm{disc}(h')$ for the Hermitian forms h and h' of any two compatible polarizations of (A, K), and if $\mathrm{disc}(h) = [-1]$, then any class in $W_K(A)$ is algebraic.

Proof. The proof is long, so we divide it into several steps.

We write $G := \mathrm{Hg}(A)$, $U := \mathrm{U}_L(V, \psi)$,

$$V_{\mathbb{C}} := V \otimes_{\mathbb{Q}} \mathbb{C} = H^1(A, \mathbb{C}),$$

and $\mathfrak{s} := \mathrm{Lie}(G^{\mathrm{der}}_{\mathbb{C}})$. Let Z° be the identity component of the center of G, and let Σ be the set of complex embeddings of L. For $\tau \in \Sigma$, we denote by $\bar{\tau}$ the complex conjugate of τ, by $V_\tau := H^1(A, \mathbb{C})_\tau$ the subspace of Definition 2.7, and by

$$\chi_\tau : (\mathrm{Res}_{L/\mathbb{Q}} \mathbb{G}_m)_{\mathbb{C}} \to \mathbb{G}_{m,\mathbb{C}}$$

the character induced by τ. We fix $\sigma_1, \sigma_2, \sigma_3 \in \Sigma$ such that

$$\Sigma = \{\sigma_1, \bar{\sigma}_1, \sigma_2, \bar{\sigma}_2, \sigma_3, \bar{\sigma}_3\}.$$

For $i = 1, 2, 3$, let $W_i := V_{\sigma_i}$ and $\overline{W}_i := V_{\bar{\sigma}_i}$, and let

$$r_i := \dim H^{1,0}(A) \cap W_i \in \{0, 1, 2\}$$

be the multiplicity of L on $H^{1,0}(A)$ at σ_i. **Steps 1–3** hold for any choice of $\sigma_1, \sigma_2, \sigma_3$.

Step 1. In this step, we show that $U_{\mathbb{C}} \cong \prod_{i=1}^3 \mathrm{GL}(W_i)$, that the spaces V_τ are pairwise non-isomorphic irreducible $G_{\mathbb{C}}$-modules, and that $\mathfrak{s}$ projects onto $\mathfrak{sl}(W_i)$ for $i = 1, 2, 3$.

Since $\dim_{\mathbb{Q}} V = 12$ and $[L : \mathbb{Q}] = 6$, we have $\dim_L V = 2$, hence

$$V_{\mathbb{C}} = \bigoplus_{\tau \in \Sigma} V_\tau \quad \text{and} \quad \dim V_\tau = 2 \text{ for any } \tau \in \Sigma.$$

The Rosati involution of ψ restricts to complex conjugation on the CM field L [Moo99, (1.19)], so $\psi(ax, y) = \psi(x, \bar{a}y)$ for any $a \in L$ and $x, y \in V$. For $x \in V_\tau$ and $y \in V_{\tau'}$, we obtain $\tau(a)\psi(x, y) = \bar{\tau}'(a)\psi(x, y)$ for any $a \in L$, so $\psi(x, y) = 0$ unless $\tau' = \bar{\tau}$. As ψ is non-degenerate, ψ induces a perfect pairing between W_i and $\overline{W}_i$, so $\overline{W}_i \cong W_i^*$. Any L-linear automorphism g of $V_{\mathbb{C}}$

preserves each V_τ, and g preserves ψ if and only if $g|_{\overline{W}_i}$ is contragredient to $g|_{W_i}$ for $i = 1, 2, 3$. Thus $g \to (g|_{W_1}, g|_{W_2}, g|_{W_3})$ induces isomorphisms

$$U_{\mathbb{C}} \cong \prod_{i=1}^{3} \mathrm{GL}(W_i) \quad \text{and} \quad U_{\mathbb{C}}^{\mathrm{der}} \cong \prod_{i=1}^{3} \mathrm{SL}(W_i).$$

On $\mathrm{U}_{L,\mathbb{C}}$ we have $\chi_{\bar{\tau}} = \chi_\tau^{-1}$, so $(\chi_{\sigma_1}, \chi_{\sigma_2}, \chi_{\sigma_3})$ induces an isomorphism $\mathrm{U}_{L,\mathbb{C}} \cong \mathbb{G}_{m,\mathbb{C}}^3$, under which

$$\det_L(g) = \big(\det(g|_{W_1}), \det(g|_{W_2}), \det(g|_{W_3})\big) \quad \text{for any } g \in U(\mathbb{C}). \tag{7.2}$$

In particular, $\ker(\det_L) = U^{\mathrm{der}}$.

For any $z \in \mathbb{U}$, $\eta_A(z)$ commutes with L, since L acts on V by endomorphisms of Hodge structures, and $\eta_A(z)$ preserves ψ, since ψ is a polarization. By the minimality in Definition 2.2, $G \subset U$. Hence $G^{\mathrm{der}} \subset U^{\mathrm{der}}$ and $\mathfrak{s} \subset \bigoplus_{i=1}^3 \mathfrak{sl}(W_i)$. Note that $x \in \mathfrak{s}$ acts on $\overline{W}_i$ by the negative transpose of $x|_{W_i}$, so x is determined by $(x|_{W_1}, x|_{W_2}, x|_{W_3})$.

By Lemma 2.3 (1), the commutant of G in $\mathrm{End}_{\mathbb{Q}}(V)$ is L. Since invariants commute with extension of scalars, the commutant of $G_{\mathbb{C}}$ in $\mathrm{End}(V_{\mathbb{C}})$ is $L \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C}^\Sigma$, where the τ-coordinate acts on V_τ by scalars and on $V_{\tau'}$ by 0 for $\tau' \neq \tau$. Since G is reductive (Lemma 2.3 (1)), $V_{\mathbb{C}}$ is a semisimple $G_{\mathbb{C}}$-module. Hence any V_τ is irreducible, and $\mathrm{Hom}_{G_{\mathbb{C}}}(V_\tau, V_{\tau'}) = 0$ for $\tau \neq \tau'$, since otherwise the commutant of $G_{\mathbb{C}}$ would be larger than $\mathbb{C}^\Sigma$.

The group G is the almost direct product of Z° and G^{der}, that is, $G = Z^\circ \cdot G^{\mathrm{der}}$ and $Z^\circ \cap G^{\mathrm{der}}$ is finite (cf. [Bor91, Proposition 14.2]). The group $Z_{\mathbb{C}}^\circ$ acts on any V_τ by $G_{\mathbb{C}}$-endomorphisms, hence by scalars. Thus the $G_{\mathbb{C}}^{\mathrm{der}}$-submodules of W_i are $G_{\mathbb{C}}$-submodules, so W_i is an irreducible $G_{\mathbb{C}}^{\mathrm{der}}$-module, hence an irreducible $\mathfrak{s}$-module. Let $p_i : \mathfrak{s} \to \mathfrak{sl}(W_i)$ be the restriction to W_i. Then $p_i(\mathfrak{s})$ is semisimple, as a quotient of $\mathfrak{s}$, and $p_i(\mathfrak{s}) \neq 0$, as W_i is irreducible and $\dim W_i = 2$. Any proper Lie subalgebra of $\mathfrak{sl}(W_i)$ is of dimension at most 2, and any Lie algebra of dimension at most 2 is solvable. Hence

$$p_i(\mathfrak{s}) = \mathfrak{sl}(W_i) \quad \text{for } i = 1, 2, 3.$$

Step 2. In this step, we show that $r_i = 1$ for some i and that $r_j \neq 1$ for some j.

Let

$$\mu : \mathbb{G}_{m,\mathbb{C}} \to \mathrm{GL}(V_{\mathbb{C}})$$

be the cocharacter acting by z on $H^{1,0}(A)$ and by z^{-1} on $H^{0,1}(A)$. Then $\mu(z) = \eta_A(z)$ for $z \in \mathbb{U}$. Since $\mu^{-1}(G_{\mathbb{C}})$ is a closed subgroup of $\mathbb{G}_{m,\mathbb{C}}$ containing the infinite set $\mathbb{U}$, μ factors through $G_{\mathbb{C}}$. Let $\xi \in \mathrm{Lie}(G_{\mathbb{C}})$ be the differential of μ at 1, which acts by 1 on $H^{1,0}(A)$ and by -1 on $H^{0,1}(A)$. Since L preserves $H^{1,0}(A)$ and $H^{0,1}(A)$, we have

$$W_i = \big(W_i \cap H^{1,0}(A)\big) \oplus \big(W_i \cap H^{0,1}(A)\big),$$

so $\mathrm{tr}(\xi|_{W_i}) = r_i - (2 - r_i) = 2r_i - 2$. Since

$$\mathrm{Lie}(G_{\mathbb{C}}) = \mathrm{Lie}(Z_{\mathbb{C}}^\circ) \oplus \mathfrak{s},$$

we may write $\xi = c + s$ with $c \in \mathrm{Lie}(Z_{\mathbb{C}}^\circ)$ and $s \in \mathfrak{s}$. By **Step 1**, c acts on W_i by a scalar c_i and on $\overline{W}_i$ by $-c_i$, while $\mathrm{tr}(p_i(s)) = 0$. Thus

$$c_i = r_i - 1 \quad \text{for } i = 1, 2, 3.$$

We claim that $s \neq 0$ and $c \neq 0$. If $s = 0$, then the composition of μ with $G_{\mathbb{C}} \to G_{\mathbb{C}}/Z_{\mathbb{C}}^\circ$ has zero differential, hence is trivial, so $\mu(\mathbb{G}_{m,\mathbb{C}}) \subset Z_{\mathbb{C}}^\circ$. Then $\eta_A(\mathbb{U}) \subset Z^\circ(\mathbb{R})$, and the minimality of G implies that $G = Z^\circ$. Thus $\mathfrak{s} = 0$, which contradicts **Step 1**. If $c = 0$, then the composition of μ with

$$G_{\mathbb{C}} \to G_{\mathbb{C}}/G_{\mathbb{C}}^{\mathrm{der}}$$

has zero differential, so $\mu(\mathbb{G}_{m,\mathbb{C}}) \subset G_{\mathbb{C}}^{\mathrm{der}}$, and the minimality of G implies that $G = G^{\mathrm{der}} \subset U^{\mathrm{der}}$. Then $G_{\mathbb{C}}$ acts on W_i through $\mathrm{SL}(W_i)$, so the perfect pairing $W_i \times W_i \to \wedge^2 W_i \cong \mathbb{C}$ is $G_{\mathbb{C}}$-invariant. Hence $W_i \cong W_i^* \cong \overline{W}_i$ as $G_{\mathbb{C}}$-modules, which contradicts **Step 1**.

Since $c \neq 0$, we have $c_j \neq 0$ for some j, that is, $r_j \neq 1$. If $r_i \neq 1$ for $i = 1, 2, 3$, then $r_i \in \{0, 2\}$, so $W_i \subset H^{1,0}(A)$ or $W_i \subset H^{0,1}(A)$, and $\xi|_{W_i}$ is a scalar. Then $p_i(s) = 0$ for $i = 1, 2, 3$, so $s = 0$, a contradiction. Hence $r_i = 1$ for some i. The argument also shows that

$$p_j(s) = 0 \quad \text{for any } j \text{ with } r_j \neq 1. \tag{7.3}$$

Step 3. In this step, we show that $G^{\mathrm{der}} = U^{\mathrm{der}}$. In particular,

$$G_{\mathbb{C}}^{\mathrm{der}} \cong \mathrm{SL}_{2,\mathbb{C}}^{3}.$$

Let $\mathfrak{s} = \mathfrak{a}_1 \oplus \cdots \oplus \mathfrak{a}_m$ be the decomposition of $\mathfrak{s}$ into simple ideals. Fix i. Since $p_i(\mathfrak{s}) = \mathfrak{sl}(W_i)$, any $p_i(\mathfrak{a}_k)$ is an ideal of $\mathfrak{sl}(W_i)$, hence $p_i(\mathfrak{a}_k) \in \{0, \mathfrak{sl}(W_i)\}$. Since $[\mathfrak{a}_k, \mathfrak{a}_l] = 0$ for $k \neq l$ and $\mathfrak{sl}(W_i)$ is not abelian, there exists exactly one k with $p_i(\mathfrak{a}_k) = \mathfrak{sl}(W_i)$, which we denote by $k(i)$. As $\mathfrak{a}_{k(i)}$ is simple, p_i is injective on $\mathfrak{a}_{k(i)}$, so

$$\ker(p_i) = \bigoplus_{l \neq k(i)} \mathfrak{a}_l.$$

As any $x \in \mathfrak{s}$ is determined by $(p_1(x), p_2(x), p_3(x))$, any k is equal to $k(i)$ for some i. Thus $\{1, 2, 3\}$ is the disjoint union of the non-empty sets $B_k := \{i \mid k(i) = k\}$ for $1 \leq k \leq m$, and i, j belong to one set B_k if and only if $\ker(p_i) = \ker(p_j)$ (cf. [MZ99, (3.1)]).

We show that $m = 1$ or $m = 3$. Let $\overline{\mathbb{Q}} \subset \mathbb{C}$ be the algebraic closure of $\mathbb{Q}$, and $\Gamma := \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Any $\tau \in \Sigma$ takes values in $\overline{\mathbb{Q}}$, so $V_\tau = V'_\tau \otimes_{\overline{\mathbb{Q}}} \mathbb{C}$, where $V'_\tau \subset V \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$ is the subspace on which any $a \in L$ acts by $\tau(a)$. The group Γ acts on $V \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$ through the second factor, and $\gamma(V'_\tau) = V'_{\gamma\circ\tau}$ for any $\gamma \in \Gamma$. Let

$$\mathfrak{g} := \mathrm{Lie}(G^{\mathrm{der}}) \subset \mathrm{End}_{\mathbb{Q}}(V),$$

and for $\tau \in \Sigma$ let

$$\mathfrak{k}_\tau \subset \mathfrak{g} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$$

be the subspace of elements vanishing on V'_τ. Then

$$\gamma(\mathfrak{k}_\tau) = \mathfrak{k}_{\gamma\circ\tau}, \quad \mathfrak{k}_{\sigma_i} \otimes_{\overline{\mathbb{Q}}} \mathbb{C} = \ker(p_i), \quad \text{and} \quad \mathfrak{k}_{\bar\tau} = \mathfrak{k}_\tau,$$

where the last equality holds because any $x \in \mathfrak{g}$ satisfies $\psi(xv, w) + \psi(v, xw) = 0$ and ψ pairs V'_τ perfectly with $V'_{\bar\tau}$. The pairs $\{\sigma_i, \bar\sigma_i\}$ correspond to the three embeddings $\sigma_i|_F$ of F, on which Γ acts transitively. Hence the partition of $\{1, 2, 3\}$ into the sets B_k, viewed as a partition of the set of embeddings of F, is Γ-stable. Thus Γ permutes the sets B_k transitively, and all B_k have equal cardinality. Since 3 is prime, $m = 1$ or $m = 3$.

Suppose that $m = 1$. Then $\ker(p_j) = 0$ for $j = 1, 2, 3$. By **Step 2**, there exists j with $r_j \neq 1$, and (7.3) implies that $s = 0$, which contradicts **Step 2**. Hence $m = 3$, and $i \to k(i)$ is a bijection. For any i, p_i maps $\mathfrak{a}_{k(i)}$ isomorphically onto $\mathfrak{sl}(W_i)$, and $p_l(\mathfrak{a}_{k(i)}) = 0$ for $l \neq i$. Thus

$$\mathfrak{s} = \bigoplus_{i=1}^{3} \mathfrak{sl}(W_i) = \mathrm{Lie}(U_{\mathbb{C}}^{\mathrm{der}}).$$

Since $G_{\mathbb{C}}^{\mathrm{der}} \subset U_{\mathbb{C}}^{\mathrm{der}}$ are connected algebraic groups with equal Lie algebras,

$$G_{\mathbb{C}}^{\mathrm{der}} = U_{\mathbb{C}}^{\mathrm{der}} \cong \mathrm{SL}_{2,\mathbb{C}}^{3},$$

and hence $G^{\mathrm{der}} = U^{\mathrm{der}}$.

Step 4. In this step, we determine the torus $T := \det_L(G)$, and prove the descriptions of G, Z°, K, and the multiplicities in (1) and (2).

The group T is a connected $\mathbb{Q}$-subtorus of U_L. We have $G \subset \det_L^{-1}(T)$. For any $g \in \det_L^{-1}(T)(\mathbb{C})$, there exists $k \in G(\mathbb{C})$ with $\det_L(k) = \det_L(g)$, so $k^{-1}g \in \ker(\det_L)(\mathbb{C}) = U^{\mathrm{der}}(\mathbb{C})$, which is contained in $G(\mathbb{C})$ by **Step 3**. Hence $G = \det_L^{-1}(T)$. Since the kernel of $\det_L|_G$ is $U^{\mathrm{der}} = G^{\mathrm{der}}$, we have

$$\dim Z^\circ = \dim G - \dim G^{\mathrm{der}} = \dim T.$$

Let

$$\lambda := \det_L \circ \mu : \mathbb{G}_{m,\mathbb{C}} \to \mathrm{U}_{L,\mathbb{C}}\,.$$

We claim that T is the smallest $\mathbb{Q}$-subtorus $T_0 \subset \mathrm{U}_L$ such that λ factors through $T_{0,\mathbb{C}}$. Let $T_0 \subset \mathrm{U}_L$ be a $\mathbb{Q}$-subtorus. Since $\mathbb{U}$ is Zariski dense in $\mathbb{G}_{m,\mathbb{C}}$ and

$$\det_L(\eta_A(\mathbb{U})) \subset \mathrm{U}_L(\mathbb{R}),$$

λ factors through $T_{0,\mathbb{C}}$ if and only if $\eta_A(\mathbb{U}) \subset \det_L^{-1}(T_0)(\mathbb{R})$. By the minimality of G, the inclusion $\eta_A(\mathbb{U}) \subset \det_L^{-1}(T_0)(\mathbb{R})$ holds if and only if $G \subset \det_L^{-1}(T_0)$, that is, $T \subset T_0$.

The characters $\chi_{\sigma_1}, \chi_{\sigma_2}, \chi_{\sigma_3}$ induce isomorphisms $X^*(\mathrm{U}_L) \cong \mathbb{Z}^3$ and $X_*(\mathrm{U}_L) \cong \mathbb{Z}^3$, under which the natural pairing is the standard pairing. By (7.2), $\det(\mu(z)|_{W_i}) = z^{r_i} z^{-(2-r_i)}$, so

$$\lambda = (2r_1 - 2, 2r_2 - 2, 2r_3 - 2) \in \mathbb{Z}^3.$$

The group Γ acts on characters by $\gamma \cdot \chi_\tau = \chi_{\gamma\circ\tau}$. Since $\chi_{\bar\tau} = \chi_\tau^{-1}$ on U_L, Γ acts on $X^*(\mathrm{U}_L)$ and on $X_*(\mathrm{U}_L)$ by signed permutation matrices, whose permutations are given by the action of Γ on the embeddings of F. In particular, the permutation action of Γ on the three coordinates is transitive. By [Moo99, (1.1)], $T_0 \to X_*(T_0)$ is a bijection between the $\mathbb{Q}$-subtori of U_L and the Γ-stable saturated sublattices of $X_*(\mathrm{U}_L)$. By the claim above, $X_*(T) \otimes \mathbb{Q}$ is the $\mathbb{Q}$-span of the orbit $\Gamma\lambda$.

By **Step 2**, λ has at least one zero entry and at least one non-zero entry, and any non-zero entry is ± 2. If λ has exactly one non-zero entry, then by transitivity $\Gamma\lambda$ contains a non-zero multiple of any coordinate vector, so $X_*(T) \otimes \mathbb{Q} = \mathbb{Q}^3$ and $T = \mathrm{U}_L$. If λ has exactly one zero entry, at the coordinate j_0, and $\Gamma\lambda \subset \mathbb{Q}\lambda$, then $\gamma\lambda = \pm\lambda$ for any $\gamma \in \Gamma$, so any γ fixes the coordinate j_0, which contradicts transitivity, so $\Gamma\lambda$ spans a subspace of dimension at least 2. Hence $\dim T \in \{2, 3\}$.

Assume that $\dim T = 2$. The annihilator of $X_*(T)$ in $X^*(\mathrm{U}_L) \otimes \mathbb{Q}$ is a Γ-stable line. Let $n = (n_1, n_2, n_3)$ be a generator of the intersection of the annihilator with $X^*(\mathrm{U}_L) = \mathbb{Z}^3$. Since Γ acts on $\mathbb{Z}n$ by automorphisms, $\gamma n = \pm n$ for any $\gamma \in \Gamma$. By transitivity, $|n_1| = |n_2| = |n_3|$, and since $n \neq 0$ is primitive, $|n_i| = 1$ for $i = 1, 2, 3$. Define $\kappa : \Sigma \to \{1, -1\}$ by $\kappa(\sigma_i) = n_i$ and $\kappa(\bar\sigma_i) = -n_i$, and let $\Sigma_+ := \kappa^{-1}(1)$. Then Σ_+ contains exactly one element of any pair $\{\tau, \bar\tau\}$, and n is the restriction of $\prod_{\tau\in\Sigma_+} \chi_\tau$ to U_L. For two subsets $S, S' \subset \Sigma$ containing exactly one element of any pair $\{\tau, \bar\tau\}$, the restrictions of $\prod_{\tau\in S} \chi_\tau$ and $\prod_{\tau\in S'} \chi_\tau$ to U_L are equal if and only if $S = S'$, and are inverse to each other if and only if $S' = \{\bar\tau \mid \tau \in S\}$. Since $\gamma n = \pm n$, there exists a homomorphism $f : \Gamma \to \{1, -1\}$, factoring through a finite quotient of Γ, such that

$$\kappa(\gamma \circ \tau) = f(\gamma)\kappa(\tau) \quad \text{for any } \gamma \in \Gamma \text{ and } \tau \in \Sigma.$$

Fix $\tau_0 \in \Sigma_+$. The stabilizer of τ_0 in Γ is contained in $\ker(f)$, so the fixed field $\overline{\mathbb{Q}}^{\ker(f)}$ is contained in $\tau_0(L)$. The complex conjugation $\iota \in \Gamma$ satisfies $\kappa(\iota \circ \tau) = \kappa(\bar\tau) = -\kappa(\tau)$, so $f(\iota) = -1$. Thus $\ker(f)$ is a subgroup of index 2 not containing ι, and

$$K := \tau_0^{-1}\big(\overline{\mathbb{Q}}^{\ker(f)}\big)$$

is an imaginary quadratic subfield of L. Let $\sigma := \tau_0|_K$. For $\gamma \in \Gamma$, the embedding $\gamma \circ \tau_0$ restricts to σ on K if and only if γ fixes $\tau_0(K)$ pointwise, that is, $f(\gamma) = 1$. As Γ acts transitively on Σ, Σ_+ is the set of embeddings of L above σ.

Replacing σ_i with $\bar\sigma_i$ interchanges W_i and $\overline{W}_i$, replaces r_i with $2 - r_i$, and changes the sign of the i-th coordinate of $\mathbb{Z}^3$, while **Steps 1–3** hold for any choice of $\sigma_1, \sigma_2, \sigma_3$. Possibly replacing σ_i with $\bar\sigma_i$ for some i, we may assume that $\sigma_1, \sigma_2, \sigma_3 \in \Sigma_+$, so $n = (1, 1, 1)$. Since $\sigma_1, \sigma_2, \sigma_3$ are the embeddings of L above σ,

$$\chi_{\sigma_1}\chi_{\sigma_2}\chi_{\sigma_3} = \chi_\sigma \circ \mathrm{Nm}_{L/K} \quad \text{on } \mathrm{U}_{L,\mathbb{C}},$$

where χ_σ is the character of $(\mathrm{Res}_{K/\mathbb{Q}}\,\mathbb{G}_m)_\mathbb{C}$ induced by σ. The norm $\mathrm{Nm}_{L/K}$ commutes with complex conjugation, so it maps U_L into U_K. Let T_K be the kernel of

$$\mathrm{Nm}_{L/K} : \mathrm{U}_L \to \mathrm{U}_K\,.$$

Then

$$T_{K,\mathbb{C}} = \{(t_1, t_2, t_3) \in \mathbb{G}_{m,\mathbb{C}}^3 \mid t_1t_2t_3 = 1\},$$

so T_K is a connected $\mathbb{Q}$-subtorus with $\dim T_K = 2$. Since n vanishes on $X_*(T)$, $\chi_\sigma \circ \mathrm{Nm}_{L/K}$ is trivial on $T_{\mathbb{C}}$, so $T \subset T_K$. Since $\dim T = \dim T_K = 2$, we deduce that $T = T_K$. In particular, $\lambda \in X_*(T_K)$, so $\sum_{i=1}^3 (2r_i - 2) = 0$, that is, $r_1 + r_2 + r_3 = 3$. Since $H^1(A, \mathbb{C})_\sigma = W_1 \oplus W_2 \oplus W_3$, the multiplicity of K on $H^{1,0}(A)$ at σ is $r_1 + r_2 + r_3 = 3$, and the multiplicity at $\bar{\sigma}$ is $\dim H^{1,0}(A) - 3 = 3$. Thus the multiplicities of K on $H^{1,0}(A)$ are $(3, 3)$. As $r_i \in \{0, 1, 2\}$, $r_1 + r_2 + r_3 = 3$, and $r_i \neq 1$ for some i (**Step 2**), (r_1, r_2, r_3) is a permutation of $(0, 1, 2)$, and so is $(2 - r_1, 2 - r_2, 2 - r_3)$, the triple of multiplicities of the embeddings of L above $\bar{\sigma}$.

Conversely, assume that L contains an imaginary quadratic subfield K whose multiplicities on $H^{1,0}(A)$ are $(3, 3)$, and let σ be an embedding of K. Possibly replacing σ_i with $\bar{\sigma}_i$ for some i, we may assume that $\sigma_1, \sigma_2, \sigma_3$ are the embeddings of L above σ. Then $H^1(A, \mathbb{C})_\sigma = W_1 \oplus W_2 \oplus W_3$, so $r_1 + r_2 + r_3 = 3$, and

$$(\chi_\sigma \circ \mathrm{Nm}_{L/K}) \circ \lambda(z) = z^{\sum_{i=1}^3 (2r_i - 2)} = 1 \quad \text{for any } z \in \mathbb{C}^*.$$

Thus λ factors through $T_{K,\mathbb{C}}$. By the claim above, $T \subset T_K$, so $\dim T \leq 2$, hence $\dim T = 2$.

Assume that L contains no imaginary quadratic subfield whose multiplicities on $H^{1,0}(A)$ are $(3, 3)$. By the case $\dim T = 2$ treated above, $\dim T \neq 2$, so $\dim T = 3$ and $T = \mathrm{U}_L$. Hence $G = \det_L^{-1}(\mathrm{U}_L) = U$ and $\dim Z^\circ = 3$.

Assume that L contains an imaginary quadratic subfield K whose multiplicities on $H^{1,0}(A)$ are $(3, 3)$. Then $\dim T = 2$. Two distinct quadratic subfields of L would generate a subfield of degree 4 of L, which is impossible as $4 \nmid 6$. Hence K is unique, and K is the imaginary quadratic subfield constructed above. Thus $T = T_K$, so

$$G = \det_L^{-1}(T_K) = \{g \in U \mid \mathrm{Nm}_{L/K}(\det_L g) = 1\},$$

$\dim Z^\circ = 2$, and the embeddings of L above a fixed embedding of K have multiplicities 0, 1, and 2 in some order. Since F is totally real, $K \not\subset F$, so $[KF : F] = 2 = [L : F]$, and $L = KF$.

Step 5. In this step, we compute the Hodge classes of A.

In case (2), we keep the choice of σ and $\sigma_1, \sigma_2, \sigma_3$ in **Step 4**, so that $\sigma_1, \sigma_2, \sigma_3$ are the embeddings of L above the embedding σ of K and $T_{\mathbb{C}} = \{t_1t_2t_3 = 1\}$. By Lemma 2.3 (2), $\mathrm{Hdg}^p(A) = (\wedge^{2p} V)^G$, and since invariants commute with extension of scalars,

$$\mathrm{Hdg}^p(A) \otimes_{\mathbb{Q}} \mathbb{C} = (\wedge^{2p} V_{\mathbb{C}})^{G_{\mathbb{C}}}.$$

Since $V_{\mathbb{C}} = \bigoplus_{i=1}^3 (W_i \oplus \overline{W}_i)$, we have an isomorphism of $U_{\mathbb{C}}$-modules

$$\wedge^* V_{\mathbb{C}} \cong \bigotimes_{i=1}^3 M_i, \quad \text{where} \quad M_i := \wedge^*(W_i \oplus \overline{W}_i) = \bigoplus_{0 \leq a,b \leq 2} \wedge^a W_i \otimes \wedge^b \overline{W}_i,$$

and $\mathrm{GL}(W_j)$ acts trivially on M_i for $j \neq i$.

Fix i, a basis x_1, x_2 of W_i, and the basis y_1, y_2 of $\overline{W}_i$ dual to x_1, x_2 under ψ. Let

$$d_i := x_1 \wedge y_1 + x_2 \wedge y_2, \quad P_i^+ := x_1 \wedge x_2, \quad \text{and} \quad P_i^- := y_1 \wedge y_2.$$

The class d_i is the image of the canonical element of $W_i \otimes W_i^*$, so d_i does not depend on the basis and is fixed by $U_{\mathbb{C}}$, while any $g \in U_{\mathbb{C}}$ acts on P_i^+ by $\det(g|_{W_i})$ and on P_i^- by $\det(g|_{W_i})^{-1}$. Moreover, $d_i^2 = 2x_1 \wedge y_1 \wedge x_2 \wedge y_2 \neq 0$. We claim that 1, d_i, d_i^2, P_i^+, and P_i^- form a basis of $M_i^{\mathrm{SL}(W_i)}$. The $\mathrm{SL}(W_i)$-modules $\wedge^0 W_i$ and $\wedge^2 W_i$ are trivial, and $\wedge^1 W_i = W_i$ and $\wedge^1 \overline{W}_i \cong W_i^*$ are irreducible and non-trivial (cf. [Ful97, Section 8.2]). Hence $\wedge^a W_i \otimes \wedge^b \overline{W}_i$ has no non-zero invariants if exactly one of a, b is 1. For $(a, b) = (0, 0), (2, 0), (0, 2), (2, 2)$, it is a trivial line, spanned by 1, P_i^+, P_i^-, and d_i^2, respectively. For $(a, b) = (1, 1)$, it is isomorphic to $W_i \otimes W_i^* \cong \mathrm{End}(W_i)$, whose invariants are the scalars by Schur's lemma, and the scalars correspond to $\mathbb{C}d_i$.

Since $U_{\mathbb{C}}^{\mathrm{der}} = \prod_{i=1}^{3} \mathrm{SL}(W_i)$ and $\mathrm{SL}(W_j)$ acts trivially on M_i for $j \neq i$, the space $(\wedge^* V_{\mathbb{C}})^{U_{\mathbb{C}}^{\mathrm{der}}}$ has a basis consisting of the products $m = m_1 m_2 m_3$ in $\wedge^* V_{\mathbb{C}}$ with $m_i \in \{1, d_i, d_i^2, P_i^+, P_i^-\}$. The 125 products m are non-zero and belong to pairwise distinct summands

$$\bigotimes_{i=1}^{3} (\wedge^{a_i} W_i \otimes \wedge^{b_i} \overline{W}_i).$$

By (7.2), $g \in U(\mathbb{C})$ acts on m by $t_1^{\epsilon_1} t_2^{\epsilon_2} t_3^{\epsilon_3}$, where $(t_1, t_2, t_3) = \det_L(g)$ and ϵ_i is 1, -1, or 0 according as $m_i = P_i^+$, $m_i = P_i^-$, or $m_i \in \{1, d_i, d_i^2\}$. Since $G_{\mathbb{C}} = \det_L^{-1}(T_{\mathbb{C}})$ contains $U_{\mathbb{C}}^{\mathrm{der}}$, the space $(\wedge^* V_{\mathbb{C}})^{G_{\mathbb{C}}}$ is spanned by the products m such that the character $(\epsilon_1, \epsilon_2, \epsilon_3)$ is trivial on $T_{\mathbb{C}}$.

In case (1), $T = \mathrm{U}_L$, so $(\epsilon_1, \epsilon_2, \epsilon_3) = (0, 0, 0)$, and $(\wedge^* V_{\mathbb{C}})^{G_{\mathbb{C}}}$ has the basis $d_1^{a_1} d_2^{a_2} d_3^{a_3}$ with $0 \leq a_i \leq 2$. In case (2), $T_{\mathbb{C}} = \{t_1 t_2 t_3 = 1\}$ contains the images of the cocharacters $(1, -1, 0)$ and $(0, 1, -1)$, so a character $(\epsilon_1, \epsilon_2, \epsilon_3)$ is trivial on $T_{\mathbb{C}}$ if and only if $\epsilon_1 = \epsilon_2 = \epsilon_3$. The only such characters in $\{-1, 0, 1\}^3$ are $(0, 0, 0)$, $(1, 1, 1)$, and $(-1, -1, -1)$. Thus $(\wedge^* V_{\mathbb{C}})^{G_{\mathbb{C}}}$ has the basis consisting of $d_1^{a_1} d_2^{a_2} d_3^{a_3}$ with $0 \leq a_i \leq 2$, $P_1^+ P_2^+ P_3^+$, and $P_1^- P_2^- P_3^-$, where the last two classes belong to $\wedge^6 V_{\mathbb{C}}$.

In both cases, $(\wedge^2 V_{\mathbb{C}})^{G_{\mathbb{C}}}$ is spanned by d_1, d_2, d_3, so $\dim \mathrm{Hdg}^1(A) = 3$. By the Lefschetz $(1,1)$ theorem (cf. [Moo99, (2.10)]), $\mathrm{Hdg}^1(A) = \mathcal{D}^1(A)$, so $\mathcal{D}^*(A) \otimes_{\mathbb{Q}} \mathbb{C}$ is the subalgebra of $\wedge^* V_{\mathbb{C}}$ generated by d_1, d_2, d_3. Since $d_i^3 \in \wedge^3 W_i \otimes \wedge^3 \overline{W}_i = 0$, the classes $d_1^{a_1} d_2^{a_2} d_3^{a_3}$ with $0 \leq a_i \leq 2$ form a basis of $\mathcal{D}^*(A) \otimes_{\mathbb{Q}} \mathbb{C}$, and

$$d_1^{a_1} d_2^{a_2} d_3^{a_3} \in \mathcal{D}^{a_1+a_2+a_3}(A) \otimes_{\mathbb{Q}} \mathbb{C}.$$

Hence $\dim \mathcal{D}^1(A) = 3$, $\dim \mathcal{D}^2(A) = 6$, with basis d_i^2 and $d_i d_j$ for $i < j$, and $\dim \mathcal{D}^3(A) = 7$, with basis $d_1 d_2 d_3$ and $d_i^2 d_j$ for $i \neq j$.

In case (1),

$$\mathrm{Hdg}^p(A) \otimes_{\mathbb{Q}} \mathbb{C} = \mathcal{D}^p(A) \otimes_{\mathbb{Q}} \mathbb{C} \quad \text{for any } p.$$

Since $\mathcal{D}^p(A) \subset \mathrm{Hdg}^p(A)$, we deduce that $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p. In case (2),

$$\mathrm{Hdg}^p(A) \otimes_{\mathbb{Q}} \mathbb{C} = \mathcal{D}^p(A) \otimes_{\mathbb{Q}} \mathbb{C} \quad \text{for } p \neq 3,$$

so $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for $p \neq 3$. In both cases, $\dim \mathrm{Hdg}^2(A) = \dim \mathcal{D}^2(A) = 6$.

Consider case (2) and $p = 3$. Since $\sigma_1, \sigma_2, \sigma_3$ are the embeddings of L above σ, we have

$$H^1(A, \mathbb{C})_\sigma = W_1 \oplus W_2 \oplus W_3 \quad \text{and} \quad H^1(A, \mathbb{C})_{\bar{\sigma}} = \overline{W}_1 \oplus \overline{W}_2 \oplus \overline{W}_3,$$

so

$$\wedge^6 H^1(A, \mathbb{C})_\sigma = \mathbb{C} P_1^+ P_2^+ P_3^+ \quad \text{and} \quad \wedge^6 H^1(A, \mathbb{C})_{\bar{\sigma}} = \mathbb{C} P_1^- P_2^- P_3^-.$$

By Definition 2.7, $W_K(A) \otimes_{\mathbb{Q}} \mathbb{C}$ is spanned by $P_1^+ P_2^+ P_3^+$ and $P_1^- P_2^- P_3^-$. Hence

$$\mathrm{Hdg}^3(A) \otimes_{\mathbb{Q}} \mathbb{C} = \big(\mathcal{D}^3(A) \otimes_{\mathbb{Q}} \mathbb{C}\big) \oplus \big(W_K(A) \otimes_{\mathbb{Q}} \mathbb{C}\big).$$

Since $\mathcal{D}^3(A)$ and $W_K(A)$ are $\mathbb{Q}$-subspaces of $\mathrm{Hdg}^3(A)$ (Lemma 2.10 (2)), we deduce that $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_K(A)$.

Finally, d_i is a sum of products $x \wedge y$ with $x \in W_i$ and $y \in \overline{W}_i$. Since $x \wedge P_i^+ \in \wedge^3 W_i = 0$ and $y \wedge P_i^- \in \wedge^3 \overline{W}_i = 0$, we have $d_i P_1^+ P_2^+ P_3^+ = 0$ and $d_i P_1^- P_2^- P_3^- = 0$ for $i = 1, 2, 3$. As $\mathcal{D}^1(A) \otimes_{\mathbb{Q}} \mathbb{C}$ is spanned by d_1, d_2, d_3, we obtain $\mathcal{D}^1(A) \cdot W_K(A) = 0$.

Step 6. In this step, we show that, in case (2), $\mathrm{disc}(h)$ does not depend on the compatible polarization, and that $W_K(A)$ is algebraic if $\mathrm{disc}(h) = [-1]$.

Let Θ and Θ' be compatible polarizations of (A, K), with Riemann forms φ and φ' on $H_1(A, \mathbb{Q})$ and Hermitian forms h and h', respectively. There exists a unique $a \in \mathrm{End}_{\mathbb{Q}}(H_1(A, \mathbb{Q}))$ such that $\varphi'(x, y) = \varphi(ax, y)$ for any $x, y \in H_1(A, \mathbb{Q})$. Since φ and φ' are morphisms of Hodge structures, a is an endomorphism of the Hodge structure $H_1(A, \mathbb{Q})$, so $a \in \mathrm{End}^0(A) = L$ (cf. [Moo99, (2.3)]). Since φ' is alternating, $\varphi(ax, y) = -\varphi(ay, x) = \varphi(x, ay)$, and since the Rosati involution of Θ restricts to complex conjugation on L, $\varphi(x, ay) = \varphi(\bar{a}x, y)$. Thus $a = \bar{a}$, so $a \in F$. Let J be the complex structure on $H_1(A, \mathbb{R})$. Then the symmetric forms $\varphi(x, Jy)$ and

$\varphi'(x, Jy) = \varphi(x, Jay)$ are positive definite (cf. [Moo99, (1.15), (2.2)]), and a acts on the non-zero summand $H_1(A, \mathbb{Q}) \otimes_{F,\rho} \mathbb{R}$ of $H_1(A, \mathbb{R})$ by $\rho(a)$ for any embedding $\rho : F \to \mathbb{R}$. Hence a is totally positive.

The form $(x, y) \to h(ax, y)$ is K-linear in y and conjugate-linear in x, since a commutes with K, and

$$\mathrm{Tr}_{K/\mathbb{Q}}\big(\delta h(ax, y)\big) = \varphi(ax, y) = \varphi'(x, y) = \mathrm{Tr}_{K/\mathbb{Q}}\big(\delta h'(x, y)\big),$$

where $\delta \in K$ with $\bar{\delta} = -\delta \neq 0$ is the element fixed in Definition 2.9. Since the trace form of $K/\mathbb{Q}$ is non-degenerate, a form on $H_1(A, \mathbb{Q})$ that is K-linear in y and conjugate-linear in x is determined by the composition of the form with $\mathrm{Tr}_{K/\mathbb{Q}}(\delta\cdot)$, so $h'(x, y) = h(ax, y)$. As $H_1(A, \mathbb{Q})$ is an L-vector space of L-dimension 2, the K-determinant of a on $H_1(A, \mathbb{Q})$ is $\mathrm{Nm}_{L/K}(a)^2$. Since $L = KF$ and $[L : K] = [F : \mathbb{Q}]$, the three embeddings of L above an embedding of K restrict to the three embeddings of F, so

$$\mathrm{Nm}_{L/K}(a) = \mathrm{Nm}_{F/\mathbb{Q}}(a) \in \mathbb{Q}.$$

Computing in a K-basis of $H_1(A, \mathbb{Q})$, we obtain

$$\det h' = \mathrm{Nm}_{F/\mathbb{Q}}(a)^2 \det h = \mathrm{Nm}_{K/\mathbb{Q}}\big(\mathrm{Nm}_{F/\mathbb{Q}}(a)\big) \det h,$$

so $\mathrm{disc}(h') = \mathrm{disc}(h)$. If the Hermitian form is read off from the dual form ψ on V instead, the discriminant does not change: the polarization ψ of V is the dual of a Riemann form on $H_1(A, \mathbb{Q})$, and passing to the dual form replaces $\det h$ with $(\det h)^{-1}$, whose class in $\mathbb{Q}^*/\mathrm{Nm}_{K/\mathbb{Q}}(K^*)$ is $\mathrm{disc}(h)$, since $(\det h)^2 = \mathrm{Nm}_{K/\mathbb{Q}}(\det h)$.

Assume that $\mathrm{disc}(h) = [-1]$. Since $\dim A = 6$ and the multiplicities of K on $H^{1,0}(A)$ are $(3, 3)$, (A, K) is of Weil type, and Θ is a compatible polarization of (A, K) whose Hermitian form h satisfies $\mathrm{disc}(h) = [-1]$. By Theorem 2.14, any class in $W_K(A)$ is algebraic. □

If $\mathrm{disc}(h) \neq [-1]$ in Proposition 7.2 (2), then Theorem 2.14 cannot be applied to A with any compatible polarization of (A, K), since $\mathrm{disc}(h)$ does not depend on the compatible polarization. When K is $\mathbb{Q}(i)$ or $\mathbb{Q}(\omega)$, this case is treated by Theorem 1.3.

7.3. **Quartic CM endomorphism field.** In this subsection, we determine the Hodge group and the Hodge classes of a simple abelian sixfold whose endomorphism algebra is a CM field of degree 4, and we prove that such a sixfold satisfies the Hodge conjecture (Theorem 7.4).

Lemma 7.3. *Let A be a simple abelian sixfold with $\mathrm{End}^0(A) = L$ a CM field of degree 4, and assume that L contains an imaginary quadratic subfield K whose multiplicities on $H^{1,0}(A)$ are $(3, 3)$. Then (A, K) admits a compatible polarization whose Hermitian form has discriminant $[-1]$.*

Proof. Let $F := L^+$, and write $F = \mathbb{Q}(\sqrt{f_0})$, where f_0 is a square-free positive integer. Since $K \neq F$ and $[L : \mathbb{Q}] = 4$, we have $L = KF = K(\sqrt{f_0})$. Let $\sigma, \bar{\sigma}$ be the complex embeddings of K as in Definition 2.7, and let ρ_1, ρ_2 be the real embeddings of F. For $j = 1, 2$, let $\tau_j : L \hookrightarrow \mathbb{C}$ be the embedding with $\tau_j|_K = \sigma$ and $\tau_j|_F = \rho_j$, and let r_j be the multiplicity of L on $H^{1,0}(A)$ at τ_j. Since $\dim_L H^1(A, \mathbb{Q}) = 3$, we have $\dim H^1(A, \mathbb{C})_\tau = 3$ for any embedding τ of L. Complex conjugation maps $H^{0,1}(A) \cap H^1(A, \mathbb{C})_\tau$ onto $H^{1,0}(A) \cap H^1(A, \mathbb{C})_{\bar{\tau}}$, so the multiplicity of L at $\bar{\tau}_j$ is $3 - r_j$. Moreover,

$$H^1(A, \mathbb{C})_\sigma = H^1(A, \mathbb{C})_{\tau_1} \oplus H^1(A, \mathbb{C})_{\tau_2},$$

so $r_1 + r_2 = 3$.

We show that $r_1, r_2 \in \{1, 2\}$. Otherwise $\{r_1, r_2\} = \{0, 3\}$, so for any embedding τ of L, the space $H^1(A, \mathbb{C})_\tau$ is contained in $H^{1,0}(A)$ or in $H^{0,1}(A)$. Then for any $z \in \mathbb{U}$, $\eta_A(z)$ acts on any $H^1(A, \mathbb{C})_\tau$ by the scalar z or $\bar{z}$, and hence $\eta_A(\mathbb{U}) \subset \mathrm{U}_L(\mathbb{R})$, where U_L acts on $H^1(A, \mathbb{Q})$ through the action of L. By the minimality in Definition 2.2, $\mathrm{Hg}(A) \subset \mathrm{U}_L$, so the commutant of $\mathrm{Hg}(A)$ in $\mathrm{End}_{\mathbb{Q}}(H^1(A, \mathbb{Q}))$ contains $\mathrm{End}_L(H^1(A, \mathbb{Q})) \cong M_3(L)$. This contradicts Lemma 2.3 (1), since $\mathrm{End}^0(A) = L$.

Let Θ be a compatible polarization of (A, K) (Lemma 2.10 (1)), let $\Lambda := H_1(A, \mathbb{Q})$, and let ψ be the Riemann form of Θ on Λ. The Rosati involution of Θ induces complex conjugation on L [Moo99, (1.19)], so $\psi(ax, y) = \psi(x, \bar{a}y)$ for any $a \in L$ and $x, y \in \Lambda$. Since the trace form of $L/\mathbb{Q}$ is non-degenerate, there exists a unique L/F-Hermitian form h_L on Λ, conjugate-linear in the first variable, such that

$$\psi(x, y) = \mathrm{Tr}_{L/\mathbb{Q}}(\delta h_L(x, y))$$

for any $x, y \in \Lambda$, where $\delta \in K$ with $\bar{\delta} = -\delta \neq 0$ is the element fixed in Definition 2.9 (cf. [Moo99, (1.22)]).

Next we compute the signatures of h_L at ρ_1 and ρ_2. Let J be the complex structure on $H_1(A, \mathbb{R}) = \Lambda \otimes \mathbb{R}$. Then J commutes with the action of L, and $H^{1,0}(A)$ is dual to the i-eigenspace of J in $H_1(A, \mathbb{C})$, compatibly with the action of L. We have $\psi(x, Jx) > 0$ for any $x \neq 0$ (cf. [Moo99, (1.15), (2.2)]). Write $\sigma(\delta) = it$ with $t \in \mathbb{R}^*$. Fix $j \in \{1, 2\}$, and let $\Lambda_j := \Lambda \otimes_{F, \rho_j} \mathbb{R}$. Then Λ_j is a complex vector space of dimension 3 through the isomorphism $L \otimes_{F, \rho_j} \mathbb{R} \cong \mathbb{C}$ induced by τ_j. We denote by I the multiplication by i on Λ_j, and by h_j the $\mathbb{C}$-Hermitian form on Λ_j induced by h_L and τ_j. We have $H_1(A, \mathbb{R}) = \Lambda_1 \oplus \Lambda_2$, and $\psi(\Lambda_1, \Lambda_2) = 0$ because $\psi(fx, y) = \psi(x, fy)$ for any $f \in F$. Moreover, J preserves Λ_j and commutes with I, so

$$\Lambda_j = \Lambda_j^+ \oplus \Lambda_j^-, \quad \text{where } J = I \text{ on } \Lambda_j^+ \text{ and } J = -I \text{ on } \Lambda_j^-.$$

The i-eigenspace of J in $\Lambda_j \otimes_{\mathbb{R}} \mathbb{C}$ is the direct sum of a subspace of complex dimension $\dim_{\mathbb{C}} \Lambda_j^+$ on which L acts through τ_j, and a subspace of complex dimension $\dim_{\mathbb{C}} \Lambda_j^-$ on which L acts through $\bar{\tau}_j$. Hence $\dim_{\mathbb{C}} \Lambda_j^+ = r_j$ and $\dim_{\mathbb{C}} \Lambda_j^- = 3 - r_j$. For $x, y \in \Lambda_j$, only τ_j and $\bar{\tau}_j$ contribute to the trace, and $\tau_j(\delta) = \sigma(\delta) = it$, so

$$\psi(x, y) = 2\,\mathrm{Re}\left(ith_j(x, y)\right) = -2t\,\mathrm{Im}\, h_j(x, y).$$

Since J preserves ψ and commutes with I, it preserves h_j, and hence Λ_j^+ and Λ_j^- are h_j-orthogonal. For $x \in \Lambda_j^{\pm}$, we have

$$\psi(x, Jx) = -2t\,\mathrm{Im}\, h_j(x, \pm Ix) = \mp 2th_j(x, x).$$

Thus th_j is negative definite on Λ_j^+ and positive definite on Λ_j^-, so the signature of h_L at ρ_j is $(3 - r_j, r_j)$ if $t > 0$ and $(r_j, 3 - r_j)$ if $t < 0$. Since t does not depend on j, $r_1 + r_2 = 3$, and $r_1, r_2 \in \{1, 2\}$, the signatures of h_L at ρ_1 and ρ_2 are $(2, 1)$ and $(1, 2)$, in some order.

By the Gram–Schmidt process, there exists an L-basis e_1, e_2, e_3 of Λ such that $h_L(e_k, e_l) = 0$ for $k \neq l$. Let $b_k := h_L(e_k, e_k) \in F^*$, and let $\beta := b_1 b_2 b_3$, the determinant of the Gram matrix of h_L in the basis e_1, e_2, e_3. Then $\rho_j(\beta) < 0$ if and only if exactly one of $\rho_j(b_1), \rho_j(b_2), \rho_j(b_3)$ is negative, that is, if and only if h_L has signature $(2, 1)$ at ρ_j. Thus $\rho_1(\beta)\rho_2(\beta) < 0$. Since $\rho_1(\sqrt{f_0})\rho_2(\sqrt{f_0}) < 0$, exactly one of $\sqrt{f_0}/\beta$ and $-\sqrt{f_0}/\beta$ is totally positive, and we denote it by a. Then

$$\mathrm{Nm}_{F/\mathbb{Q}}(a) = -\frac{f_0}{\mathrm{Nm}_{F/\mathbb{Q}}(\beta)}.$$

Let $\psi'(x, y) := \psi(ax, y)$ for $x, y \in \Lambda$. Since $\bar{a} = a$, we have $\psi'(y, x) = -\psi(x, ay) = -\psi(ax, y) = -\psi'(x, y)$, so ψ' is a rational alternating form. As a commutes with J, $\psi'(Jx, Jy) = \psi'(x, y)$. Since a acts on Λ_j by $\rho_j(a) > 0$ and $\psi'(\Lambda_1, \Lambda_2) = 0$, we have $\psi'(x, Jx) > 0$ for any $x \neq 0$. By the Riemann conditions (cf. [Moo99, (1.15), (2.2)]), a positive integral multiple of ψ' is the Riemann form of an ample line bundle on A, so ψ' is the Riemann form of an ample class $\Theta' \in NS(A) \otimes \mathbb{Q}$. For $k \in K$, we have $\psi'(kx, y) = \psi(kax, y) = \psi(ax, \bar{k}y) = \psi'(x, \bar{k}y)$, so Θ' is a compatible polarization of (A, K). Since $\delta \in K$ and $\bar{a} = a$,

$$\psi'(x, y) = \mathrm{Tr}_{L/\mathbb{Q}}(\delta a h_L(x, y)) = \mathrm{Tr}_{K/\mathbb{Q}}\left(\delta\,\mathrm{Tr}_{L/K}(a h_L(x, y))\right),$$

and $h' := \mathrm{Tr}_{L/K}(a h_L)$ is a K-Hermitian form, conjugate-linear in the first variable. Thus h' is the Hermitian form of Θ'.

We compute $\det h'$ in the K-basis $e_1, \sqrt{f_0}e_1, e_2, \sqrt{f_0}e_2, e_3, \sqrt{f_0}e_3$ of Λ. The K-spans of $e_k, \sqrt{f_0}e_k$ for $k = 1, 2, 3$ are h'-orthogonal. For $u, w \in \mathbb{Q}$, we have $\mathrm{Tr}_{L/K}(u + w\sqrt{f_0}) = 2u$. Thus if $ab_k = u + w\sqrt{f_0}$, then the Gram matrix of h' in the basis $e_k, \sqrt{f_0}e_k$ is

$$2\begin{pmatrix} u & f_0 w \\ f_0 w & f_0 u \end{pmatrix},$$

whose determinant is $4f_0(u^2 - f_0 w^2) = 4f_0 \,\mathrm{Nm}_{F/\mathbb{Q}}(ab_k)$. Hence

$$\det h' = (4f_0)^3 \,\mathrm{Nm}_{F/\mathbb{Q}}(a^3\beta) = 64 f_0^3 \Big(- \frac{f_0}{\mathrm{Nm}_{F/\mathbb{Q}}(\beta)}\Big)^3 \mathrm{Nm}_{F/\mathbb{Q}}(\beta) = -\Big(\frac{8f_0^3}{\mathrm{Nm}_{F/\mathbb{Q}}(\beta)}\Big)^2.$$

Since $q^2 = \mathrm{Nm}_{K/\mathbb{Q}}(q)$ for any $q \in \mathbb{Q}^*$, we have $\mathrm{disc}(h') = [-1]$. □

Theorem 7.4. *Let A be a simple abelian sixfold with $\mathrm{End}^0(A) = L$ a CM field of degree 4, let $F := L^+$, $V := H^1(A, \mathbb{Q})$, and ψ a polarization of V. Then*

$$\mathrm{Hg}(A)^{\mathrm{der}} = \mathrm{U}_L(V, \psi)^{\mathrm{der}}, \quad \mathrm{Hg}(A)^{\mathrm{der}}_{\mathbb{C}} \cong \mathrm{SL}_{3,\mathbb{C}} \times \mathrm{SL}_{3,\mathbb{C}},$$

and exactly one of the following holds.

1. *L contains no imaginary quadratic subfield whose multiplicities on $H^{1,0}(A)$ are $(3, 3)$. Then $\mathrm{Hg}(A) = \mathrm{U}_L(V, \psi)$, $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p, $\dim \mathrm{Hdg}^2(A) = 3$, and $\dim \mathrm{Hdg}^3(A) = 4$.*
2. *L contains an imaginary quadratic subfield K whose multiplicities on $H^{1,0}(A)$ are $(3, 3)$. Then K is unique, $L = KF$ is biquadratic,*

   $$\mathrm{Hg}(A) = \{g \in \mathrm{U}_L(V, \psi) \mid \mathrm{Nm}_{L/K}(\det_L g) = 1\}$$

   has one-dimensional connected center, $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for $p \neq 3$, $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_K(A)$, $\dim \mathrm{Hdg}^2(A) = 3$, and $\dim \mathrm{Hdg}^3(A) = 6$.

In both cases, $\mathrm{HC}(A)$ holds.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix notation and show three facts: the spaces V_τ are irreducible and pairwise non-isomorphic $\mathrm{Hg}(A)_{\overline{\mathbb{Q}}}$-modules, each W_i is an irreducible and non-trivial $\mathrm{Hg}(A)^{\mathrm{der}}_{\overline{\mathbb{Q}}}$-module, and r_1, r_2 do not both belong to $\{0, 3\}$.

Since $\dim_{\mathbb{Q}} V = 12$, we have $\dim_L V = 3$. The Rosati involution induces complex conjugation on L [Moo99, (1.19)], so $\psi(ax, y) = \psi(x, \bar{a}y)$ for any $a \in L$ and $x, y \in V$.

Let $\overline{\mathbb{Q}}$ be the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, and $V_{\overline{\mathbb{Q}}} := V \otimes \overline{\mathbb{Q}}$. For $\tau \in \mathrm{Hom}(L, \mathbb{C})$, let $V_\tau \subset V_{\overline{\mathbb{Q}}}$ be the subspace on which L acts through τ. Then $V_{\overline{\mathbb{Q}}} = \bigoplus_\tau V_\tau$, $\dim V_\tau = 3$, and

$$V_\tau \otimes_{\overline{\mathbb{Q}}} \mathbb{C} = H^1(A, \mathbb{C})_\tau.$$

For $a \in L$, $x \in V_\tau$, and $y \in V_{\tau'}$, we have $\tau(a)\psi(x, y) = \psi(x, \bar{a}y) = \tau'(\bar{a})\psi(x, y)$, so ψ pairs V_τ perfectly with $V_{\bar{\tau}}$, and $\psi(V_\tau, V_{\tau'}) = 0$ for $\tau' \neq \bar{\tau}$. We choose $\tau_1, \tau_2 \in \mathrm{Hom}(L, \mathbb{C})$ with $\mathrm{Hom}(L, \mathbb{C}) = \{\tau_1, \bar{\tau}_1, \tau_2, \bar{\tau}_2\}$, and if L contains an imaginary quadratic subfield K whose multiplicities are $(3, 3)$, we choose τ_1, τ_2 such that $\tau_1|_K = \tau_2|_K =: \sigma$. Let $W_i := V_{\tau_i}$. Then ψ induces isomorphisms $V_{\bar{\tau}_i} \cong W_i^*$, and restriction to $W_1 \oplus W_2$ induces an isomorphism

$$\mathrm{U}_L(V, \psi)_{\overline{\mathbb{Q}}} \cong \mathrm{GL}(W_1) \times \mathrm{GL}(W_2),$$

under which $\mathrm{GL}(W_i)$ acts on $V_{\bar{\tau}_i} \cong W_i^*$ by the contragredient action. Here $(\mathrm{U}_L)_{\overline{\mathbb{Q}}}$ corresponds to the torus of pairs of scalars, $\det_L$ corresponds to $(g_1, g_2) \to (\det g_1, \det g_2)$, and

$$\mathrm{SU}_L(V, \psi)_{\overline{\mathbb{Q}}} = \mathrm{U}_L(V, \psi)^{\mathrm{der}}_{\overline{\mathbb{Q}}} = \mathrm{SL}(W_1) \times \mathrm{SL}(W_2).$$

Let r_i be the multiplicity of L on $H^{1,0}(A)$ at τ_i. Since complex conjugation maps $H^{0,1}(A) \cap H^1(A, \mathbb{C})_{\tau_i}$ onto $H^{1,0}(A) \cap H^1(A, \mathbb{C})_{\bar{\tau}_i}$, the multiplicity of L at $\bar{\tau}_i$ is $3 - r_i$.

The polarization ψ is a morphism of Hodge structures, so it vanishes on $H^{1,0}(A)\times H^{1,0}(A)$ and on $H^{0,1}(A)\times H^{0,1}(A)$. Thus $\eta_A(\mathbb{U})$ preserves ψ, and it commutes with L because L acts by endomorphisms of Hodge structures. By the minimality in Definition 2.2, $\mathrm{Hg}(A)\subset \mathrm{U}_L(V,\psi)$. By Lemma 2.3 (1), $\mathrm{Hg}(A)$ is connected and reductive, and the commutant of $\mathrm{Hg}(A)_{\overline{\mathbb{Q}}}$ in $\mathrm{End}(V_{\overline{\mathbb{Q}}})$ is $L\otimes\overline{\mathbb{Q}}\cong\overline{\mathbb{Q}}^4$, acting on V_τ through the τ-coordinate. Since $V_{\overline{\mathbb{Q}}}$ is a semisimple $\mathrm{Hg}(A)_{\overline{\mathbb{Q}}}$-module, the spaces V_τ are irreducible and pairwise non-isomorphic $\mathrm{Hg}(A)_{\overline{\mathbb{Q}}}$-modules: otherwise the commutant would contain a non-scalar endomorphism of some V_τ or a non-zero homomorphism $V_\tau\to V_{\tau'}$ with $\tau\neq\tau'$.

Let Z be the connected center of $\mathrm{Hg}(A)$. Then $Z_{\overline{\mathbb{Q}}}$ is contained in the commutant $L\otimes\overline{\mathbb{Q}}$, so $Z\subset \mathrm{U}_L$, and Z acts on each V_τ by scalars. Moreover, $\mathrm{Hg}(A)=Z\cdot\mathrm{Hg}(A)^{\mathrm{der}}$ and $Z\cap\mathrm{Hg}(A)^{\mathrm{der}}$ is finite (cf. [Bor91, Proposition 14.2]). Thus each W_i is an irreducible $\mathrm{Hg}(A)^{\mathrm{der}}_{\overline{\mathbb{Q}}}$-module, and the action of $\mathrm{Hg}(A)^{\mathrm{der}}_{\overline{\mathbb{Q}}}$ on W_i is non-trivial, since otherwise $\mathrm{Hg}(A)_{\overline{\mathbb{Q}}}$ would act on W_i by scalars.

Finally, we show that r_1 and r_2 do not both belong to $\{0,3\}$. Otherwise, for any $\tau\in\mathrm{Hom}(L,\mathbb{C})$, the space $H^1(A,\mathbb{C})_\tau$ is contained in $H^{1,0}(A)$ or in $H^{0,1}(A)$. Then for any $z\in\mathbb{U}$, $\eta_A(z)$ acts on any $H^1(A,\mathbb{C})_\tau$ by z or $\bar z$, so $\eta_A(\mathbb{U})\subset\mathrm{U}_L(\mathbb{R})$. By the minimality in Definition 2.2, $\mathrm{Hg}(A)\subset\mathrm{U}_L$, so the commutant of $\mathrm{Hg}(A)$ contains $\mathrm{End}_L(V)\cong M_3(L)$, which contradicts Lemma 2.3 (1).

Step 2. In this step, we show that

$$\mathrm{Hg}(A)^{\mathrm{der}}=\mathrm{U}_L(V,\psi)^{\mathrm{der}}.$$

Let

$$\mathfrak{g}:=\mathrm{Lie}(\mathrm{Hg}(A)^{\mathrm{der}})\otimes\overline{\mathbb{Q}}.$$

The action of $\mathfrak{g}$ on $V_{\overline{\mathbb{Q}}}$ is faithful and determined by the action on $W_1\oplus W_2$, so

$$\mathfrak{g}\subset\mathfrak{sl}(W_1)\oplus\mathfrak{sl}(W_2).$$

By **Step 1**, $\mathfrak{g}$ acts on W_i irreducibly and non-trivially. By [Moo99, Theorem (3.11), Corollary (3.13)], W_i is a minuscule representation of $\mathfrak{g}$. Since $\dim W_i=3$ is a prime, [Moo99, Example (3.6), Table 3] implies that the image of $\mathfrak{g}$ in $\mathfrak{gl}(W_i)$ is $\mathfrak{sl}(W_i)$, acting by the standard representation. Thus $\mathfrak{g}$ surjects onto $\mathfrak{sl}(W_1)$ and onto $\mathfrak{sl}(W_2)$. Since $\mathfrak{sl}_3$ is simple, the kernel of either projection is 0 or the other factor, so by Goursat's lemma (cf. [MZ99, (3.1)]), either $\mathfrak{g}=\mathfrak{sl}(W_1)\oplus\mathfrak{sl}(W_2)$, or $\mathfrak{g}$ is the graph of an isomorphism $\mathfrak{sl}(W_1)\to\mathfrak{sl}(W_2)$. Any automorphism of $\mathfrak{sl}_3$ is inner, or the composite of an inner automorphism with $X\to -X^t$ [Hum72, Section 14.3]. Thus in the graph case, W_2 is isomorphic to W_1 or to W_1^* as a $\mathfrak{g}$-module.

Suppose that $\mathfrak{g}$ is a graph. Replacing τ_2 with $\bar\tau_2$ interchanges W_2 and W_2^*, so possibly replacing τ_2 with $\bar\tau_2$, we may assume that there exists a $\mathfrak{g}$-isomorphism $f:W_1\to W_2$. Then $W_1^*\cong W_2^*$ as $\mathfrak{g}$-modules, and $W_1\ncong W_1^*$ as $\mathfrak{g}$-modules, because the standard and the dual representations of $\mathfrak{sl}_3$ are not isomorphic. Thus the $\mathfrak{g}$-isomorphism classes of the spaces V_τ are $\{V_{\tau_1},V_{\tau_2}\}$ and $\{V_{\bar\tau_1},V_{\bar\tau_2}\}$.

Let $\gamma\in\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ act on $V_{\overline{\mathbb{Q}}}$ through the second factor. Then $\gamma(V_\tau)=V_{\gamma\circ\tau}$ for any τ. Since $\mathrm{Hg}(A)^{\mathrm{der}}$ is defined over $\mathbb{Q}$, γ acts on $\mathfrak{g}$, and $\gamma(Xx)=\gamma(X)\gamma(x)$ for any $X\in\mathfrak{g}$ and $x\in V_{\overline{\mathbb{Q}}}$. Thus for any $\mathfrak{g}$-isomorphism $f':V_\tau\to V_{\tau'}$, the map

$$\gamma\circ f'\circ\gamma^{-1}:V_{\gamma\circ\tau}\to V_{\gamma\circ\tau'}$$

is a $\mathfrak{g}$-isomorphism. Hence γ maps $\{\tau_1,\tau_2\}$ onto $\{\tau_1,\tau_2\}$ or onto $\{\bar\tau_1,\bar\tau_2\}$. For $\tau\in\mathrm{Hom}(L,\mathbb{C})$, let t_τ be the character of $(\mathrm{U}_L)_{\overline{\mathbb{Q}}}$ by which U_L acts on V_τ. Then $t_{\bar\tau}=t_\tau^{-1}$, γ maps t_τ to $t_{\gamma\circ\tau}$, and $t_{\tau_i}(\det_L g)=\det(g|_{W_i})$ for any $g\in\mathrm{U}_L(V,\psi)_{\overline{\mathbb{Q}}}$. Let $\chi:=t_{\tau_1}t_{\tau_2}^{-1}$. Then γ maps χ to χ or to χ^{-1}, so $\ker\chi\subset(\mathrm{U}_L)_{\overline{\mathbb{Q}}}$ is stable under $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and is defined over $\mathbb{Q}$ (cf. [Bor91, Chapter AG, §14]). Thus $\det_L^{-1}(\ker\chi)$ is an algebraic subgroup of $\mathrm{U}_L(V,\psi)$ defined over $\mathbb{Q}$.

We show that $r_1=r_2$. Since $\eta_A(\mathbb{U})\subset\mathrm{Hg}(A)(\mathbb{R})$, the Lie algebra $\mathrm{Lie}(\mathrm{Hg}(A))\otimes\mathbb{C}$ contains the element ξ acting by 1 on $H^{1,0}(A)$ and by -1 on $H^{0,1}(A)$. Write $\xi=\xi_Z+\xi'$ with $\xi_Z\in\mathrm{Lie}(Z)\otimes\mathbb{C}$ and $\xi'\in\mathfrak{g}\otimes_{\overline{\mathbb{Q}}}\mathbb{C}$. Then ξ_Z acts on $W_i\otimes\mathbb{C}$ by a scalar c_i, and ξ' acts on $W_i\otimes\mathbb{C}$ with trace

0. Since ξ acts on $W_i \otimes \mathbb{C} = H^1(A,\mathbb{C})_{\tau_i}$ with eigenvalue 1 of multiplicity r_i and eigenvalue -1 of multiplicity $3-r_i$, we have $c_i = (2r_i-3)/3$, and $\xi'|_{W_i} = \xi|_{W_i} - c_i$ is diagonalizable with eigenvalues in $\{1-c_i, -1-c_i\}$. Moreover, $\xi'|_{W_2} = f \circ \xi'|_{W_1} \circ f^{-1}$. If $r_1 \in \{0,3\}$, then $\xi'|_{W_1} = 0$, so $\xi'|_{W_2} = 0$ and $r_2 \in \{0,3\}$, which contradicts **Step 1**. Thus $r_1 \in \{1,2\}$, and similarly $r_2 \in \{1,2\}$. Then the largest eigenvalues of $\xi'|_{W_1}$ and $\xi'|_{W_2}$ are $1-c_1$ and $1-c_2$, so $c_1 = c_2$ and $r_1 = r_2$.

For $z \in \mathbb{U}$, $\eta_A(z)$ acts on $W_i \otimes \mathbb{C}$ with determinant $z^{r_i}\bar{z}^{3-r_i} = z^{2r_i-3}$, so $\chi(\det_L \eta_A(z)) = z^{2r_1-2r_2} = 1$. Thus $\eta_A(\mathbb{U}) \subset \det_L^{-1}(\ker\chi)(\mathbb{R})$, and the minimality in Definition 2.2 implies that $\mathrm{Hg}(A) \subset \det_L^{-1}(\ker\chi)$. Let λ_i be the character of $Z_{\overline{\mathbb{Q}}}$ by which Z acts on W_i. Then $\chi(\det_L g) = \lambda_1(g)^3\lambda_2(g)^{-3} = 1$ for any $g \in Z_{\overline{\mathbb{Q}}}$, so the character $\lambda_1\lambda_2^{-1}$ of the connected group $Z_{\overline{\mathbb{Q}}}$ has finite image, and hence $\lambda_1 = \lambda_2$. Since $\mathrm{Hg}(A)^{\mathrm{der}}$ is connected, f is $\mathrm{Hg}(A)^{\mathrm{der}}_{\overline{\mathbb{Q}}}$-equivariant, and since $\lambda_1 = \lambda_2$, f is $Z_{\overline{\mathbb{Q}}}$-equivariant. As

$$\mathrm{Hg}(A)_{\overline{\mathbb{Q}}} = Z_{\overline{\mathbb{Q}}} \cdot \mathrm{Hg}(A)^{\mathrm{der}}_{\overline{\mathbb{Q}}},$$

f is an isomorphism of $\mathrm{Hg}(A)_{\overline{\mathbb{Q}}}$-modules, which contradicts **Step 1**.

Thus $\mathfrak{g} = \mathfrak{sl}(W_1) \oplus \mathfrak{sl}(W_2)$. Since $\mathrm{Hg}(A)^{\mathrm{der}}_{\overline{\mathbb{Q}}}$ is a connected subgroup of $\mathrm{SL}(W_1) \times \mathrm{SL}(W_2)$ with Lie algebra $\mathfrak{g}$, we have

$$\mathrm{Hg}(A)^{\mathrm{der}}_{\overline{\mathbb{Q}}} = \mathrm{SL}(W_1) \times \mathrm{SL}(W_2).$$

Hence, by **Step 1**,

$$\mathrm{Hg}(A)^{\mathrm{der}} = \mathrm{U}_L(V,\psi)^{\mathrm{der}} \quad \text{and} \quad \mathrm{Hg}(A)^{\mathrm{der}}_{\mathbb{C}} \cong \mathrm{SL}_{3,\mathbb{C}} \times \mathrm{SL}_{3,\mathbb{C}}\,.$$

Step 3. In this step, we determine $\mathrm{Hg}(A)$ by the torus $T := \det_L(\mathrm{Hg}(A)) \subset \mathrm{U}_L$, and we prove the descriptions of $\mathrm{Hg}(A)$, K, and Z in (1) and (2).

Since $\mathrm{Hg}(A)$ is connected, T is a subtorus of U_L defined over $\mathbb{Q}$. By **Steps 1 and 2**,

$$\ker(\det_L) = \mathrm{SU}_L(V,\psi) = \mathrm{Hg}(A)^{\mathrm{der}} \subset \mathrm{Hg}(A),$$

so

$$\mathrm{Hg}(A) = \det_L^{-1}(T).$$

If $T' \subset \mathrm{U}_L$ is a subtorus defined over $\mathbb{Q}$ with $\det_L(\eta_A(\mathbb{U})) \subset T'(\mathbb{R})$, then $\eta_A(\mathbb{U}) \subset \det_L^{-1}(T')(\mathbb{R})$, so $\mathrm{Hg}(A) \subset \det_L^{-1}(T')$ and $T \subset T'$. Thus T is the smallest subtorus T' of U_L defined over $\mathbb{Q}$ with $\det_L(\eta_A(\mathbb{U})) \subset T'(\mathbb{R})$.

The characters t_{τ_1}, t_{τ_2} of **Step 2** induce an isomorphism

$$(\mathrm{U}_L)_{\overline{\mathbb{Q}}} \cong \mathbb{G}_m^2,$$

and we let e_{τ_1}, e_{τ_2} be the corresponding basis of the cocharacter lattice $X_*(\mathrm{U}_L)$, and $e_{\bar{\tau}_i} := -e_{\tau_i}$. Then $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ maps e_τ to $e_{\gamma\circ\tau}$. By **Step 2**, the complexification of $\det_L \circ \eta_A$ is the cocharacter

$$v := (2r_1-3)e_{\tau_1} + (2r_2-3)e_{\tau_2}.$$

By [Moo99, (1.1)], the subtori T' of U_L defined over $\mathbb{Q}$ correspond to the $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable saturated sublattices $X_*(T') \subset X_*(\mathrm{U}_L)$, and $\det_L(\eta_A(\mathbb{U})) \subset T'(\mathbb{R})$ if and only if $v \in X_*(T')$. Thus $X_*(T) \otimes \mathbb{Q}$ is the $\mathbb{Q}$-span of the $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-orbit of v. Since $2r_i - 3$ is odd, $v \neq 0$, so $\dim T \in \{1,2\}$.

We show that if $\dim T = 1$, then L contains an imaginary quadratic subfield whose multiplicities are $(3,3)$. Since L is a field, $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts transitively on $\mathrm{Hom}(L,\mathbb{C})$, and since $\gamma \circ \bar{\tau} = \overline{\gamma \circ \tau}$ for any γ and τ, there exists γ mapping $\{\tau_1, \bar{\tau}_1\}$ onto $\{\tau_2, \bar{\tau}_2\}$ and $\{\tau_2, \bar{\tau}_2\}$ onto $\{\tau_1, \bar{\tau}_1\}$. Since $\dim T = 1$, the line $\mathbb{Q}v = X_*(T) \otimes \mathbb{Q}$ is stable under $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, and since γ has finite order on it, $\gamma(v) = \pm v$, so $|2r_1 - 3| = |2r_2 - 3|$. By **Step 1**, r_1 and r_2 do not both belong to $\{0,3\}$, so $r_1, r_2 \in \{1,2\}$. Let $\Sigma_+ \subset \mathrm{Hom}(L,\mathbb{C})$ be the set of τ such that e_τ has coefficient 1 in v, that is, $\tau_i \in \Sigma_+$ if $r_i = 2$ and $\bar{\tau}_i \in \Sigma_+$ if $r_i = 1$. Then $v = \sum_{\tau\in\Sigma_+} e_\tau$, the complement of Σ_+ is $\bar{\Sigma}_+ := \{\bar{\tau} \mid \tau \in \Sigma_+\}$, and the multiplicity of L at any $\tau \in \Sigma_+$ is 2. Since $\gamma(v) = \pm v$ for any $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, γ maps Σ_+ onto Σ_+ or onto $\bar{\Sigma}_+$. Let Γ_+ be the stabilizer of Σ_+ in $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$,

which has index 2 since complex conjugation maps Σ_+ onto $\bar{\Sigma}_+$. Fix $\tau_0 \in \Sigma_+$. Any γ fixing τ_0 belongs to Γ_+, so the fixed field $\overline{\mathbb{Q}}^{\Gamma_+}$ is a quadratic subfield of $\tau_0(L)$, and it is imaginary since complex conjugation does not belong to Γ_+. Let $K' := \tau_0^{-1}(\overline{\mathbb{Q}}^{\Gamma_+})$. For $\tau = \gamma \circ \tau_0$, we have $\tau|_{K'} = \tau_0|_{K'}$ if and only if $\gamma \in \Gamma_+$, if and only if $\tau \in \Sigma_+$. Thus the multiplicities of K' are $(4,2)$ or $(2,4)$. Since $K' \neq F$, $L = K'F$ is biquadratic, $\mathrm{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/2)^2$, and L contains a unique imaginary quadratic subfield $K \neq K'$. The fibers of $\mathrm{Hom}(L,\mathbb{C}) \to \mathrm{Hom}(K,\mathbb{C})$ and of $\mathrm{Hom}(L,\mathbb{C}) \to \mathrm{Hom}(K',\mathbb{C})$ are the orbits of the distinct subgroups $\mathrm{Gal}(L/K)$ and $\mathrm{Gal}(L/K')$ of order 2, which act on $\mathrm{Hom}(L,\mathbb{C})$ by precomposition. Thus any fiber over $\mathrm{Hom}(K,\mathbb{C})$ contains one element of Σ_+ and one element of $\bar{\Sigma}_+$, at which the multiplicities of L are 2 and 1. Hence the multiplicities of K are $(3,3)$.

We show that if L contains an imaginary quadratic subfield K whose multiplicities are $(3,3)$, then

$$T = \ker(\mathrm{Nm}_{L/K} : \mathrm{U}_L \to \mathrm{U}_K).$$

For an L-linear endomorphism g of V, the K-determinant of g is $\mathrm{Nm}_{L/K}(\det_L g)$. Since the multiplicities of K are $(3,3)$, $\eta_A(z)$ acts on $H^1(A,\mathbb{C})_\sigma$ with determinant $z^3\bar{z}^3 = 1$ for any $z \in \mathbb{U}$, so

$$\det_L(\eta_A(\mathbb{U})) \subset \ker(\mathrm{Nm}_{L/K})(\mathbb{R}).$$

Since $\tau_1|_K = \tau_2|_K = \sigma$, the subgroup $\ker(\mathrm{Nm}_{L/K})_{\overline{\mathbb{Q}}}$ of

$$(\mathrm{U}_L)_{\overline{\mathbb{Q}}} \cong \mathbb{G}_m^2$$

is $\{(t,t^{-1})\}$, which is a one-dimensional torus. As $\dim T \geq 1$ and $T \subset \ker(\mathrm{Nm}_{L/K})$ by the minimality of T, we have $T = \ker(\mathrm{Nm}_{L/K})$.

We now determine $\mathrm{Hg}(A)$ and Z. In case (1), $\dim T = 2$, since $\dim T = 1$ would imply that L contains an imaginary quadratic subfield whose multiplicities are $(3,3)$. Thus $T = \mathrm{U}_L$, and

$$\mathrm{Hg}(A) = \det_L^{-1}(\mathrm{U}_L) = \mathrm{U}_L(V,\psi).$$

In this case, $Z = \mathrm{U}_L$ and $Z_{\overline{\mathbb{Q}}}$ is the torus of pairs of scalars (t_1,t_2). In case (2),

$$\mathrm{Hg}(A) = \det_L^{-1}(\ker \mathrm{Nm}_{L/K}) = \{g \in \mathrm{U}_L(V,\psi) \mid \mathrm{Nm}_{L/K}(\det_L g) = 1\},$$

and

$$\dim Z = \dim \mathrm{Hg}(A) - \dim \mathrm{Hg}(A)^{\mathrm{der}} = \dim T = 1.$$

Since $Z \subset \mathrm{U}_L$ acts on W_i by a scalar t_i, and $\det_L(Z) \subset T$, we have $(t_1t_2)^3 = 1$ on $Z_{\overline{\mathbb{Q}}}$. As Z is connected, $Z_{\overline{\mathbb{Q}}} = \{(t,t^{-1})\}$. Moreover, K and F are distinct quadratic subfields of L, so $L = KF$ is biquadratic.

It remains to show the uniqueness of K in case (2). Let K' be the imaginary quadratic subfield of L different from K. Let g_K and $g_{K'}$ be the generators of $\mathrm{Gal}(L/K)$ and $\mathrm{Gal}(L/K')$. Then $\tau_2 = \tau_1 \circ g_K$, and $g_{K'}$ is the composite of g_K with complex conjugation, so $\tau_1 \circ g_{K'} = \bar{\tau}_2$. Thus the embeddings of L restricting to $\tau_1|_{K'}$ are τ_1 and $\bar{\tau}_2$, and the multiplicities of K' are

$$(r_1 + 3 - r_2,\ 3 - r_1 + r_2).$$

Since $\tau_1|_K = \tau_2|_K = \sigma$ and the multiplicity of K at σ is 3, we have $r_1 + r_2 = 3$. Thus the multiplicities of K' are $(3,3)$ if and only if $r_1 = r_2$, which contradicts $r_1 + r_2 = 3$. As F is real, K is the unique imaginary quadratic subfield of L whose multiplicities are $(3,3)$.

Step 4. In this step, we compute $\mathrm{Hdg}^p(A)$ for any p.

Let W be a $\overline{\mathbb{Q}}$-vector space with $\dim W = 3$. By [Ful97, Section 8.2], $\wedge^a W$ is an irreducible representation of $\mathrm{GL}(W)$ for $0 \leq a \leq 3$, and restricted to $\mathrm{SL}(W)$, the representations $\wedge^0 W$ and $\wedge^3 W$ are trivial, while $\wedge^1 W$ and $\wedge^2 W$ are non-trivial and not isomorphic. Since $\wedge^a W \otimes \wedge^b W^* \cong \mathrm{Hom}(\wedge^b W, \wedge^a W)$, Schur's lemma implies that the $\mathrm{SL}(W)$-invariants in

$$\wedge^\bullet(W \oplus W^*) = \bigoplus_{a,b} \wedge^a W \otimes \wedge^b W^*$$

have a basis

$$1,\ d,\ d^2,\ d^3,\ \wedge^3 W,\ \wedge^3 W^*,$$

where $d := \sum_k w_k \wedge w_k^* \in \wedge^2(W \oplus W^*)$ for a basis w_1, w_2, w_3 of W with dual basis w_1^*, w_2^*, w_3^*, and where $\wedge^3 W$ and $\wedge^3 W^*$ also denote generators of the lines $\wedge^3 W$ and $\wedge^3 W^*$. Here d does not depend on the basis, $d^a = a! \sum_{|S|=a} \bigwedge_{k\in S}(w_k \wedge w_k^*)$ is a non-zero element of $\wedge^a W \otimes \wedge^a W^*$ for $0 \le a \le 3$, and a scalar $t \in \mathrm{GL}(W)$ acts on $\wedge^a W \otimes \wedge^b W^*$ by t^{a-b}.

We have

$$\wedge^\bullet V_{\overline{\mathbb{Q}}} = \wedge^\bullet(W_1 \oplus W_1^*) \otimes \wedge^\bullet(W_2 \oplus W_2^*).$$

Let $d_i \in \wedge^2(W_i \oplus W_i^*)$ be the element d for $W = W_i$. Hence the $\mathrm{SL}(W_1) \times \mathrm{SL}(W_2)$-invariants in $\wedge^\bullet V_{\overline{\mathbb{Q}}}$ have a basis consisting of the products $x_1 x_2$, where $x_i \in \{1, d_i, d_i^2, d_i^3, \wedge^3 W_i, \wedge^3 W_i^*\}$. By **Steps 2 and 3**,

$$\mathrm{Hg}(A)_{\overline{\mathbb{Q}}} = Z_{\overline{\mathbb{Q}}} \cdot (\mathrm{SL}(W_1) \times \mathrm{SL}(W_2)),$$

where $Z_{\overline{\mathbb{Q}}}$ is the torus of pairs of scalars (t_1, t_2) in case (1) and the torus $\{(t, t^{-1})\}$ in case (2). The element (t_1, t_2) acts trivially on d_1, d_2, by t_i^3 on $\wedge^3 W_i$, and by t_i^{-3} on $\wedge^3 W_i^*$. Thus the $\mathrm{Hg}(A)_{\overline{\mathbb{Q}}}$-invariants in $\wedge^\bullet V_{\overline{\mathbb{Q}}}$ have a basis consisting of

- the monomials $d_1^{a_1} d_2^{a_2}$ with $0 \le a_1, a_2 \le 3$ in case (1), and
- the monomials $d_1^{a_1} d_2^{a_2}$ with $0 \le a_1, a_2 \le 3$, together with $\wedge^3 W_1 \cdot \wedge^3 W_2$ and $\wedge^3 W_1^* \cdot \wedge^3 W_2^*$, in case (2).

By Lemma 2.3 (2), $\mathrm{Hdg}^p(A)$ is the space of $\mathrm{Hg}(A)$-invariants in $\wedge^{2p} V$, and taking invariants commutes with extension of scalars. Thus $\mathrm{Hdg}^1(A) \otimes \overline{\mathbb{Q}}$ is spanned by d_1, d_2. By the Lefschetz $(1,1)$ theorem [Moo99, (2.10)], $\mathrm{Hdg}^1(A) = \mathcal{D}^1(A)$. Since the cup product is defined over $\mathbb{Q}$, $\mathcal{D}^p(A) \otimes \overline{\mathbb{Q}}$ is spanned by the monomials $d_1^{a_1} d_2^{a_2}$ with $a_1 + a_2 = p$. In case (1), we deduce that $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p, $\dim \mathrm{Hdg}^2(A) = 3$, and $\dim \mathrm{Hdg}^3(A) = 4$. In case (2), we deduce that $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for $p \ne 3$ and $\dim \mathrm{Hdg}^2(A) = 3$. Since $\tau_1|_K = \tau_2|_K = \sigma$, we have $(W_1 \oplus W_2) \otimes \mathbb{C} = H^1(A, \mathbb{C})_\sigma$, and hence

$$\wedge^3 W_1 \cdot \wedge^3 W_2 = \wedge^6 H^1(A, \mathbb{C})_\sigma \quad \text{and} \quad \wedge^3 W_1^* \cdot \wedge^3 W_2^* = \wedge^6 H^1(A, \mathbb{C})_{\bar\sigma}$$

after extension of scalars to $\mathbb{C}$. As (A, K) is of Weil type, the elements $\wedge^6 H^1(A, \mathbb{C})_\sigma$ and $\wedge^6 H^1(A, \mathbb{C})_{\bar\sigma}$ span $W_K(A) \otimes \mathbb{C}$ (Definition 2.7), and $W_K(A) \subset \mathrm{Hdg}^3(A)$. The monomials $d_1^{a_1} d_2^{a_2}$ are contained in

$$\bigoplus_{a,b} \wedge^a W_1 \otimes \wedge^a W_1^* \otimes \wedge^b W_2 \otimes \wedge^b W_2^*,$$

while $\wedge^3 W_1 \cdot \wedge^3 W_2$ and $\wedge^3 W_1^* \cdot \wedge^3 W_2^*$ are not, so $\mathcal{D}^3(A) \cap W_K(A) = 0$. Thus $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_K(A)$ and $\dim \mathrm{Hdg}^3(A) = 4 + 2 = 6$.

Step 5. In this step, we prove that $\mathrm{HC}(A)$ holds.

In case (1), any Hodge class on A is a polynomial in classes of divisors by **Step 4**, so it is algebraic, and $\mathrm{HC}(A)$ holds. In case (2), (A, K) is of Weil type. The form ψ is dual to the Riemann form of an ample class on $H_1(A, \mathbb{Q})$, and replacing (V, ψ) with $H_1(A, \mathbb{Q})$ and the dual form of ψ does not change $\mathrm{U}_L(V, \psi)$, up to the contragredient isomorphism. By Lemma 7.3, which works on $H_1(A, \mathbb{Q})$, (A, K) admits a compatible polarization whose Hermitian form has discriminant $[-1]$. By Theorem 2.14, any class in $W_K(A)$ is algebraic. By **Step 4**, $\mathrm{Hdg}^2(A) = \mathcal{D}^2(A)$ and $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_K(A)$, so $\mathrm{HC}^2(A)$ and $\mathrm{HC}^3(A)$ hold. Thus $\mathrm{HC}(A)$ holds by Lemma 2.6. □

7.4. **Definite quaternion endomorphism algebra.** In this subsection, we determine the Hodge classes on a simple abelian sixfold A whose endomorphism algebra $D := \mathrm{End}^0(A)$ is of Albert type III. The quaternionic Weil space $W_D(A)$ of Definition-Theorem 7.6 (cf. [vGV03, Sections 4.5–4.9]) is irreducible under the operators u^* with $u \in D^*$, so $\mathrm{HC}(A)$ holds as soon as any class in $W_K(A)$ is algebraic for one imaginary quadratic subfield $K \subset D$ (Corollary 7.7).

Definition 7.5. For $a, b \in \mathbb{Q}^*$, we denote by $(a,b)_{\mathbb{Q}}$ the quaternion algebra over $\mathbb{Q}$ generated by i, j with $i^2 = a$, $j^2 = b$, and $ij = -ji$. For a quaternion algebra D over $\mathbb{Q}$, we denote by $x \to \bar{x}$ the canonical involution of D, by $\mathrm{Trd}_{D/\mathbb{Q}}(x) = x + \bar{x}$ the reduced trace, and by $\mathrm{Nrd}_{D/\mathbb{Q}}(x) = x\bar{x}$ the reduced norm. We say that $x \in D$ is a *pure quaternion* if $\mathrm{Trd}_{D/\mathbb{Q}}(x) = 0$. We say that D is *totally definite* if $D \otimes_{\mathbb{Q}} \mathbb{R}$ is the algebra of Hamilton quaternions.

Definition-Theorem 7.6. Let A be a simple abelian sixfold such that $D := \mathrm{End}^0(A)$ is of Albert type III, and let Θ be an ample class. Then D is a totally definite quaternion algebra over $\mathbb{Q}$, and $\mathrm{Hg}(A)$ is the identity component of the group of D-linear automorphisms of $H_1(A, \mathbb{Q})$ preserving the Riemann form of Θ. Moreover,

$$\mathrm{Hg}(A)_{\mathbb{C}} \cong \mathrm{SO}_{6,\mathbb{C}},$$

and $H^1(A, \mathbb{C})$ is the sum of two copies of the standard representation of $\mathrm{Hg}(A)_{\mathbb{C}}$. Any imaginary quadratic subfield $K \subset D$ has multiplicities $(3,3)$ on $H^{1,0}(A)$. For $u \in D^*$, write $u^* := m^{-6}(mu)^*$ on $H^6(A, \mathbb{Q})$ for any positive integer m with $mu \in \mathrm{End}(A)$. Fix an imaginary quadratic subfield $K \subset D$, and let $W_D(A)$ be the $\mathbb{Q}$-span of $u^* W_K(A)$ for $u \in D^*$. Then $W_D(A)$ does not depend on K, $\dim W_D(A) = 7$, $W_D(A)$ is stable under u^* for any $u \in D^*$ and irreducible under the operators u^* with $u \in D^*$, $\mathrm{Hdg}^p(A) = \mathbb{Q}\Theta^p$ for $p \neq 3$, $\mathrm{Hdg}^3(A) = \mathbb{Q}\Theta^3 \oplus W_D(A)$, and $\Theta \cdot W_D(A) = 0$. We call $W_D(A)$ the *quaternionic Weil space* of A. In particular, $\mathrm{HC}^p(A)$ holds for $p \neq 3$, and $\mathrm{HC}(A)$ holds if and only if $W_D(A)$ contains a non-zero algebraic class, if and only if $\mathrm{Hdg}^3(A)$ contains an algebraic class not in $\mathbb{Q}\Theta^3$.

Proof. The proof is long, so we divide it into several steps. Let ψ be the Riemann form of Θ on $H_1(A, \mathbb{Q})$, on which D acts through endomorphisms, and let $\mathcal{G}$ be the identity component of the group of D-linear automorphisms of $H_1(A, \mathbb{Q})$ preserving ψ. Through the duality between $H_1(A, \mathbb{Q})$ and $H^1(A, \mathbb{Q})$, we regard $\mathrm{Hg}(A)$ as a subgroup of $\mathrm{GL}(H_1(A, \mathbb{Q}))$. By [Moo99, (1.19)], D is a totally definite quaternion algebra over a totally real field F (that is, $D \otimes_{F,\rho} \mathbb{R}$ is the algebra of Hamilton quaternions for any real embedding ρ of F), and the Rosati involution of Θ is the canonical involution of D, that is,

$$\psi(ux, y) = \psi(x, \bar{u}y) \quad \text{for any } u \in D \text{ and } x, y \in H_1(A, \mathbb{Q}).$$

By Lemma 2.3 (1), the commutant of $\mathrm{Hg}(A)$ in $\mathrm{End}_{\mathbb{Q}}(H_1(A, \mathbb{Q}))$ is D. Since $\Theta \in \mathrm{Hdg}^1(A)$, $\mathrm{Hg}(A)$ preserves ψ (Lemma 2.3 (2)). Since $\mathrm{Hg}(A)$ is connected, $\mathrm{Hg}(A) \subset \mathcal{G}$.

Step 1. In this step, we show that $F = \mathbb{Q}$ and describe $\mathcal{G}_{\mathbb{C}}$.

Let $e := [F : \mathbb{Q}]$. Since $H_1(A, \mathbb{Q})$ is a D-vector space, $\dim_{\mathbb{Q}} H_1(A, \mathbb{Q}) = 12$, and $\dim_{\mathbb{Q}} D = 4e$, we have $e \in \{1, 3\}$. Since $D \otimes_{\mathbb{Q}} \mathbb{C} \cong \prod_\tau M_2(\mathbb{C})$, where τ runs through the embeddings $F \hookrightarrow \mathbb{R}$, we have

$$H_1(A, \mathbb{C}) = \bigoplus_\tau N_\tau \otimes T,$$

where $T := \mathbb{C}^2$, $\dim N_\tau = 6/e$, the factor of $D \otimes_{\mathbb{Q}} \mathbb{C}$ indexed by τ acts on $N_\tau \otimes T$ through T, and the other factors act on $N_\tau \otimes T$ by 0. Since $\bar{f} = f$ for any $f \in F$, the subspaces $N_\tau \otimes T$ are pairwise orthogonal for ψ. Let ϖ be a non-zero alternating form on T. The canonical involution of $M_2(\mathbb{C})$ is $g \to \mathrm{adj}(g)$, and $\varpi(gx, y) = \varpi(x, \mathrm{adj}(g)y)$ for any $g \in M_2(\mathbb{C})$. Through ϖ, a bilinear form b on $N_\tau \otimes T$ with $b(gx, y) = b(x, \mathrm{adj}(g)y)$ for any $g \in M_2(\mathbb{C})$ corresponds to an $M_2(\mathbb{C})$-linear map $N_\tau \otimes T \to N_\tau^* \otimes T$, so $b = q \otimes \varpi$ for a bilinear form q on N_τ by Schur's lemma. Thus the restriction of ψ to $N_\tau \otimes T$ is $q_\tau \otimes \varpi$, where q_τ is symmetric and non-degenerate because ψ is alternating and non-degenerate. Since the D-linear automorphisms of $H_1(A, \mathbb{C})$ are the elements of $\prod_\tau \mathrm{GL}(N_\tau)$ acting through the factors N_τ, we have

$$\mathcal{G}_{\mathbb{C}} = \prod_\tau \mathrm{SO}(N_\tau, q_\tau).$$

Suppose that $e = 3$. Then

$$\mathcal{G}_{\mathbb{C}} \cong \mathrm{SO}^3_{2,\mathbb{C}}$$

is a torus, so $\mathrm{Hg}(A)$ is a torus. A $\mathrm{Hg}(A)$-stable subspace of $H_1(A,\mathbb{Q})$ is a Hodge substructure, which corresponds to an abelian subvariety of A up to isogeny [Moo99, Theorem (2.2)]. Since A is simple, $H_1(A,\mathbb{Q})$ is an irreducible representation of $\mathrm{Hg}(A)$. Let $R \subset \mathrm{End}_{\mathbb{Q}}(H_1(A,\mathbb{Q}))$ be the $\mathbb{Q}$-subalgebra such that $R \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$ is spanned by $\mathrm{Hg}(A)(\overline{\mathbb{Q}})$. Since $\mathrm{Hg}(A)$ is a torus, R is commutative and semisimple, and $H_1(A,\mathbb{Q})$ is an irreducible faithful R-module. Thus R is a field, $\dim_R H_1(A,\mathbb{Q}) = 1$, and the commutant of R in $\mathrm{End}_{\mathbb{Q}}(H_1(A,\mathbb{Q}))$ is R. Since the commutant of R is the commutant D of $\mathrm{Hg}(A)$, D is commutative, a contradiction. Thus $e = 1$, and D is a totally definite quaternion algebra over $\mathbb{Q}$.

From now on, we write $H_1(A,\mathbb{C}) = N \otimes T$ with $\dim N = 6$, $\psi = q \otimes \varpi$, and $\mathcal{G}_{\mathbb{C}} = \mathrm{SO}(N,q)$.

Step 2. In this step, we show that $\mathrm{Hg}(A) = \mathcal{G}$.

The commutant of $\mathrm{Hg}(A)_{\mathbb{C}}$ in $\mathrm{End}(N \otimes T)$ is $D \otimes_{\mathbb{Q}} \mathbb{C} = \mathrm{End}(T)$, so $\mathrm{Hg}(A)_{\mathbb{C}}$ acts on $N \otimes T$ through N, and

$$\mathrm{End}_{\mathrm{Hg}(A)_{\mathbb{C}}}(N) \otimes \mathrm{End}(T) = \mathrm{End}(T).$$

Thus

$$\mathrm{End}_{\mathrm{Hg}(A)_{\mathbb{C}}}(N) = \mathbb{C},$$

and N is an irreducible faithful representation of $\mathrm{Hg}(A)_{\mathbb{C}}$ because $\mathrm{Hg}(A)$ is reductive. The connected center Z of $\mathrm{Hg}(A)$ commutes with $\mathrm{Hg}(A)$, so Z is contained in the commutant D of $\mathrm{Hg}(A)$. Since $Z \subset \mathrm{Hg}(A)$ commutes with D, Z consists of scalars. A scalar preserving ψ is ± 1, so Z is trivial, and $\mathrm{Hg}(A)$ is semisimple.

Let $\mathfrak{h}$ be the Lie algebra of $\mathrm{Hg}(A)_{\mathbb{C}}$. By [Moo99, Theorem (3.11), Corollary (3.13)], $\mathfrak{h}$ acts on N as a tensor product of minuscule representations of the simple factors of $\mathfrak{h}$, and any simple factor of $\mathfrak{h}$ acts non-trivially because N is faithful. By [Moo99, Table 3], the tensor products of minuscule representations with non-trivial factors and $\dim N = 6$ are as follows, where $\mathbb{C}^n$ denotes the standard representation.

(1) $\mathfrak{sl}_6$: $\mathbb{C}^6$ or $(\mathbb{C}^6)^*$.
(2) $\mathfrak{sl}_4$: $\wedge^2\mathbb{C}^4$.
(3) $\mathfrak{sp}_6$: $\mathbb{C}^6$.
(4) $\mathfrak{sl}_2 \times \mathfrak{sl}_3$: $\mathbb{C}^2 \otimes \mathbb{C}^3$ or $\mathbb{C}^2 \otimes (\mathbb{C}^3)^*$.

The list above follows from the classification of minuscule representations: the minuscule representations of type A_r are the $\wedge^k\mathbb{C}^{r+1}$, of dimension $\binom{r+1}{k}$, those of types B_r and D_r other than the standard representation of $D_3 = A_3$ are of dimension a power of 2 or $2r \geq 8$, the minuscule representation of C_r is the standard one of dimension $2r$, those of types E_6 and E_7 are of dimensions 27 and 56, and a tensor product of non-trivial factors with total dimension 6 has two factors of dimensions 2 and 3. Since q is a non-degenerate symmetric $\mathrm{Hg}(A)_{\mathbb{C}}$-invariant form on N, cases (1) and (4) are excluded because they are not self-dual, and case (3) is excluded because any invariant bilinear form on the standard representation of $\mathfrak{sp}_6$ is alternating. Thus $\mathfrak{h} \cong \mathfrak{sl}_4$ acting on $N \cong \wedge^2\mathbb{C}^4$, and $\mathfrak{h} \to \mathfrak{so}(N,q)$ is injective. Since $\dim \mathfrak{sl}_4 = 15 = \dim \mathfrak{so}(N,q)$, the Lie algebras of $\mathrm{Hg}(A)$ and $\mathcal{G}$ coincide. Since $\mathrm{Hg}(A) \subset \mathcal{G}$ and both groups are connected, $\mathrm{Hg}(A) = \mathcal{G}$, and

$$\mathrm{Hg}(A)_{\mathbb{C}} \cong \mathrm{SO}_{6,\mathbb{C}}$$

with N the standard representation. Since $H^1(A,\mathbb{C}) = N^* \otimes T^*$ and $N^* \cong N$ through q, $H^1(A,\mathbb{C})$ is the sum of two copies of the standard representation.

Step 3. In this step, we show that any imaginary quadratic subfield $K \subset D$ has multiplicities $(3,3)$, and we prove the properties of $W_D(A)$.

Any $x \in D \setminus \mathbb{Q}$ generates a quadratic subfield of D, which is imaginary because $D \otimes_{\mathbb{Q}} \mathbb{R}$ is a division algebra. Thus D contains an imaginary quadratic subfield. Let $K \subset D$ be an imaginary quadratic subfield. For $a \in K \setminus \mathbb{Q}$, $\bar{a} = \mathrm{Trd}_{D/\mathbb{Q}}(a) - a$ is the conjugate of a in K, so by the Skolem–Noether theorem there exists $\gamma \in D^*$ with $\gamma a = \bar{a}\gamma$ for any $a \in K$. For $u \in D^*$ and $0 \leq k \leq 12$, we also write $u^* := m^{-k}(mu)^*$ on $H^k(A,\mathbb{Q})$, so that u^* on $H^k(A,\mathbb{Q}) = \wedge^k H^1(A,\mathbb{Q})$ is the k-th exterior power of u^* on $H^1(A,\mathbb{Q})$ [Moo99, (2.1)], and $(uv)^* = v^*u^*$. For $a \in K$,

$a^*\gamma^* = \gamma^*\bar{a}^*$, so γ^* maps $H^1(A,\mathbb{C})_\sigma$ isomorphically onto $H^1(A,\mathbb{C})_{\bar\sigma}$. Since γ^* preserves $H^{1,0}(A)$ and $\dim H^{1,0}(A) = 6$, the multiplicities of K on $H^{1,0}(A)$ are $(3,3)$. In particular, (A,K) is of Weil type, and $W_K(A)$ is defined.

On $H^1(A,\mathbb{C}) = N^*\otimes T^*$, the operator u^* for $u \in D^*$ acts as $\mathrm{id}_{N^*}\otimes {}^t u$, where ${}^t u$ is the transpose of the action of u on T. Since D^* is the complement of the zero locus of $\mathrm{Nrd}_{D/\mathbb{Q}}$ in the affine space D, D^* is Zariski dense in $(D\otimes_{\mathbb{Q}}\mathbb{C})^* = \mathrm{GL}(T)$. Thus the operators ${}^t u$ with $u \in D^*$ form a Zariski dense subset of $\mathrm{GL}(T^*)$. Let

$$\mu : \wedge^6 N^* \otimes \mathrm{Sym}^6 T^* \to \wedge^6(N^*\otimes T^*) = H^6(A,\mathbb{C})$$

be the linear map such that

$$\mu\big((n_1\wedge\cdots\wedge n_6)\otimes \ell^6\big) = (n_1\otimes\ell)\wedge\cdots\wedge(n_6\otimes\ell)$$

for any $n_1,\dots,n_6 \in N^*$ and $\ell \in T^*$. The map μ exists because the right-hand side is alternating multilinear in $n_1,\dots,n_6$ and homogeneous of degree 6 in ℓ. It is equivariant for $\mathrm{GL}(N^*)\times\mathrm{GL}(T^*)$ and non-zero. Since $\mathrm{Sym}^6 T^*$ is an irreducible representation of $\mathrm{GL}(T^*)$ [Ful97, Section 8.2, Theorem 2] and $\dim \wedge^6 N^* = 1$, μ is injective. Let Σ be the image of μ. Then $\dim\Sigma = 7$, Σ is stable and irreducible under $\mathrm{GL}(T^*)$, and $\mathrm{Hg}(A)_{\mathbb{C}}$ acts trivially on Σ because $\mathrm{Hg}(A)_{\mathbb{C}}$ acts on $\wedge^6 N^*$ by the determinant, which is 1.

Since $K\otimes_{\mathbb{Q}}\mathbb{C} \cong \mathbb{C}\times\mathbb{C}$ acts faithfully on T^*, there exist non-zero $\ell_\sigma, \ell_{\bar\sigma} \in T^*$ with $T^* = \mathbb{C}\ell_\sigma\oplus\mathbb{C}\ell_{\bar\sigma}$ such that ${}^t a$ acts on ℓ_τ by $\tau(a)$ for any $a \in K$ and $\tau \in \{\sigma,\bar\sigma\}$. Thus $H^1(A,\mathbb{C})_\tau = N^*\otimes\ell_\tau$, and

$$W_K(A)\otimes_{\mathbb{Q}}\mathbb{C} = \mu(\wedge^6 N^*\otimes \ell_\sigma^6)\oplus\mu(\wedge^6 N^*\otimes\ell_{\bar\sigma}^6) \subset \Sigma.$$

Since u^* preserves Σ for any $u \in D^*$, $W_D(A)\otimes_{\mathbb{Q}}\mathbb{C} \subset \Sigma$. Since $u^*v^* = (vu)^*$, $W_D(A)$ is stable under u^* for any $u \in D^*$. By Zariski density, $W_D(A)\otimes_{\mathbb{Q}}\mathbb{C}$ is stable under $\mathrm{GL}(T^*)$, so $W_D(A)\otimes_{\mathbb{Q}}\mathbb{C} = \Sigma$ by the irreducibility of Σ. Thus $\dim W_D(A) = 7$, and $W_D(A)$ does not depend on K because Σ does not depend on K. If $Y \subset W_D(A)$ is a non-zero subspace stable under u^* for any $u \in D^*$, then $Y\otimes_{\mathbb{Q}}\mathbb{C}$ is stable under $\mathrm{GL}(T^*)$ by Zariski density, so $Y = W_D(A)$. Thus $W_D(A)$ is irreducible under the operators u^* with $u \in D^*$.

Step 4. In this step, we compute $\mathrm{Hdg}^p(A)$ for $0 \le p \le 6$ and show that $\Theta\cdot W_D(A) = 0$.

By Lemma 2.3 (2), $\mathrm{Hdg}^p(A)$ is the subspace of $\mathrm{Hg}(A)$-invariants in $\wedge^{2p}H^1(A,\mathbb{Q})$. A basis of T^* induces an isomorphism $H^1(A,\mathbb{C}) \cong N^*\oplus N^*$ of representations of $\mathrm{Hg}(A)_{\mathbb{C}} = \mathrm{SO}(N,q)$, so

$$\wedge^{2p}H^1(A,\mathbb{C}) \cong \bigoplus_{a+b=2p} \wedge^a N^*\otimes\wedge^b N^*.$$

Through $\mathfrak{so}(N,q)\cong\mathfrak{sl}_4$ and $N\cong\wedge^2\mathbb{C}^4$, $\wedge^2 N$ is the adjoint representation, which is irreducible of dimension 15, and

$$\wedge^3 N \cong \mathrm{Sym}^2\mathbb{C}^4\oplus\mathrm{Sym}^2(\mathbb{C}^4)^*$$

is the sum of two non-isomorphic irreducible representations of dimension 10 [FH91, Lectures 19–20]. Since $N\cong N^*$ through q and $\mathrm{SO}(N,q)$ acts trivially on $\wedge^6 N$, we have $\wedge^a N \cong \wedge^{6-a}N$, and any $\wedge^a N$ is self-dual. Moreover, $\wedge^0 N$ and $\wedge^6 N$ are trivial, and $\wedge^1 N$ and $\wedge^5 N$ are the standard representation. Thus the subspace of invariants in $\wedge^a N^*\otimes\wedge^b N^* \cong \mathrm{Hom}(\wedge^a N,\wedge^b N)$ is of dimension 2 if $a=b=3$, of dimension 1 if $a=b\neq 3$ or $a+b=6$ with $a\neq 3$, and 0 otherwise. Hence $\dim\mathrm{Hdg}^p(A) = 1$ for $p\neq 3$, and $\dim\mathrm{Hdg}^3(A) = 8$. Since Θ is ample, $\Theta^6 \neq 0$, so Θ^p is a non-zero element of $\mathrm{Hdg}^p(A)$, and $\mathrm{Hdg}^p(A) = \mathbb{Q}\Theta^p$ for $p\neq 3$.

Since $\mathrm{Hg}(A)_{\mathbb{C}}$ acts trivially on $\Sigma = W_D(A)\otimes_{\mathbb{Q}}\mathbb{C}$, we have $W_D(A)\subset\mathrm{Hdg}^3(A)$. For $u\in D^*$ and $x,y\in H_1(A,\mathbb{Q})$,

$$\psi(ux,uy) = \psi(x,\bar u uy) = \mathrm{Nrd}_{D/\mathbb{Q}}(u)\psi(x,y),$$

so $u^*\Theta = \mathrm{Nrd}_{D/\mathbb{Q}}(u)\Theta$ and $u^*\Theta^p = \mathrm{Nrd}_{D/\mathbb{Q}}(u)^p\Theta^p$. If $\Theta^3 \in W_D(A)$, then $\mathbb{Q}\Theta^3$ is a non-zero proper subspace of $W_D(A)$ stable under u^* for any $u\in D^*$, which contradicts the irreducibility of $W_D(A)$. Thus $\mathbb{Q}\Theta^3\cap W_D(A) = 0$, and since $1+7=8$, we have

$$\mathrm{Hdg}^3(A) = \mathbb{Q}\Theta^3\oplus W_D(A).$$

The map $W_D(A) \to \mathrm{Hdg}^4(A) = \mathbb{Q}\Theta^4$, $w \to \Theta \cdot w$, has a kernel Y with $\dim Y \geq 6$. Since

$$\Theta \cdot u^* w = \mathrm{Nrd}_{D/\mathbb{Q}}(u)^{-1} u^*(\Theta \cdot w),$$

Y is stable under u^* for any $u \in D^*$, so $Y = W_D(A)$ by the irreducibility of $W_D(A)$. Thus $\Theta \cdot W_D(A) = 0$.

Step 5. In this step, we prove the criterion for $\mathrm{HC}(A)$.

By the Lefschetz $(1,1)$ theorem (cf. [Moo99, (2.10)]), Θ is algebraic, so Θ^p is algebraic for any p, and $\mathrm{HC}^p(A)$ holds for $p \neq 3$ by **Step 4**. For $u \in D^*$ and a positive integer m with $mu \in \mathrm{End}(A)$, $(mu)^*$ maps algebraic classes to algebraic classes, so u^* maps algebraic classes in $W_D(A)$ to algebraic classes. Let $Y \subset W_D(A)$ be the subspace of algebraic classes. If $W_D(A)$ contains a non-zero algebraic class, then Y is non-zero and stable under u^* for any $u \in D^*$, so $Y = W_D(A)$ by the irreducibility of $W_D(A)$. Since Θ^3 is algebraic, $\mathrm{HC}^3(A)$ holds by **Step 4**, and since $\mathrm{HC}^2(A)$ holds, $\mathrm{HC}(A)$ holds (Lemma 2.6). If $\beta \in \mathrm{Hdg}^3(A)$ is an algebraic class not in $\mathbb{Q}\Theta^3$, then $\beta = r\Theta^3 + w$ with $r \in \mathbb{Q}$ and $0 \neq w \in W_D(A)$, and $w = \beta - r\Theta^3$ is algebraic. Conversely, if $\mathrm{HC}(A)$ holds, then any non-zero $w \in W_D(A)$ is algebraic and not in $\mathbb{Q}\Theta^3$. □

Corollary 7.7. *Let A and D be as in Definition-Theorem 7.6. If there exists an imaginary quadratic subfield $K \subset D$ such that any class in $W_K(A)$ is algebraic, then $\mathrm{HC}(A)$ holds.*

Proof. By Definition-Theorem 7.6, K has multiplicities $(3,3)$ on $H^{1,0}(A)$, so (A,K) is of Weil type and $W_K(A)$ is two-dimensional (Definition 2.7). Taking $u = 1$, we have $W_K(A) \subset W_D(A)$, so $W_D(A)$ contains a non-zero algebraic class, and $\mathrm{HC}(A)$ holds by Definition-Theorem 7.6. □

In Proposition 7.8, for a D-basis e_1, e_2, e_3 of $H_1(A,\mathbb{Q})$ we write $x = \sum_l e_l \cdot x_l$ and $y = \sum_l e_l \cdot y_l$ with $x_l, y_l \in D$, and we say that the basis is orthogonal for ψ if $\psi(e_l \cdot D, e_m \cdot D) = 0$ for $l \neq m$.

Proposition 7.8. *Let A and D be as in Definition-Theorem 7.6. Regard $H_1(A,\mathbb{Q})$ as a right D-vector space of dimension 3 through $x \cdot a := \bar{a}x$, and write the Riemann form of a polarization as*

$$\psi(x,y) = \mathrm{Trd}_{D/\mathbb{Q}}\Big(\sum_{l=1}^{3} \bar{x}_l t_l y_l\Big)$$

in a D-basis of $H_1(A,\mathbb{Q})$ orthogonal for ψ, where each t_l is a non-zero pure quaternion and $\bar{x}$ denotes the conjugate of $x \in D$. Let $\lambda := \prod_{l=1}^3 \mathrm{Nrd}_{D/\mathbb{Q}}(t_l)$. Then the class of λ in $\mathbb{Q}^*/(\mathbb{Q}^*)^2$ does not depend on the basis or on the polarization, and the following are equivalent.

(1) There exist an imaginary quadratic subfield $K \subset D$ and a compatible polarization of (A,K) whose Hermitian form has discriminant $[-1]$.
(2) $\mathbb{Q}(\sqrt{-\lambda})$ embeds into D.

If condition (1) holds, then $\mathrm{HC}(A)$ holds. Moreover, there exist simple abelian sixfolds A with $\mathrm{End}^0(A) = (-1,-1)_{\mathbb{Q}}$ and a polarization with $(t_1,t_2,t_3) = (i, i+j, i+2j+3k)$, where $i^2 = j^2 = -1$ and $k = ij$. For such A, $\lambda = 28$, and conditions (1) and (2) fail.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that any two polarizations of A are proportional, and that any polarization is compatible for (A,K) for any imaginary quadratic subfield $K \subset D$.

By Definition-Theorem 7.6 with $p = 1$, $\mathrm{Hdg}^1(A) = \mathbb{Q}\Theta$. By the Lefschetz $(1,1)$ theorem (cf. [Moo99, (2.10)]), $NS(A) \otimes \mathbb{Q} = \mathbb{Q}\Theta$, so any polarization is a positive rational multiple of Θ. By [Moo99, (1.19)], the Rosati involution of any polarization is the canonical involution of D. For $a \in K \setminus \mathbb{Q}$, $\bar{a} = \mathrm{Trd}_{D/\mathbb{Q}}(a) - a$ is the conjugate of a in K, so the canonical involution restricts to the complex conjugation on K, and any polarization is a compatible polarization of (A,K). By Lemma 2.10 (6), condition (1) holds if and only if there exists an imaginary quadratic subfield $K \subset D$ such that the Hermitian form of one fixed polarization of (A,K) has discriminant $[-1]$.

Step 2. In this step, we show that the orthogonal D-basis exists and that the class of λ is well defined.

For $x \in H_1(A,\mathbb{Q})$ and $a,b \in D$, $(x\cdot a)\cdot b = \bar{b}\bar{a}x = x\cdot(ab)$, and the relation $\psi(ux,y) = \psi(x,\bar{u}y)$ for $u \in D$ implies $\psi(x\cdot a,y) = \psi(x,y\cdot\bar{a})$. Since the reduced trace pairing on D is non-degenerate, there exists a unique map $\Phi : H_1(A,\mathbb{Q}) \times H_1(A,\mathbb{Q}) \to D$ such that

$$\mathrm{Trd}_{D/\mathbb{Q}}(\Phi(x,y)a) = \psi(x,y\cdot a) \quad \text{for any } a \in D.$$

Then $\psi = \mathrm{Trd}_{D/\mathbb{Q}} \circ \Phi$, and a direct computation implies

$$\Phi(x\cdot a, y\cdot b) = \bar{a}\Phi(x,y)b, \qquad \Phi(y,x) = -\overline{\Phi(x,y)}$$

for any $a,b \in D$. Since ψ is non-degenerate, Φ is non-degenerate. Since

$$\psi(e_l\cdot a, e_m\cdot b) = \mathrm{Trd}_{D/\mathbb{Q}}(\bar{a}\Phi(e_l,e_m)b)$$

and the reduced trace pairing is non-degenerate, a D-basis e_1,e_2,e_3 is orthogonal for ψ in the sense fixed before Proposition 7.8 if and only if $\Phi(e_l,e_m) = 0$ for $l \neq m$, and in this case

$$\Phi(x,y) = \sum_{l=1}^{3} \bar{x}_l t_l y_l \quad \text{with } t_l = \Phi(e_l,e_l) = -\bar{t}_l,$$

with x_l, y_l the coordinates fixed before Proposition 7.8. If $\Phi(v,v) = 0$ for any v, then $\Phi(x,y) + \Phi(y,x) = 0$, so $\Phi(x,y) = \overline{\Phi(x,y)} \in \mathbb{Q}$ for any x,y. Then $\bar{a}\Phi(x,y) = \Phi(x\cdot a,y) \in \mathbb{Q}$ for any $a \in D$, so $\Phi = 0$, a contradiction. Thus there exists v with $\Phi(v,v) \neq 0$, and $H_1(A,\mathbb{Q})$ is the direct sum of $v\cdot D$ and the right D-subspace $\{y \mid \Phi(v,y) = 0\}$, on which Φ is non-degenerate. By induction on the dimension, an orthogonal D-basis exists.

Let $\Phi(e) \in M_3(D)$ be the matrix $(\Phi(e_l,e_m))_{l,m}$ of a D-basis $e = (e_1,e_2,e_3)$. If $e'_m = \sum_l e_l\cdot s_{lm}$ with $s = (s_{lm}) \in \mathrm{GL}_3(D)$, then $\Phi(e') = \bar{s}^t\Phi(e)s$. Since the involution $s \to \bar{s}^t$ of $M_3(D)$ preserves the reduced norm $\mathrm{Nrd}_{M_3(D)/\mathbb{Q}}$, we have

$$\mathrm{Nrd}_{M_3(D)/\mathbb{Q}}(\Phi(e')) = \mathrm{Nrd}_{M_3(D)/\mathbb{Q}}(s)^2\,\mathrm{Nrd}_{M_3(D)/\mathbb{Q}}(\Phi(e)).$$

For an orthogonal basis,

$$\mathrm{Nrd}_{M_3(D)/\mathbb{Q}}(\Phi(e)) = \prod_l \mathrm{Nrd}_{D/\mathbb{Q}}(t_l) = \lambda.$$

Thus the class of λ in $\mathbb{Q}^*/(\mathbb{Q}^*)^2$ does not depend on the basis. By **Step 1**, another polarization has Riemann form $r\psi$ with $r \in \mathbb{Q}_{>0}$, which replaces t_l with rt_l and λ with $r^6\lambda$. Thus the class of λ does not depend on the polarization.

Step 3. In this step, we compute the discriminant of the Hermitian form of ψ for (A,K), where $K \subset D$ is an imaginary quadratic subfield.

Let $K \subset D$ be an imaginary quadratic subfield. Then $K = \mathbb{Q}(\alpha)$ for a pure quaternion α with $\alpha^2 = -c$, where $c = \mathrm{Nrd}_{D/\mathbb{Q}}(\alpha) > 0$. For pure quaternions y, $\alpha y + y\alpha = (\alpha+y)^2 - \alpha^2 - y^2 \in \mathbb{Q}$, and $y \to \alpha y + y\alpha$ is a non-zero linear form on the three-dimensional space of pure quaternions because $\alpha\cdot\alpha + \alpha\cdot\alpha = -2c$. Thus there exists a non-zero pure quaternion β with $\alpha\beta = -\beta\alpha$, and $\beta^2 = -c'$ with $c' = \mathrm{Nrd}_{D/\mathbb{Q}}(\beta) > 0$. Then $D = K \oplus \beta K$, $\beta a = \bar{a}\beta$ for any $a \in K$, and the pure quaternions are the elements $x_1\alpha + x_2\beta + x_3\alpha\beta$ with $x_1,x_2,x_3 \in \mathbb{Q}$.

Let $t = x_1\alpha + x_2\beta + x_3\alpha\beta$ be a non-zero pure quaternion, and let $\psi_t(y,z) := \mathrm{Trd}_{D/\mathbb{Q}}(\bar{y}tz)$ for $y,z \in D$. Regard D as a right K-vector space with basis $1,\beta$, and let

$$h_t(y,z) := \frac{\psi_t(y\alpha,z) - \alpha\psi_t(y,z)}{2c}.$$

Then h_t is a K-Hermitian form, conjugate-linear in the first variable, and $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha h_t) = \psi_t$. From $\psi_t(1,1) = \psi_t(\beta,\beta) = 0$, $\psi_t(\alpha,1) = 2cx_1$, $\psi_t(1,\beta) = -2c'x_2$, $\psi_t(\alpha,\beta) = -2cc'x_3$, and $\psi_t(\beta\alpha,\beta) = -2cc'x_1$, which follow from $\alpha^2 = -c$, $\beta^2 = -c'$, and $\alpha\beta = -\beta\alpha$, the Gram matrix of h_t in the basis $1,\beta$ is

$$\begin{pmatrix} x_1 & -c'x_3 + \frac{c'x_2}{c}\alpha \\ -c'x_3 - \frac{c'x_2}{c}\alpha & -c'x_1 \end{pmatrix}.$$

Since

$$\mathrm{Nrd}_{D/\mathbb{Q}}(t) = -t^2 = cx_1^2 + c'x_2^2 + cc'x_3^2,$$

we have

$$\det h_t = -c'x_1^2 - c'^2x_3^2 - \frac{c'^2}{c}x_2^2 = -\frac{c'}{c}\,\mathrm{Nrd}_{D/\mathbb{Q}}(t).$$

Let e_1, e_2, e_3 be a D-basis of $H_1(A,\mathbb{Q})$ orthogonal for ψ. Then $e_l \cdot 1, e_l \cdot \beta$ for $1 \le l \le 3$ form a basis of $H_1(A,\mathbb{Q})$ for the K-structure $x \cdot a$, the K-Hermitian form h with $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha h) = \psi$ is the orthogonal sum of the forms h_{t_l}, and

$$\det h = -\Big(\frac{c'}{c}\Big)^3\lambda.$$

The K-structure of (A,K) is $ax = x \cdot \bar{a}$, the conjugate structure. Let $\bar{h}(x,y) := \overline{h(x,y)}$. For the K-structure ax, $\bar{h}$ is the form of Definition 2.9 with $-\alpha$ in place of δ, because $\bar{h}$ is K-Hermitian, conjugate-linear in the first variable, and $\mathrm{Tr}_{K/\mathbb{Q}}(-\alpha\bar{h}) = \psi$. Moreover, $\det \bar{h} = \det h$ because $\det h \in \mathbb{Q}$. By Lemma 2.10 (6), replacing δ with $-\alpha$ does not change the discriminant, so the discriminant of the Hermitian form of ψ for (A,K) is the class of $-(c'/c)^3\lambda$ in $\mathbb{Q}^*/\,\mathrm{Nm}_{K/\mathbb{Q}}(K^*)$. Since $c = \mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$ and rational squares are norms, the discriminant is $[-c'\lambda]$.

Step 4. In this step, we show that conditions (1) and (2) are equivalent, and that condition (1) implies HC(A).

By **Steps 1 and 3**, condition (1) holds if and only if there exists an imaginary quadratic subfield $K \subset D$, written as in **Step 3**, such that $c'\lambda \in \mathrm{Nm}_{K/\mathbb{Q}}(K^*)$. Suppose that $c'\lambda = \mathrm{Nm}_{K/\mathbb{Q}}(z)$ with $z \in K^*$, and let $\beta' := z\beta/c'$. Since $\beta z = \bar{z}\beta$,

$$\beta'^2 = \frac{z\bar{z}\beta^2}{c'^2} = -\frac{c'\lambda}{c'} = -\lambda,$$

so $\mathbb{Q}(\beta') \cong \mathbb{Q}(\sqrt{-\lambda})$ embeds into D. Conversely, suppose that $\gamma \in D$ satisfies $\gamma^2 = -\lambda$. Since $\lambda > 0$, $\gamma \notin \mathbb{Q}$, and since

$$\mathrm{Trd}_{D/\mathbb{Q}}(\gamma)\gamma = \gamma^2 + \mathrm{Nrd}_{D/\mathbb{Q}}(\gamma) \in \mathbb{Q},$$

γ is a pure quaternion. As in **Step 3**, there exists a non-zero pure quaternion α with $\alpha\gamma = -\gamma\alpha$. Let $K := \mathbb{Q}(\alpha)$. Taking $\beta := \gamma$ and $c' := \lambda$ in **Step 3**, the discriminant for (A,K) is $[-\lambda^2] = [-1]$. Thus conditions (1) and (2) are equivalent.

If condition (1) holds for K, then any class in $W_K(A)$ is algebraic by Theorem 2.14, so HC(A) holds by Corollary 7.7.

Step 5. In this step, we construct a simple abelian sixfold A with $\mathrm{End}^0(A) = (-1,-1)_{\mathbb{Q}}$ and a polarization with $(t_1,t_2,t_3) = (i, i+j, i+2j+3k)$.

Let $D := (-1,-1)_{\mathbb{Q}}$, which is totally definite, and let $n_l := \mathrm{Nrd}_{D/\mathbb{Q}}(t_l)$. On the right D-vector space D^3, let

$$\psi(x,y) := \mathrm{Trd}_{D/\mathbb{Q}}\Big(\sum_{l=1}^{3}\bar{x}_l t_l y_l\Big),$$

and let $\rho : D \to \mathrm{End}_{\mathbb{Q}}(D^3)$ be the homomorphism with $\rho(u)x = (x_l\bar{u})_l$. Then ψ is alternating and non-degenerate, $\rho(\bar{a})x = x \cdot a$ is the right multiplication, and $\psi(\rho(u)x,y) = \psi(x,\rho(\bar{u})y)$ for any $u \in D$. On $D^3 \otimes_{\mathbb{Q}} \mathbb{R}$, let J be the endomorphism with $(Jx)_l = s_l x_l$, where $s_l := -t_l/\sqrt{n_l}$. Since $s_l^2 = -1$ and $t_l s_l = \sqrt{n_l}$, $J^2 = -1$, J commutes with $\rho(D)$, J preserves ψ, and

$$\psi(x,Jy) = \sum_{l=1}^{3}\sqrt{n_l}\,\mathrm{Trd}_{D/\mathbb{Q}}(\bar{x}_l y_l)$$

is symmetric and positive definite.

Let $\mathcal{G}$ be the identity component of the group of $\rho(D)$-linear automorphisms of D^3 preserving ψ. Since $\rho(D) \otimes_{\mathbb{Q}} \mathbb{C} \cong M_2(\mathbb{C})$ and $\psi(\rho(u)x,y) = \psi(x,\rho(\bar{u})y)$ for any $u \in D$, Schur's lemma implies that $D^3 \otimes_{\mathbb{Q}} \mathbb{C} = N \otimes T$ with $\dim N = 6$ and $\dim T = 2$, that $\rho(D)$ acts through T, and

that $\psi = q \otimes \varpi$ for a non-degenerate symmetric form q on N and a non-zero alternating form ϖ on T. Thus $\mathcal{G}_{\mathbb{C}} = \mathrm{SO}(N, q)$, the Lie algebra of $\mathcal{G}_{\mathbb{C}}$ is simple, the center of $\mathcal{G}$ is finite, and the commutant of $\mathcal{G}$ in $\mathrm{End}_{\mathbb{Q}}(D^3)$ is $\rho(D)$ because the commutant of $\mathrm{SO}(N, q)$ in $\mathrm{End}(N \otimes T)$ is $\mathrm{End}(T)$. For $g \in \mathcal{G}(\mathbb{R})^0$, let $J_g := gJg^{-1}$. Then $J_g^2 = -1$, J_g commutes with $\rho(D)$, and $(x, y) \to \psi(x, J_g y)$ is symmetric and positive definite. Let $\Lambda \subset D^3$ be a lattice with $\psi(\Lambda, \Lambda) \subset \mathbb{Z}$. By [Moo99, Theorem (2.2)], (Λ, J_g, ψ) defines a polarized abelian sixfold A_g with $H_1(A_g, \mathbb{Q}) = D^3$ and Riemann form ψ, and $\mathrm{End}^0(A_g)$ is the algebra of $\mathbb{Q}$-linear endomorphisms of D^3 commuting with J_g [Moo99, (2.3)]. In particular, $\rho(D) \subset \mathrm{End}^0(A_g)$.

For $f \in \mathrm{End}_{\mathbb{Q}}(D^3) \setminus \rho(D)$, let $Z_f := \{g \in \mathcal{G}(\mathbb{R})^0 \mid fJ_g = J_g f\}$, which is a closed real analytic subset of the connected real analytic manifold $\mathcal{G}(\mathbb{R})^0$. For $g \in \mathcal{G}(\mathbb{R})^0$, let

$$C_g := \{\cos(\theta) + \sin(\theta) J_g \mid \theta \in \mathbb{R}\} \subset \mathcal{G}(\mathbb{R})^0,$$

and let $\mathcal{S} \subset \mathcal{G}_{\mathbb{R}}$ be the Zariski closure of the subgroup generated by $\bigcup_g C_g$. Then $\mathcal{S}$ is a connected normal subgroup of $\mathcal{G}_{\mathbb{R}}$, and $\dim \mathcal{S} > 0$ because C_1 is infinite and the center of $\mathcal{G}$ is finite. Since the Lie algebra of $\mathcal{G}_{\mathbb{C}}$ is simple, $\mathcal{S} = \mathcal{G}_{\mathbb{R}}$. Suppose that $Z_f = \mathcal{G}(\mathbb{R})^0$. Then f commutes with C_g for any g, hence with $\mathcal{S} = \mathcal{G}_{\mathbb{R}}$, so $f \in \rho(D)$, a contradiction. Thus Z_f is a proper closed real analytic subset of $\mathcal{G}(\mathbb{R})^0$, and Z_f has empty interior. Since $\mathrm{End}_{\mathbb{Q}}(D^3)$ is countable, the Baire category theorem implies that there exists $g \in \mathcal{G}(\mathbb{R})^0$ not contained in $\bigcup_f Z_f$. Let $A := A_g$. Then $\mathrm{End}^0(A) = \rho(D) \cong D$. Since D is a division algebra, $\mathrm{End}^0(A)$ has no non-trivial idempotent, so A is simple [Moo99, (2.3)]. Since D is a totally definite quaternion algebra over $\mathbb{Q}$, $\mathrm{End}^0(A)$ is of Albert type III [Moo99, (1.19)]. The right D-structure $x \cdot a = \rho(\bar{a})x$ of the statement is the right multiplication on D^3, the standard basis of D^3 is orthogonal for ψ, and $(t_1, t_2, t_3) = (i, i + j, i + 2j + 3k)$.

Step 6. In this step, we show that $\lambda = 28$ for any A as in **Step 5**, and that conditions (1) and (2) fail.

Since

$$\mathrm{Nrd}_{D/\mathbb{Q}}(x_1 i + x_2 j + x_3 k) = x_1^2 + x_2^2 + x_3^2,$$

we have $\mathrm{Nrd}_{D/\mathbb{Q}}(t_1) = 1$, $\mathrm{Nrd}_{D/\mathbb{Q}}(t_2) = 2$, and $\mathrm{Nrd}_{D/\mathbb{Q}}(t_3) = 14$, so $\lambda = 28$ and $\mathbb{Q}(\sqrt{-\lambda}) = \mathbb{Q}(\sqrt{-7})$. If $D \otimes_{\mathbb{Q}} \mathbb{Q}_2$ were not a division algebra, then $x_0^2 + x_1^2 + x_2^2 + x_3^2 = 0$ for some non-zero $(x_0, x_1, x_2, x_3) \in \mathbb{Q}_2^4$, and since the non-zero sums of two squares form a subgroup of $\mathbb{Q}_2^*$ by the identity $(a^2+b^2)(a'^2+b'^2) = (aa'-bb')^2+(ab'+ba')^2$, -1 would be a sum of two squares a^2+b^2 in $\mathbb{Q}_2$. This is impossible: if $a, b \in \mathbb{Z}_2$, then $a^2+b^2 \not\equiv 3 \pmod 4$, and if $\min\{v_2(a), v_2(b)\} = -m < 0$, then $(2^m a)^2 + (2^m b)^2 = -4^m \equiv 0 \pmod 4$, whereas a sum of two squares of 2-adic integers, at least one of which is a unit, is congruent to 1 or 2 modulo 4. Thus $D \otimes_{\mathbb{Q}} \mathbb{Q}_2$ is a division algebra, and D is ramified at 2. Since $-7 \equiv 1 \pmod 8$, -7 is a square in $\mathbb{Q}_2$ by Hensel's lemma, so 2 splits in $\mathbb{Q}(\sqrt{-7})$. By [Voi21, Proposition 14.6.7], $\mathbb{Q}(\sqrt{-7})$ does not embed into D, so condition (2) fails, and condition (1) fails by **Step 4**. □

Remark 7.9. The sixfolds A of Proposition 7.8 with $\mathrm{End}^0(A) = (-1, -1)_{\mathbb{Q}}$ satisfy $\mathrm{HC}(A)$ by Theorem 1.2. Indeed, since i, j, k pairwise anticommute and $i^2 = j^2 = k^2 = -1$, we have $(i+j+k)^2 = -3$, so $(-1, -1)_{\mathbb{Q}}$ contains the subfields $\mathbb{Q}(i) \cong \mathbb{Q}(\sqrt{-1})$ and $\mathbb{Q}(i+j+k) \cong \mathbb{Q}(\sqrt{-3})$.

8. Nonsimple sixfolds

In this section, we prove Theorem 1.4, which treats the nonsimple case in the proof of Theorem 1.2. We prove Theorem 1.4 at the end of Subsection 8.6.

8.1. Weil planes on nonsimple sixfolds of Weil type.

In this subsection, we prove that the Weil plane of any nonsimple abelian sixfold of Weil type is algebraic.

By Lemma 2.10 (6), rescaling a compatible polarization by $q \in \mathbb{Q}_{>0}$ does not change the discriminant, and Lemma 8.1 rescales one K-stable factor of odd dimension instead.

Lemma 8.1. *Let A be an abelian sixfold, K an imaginary quadratic field, and $\iota : K \hookrightarrow \mathrm{End}^0(A)$ a unital embedding of multiplicities $(3,3)$. Assume that A contains a K-stable abelian subvariety A' of odd dimension $d' < 6$. Then (A, ι) admits a compatible polarization whose Hermitian form has discriminant $[-1]$.*

More precisely, let $f : A' \times A'' \to A$ be a K-linear isogeny with $A'' \subset A$ a K-stable abelian subvariety, let Θ' and Θ'' be compatible polarizations of $(A', \iota|_{A'})$ and $(A'', \iota|_{A''})$, let h_1 be the Hermitian form of the compatible polarization $\mathrm{pr}_1^*\,\Theta' + \mathrm{pr}_2^*\,\Theta''$ of $A' \times A''$, and compute determinants in a K-basis of $H_1(A', \mathbb{Q}) \oplus H_1(A'', \mathbb{Q})$ compatible with the direct sum. Then $\det h_1 < 0$, and the Hermitian form of

$$\mathrm{pr}_1^*(q\Theta') + \mathrm{pr}_2^*\,\Theta''$$

with $q := -1/\det h_1$ has determinant $-(\det h_1)^{1-d'}$, whose class is $[-1]$. In particular, any class in $W_K(A)$ is algebraic.

Proof. We first prove the second paragraph of the lemma. Let ψ' and ψ'' be the Riemann forms, and h' and h'' the Hermitian forms, of Θ' and Θ''. The Riemann form of $\mathrm{pr}_1^*\,\Theta' + \mathrm{pr}_2^*\,\Theta''$ is $\psi' \oplus \psi''$, and the Hermitian form depends $\mathbb{Q}$-linearly on the Riemann form (Definition 2.9). Hence $h_1 = h' \oplus h''$, and for any $q \in \mathbb{Q}_{>0}$, the Hermitian form of

$$\mathrm{pr}_1^*(q\Theta') + \mathrm{pr}_2^*\,\Theta''$$

is $qh' \oplus h''$.

Since f is a K-linear isogeny,

$$f^* : H^1(A, \mathbb{Q}) \to H^1(A' \times A'', \mathbb{Q})$$

is a K-linear isomorphism of Hodge structures, so the multiplicities of K on $H^{1,0}(A' \times A'')$ are $(3,3)$. By Lemma 2.10 (2), h_1 has signature $(3,3)$. Let M be the matrix of h_1 in the given K-basis. Then $M^T = \bar{M}$, so $\det h_1 = \overline{\det h_1} \in \mathbb{Q}$. Moreover, $\sigma(M)$ is a complex Hermitian matrix with three positive and three negative eigenvalues. Thus $\det h_1 < 0$, and $q := -1/\det h_1$ is positive. Since $\dim_K H_1(A', \mathbb{Q}) = d'$ and d' is odd, we have

$$\begin{aligned}\det(qh' \oplus h'') &= q^{d'} \det h' \det h'' = q^{d'} \det h_1 \\ &= (-1)^{d'} (\det h_1)^{1-d'} = -(\det h_1)^{1-d'}.\end{aligned}$$

Since $1 - d'$ is even,

$$(\det h_1)^{1-d'} = \mathrm{Nm}_{K/\mathbb{Q}}\left((\det h_1)^{(1-d')/2}\right).$$

Hence the class of $-(\det h_1)^{1-d'}$ in $\mathbb{Q}^*/\mathrm{Nm}_{K/\mathbb{Q}}(K^*)$ is $[-1]$.

Now we prove the first paragraph of the lemma. By Lemma 2.10 (1), (A, ι) admits a compatible polarization Θ with Riemann form ψ. By Lemma 2.10 (3), there exists a K-stable abelian subvariety $A'' \subset A$ such that the addition map $f : A' \times A'' \to A$ is a K-linear isogeny and

$$f^*\Theta = \mathrm{pr}_1^*\,\Theta' + \mathrm{pr}_2^*\,\Theta'',$$

where $\Theta' := \Theta|_{A'}$ and $\Theta'' := \Theta|_{A''}$. The class Θ' is a compatible polarization of $(A', \iota|_{A'})$, since the Riemann form of Θ' is the restriction of ψ to the K-stable subspace $H_1(A', \mathbb{Q})$, on which $\psi(kx, y) = \psi(x, \bar{k}y)$ still holds. Likewise, Θ'' is a compatible polarization of $(A'', \iota|_{A''})$. Let h_1 be the Hermitian form of $\mathrm{pr}_1^*\,\Theta' + \mathrm{pr}_2^*\,\Theta''$, let $q := -1/\det h_1$, and let

$$\Theta_1 := \mathrm{pr}_1^*(q\Theta') + \mathrm{pr}_2^*\,\Theta''.$$

Since $q > 0$, Θ_1 is the class of an ample $\mathbb{Q}$-divisor, and the Riemann form $\psi_1 := q\psi' \oplus \psi''$ of Θ_1 satisfies $\psi_1(kx, y) = \psi_1(x, \bar{k}y)$. Thus Θ_1 is a compatible polarization of $A' \times A''$, and the Hermitian form h_2 of Θ_1 has discriminant $[-1]$ by the second paragraph of the lemma.

Since f is an isogeny, there exist a positive integer m and an isogeny $g : A \to A' \times A''$ such that $f \circ g = m \cdot \mathrm{id}_A$. The isogeny g is K-linear, since $g = mf^{-1}$ in $\mathrm{Hom}(A, A' \times A'') \otimes_{\mathbb{Z}} \mathbb{Q}$ and f is K-linear. By Lemma 2.10 (4), $g^*\Theta_1$ is a compatible polarization of (A, ι), and g_* is an isometry between the Hermitian form h_3 of $g^*\Theta_1$ and h_2. For a K-basis (e_i) of $H_1(A, \mathbb{Q})$, the determinant

of h_3 in (e_i) is equal to the determinant of h_2 in the K-basis (g_*e_i). A change of K-basis with matrix P multiplies the determinant of h_2 by $\mathrm{Nm}_{K/\mathbb{Q}}(\det P)$. Hence $\mathrm{disc}(h_3) = \mathrm{disc}(h_2) = [-1]$. Since (A, ι) is of Weil type, Theorem 2.14 implies that any class in $W_K(A)$ is algebraic. □

Theorem 8.2. *Let A be an abelian sixfold, K an imaginary quadratic field, and $\iota : K \hookrightarrow \mathrm{End}^0(A)$ a unital embedding of multiplicities $(3,3)$. If A is not simple, then any class in $W_K(A)$ is algebraic.*

Proof. The proof is long, so we divide it into several steps.

Step 1. We show that any class in $W_K(A)$ is algebraic if A contains a K-stable abelian subvariety A' with $0 < \dim A' < 6$.

By Lemma 2.10 (1) and (3), there exist a K-stable abelian subvariety $A'' \subset A$ and a K-linear isogeny $f : A' \times A'' \to A$. Since f^* is a K-linear isomorphism of Hodge structures, the sum of the multiplicities of K on $H^{1,0}(A')$ and on $H^{1,0}(A'')$ is $(3,3)$. If $\dim A'$ is odd, then Lemma 8.1 implies the claim. Thus we may assume that $\dim A'$ is even. Since $\dim A'' = 6 - \dim A'$ and the roles of A' and A'' are symmetric, possibly exchanging A' and A'', we may assume that $\dim A' = 2$ and $\dim A'' = 4$. Let $(a, 2-a)$ be the multiplicities of K on $H^{1,0}(A')$.

First assume that $a \in \{0, 2\}$. Then

$$\{H^1(A', \mathbb{C})_\sigma, H^1(A', \mathbb{C})_{\bar\sigma}\} = \{H^{1,0}(A'), H^{0,1}(A')\},$$

so for any $z \in \mathbb{U}$, $\eta_{A'}(z)$ acts on $H^1(A', \mathbb{C})_\sigma$ and on $H^1(A', \mathbb{C})_{\bar\sigma}$ by scalars. Hence any K-linear endomorphism φ of $H^1(A', \mathbb{Q})$ commutes with $\eta_{A'}(\mathbb{U})$. The elements of $\mathrm{GL}(H^1(A', \mathbb{Q}))$ commuting with φ form an algebraic subgroup defined over $\mathbb{Q}$, so φ commutes with $\mathrm{Hg}(A')$ (Definition 2.2), and φ is the image of an element of $\mathrm{End}^0(A')$ by Lemma 2.3 (1). Choose a K-linear idempotent ϵ of the two-dimensional K-vector space $H^1(A', \mathbb{Q})$ with $\dim_K \epsilon(H^1(A', \mathbb{Q})) = 1$, and denote the corresponding element of $\mathrm{End}^0(A')$ also by ϵ. Since

$$\mathrm{End}^0(A') \to \mathrm{End}_{\mathbb{Q}}(H^1(A', \mathbb{Q}))$$

is injective, ϵ commutes with $\iota|_{A'}(K)$. Choose a positive integer m with $m\epsilon \in \mathrm{End}(A')$ and let $E' := m\epsilon(A')$. Then $\dim E' = 1$, and $H_1(E', \mathbb{Q}) = \epsilon_* H_1(A', \mathbb{Q})$ is stable under $\iota|_{A'}(K)$. Since f is K-linear, $f(E' \times 0) \subset A$ is a K-stable elliptic curve. Thus Lemma 8.1 with $d' = 1$ implies the claim.

Next assume that $a = 1$. Then $(A', \iota|_{A'})$ and $(A'', \iota|_{A''})$ are of Weil type, of multiplicities $(1,1)$ and $(2,2)$, respectively. By Lemma 2.10 (2), $W_K(A') \subset \mathrm{Hdg}^1(A')$ and $W_K(A'') \subset \mathrm{Hdg}^2(A'')$, so any class in $W_K(A')$ or in $W_K(A'')$ is algebraic by Theorem 2.16. Since

$$H^1(A' \times A'', \mathbb{C})_\sigma = \mathrm{pr}_1^* H^1(A', \mathbb{C})_\sigma \oplus \mathrm{pr}_2^* H^1(A'', \mathbb{C})_\sigma,$$

where the summands are of dimensions 2 and 4, we have

$$\wedge^6 H^1(A' \times A'', \mathbb{C})_\sigma = \mathrm{pr}_1^* \wedge^2 H^1(A', \mathbb{C})_\sigma \cup \mathrm{pr}_2^* \wedge^4 H^1(A'', \mathbb{C})_\sigma,$$

and similarly for $\bar\sigma$. Let $W_0 \subset H^6(A' \times A'', \mathbb{Q})$ be the $\mathbb{Q}$-span of the classes $\mathrm{pr}_1^* w' \cup \mathrm{pr}_2^* w''$ with $w' \in W_K(A')$ and $w'' \in W_K(A'')$. Then $W_K(A' \times A'')_{\mathbb{C}} \subset W_{0,\mathbb{C}}$. Since $W_K(A' \times A'')$ and W_0 are $\mathbb{Q}$-subspaces of $H^6(A' \times A'', \mathbb{Q})$, we have $W_K(A' \times A'') \subset W_0$. Any class $\mathrm{pr}_1^* w' \cup \mathrm{pr}_2^* w''$ with $w' \in W_K(A')$ and $w'' \in W_K(A'')$ is the class of an exterior product of algebraic cycles, so any class in $W_K(A' \times A'')$ is algebraic. By Lemma 2.10 (4), f^* maps $W_K(A)$ isomorphically onto $W_K(A' \times A'')$, and $w \in W_K(A)$ is algebraic if and only if f^*w is algebraic. This proves the claim.

Step 2. We prove the theorem.

By [Moo99, (2.3)], there exist pairwise non-isogenous simple abelian varieties $Y_1, \dots, Y_l$ and positive integers $r_1, \dots, r_l$ such that A is isogenous to $\prod_{i=1}^l Y_i^{r_i}$ and

$$\mathrm{End}^0(A) \cong \prod_{i=1}^{l} M_{r_i}(\mathrm{End}^0(Y_i)).$$

Let $e_1 \in \mathrm{End}^0(A)$ be the central idempotent corresponding to the factor $M_{r_1}(\mathrm{End}^0(Y_1))$.

First assume that $l \geq 2$. Then $e_1 \neq 0, 1$. Choose a positive integer m with $me_1 \in \mathrm{End}(A)$ and let $A_1 := me_1(A)$. Then $H_1(A_1, \mathbb{Q}) = e_{1*}H_1(A, \mathbb{Q})$, so $0 < \dim A_1 < 6$. Since e_1 is central, e_1 commutes with $\iota(K)$, so A_1 is K-stable. Thus **Step 1** implies that any class in $W_K(A)$ is algebraic.

Next assume that $l = 1$. Since A is not simple, $r_1 \geq 2$, so $\dim Y_1 = 6/r_1 \leq 3$. By [MZ99, Theorem (0.1)(iv)], the Hodge algebra of $Y_1^{r_1}$ is equal to the divisor algebra of $Y_1^{r_1}$, so $\mathrm{HC}(Y_1^{r_1})$ holds. By Lemma 2.6, $\mathrm{HC}(A)$ holds. Since $W_K(A) \subset \mathrm{Hdg}^3(A)$ (Lemma 2.10 (2)), any class in $W_K(A)$ is algebraic. □

8.2. **A simple fivefold times an elliptic curve.** In this subsection, we prove the Hodge conjecture for any abelian sixfold isogenous to the product of a simple abelian fivefold and an elliptic curve.

Lemma 8.3. *Let K be an imaginary quadratic field, B a simple abelian fivefold with $\mathrm{End}^0(B) = K$ and multiplicities $(3, 2)$ on $H^{1,0}(B)$, and E an elliptic curve with a K-action of multiplicities $(0, 1)$. Let $A := B \times E$ with the diagonal K-action, $V := H^1(B, \mathbb{Q})$, ψ the alternating form on V induced by a polarization of B, and Θ_B, Θ_E ample classes on B, E, pulled back to A. Then $\mathrm{Hg}(B) = \mathrm{U}_K(V, \psi)$ and*

$$\mathrm{Hg}(A) = \{(g, t) \in \mathrm{U}_K(V, \psi) \times \mathrm{U}_K \mid t = \det_K(g)^{-1}\}.$$

For $p \neq 3$,

$$\mathrm{Hdg}^p(A) = \mathbb{Q}\Theta_B^p + \mathbb{Q}\Theta_B^{p-1}\Theta_E,$$

with the convention $\Theta_B^{-1} := 0$, and

$$\mathrm{Hdg}^3(A) = \mathbb{Q}\Theta_B^3 \oplus \mathbb{Q}\Theta_B^2\Theta_E \oplus W_K(A).$$

Proof. We first compute the Hodge groups. Since the Rosati involution of the polarization of B is positive, it restricts to complex conjugation on $\mathrm{End}^0(B) = K$, so $\psi(kx, y) = \psi(x, \bar{k}y)$ for any $k \in K$ and $x, y \in V$. Since $\dim B = 5$ is prime, [MZ99, Theorem (2.7)] shows that $\mathrm{Hg}(B)$ is the centralizer of $\mathrm{End}^0(B)$ in the symplectic group of ψ, that is, $\mathrm{Hg}(B) = \mathrm{U}_K(V, \psi)$. Since K acts on E, $\mathrm{End}^0(E) = K$, and $\mathrm{Hg}(E) = \mathrm{U}_K$ [Moo99, (2.6)], acting on the one-dimensional K-vector space $H^1(E, \mathbb{Q})$ by scalars.

Let $V_\sigma := H^1(B, \mathbb{C})_\sigma$ and $V_{\bar{\sigma}} := H^1(B, \mathbb{C})_{\bar{\sigma}}$. For any $k \in K$ and $x, y \in V_\sigma$, we have

$$\sigma(k)\psi(x, y) = \psi(kx, y) = \psi(x, \bar{k}y) = \bar{\sigma}(k)\psi(x, y),$$

so ψ vanishes on $V_\sigma \times V_\sigma$. Likewise, ψ vanishes on $V_{\bar{\sigma}} \times V_{\bar{\sigma}}$, so ψ restricts to a perfect pairing between V_σ and $V_{\bar{\sigma}}$. Hence $g \to g|_{V_\sigma}$ is an isomorphism

$$\mathrm{U}_K(V, \psi)_{\mathbb{C}} \to \mathrm{GL}(V_\sigma) \cong \mathrm{GL}_5,$$

under which V_σ and $V_{\bar{\sigma}}$ are the standard representation $\mathbb{C}^5$ and the dual representation $(\mathbb{C}^5)^*$, and $\sigma \circ \det_K$ is the determinant. Similarly, $t \to \sigma(t)$ is an isomorphism $\mathrm{U}_{K,\mathbb{C}} \to \mathbb{G}_m$, and $t \in \mathrm{U}_K$ acts on $H^1(E, \mathbb{C})_\sigma$ by $\sigma(t)$ and on $H^1(E, \mathbb{C})_{\bar{\sigma}}$ by $\sigma(t)^{-1}$.

Let

$$H := \{(g, t) \in \mathrm{U}_K(V, \psi) \times \mathrm{U}_K \mid t = \det_K(g)^{-1}\}.$$

For $z \in \mathbb{U}$, $\eta_B(z)$ acts on V_σ by z on the three-dimensional subspace $H^{1,0}(B) \cap V_\sigma$ and by z^{-1} on the two-dimensional subspace $H^{0,1}(B) \cap V_\sigma$, so $\sigma(\det_K(\eta_B(z))) = z^3 z^{-2} = z$. Since the multiplicity of K on $H^{1,0}(E)$ at σ is 0, $H^1(E, \mathbb{C})_\sigma = H^{0,1}(E)$, so $\eta_E(z)$ acts on $H^1(E, \mathbb{C})_\sigma$ by z^{-1}. Hence $\eta_A(z) = (\eta_B(z), \eta_E(z)) \in H(\mathbb{R})$. By Definition 2.2 and Lemma 2.3 (3), $\mathrm{Hg}(A) \subset H$. The projection $H \to \mathrm{U}_K(V, \psi)$ is an isomorphism with inverse $g \to (g, \det_K(g)^{-1})$, and the projection

$$\mathrm{Hg}(A) \to \mathrm{Hg}(B) = \mathrm{U}_K(V, \psi)$$

is surjective by Lemma 2.3 (3). Thus $\mathrm{Hg}(A) = H$.

Now we compute the Hodge classes. By Lemma 2.3 (2), $\mathrm{Hdg}^p(A)$ is the subspace of $\mathrm{Hg}(A)$-invariants in $\wedge^{2p}H^1(A, \mathbb{Q})$. The invariants of a representation of an algebraic group over $\mathbb{Q}$ are

the solutions of linear equations over $\mathbb{Q}$, so $\mathrm{Hdg}^p(A)_{\mathbb{C}}$ is the subspace of $\mathrm{Hg}(A)_{\mathbb{C}}$-invariants in $\wedge^{2p}H^1(A,\mathbb{C})$. Under the isomorphism

$$\mathrm{Hg}(A)_{\mathbb{C}} \cong \mathrm{U}_K(V,\psi)_{\mathbb{C}} \cong \mathrm{GL}_5$$

above, we have

$$H^1(A,\mathbb{C}) = V_\sigma \oplus V_{\bar\sigma} \oplus H^1(E,\mathbb{C})_\sigma \oplus H^1(E,\mathbb{C})_{\bar\sigma} \cong \mathbb{C}^5 \oplus (\mathbb{C}^5)^* \oplus \det{}^{-1} \oplus \det.$$

Hence $\wedge^{2p}H^1(A,\mathbb{C})$ is the direct sum of the representations

$$\begin{aligned} M_{a,b,c,d} &:= \wedge^a V_\sigma \otimes \wedge^b V_{\bar\sigma} \otimes \wedge^c H^1(E,\mathbb{C})_\sigma \otimes \wedge^d H^1(E,\mathbb{C})_{\bar\sigma} \\ &\cong \wedge^a\mathbb{C}^5 \otimes \wedge^b(\mathbb{C}^5)^* \otimes \det{}^{d-c} \end{aligned}$$

over $0 \le a,b \le 5$ and $c,d \in \{0,1\}$ with $a+b+c+d = 2p$. A scalar matrix $\lambda I_5 \in \mathrm{GL}_5$ acts on $M_{a,b,c,d}$ by $\lambda^{a-b+5(d-c)}$. Thus $M_{a,b,c,d}$ contains a non-zero invariant only if $a-b+5(d-c)=0$. Since $|a-b| \le 5$ and $|d-c| \le 1$, the equality $a-b+5(d-c)=0$ holds exactly in the following cases:

(1) $a=b$ and $c=d$;
(2) $(a,b,c,d) = (5,0,1,0)$;
(3) $(a,b,c,d) = (0,5,0,1)$.

In case (1),

$$M_{a,a,c,c} \cong \wedge^a\mathbb{C}^5 \otimes (\wedge^a\mathbb{C}^5)^*$$

is the space of endomorphisms of $\wedge^a\mathbb{C}^5$, and the invariants are the GL_5-equivariant endomorphisms. Since $\wedge^a\mathbb{C}^5$ is irreducible [Ful97, Section 8.2, Theorem 2], the invariants of $M_{a,a,c,c}$ form a line by Schur's lemma. Since Θ_B is ample, $\Theta_B^5 \neq 0$, so $\Theta_B^a \neq 0$ for $0 \le a \le 5$, and $\Theta_B^a\Theta_E^c \neq 0$ by the Künneth formula. The class

$$\Theta_B^a\Theta_E^c \in H^{2a}(B,\mathbb{Q}) \otimes H^{2c}(E,\mathbb{Q})$$

is a Hodge class, so $\Theta_B^a\Theta_E^c \otimes 1$ is contained in the sum of the invariants of $M_{a',b',c',d'}$ with $a'+b' = 2a$ and $c'+d' = 2c$. By the list above, the only such summand with non-zero invariants is $M_{a,a,c,c}$. Hence the invariant line of $M_{a,a,c,c}$ is spanned by $\Theta_B^a\Theta_E^c$.

In cases (2) and (3), $\dim M_{a,b,c,d} = 1$ and $M_{a,b,c,d}$ is a trivial representation. Since

$$H^1(A,\mathbb{C})_\sigma = V_\sigma \oplus H^1(E,\mathbb{C})_\sigma$$

is six-dimensional, $M_{5,0,1,0} = \wedge^6 H^1(A,\mathbb{C})_\sigma$, and likewise $M_{0,5,0,1} = \wedge^6 H^1(A,\mathbb{C})_{\bar\sigma}$. The multiplicities of K on $H^{1,0}(A)$ are $(3,3)$, so A is of Weil type and $M_{5,0,1,0} \oplus M_{0,5,0,1} = W_K(A)_{\mathbb{C}}$.

Hence, with $\Theta_B^{-1} := 0$ and $\Theta_B^6 = 0$, the space $\mathrm{Hdg}^p(A)_{\mathbb{C}}$ is spanned by Θ_B^p and $\Theta_B^{p-1}\Theta_E$ if $p \neq 3$, and by Θ_B^3, $\Theta_B^2\Theta_E$, and $W_K(A)_{\mathbb{C}}$ if $p=3$. Since Θ_B^p, $\Theta_B^{p-1}\Theta_E$ are Hodge classes and $W_K(A) \subset \mathrm{Hdg}^3(A)$ (Lemma 2.10 (2)), $\mathrm{Hdg}^p(A)$ is spanned by Θ_B^p and $\Theta_B^{p-1}\Theta_E$ if $p \neq 3$, and by Θ_B^3, $\Theta_B^2\Theta_E$, and $W_K(A)$ if $p=3$. For $p=3$, the sum is direct, since Θ_B^3, $\Theta_B^2\Theta_E$, and $W_K(A)$ are contained in the Künneth components $H^6(B,\mathbb{Q})$, $H^4(B,\mathbb{Q}) \otimes H^2(E,\mathbb{Q})$, and $H^5(B,\mathbb{Q}) \otimes H^1(E,\mathbb{Q})$, respectively. □

Proposition 8.4. *Let B be a simple abelian fivefold and E an elliptic curve. Then* $\mathrm{HC}(A)$ *holds for any abelian sixfold A isogenous to $B \times E$. More precisely, $F := \mathrm{End}^0(B)$ is $\mathbb{Q}$, a totally real field of degree 5, an imaginary quadratic field, or a CM field of degree 10, and $\mathrm{Hdg}^p(B\times E) = \mathcal{D}^p(B\times E)$ for any p, except in the following case: E has complex multiplication by an imaginary quadratic field K which embeds into F, and the multiplicities of K on $H^{1,0}(B)$ are 3 and 2 in some order. In this case, let K act on E so that the multiplicity of K on $H^{1,0}(E)$ is 0 at the embedding at which the multiplicity on $H^{1,0}(B)$ is 3. Then $B\times E$ is of Weil type, $\mathrm{Hdg}^p(B\times E) = \mathcal{D}^p(B\times E)$ for $p \neq 3$,*

$$\mathrm{Hdg}^3(B\times E) = \mathcal{D}^3(B\times E) \oplus W_K(B\times E),$$

and any class in $W_K(B\times E)$ is algebraic.

Proof. The proof is long, so we divide it into several steps.

Step 1. We show that $\mathrm{Hdg}^p(B\times E)=\mathcal{D}^p(B\times E)$ for any p unless E has complex multiplication by an imaginary quadratic field K which embeds into F.

Since $\dim B=5$ is prime, F is $\mathbb{Q}$, a totally real field of degree 5, an imaginary quadratic field, or a CM field of degree 10 [Moo99, (2.16) and the paragraph after it]. Since B is simple and $\dim B>1$, $\mathrm{Hom}(E,B)=0$. By [MZ99, Proposition (3.8)], either

$$\mathrm{Hg}(B\times E)=\mathrm{Hg}(B)\times\mathrm{Hg}(E),$$

or $\mathrm{End}^0(E)$ is an imaginary quadratic field which embeds into F. Assume that

$$\mathrm{Hg}(B\times E)=\mathrm{Hg}(B)\times\mathrm{Hg}(E).$$

By the Künneth formula,

$$H^*(B\times E,\mathbb{Q})=H^*(B,\mathbb{Q})\otimes H^*(E,\mathbb{Q}),$$

and the invariants of $\mathrm{Hg}(B)\times\mathrm{Hg}(E)$ in $H^*(B,\mathbb{Q})\otimes H^*(E,\mathbb{Q})$ are the tensor product of the $\mathrm{Hg}(B)$-invariants in $H^*(B,\mathbb{Q})$ and the $\mathrm{Hg}(E)$-invariants in $H^*(E,\mathbb{Q})$. The Hodge circle acts on $H^{i,j}$ by z^{i-j}, so $\mathrm{Hg}(B)$ and $\mathrm{Hg}(E)$ have no non-zero invariants in odd degrees. Thus, by Lemma 2.3 (2), the Hodge algebra of $B\times E$ is the tensor product of the Hodge algebras of B and of E. By [MZ99, Theorem (0.2)(iv)], the Hodge algebra of B is the divisor algebra of B, and the Hodge algebra of E is spanned by 1 and Θ_E, where Θ_E is an ample class on E. Hence $\mathrm{Hdg}^p(B\times E)=\mathcal{D}^p(B\times E)$ for any p.

In the rest of the proof, we assume that $\mathrm{End}^0(E)=K$ is an imaginary quadratic field and that K embeds into F. In particular, F is an imaginary quadratic field or a CM field of degree 10. We let K act on B through an embedding $K\hookrightarrow F$.

Step 2. We show that the multiplicities of K on $H^{1,0}(B)$ are 4 and 1, or 3 and 2, in some order, and we bound $\mathrm{Hg}(B\times E)$.

Let $V:=H^1(B,\mathbb{Q})$. If the multiplicities of K on $H^{1,0}(B)$ were $(5,0)$ or $(0,5)$, then

$$\{H^1(B,\mathbb{C})_\sigma,H^1(B,\mathbb{C})_{\bar\sigma}\}=\{H^{1,0}(B),H^{0,1}(B)\},$$

so any K-linear endomorphism of V would commute with $\eta_B(\mathbb{U})$, hence with $\mathrm{Hg}(B)$. By Lemma 2.3 (1), $\mathrm{End}_K(V)\cong M_5(K)$ would be contained in the image of the commutative algebra F, which is impossible. The statement of the proposition does not distinguish σ from $\bar\sigma$. Thus, possibly exchanging σ and $\bar\sigma$, we may assume that the multiplicities of K on $H^{1,0}(B)$ are $(r_\sigma,r_{\bar\sigma})\in\{(4,1),(3,2)\}$. Let $r:=r_\sigma-r_{\bar\sigma}\in\{3,1\}$. We let K act on E so that the multiplicities of K on $H^{1,0}(E)$ are $(0,1)$.

Since the Rosati involution of a polarization of B is positive, it is complex conjugation on F [Moo99, (1.19)], so the alternating form ψ on V induced by a polarization of B satisfies $\psi(ax,y)=\psi(x,\bar a y)$ for any $a\in F$. By [MZ99, Theorem (2.7)], $\mathrm{Hg}(B)$ is the centralizer of F in the symplectic group of ψ, that is, $\mathrm{Hg}(B)=\mathrm{U}_F(V,\psi)$. In particular, $\mathrm{Hg}(B)\subset\mathrm{U}_K(V,\psi)$. By [Moo99, (2.6)], $\mathrm{Hg}(E)=\mathrm{U}_K$. Let

$$H_r:=\{(g,t)\in\mathrm{Hg}(B)\times\mathrm{U}_K\mid \det_K(g)t^r=1\}.$$

For $z\in\mathbb{U}$, $\eta_B(z)$ acts on $H^1(B,\mathbb{C})_\sigma$ by z on a subspace of dimension r_σ and by z^{-1} on a subspace of dimension $r_{\bar\sigma}$, so $\sigma(\det_K(\eta_B(z)))=z^r$. Since $H^1(E,\mathbb{C})_\sigma=H^{0,1}(E)$, $\eta_E(z)$ acts on $H^1(E,\mathbb{C})_\sigma$ by z^{-1}. Thus $\eta_{B\times E}(z)\in H_r(\mathbb{R})$, and $\mathrm{Hg}(B\times E)\subset H_r$ by Definition 2.2 and Lemma 2.3 (3).

Let $V_\sigma:=H^1(B,\mathbb{C})_\sigma$ and $V_{\bar\sigma}:=H^1(B,\mathbb{C})_{\bar\sigma}$, and identify $\mathrm{U}_{K,\mathbb{C}}$ with $\mathbb{G}_m$ through σ, so that t acts on $H^1(E,\mathbb{C})_\sigma$ by t and on $H^1(E,\mathbb{C})_{\bar\sigma}$ by t^{-1}.

Step 3. We prove the proposition when $F=K$, except for the algebraicity of $W_K(B\times E)$.

If $r = 1$, then the multiplicities of K on $H^{1,0}(B)$ and on $H^{1,0}(E)$ are $(3,2)$ and $(0,1)$, and Lemma 8.3 implies that

$$\mathrm{Hdg}^p(B \times E) = \mathbb{Q}\Theta_B^p + \mathbb{Q}\Theta_B^{p-1}\Theta_E \quad \text{for } p \neq 3,$$
$$\mathrm{Hdg}^3(B \times E) = \mathbb{Q}\Theta_B^3 \oplus \mathbb{Q}\Theta_B^2\Theta_E \oplus W_K(B \times E).$$

For $p = 1$, this implies that $\mathcal{D}^1(B \times E) = \mathbb{Q}\Theta_B + \mathbb{Q}\Theta_E$. Since $\Theta_E^2 = 0$,

$$\mathcal{D}^p(B \times E) = \mathbb{Q}\Theta_B^p + \mathbb{Q}\Theta_B^{p-1}\Theta_E$$

for any p, and Proposition 8.4 follows in this case.

Assume that $r = 3$. Then $\mathrm{Hg}(B) = \mathrm{U}_K(V, \psi)$. For any $k \in K$ and $x, y \in V_\sigma$, we have

$$\sigma(k)\psi(x,y) = \psi(kx,y) = \psi(x,\bar{k}y) = \bar{\sigma}(k)\psi(x,y),$$

so ψ vanishes on $V_\sigma \times V_\sigma$, and likewise on $V_{\bar{\sigma}} \times V_{\bar{\sigma}}$. Hence $g \to g|_{V_\sigma}$ is an isomorphism

$$\mathrm{Hg}(B)_{\mathbb{C}} \to \mathrm{GL}(V_\sigma) \cong \mathrm{GL}_5,$$

under which $V_\sigma \cong \mathbb{C}^5$, $V_{\bar{\sigma}} \cong (\mathbb{C}^5)^*$, and $\sigma \circ \det_K = \det$. Thus

$$H_{3,\mathbb{C}} = \{(g,t) \in \mathrm{GL}_5 \times \mathbb{G}_m \mid \det(g)t^3 = 1\}.$$

The morphism

$$\mathrm{SL}_5 \times \mathbb{G}_m \to H_{3,\mathbb{C}}, \quad (s,z) \to (z^{-3}s, z^5)$$

is well defined, since $\det(z^{-3}s)z^{15} = 1$, and surjective: for $(g,t) \in H_{3,\mathbb{C}}$, choose z with $z^5 = t$ and let $s := z^3 g$, so that $\det s = z^{15} \det g = t^3 \det g = 1$ and $(z^{-3}s, z^5) = (g,t)$. Hence $H_{3,\mathbb{C}}$ is connected. Since $H_{3,\mathbb{C}}$ is the kernel of the surjective character $(g,t) \to \det(g)t^3$ of $\mathrm{GL}_5 \times \mathbb{G}_m$, $\dim H_3 = 25$. By Lemma 2.3 (1) and (3), $\mathrm{Hg}(B \times E)$ is connected and surjects onto $\mathrm{Hg}(B)$, and $\dim \mathrm{Hg}(B) = 25$. Thus $\mathrm{Hg}(B \times E) = H_3$.

Through the surjection

$$\mathrm{SL}_5 \times \mathbb{G}_m \to H_{3,\mathbb{C}},$$

the factor $\mathbb{G}_m$ acts on V_σ, $V_{\bar{\sigma}}$, $H^1(E,\mathbb{C})_\sigma$, and $H^1(E,\mathbb{C})_{\bar{\sigma}}$ with weights -3, 3, 5, and -5, respectively. The space $\wedge^{2p}H^1(B \times E, \mathbb{C})$ is the direct sum of the representations

$$M_{a,b,c,d} := \wedge^a V_\sigma \otimes \wedge^b V_{\bar{\sigma}} \otimes \wedge^c H^1(E,\mathbb{C})_\sigma \otimes \wedge^d H^1(E,\mathbb{C})_{\bar{\sigma}}$$

over $0 \le a, b \le 5$ and $c, d \in \{0,1\}$ with $a + b + c + d = 2p$, and $\mathbb{G}_m$ acts on $M_{a,b,c,d}$ with weight $-3a + 3b + 5c - 5d$. The SL_5-invariants of $\wedge^a\mathbb{C}^5 \otimes \wedge^b(\mathbb{C}^5)^*$ are the SL_5-equivariant maps $\wedge^b\mathbb{C}^5 \to \wedge^a\mathbb{C}^5$. By [Ful97, Section 8.2, Theorem 2 and the paragraph on the special linear group after it], $\wedge^a\mathbb{C}^5$ and $\wedge^b\mathbb{C}^5$ are irreducible SL_5-representations, and they are isomorphic if and only if $a = b$ or $\{a,b\} = \{0,5\}$. If $a = b$, then the weight is $5(c-d)$, so a non-zero invariant exists only if $c = d$. If $(a,b) = (5,0)$ or $(0,5)$, then the weight vanishes only if $c - d = 3$ or $c - d = -3$, which is impossible since $c, d \in \{0,1\}$. Thus the invariants are contained in $\bigoplus_{a,c} M_{a,a,c,c}$. By Schur's lemma, the invariants of $M_{a,a,c,c}$ form a line, and the non-zero Hodge class $\Theta_B^a\Theta_E^c$ spans the invariant line of $M_{a,a,c,c}$. Hence

$$\mathrm{Hdg}^p(B \times E) = \mathbb{Q}\Theta_B^p + \mathbb{Q}\Theta_B^{p-1}\Theta_E = \mathcal{D}^p(B \times E)$$

for any p, and Proposition 8.4 follows in this case.

Step 4. We prove the proposition when F is a CM field of degree 10, except for the algebraicity of $W_K(B \times E)$.

Since $\dim_{\mathbb{Q}} V = 10 = [F : \mathbb{Q}]$, V is a one-dimensional F-vector space. Thus an F-linear automorphism of V is the multiplication by some $g \in F^*$, and it preserves ψ if and only if $g\bar{g} = 1$. Hence $\mathrm{Hg}(B) = \mathrm{U}_F$, and for $g \in \mathrm{U}_F$, $\det_K(g) = \mathrm{Nm}_{F/K}(g)$, which is contained in U_K since

$$\mathrm{Nm}_{F/K}(g)\overline{\mathrm{Nm}_{F/K}(g)} = \mathrm{Nm}_{F/K}(g\bar{g}) = 1.$$

Thus

$$H_r = \{(g,t) \in \mathrm{U}_F \times \mathrm{U}_K \mid \mathrm{Nm}_{F/K}(g)t^r = 1\}.$$

Let $\tau_1, \dots, \tau_5$ be the complex embeddings of F extending σ. Then $\bar{\tau}_1, \dots, \bar{\tau}_5$ are the complex embeddings of F extending $\bar{\sigma}$. The map $g \to (\tau_1(g), \dots, \tau_5(g))$ is an isomorphism $\mathrm{U}_{F,\mathbb{C}} \to \mathbb{G}_m^5$, and $\bar{\tau}_i(g) = \tau_i(g)^{-1}$ on U_F. Write $x_i := \tau_i(g)$. Then $\sigma(\mathrm{Nm}_{F/K}(g)) = x_1 \cdots x_5$, so

$$H_{r,\mathbb{C}} = \{(x_1, \dots, x_5, t) \in \mathbb{G}_m^6 \mid x_1x_2x_3x_4x_5t^r = 1\}.$$

The map $(x_1, \dots, x_5, t) \to (x_1, \dots, x_4, t)$ is an isomorphism $H_{r,\mathbb{C}} \to \mathbb{G}_m^5$, with inverse given by $x_5 = (x_1x_2x_3x_4t^r)^{-1}$. In particular, H_r is connected and $\dim H_r = 5$. By Lemma 2.3 (1) and (3), $\mathrm{Hg}(B \times E)$ is connected and surjects onto $\mathrm{Hg}(B) = \mathrm{U}_F$, and $\dim \mathrm{U}_F = 5$. Thus $\mathrm{Hg}(B \times E) = H_r$.

Let $L_i := H^1(B, \mathbb{C})_{\tau_i}$ and $L_i' := H^1(B, \mathbb{C})_{\bar{\tau}_i}$ for $1 \le i \le 5$, and let $L_0 := H^1(E, \mathbb{C})_\sigma$ and $L_0' := H^1(E, \mathbb{C})_{\bar{\sigma}}$. Then L_i, L_i', L_0, and L_0' are lines on which $H_{r,\mathbb{C}}$ acts by the characters x_i, x_i^{-1}, t, and t^{-1}, respectively, and

$$H^1(B \times E, \mathbb{C}) = \bigoplus_{i=0}^{5} (L_i \oplus L_i').$$

For $a_i, b_i, c, d \in \{0, 1\}$, the wedge product of the lines L_i with $a_i = 1$, L_i' with $b_i = 1$, L_0 if $c = 1$, and L_0' if $d = 1$ spans a line on which $\mathbb{G}_m^6$ acts by the character $\prod_{i=1}^5 x_i^{k_i} \cdot t^f$, where $k_i := a_i - b_i$ and $f := c - d$. Such lines form a basis of $\wedge^* H^1(B \times E, \mathbb{C})$, so the invariants of $H_{r,\mathbb{C}}$ are spanned by the lines whose characters are trivial on $H_{r,\mathbb{C}}$. By the isomorphism $H_{r,\mathbb{C}} \cong \mathbb{G}_m^5$ above, the restriction of $\prod_i x_i^{k_i} t^f$ to $H_{r,\mathbb{C}}$ is

$$\prod_{i=1}^{4} x_i^{k_i - k_5} \cdot t^{f - rk_5},$$

which is trivial if and only if $k_1 = \cdots = k_5$ and $f = rk_1$.

If $k_1 = 0$, then $a_i = b_i$ for any i and $c = d$, so the line is a wedge product of some of the lines $L_i \wedge L_i'$ with $0 \le i \le 5$. The lines $L_i \wedge L_i'$ are invariant lines in degree 2, so they are contained in $\mathrm{Hdg}^1(B \times E)_\mathbb{C}$, which is equal to $\mathcal{D}^1(B \times E)_\mathbb{C}$ by the Lefschetz $(1,1)$ theorem [Moo99, (2.10)]. Hence the invariants with $k_1 = 0$ in degree $2p$ are contained in $\mathcal{D}^p(B \times E)_\mathbb{C}$. If $k_1 = \pm 1$, then $|f| = r$, so $r = 1$. For $r = 1$ and $k_1 = 1$, $a_i = 1$, $b_i = 0$, $c = 1$, and $d = 0$, so the line is

$$\wedge^6(L_0 \oplus L_1 \oplus \cdots \oplus L_5) = \wedge^6 H^1(B \times E, \mathbb{C})_\sigma.$$

For $r = 1$ and $k_1 = -1$, the line is $\wedge^6 H^1(B \times E, \mathbb{C})_{\bar{\sigma}}$.

If $r = 3$, it follows that

$$\mathrm{Hdg}^p(B \times E)_\mathbb{C} \subset \mathcal{D}^p(B \times E)_\mathbb{C},$$

so $\mathrm{Hdg}^p(B \times E) = \mathcal{D}^p(B \times E)$ for any p. If $r = 1$, then the multiplicities of K on $H^{1,0}(B \times E)$ are $(3, 3)$, so $B \times E$ is of Weil type, and $\mathrm{Hdg}^p(B \times E)_\mathbb{C}$ is contained in $\mathcal{D}^p(B \times E)_\mathbb{C}$ for $p \neq 3$ and in

$$\mathcal{D}^3(B \times E)_\mathbb{C} + W_K(B \times E)_\mathbb{C}$$

for $p = 3$. Since

$$\mathcal{D}^p(B \times E) \subset \mathrm{Hdg}^p(B \times E)$$

and $W_K(B \times E) \subset \mathrm{Hdg}^3(B \times E)$ (Lemma 2.10 (2)), the inclusions are equalities over $\mathbb{Q}$. The space $\mathcal{D}^1(B \times E)_\mathbb{C}$ is spanned by the lines $L_i \wedge L_i'$ with $0 \le i \le 5$, so

$$\mathcal{D}^3(B \times E) \subset H^6(B, \mathbb{Q}) \oplus H^4(B, \mathbb{Q}) \otimes H^2(E, \mathbb{Q}),$$

while

$$W_K(B \times E) \subset H^5(B, \mathbb{Q}) \otimes H^1(E, \mathbb{Q}).$$

Thus the sum $\mathcal{D}^3(B \times E) + W_K(B \times E)$ is direct, and Proposition 8.4 follows in this case.

Step 5. We show that any class in $W_K(B \times E)$ is algebraic in the exceptional case, and we conclude the proof.

By **Steps 1–4**, it remains to consider the case where E has complex multiplication by K, K embeds into F, and $r = 1$. With the K-action on E in the statement, the multiplicities of K on

$H^{1,0}(B \times E)$ are $(3,2) + (0,1) = (3,3)$. Since $B \times E$ is not simple, $W_K(B \times E)$ is algebraic by Theorem 8.2.

By **Steps 1–4**, any Hodge class on $B \times E$ is the sum of an element of $\mathcal{D}^*(B \times E)$ and, in the exceptional case, of an element of $W_K(B \times E)$. Products of classes of divisors are algebraic, so $\mathrm{HC}(B \times E)$ holds. By Lemma 2.6, $\mathrm{HC}(A)$ holds for any abelian sixfold A isogenous to $B \times E$. □

8.3. **A product criterion for Hodge groups.** In this subsection, we describe the character groups of the norm-one tori of CM fields, and we prove a criterion for a connected subgroup of the product of a reductive group and several tori to be equal to the product.

Lemmas 8.5 and 8.6 below are used to compute the Hodge groups of products of an abelian variety with abelian varieties of CM type, such as elliptic curves with complex multiplication by pairwise non-isomorphic imaginary quadratic fields or simple abelian surfaces of CM type. For a torus T over $\mathbb{Q}$, we write $X^*(T)$ and $X_*(T)$ for the character and cocharacter groups of $T_{\overline{\mathbb{Q}}}$, with their actions of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. They are dual to each other, and we write $X^*(T)_{\mathbb{Q}} := X^*(T) \otimes \mathbb{Q}$ and $X_*(T)_{\mathbb{Q}} := X_*(T) \otimes \mathbb{Q}$. For a connected reductive group H over $\mathbb{Q}$, we denote by H^{der} the derived group of H, so that H/H^{der} is a torus.

Lemma 8.5. *Let L be a CM field, fix $\overline{\mathbb{Q}} \subset \mathbb{C}$, and for a complex embedding τ of L denote by $[\tau]$ the character of $\mathrm{U}_{L,\mathbb{C}}$ induced by τ. Then $X^*(\mathrm{U}_L)$ is generated by the characters $[\tau]$ subject to the relations $[\bar{\tau}] = -[\tau]$, and $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ maps $[\tau]$ to $[\gamma \circ \tau]$. An element $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts trivially on $X^*(\mathrm{U}_L)_{\mathbb{Q}}$ if and only if γ is trivial on the normal closure $\widetilde{L}$ of L in $\overline{\mathbb{Q}}$, so the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ in $\mathrm{GL}(X^*(\mathrm{U}_L)_{\mathbb{Q}})$ is isomorphic to $\mathrm{Gal}(\widetilde{L}/\mathbb{Q})$. If L is an imaginary quadratic field, then $X^*(\mathrm{U}_L)_{\mathbb{Q}}$ is one-dimensional, and γ acts on it by -1 if and only if γ acts non-trivially on L. If L is a quartic CM field which contains no imaginary quadratic subfield, then either L is Galois over $\mathbb{Q}$ with cyclic Galois group of order 4, or $[\widetilde{L} : \mathbb{Q}] = 8$ and $\mathrm{Gal}(\widetilde{L}/\mathbb{Q})$ acts on the four complex embeddings of L as the group of all permutations which commute with $\tau \to \bar{\tau}$, a dihedral group of order 8. In both cases, $X^*(\mathrm{U}_L)_{\mathbb{Q}}$ is an irreducible representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ of dimension 2.*

Proof. We have

$$X^*(\mathrm{Res}_{L/\mathbb{Q}}\, \mathbb{G}_m) = \mathbb{Z}[\mathrm{Hom}(L, \overline{\mathbb{Q}})],$$

on which $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts by composition. Since U_L is the kernel of Nm_{L/L^+}, $X^*(\mathrm{U}_L)$ is the quotient of $\mathbb{Z}[\mathrm{Hom}(L, \overline{\mathbb{Q}})]$ by the subgroup generated by the elements $[\tau]+[\bar{\tau}]$ (cf. [Moo99, (1.23)]). Hence $X^*(\mathrm{U}_L)$ is generated by the characters $[\tau]$ subject to the relations $[\bar{\tau}] = -[\tau]$, and γ maps $[\tau]$ to $[\gamma \circ \tau]$. Let Φ be a CM type of L. Then the characters $[\phi]$ with $\phi \in \Phi$ form a basis of $X^*(\mathrm{U}_L)$, and any $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on the basis $\{[\phi]\}_{\phi \in \Phi}$ by a signed permutation. Since $\bar{\tau}$ is the composite of τ with the complex conjugation of L, we have $\gamma \circ \bar{\tau} = \overline{\gamma \circ \tau}$ for any τ. Since $[\bar{\phi}] = -[\phi] \neq [\phi]$ in $X^*(\mathrm{U}_L)_{\mathbb{Q}}$, γ acts trivially on $X^*(\mathrm{U}_L)_{\mathbb{Q}}$ if and only if $\gamma \circ \phi = \phi$ for any $\phi \in \Phi$, that is, $\gamma \circ \tau = \tau$ for any complex embedding τ of L. Since $\widetilde{L}$ is generated by the fields $\tau(L)$, this holds if and only if γ is trivial on $\widetilde{L}$. Hence the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ in $\mathrm{GL}(X^*(\mathrm{U}_L)_{\mathbb{Q}})$ is the quotient of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ by $\mathrm{Gal}(\overline{\mathbb{Q}}/\widetilde{L})$, which is isomorphic to $\mathrm{Gal}(\widetilde{L}/\mathbb{Q})$.

If L is an imaginary quadratic field with complex embeddings $\sigma, \bar{\sigma}$, then $X^*(\mathrm{U}_L)_{\mathbb{Q}}$ is spanned by $[\sigma]$, and γ maps $[\sigma]$ to $[\sigma]$ if γ acts trivially on L, and to $[\bar{\sigma}] = -[\sigma]$ otherwise.

Assume that L is a quartic CM field which contains no imaginary quadratic subfield. Then $\dim_{\mathbb{Q}} X^*(\mathrm{U}_L)_{\mathbb{Q}} = 2$. By [MZ99, (4.1)], either L is Galois over $\mathbb{Q}$ with cyclic Galois group of order 4, or $[\widetilde{L} : \mathbb{Q}] = 8$. The group $\mathrm{Gal}(\widetilde{L}/\mathbb{Q})$ acts faithfully on the four complex embeddings of L, commuting with $\tau \to \bar{\tau}$. The permutations of a set of four elements which commute with a fixed-point-free involution form a dihedral group of order 8. Hence, if $[\widetilde{L} : \mathbb{Q}] = 8$, then $\mathrm{Gal}(\widetilde{L}/\mathbb{Q})$ is the group of all permutations of the four complex embeddings of L which commute with $\tau \to \bar{\tau}$, since both groups have order 8. If L is Galois over $\mathbb{Q}$ with cyclic Galois group, let $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ restrict to a generator of $\mathrm{Gal}(\widetilde{L}/\mathbb{Q})$. Since complex conjugation induces the unique element of order 2 of $\mathrm{Gal}(\widetilde{L}/\mathbb{Q})$, $\gamma^2 \circ \tau = \bar{\tau}$ for any τ. If $[\widetilde{L} : \mathbb{Q}] = 8$, fix a complex embedding τ of L and a

complex embedding $\tau' \notin \{\tau, \bar{\tau}\}$ of L. Then the 4-cycle

$$\tau \to \tau' \to \bar{\tau} \to \bar{\tau}' \to \tau$$

commutes with $\tau \to \bar{\tau}$, so some $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on the complex embeddings of L by this 4-cycle, and γ^2 maps any τ'' to $\bar{\tau}''$. In both cases, γ^2 acts on $X^*(\mathrm{U}_L)_{\mathbb{Q}}$ by -1. If $\ell \subset X^*(\mathrm{U}_L)_{\mathbb{Q}}$ were a Galois-stable line, γ would act on ℓ by a rational number λ with $\lambda^2 = -1$, which is impossible. Hence $X^*(\mathrm{U}_L)_{\mathbb{Q}}$ is irreducible. □

Lemma 8.6. *Let H be a connected reductive group over $\mathbb{Q}$, let $T_1, \dots, T_k$ be tori over $\mathbb{Q}$, and let $\Gamma \subset H \times \prod_{j=1}^k T_j$ be a connected algebraic subgroup defined over $\mathbb{Q}$ whose projection to each factor is surjective. Assume that the representations $X^*(T_1)_{\mathbb{Q}}, \dots, X^*(T_k)_{\mathbb{Q}}$ of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ are irreducible and pairwise non-isomorphic, and that for no j is $X^*(T_j)_{\mathbb{Q}}$ isomorphic to a subrepresentation of $X^*(H/H^{\mathrm{der}})_{\mathbb{Q}}$. Then $\Gamma = H \times \prod_{j=1}^k T_j$. In particular, if $K_1, \dots, K_k$ are pairwise non-isomorphic imaginary quadratic fields, $T_j = \mathrm{U}_{K_j}$ for any j, and for no j is $X^*(\mathrm{U}_{K_j})_{\mathbb{Q}}$ isomorphic to a subrepresentation of $X^*(H/H^{\mathrm{der}})_{\mathbb{Q}}$, then $\Gamma = H \times \prod_{j=1}^k \mathrm{U}_{K_j}$.*

Proof. Let $G := H \times \prod_{j=1}^k T_j$. Since tori are commutative, $G^{\mathrm{der}} = H^{\mathrm{der}} \times 1$, so $\Gamma^{\mathrm{der}} \subset H^{\mathrm{der}} \times 1$. Since the projection $\Gamma \to H$ is surjective and a surjective homomorphism of algebraic groups maps the derived group onto the derived group of the image, the projection maps Γ^{der} onto H^{der}. Since the projection is injective on $H^{\mathrm{der}} \times 1$, we have $\Gamma^{\mathrm{der}} = H^{\mathrm{der}} \times 1$. In particular, $H^{\mathrm{der}} \times 1 \subset \Gamma$.

Let $S := G/G^{\mathrm{der}} = H/H^{\mathrm{der}} \times \prod_{j=1}^k T_j$ and let $\pi : G \to S$ be the quotient map. Since $\ker \pi = H^{\mathrm{der}} \times 1 \subset \Gamma$, we have $\Gamma = \pi^{-1}(\pi(\Gamma))$. The image $\pi(\Gamma)$ is a connected closed subgroup of the torus S, so $\pi(\Gamma)$ is a subtorus $S_\Gamma \subset S$, and the projection of S_Γ to each factor H/H^{der} and T_j is surjective. Thus it suffices to show that $S_\Gamma = S$.

Let $M := X^*(S)_{\mathbb{Q}}$, $M_0 := X^*(H/H^{\mathrm{der}})_{\mathbb{Q}}$, and $M_j := X^*(T_j)_{\mathbb{Q}}$ for $1 \le j \le k$, so that $M = M_0 \oplus \bigoplus_{j=1}^k M_j$. By the anti-equivalence between tori over $\mathbb{Q}$ and their character groups [Moo99, (1.1)], the restriction $X^*(S) \to X^*(S_\Gamma)$ is surjective, and $X^*(S_\Gamma)_{\mathbb{Q}} = M/R$ for a $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-subrepresentation $R \subset M$. Moreover, $S_\Gamma = S$ if and only if $R = 0$, and the projection of S_Γ to the factor with character module M_i is surjective if and only if $M_i \to M/R$ is injective, that is, $R \cap M_i = 0$. Hence $R \cap M_i = 0$ for $0 \le i \le k$.

The action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on $X^*(S)$ factors through a finite quotient, so any rational representation contained in M is completely reducible, and R is the direct sum of the intersections of R with the isotypic components of M. By assumption, $M_1, \dots, M_k$ are irreducible and pairwise non-isomorphic, and M_0 contains no subrepresentation isomorphic to any M_j. Hence the M_j-isotypic component of M is M_j, and the M_j-isotypic component of R is contained in $R \cap M_j = 0$. Thus R is contained in the sum of the isotypic components of M of types different from $M_1, \dots, M_k$, which is M_0. Since $R \cap M_0 = 0$, we have $R = 0$, so $S_\Gamma = S$ and $\Gamma = G$.

Assume now that $K_1, \dots, K_k$ are pairwise non-isomorphic imaginary quadratic fields and $T_j = \mathrm{U}_{K_j}$ for any j. By Lemma 8.5, $X^*(\mathrm{U}_{K_j})_{\mathbb{Q}}$ is one-dimensional, and the kernel of the action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on it is $\mathrm{Gal}(\overline{\mathbb{Q}}/K_j)$. Since $K_1, \dots, K_k$ are pairwise non-isomorphic, the subgroups $\mathrm{Gal}(\overline{\mathbb{Q}}/K_j)$ of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ are pairwise distinct. Hence the representations

$$X^*(\mathrm{U}_{K_1})_{\mathbb{Q}}, \dots, X^*(\mathrm{U}_{K_k})_{\mathbb{Q}}$$

are irreducible and pairwise non-isomorphic, and we have proved above that $\Gamma = H \times \prod_{j=1}^k \mathrm{U}_{K_j}$. □

8.4. **Products with a simple threefold.** In this subsection, we prove $\mathrm{HC}(A)$ for any abelian sixfold A which contains a simple abelian threefold. We start with the invariants of the kernel of the K-determinant on the cohomology of a simple threefold.

Lemma 8.7. *Let X be a simple abelian threefold, K an imaginary quadratic field with a unital embedding $K \hookrightarrow \mathrm{End}^0(X)$ of multiplicities $(2, 1)$, $\chi : \mathrm{Hg}(X) \to \mathrm{U}_K$ the K-determinant on*

$H^1(X,\mathbb{Q})$, and $N := \ker\chi$. Then $\mathrm{End}^0(X)$ is K or a CM field of degree 6 containing K, and the subspace of N-invariants in $\wedge^* H^1(X,\mathbb{C})$ is

$$\Big(\bigoplus_p \mathrm{Hdg}^p(X)\otimes\mathbb{C}\Big)\oplus\wedge^3 H^1(X,\mathbb{C})_\sigma\oplus\wedge^3 H^1(X,\mathbb{C})_{\bar\sigma},$$

where $\mathrm{Hg}(X)$ acts on $\wedge^3 H^1(X,\mathbb{C})_\sigma$ through $\sigma\circ\chi$ and on $\wedge^3 H^1(X,\mathbb{C})_{\bar\sigma}$ through $\bar\sigma\circ\chi$.

Proof. By [MZ99, (2.3)], $\mathrm{End}^0(X)$ is $\mathbb{Q}$, a totally real cubic field, an imaginary quadratic field, or a CM field of degree 6. Since $\mathrm{End}^0(X)$ contains K, $\mathrm{End}^0(X)$ is K or a CM field of degree 6 containing K.

Let $V_X := H^1(X,\mathbb{Q})$, $V_{X,\sigma} := H^1(X,\mathbb{C})_\sigma$, and $V_{X,\bar\sigma} := H^1(X,\mathbb{C})_{\bar\sigma}$. Then $\dim V_{X,\sigma} = \dim V_{X,\bar\sigma} = 3$ and

$$\wedge^* H^1(X,\mathbb{C}) = \bigoplus_{0\le a,b\le 3} \wedge^a V_{X,\sigma}\otimes\wedge^b V_{X,\bar\sigma}.$$

By Lemma 2.3 (1), $\mathrm{Hg}(X)$ commutes with K. Hence $\mathrm{Hg}(X)$ preserves $V_{X,\sigma}$ and $V_{X,\bar\sigma}$, and by the definition of the K-determinant, $\mathrm{Hg}(X)$ acts on $\wedge^3 V_{X,\sigma}$ through $\sigma\circ\chi$ and on $\wedge^3 V_{X,\bar\sigma}$ through $\bar\sigma\circ\chi$. Since K acts on $H^1(X,\mathbb{C})$ by endomorphisms of Hodge structures,

$$V_{X,\sigma} = (V_{X,\sigma}\cap H^{1,0}(X))\oplus(V_{X,\sigma}\cap H^{0,1}(X)).$$

Complex conjugation exchanges $V_{X,\sigma}$ with $V_{X,\bar\sigma}$ and $H^{1,0}(X)$ with $H^{0,1}(X)$, so $\dim V_{X,\sigma}\cap H^{1,0}(X) = 2$ and

$$\dim V_{X,\sigma}\cap H^{0,1}(X) = \dim V_{X,\bar\sigma}\cap H^{1,0}(X) = 1.$$

Thus

$$\sigma(\chi(\eta_X(z))) = z^2\bar z = z \quad\text{and}\quad \bar\sigma(\chi(\eta_X(z))) = \bar z$$

for any $z\in\mathbb{U}$. In particular, the characters 1, $\sigma\circ\chi$, and $\bar\sigma\circ\chi$ of $\mathrm{Hg}(X)$ are pairwise distinct.

Let I be the subspace of N-invariants in $\wedge^* H^1(X,\mathbb{C})$. Since taking invariants commutes with extension of scalars, Lemma 2.3 (2) implies that $\bigoplus_p \mathrm{Hdg}^p(X)\otimes\mathbb{C}$ is the subspace of $\mathrm{Hg}(X)$-invariants in $\wedge^* H^1(X,\mathbb{C})$. It is contained in I, and $\wedge^3 V_{X,\sigma}$ and $\wedge^3 V_{X,\bar\sigma}$ are contained in I since χ is trivial on N. Since 1, $\sigma\circ\chi$, and $\bar\sigma\circ\chi$ are pairwise distinct, it suffices to show that I is spanned by $\wedge^3 V_{X,\sigma}$, $\wedge^3 V_{X,\bar\sigma}$, and vectors fixed by $\mathrm{Hg}(X)$.

First suppose that $\mathrm{End}^0(X) = K$. By [MZ99, (2.3), Theorem (2.7)], $\mathrm{Hg}(X) = \mathrm{U}_K(V_X,\psi)$, where ψ is the Riemann form of a polarization of X. The subspaces $V_{X,\sigma}$ and $V_{X,\bar\sigma}$ are isotropic for ψ, and ψ induces a perfect pairing between $V_{X,\sigma}$ and $V_{X,\bar\sigma}$. Hence restriction to $V_{X,\sigma}$ induces an isomorphism

$$\mathrm{Hg}(X)_{\mathbb{C}} \cong \mathrm{GL}(V_{X,\sigma}),$$

under which $N_{\mathbb{C}} \cong \mathrm{SL}(V_{X,\sigma})$ and $V_{X,\bar\sigma}\cong V_{X,\sigma}^*$. Thus

$$\wedge^a V_{X,\sigma}\otimes\wedge^b V_{X,\bar\sigma} \cong \mathrm{Hom}(\wedge^b V_{X,\sigma},\wedge^a V_{X,\sigma})$$

as $\mathrm{GL}(V_{X,\sigma})$-modules. By [Ful97, Section 8.2, Theorem 2 and the paragraph on the special linear group after it], the $\mathrm{SL}(V_{X,\sigma})$-modules $\wedge^a V_{X,\sigma}$ with $0\le a\le 3$ are irreducible, and $\wedge^a V_{X,\sigma}\cong\wedge^b V_{X,\sigma}$ if and only if $a=b$ or $\{a,b\}=\{0,3\}$. By Schur's lemma, the subspace of $\mathrm{SL}(V_{X,\sigma})$-invariants in $\mathrm{Hom}(\wedge^b V_{X,\sigma},\wedge^a V_{X,\sigma})$ is $\mathbb{C}\cdot\mathrm{id}_{\wedge^a V_{X,\sigma}}$ if $a=b$, $\wedge^3 V_{X,\sigma}$ if $(a,b)=(3,0)$, $\wedge^3 V_{X,\bar\sigma}$ if $(a,b)=(0,3)$, and zero otherwise. Since $\mathrm{GL}(V_{X,\sigma})$ fixes $\mathrm{id}_{\wedge^a V_{X,\sigma}}$, the claim follows in this case.

Now suppose that $F := \mathrm{End}^0(X)$ is a CM field of degree 6. By [MZ99, (2.3), Theorem (2.7)], $\mathrm{Hg}(X) = \mathrm{U}_F$. Since $\dim_F V_X = 1$, $H^1(X,\mathbb{C})$ is the direct sum of the lines $H^1(X,\mathbb{C})_\tau$ over the six complex embeddings τ of F. Let τ_1,τ_2,τ_3 be the complex embeddings of F extending σ, and choose non-zero vectors $x_j\in H^1(X,\mathbb{C})_{\tau_j}$ and $y_j\in H^1(X,\mathbb{C})_{\bar\tau_j}$ for $j=1,2,3$. Then x_1,x_2,x_3 span $V_{X,\sigma}$, and y_1,y_2,y_3 span $V_{X,\bar\sigma}$. Under the isomorphism $\mathrm{U}_{F,\mathbb{C}}\cong\mathbb{G}_m^3$ induced by τ_1,τ_2,τ_3, an element (t_1,t_2,t_3) acts on x_j by t_j and on y_j by t_j^{-1}, and $\sigma\circ\chi = t_1t_2t_3$. We write $\theta_1,\theta_2,\theta_3$ for the coordinate characters of $\mathbb{G}_m^3$ and write characters additively. The exterior monomials in

$x_1, x_2, x_3, y_1, y_2, y_3$ form a basis of $\wedge^* H^1(X, \mathbb{C})$ consisting of weight vectors, and the weight of such a monomial is $\sum_j (a_j - b_j)\theta_j$, where a_j (resp. b_j) is 1 if x_j (resp. y_j) occurs in the monomial and 0 otherwise. A character $\sum_j c_j\theta_j$ is trivial on $N_{\mathbb{C}} = \{t_1t_2t_3 = 1\}$ if and only if $c_1 = c_2 = c_3$. Hence I is spanned by the monomials with

$$a_1 - b_1 = a_2 - b_2 = a_3 - b_3 =: c,$$

where $c \in \{-1, 0, 1\}$. If $c = 0$, then the monomial is fixed by $\mathrm{Hg}(X)$. If $c = 1$, then $a_j = 1$ and $b_j = 0$ for $j = 1, 2, 3$, so the monomial is $x_1 \wedge x_2 \wedge x_3$, which spans $\wedge^3 V_{X,\sigma}$. If $c = -1$, then the monomial is $y_1 \wedge y_2 \wedge y_3$, which spans $\wedge^3 V_{X,\bar{\sigma}}$. □

We now treat products of two simple threefolds.

Theorem 8.8. *Let B and C be simple abelian threefolds and $A := B \times C$. Then $\mathrm{HC}(A')$ holds for any abelian sixfold A' isogenous to A. More precisely, the following hold.*

(1) If B and C are isogenous, then $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p.

(2) If B and C are not isogenous, then $F_B := \mathrm{End}^0(B)$ and $F_C := \mathrm{End}^0(C)$ are fields. If there exists no imaginary quadratic field which embeds into both F_B and F_C, then

$$\mathrm{Hg}(A) = \mathrm{Hg}(B) \times \mathrm{Hg}(C)$$

and $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p.

(3) If B and C are not isogenous and there exists an imaginary quadratic field K which embeds into both F_B and F_C, then K is unique up to isomorphism. Embed K into F_B and F_C so that the multiplicities on $H^{1,0}(B)$ and $H^{1,0}(C)$ are $(2,1)$ and $(1,2)$, respectively, let K act on A diagonally, so that the multiplicities of K on $H^{1,0}(A)$ are $(3,3)$, and let $\chi_B : \mathrm{Hg}(B) \to \mathrm{U}_K$ and $\chi_C : \mathrm{Hg}(C) \to \mathrm{U}_K$ be the K-determinants. Then the group

$$\{(g_B, g_C) \in \mathrm{Hg}(B) \times \mathrm{Hg}(C) \mid \chi_B(g_B)\chi_C(g_C) = 1\}$$

is connected and equal to $\mathrm{Hg}(A)$, $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for $p \neq 3$, $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_K(A)$, and any class in $W_K(A)$ is algebraic.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we recall the Hodge groups of simple abelian threefolds. Let X be a simple abelian threefold, $V_X := H^1(X, \mathbb{Q})$, $F_X := \mathrm{End}^0(X)$, and ψ_X the Riemann form of a polarization of X. By [MZ99, (2.3), Proposition (2.4), Theorem (2.7)], F_X is a field, $\mathrm{Hg}(X)$ is the centralizer of F_X in $\mathrm{Sp}(V_X, \psi_X)$, the Hodge algebra of X^n is the divisor algebra of X^n for any $n \geq 1$, and exactly one of the following holds.

(a) F_X is $\mathbb{Q}$ or a totally real cubic field. In this case,

$$\mathrm{Hg}(Y \times X) = \mathrm{Hg}(Y) \times \mathrm{Hg}(X)$$

for any abelian variety Y with $\mathrm{Hom}(X, Y) = 0$ [MZ99, Proposition (2.4), Lemma (3.4)].

(b) F_X is an imaginary quadratic field, and $\mathrm{Hg}(X) = \mathrm{U}_{F_X}(V_X, \psi_X)$.

(c) F_X is a CM field of degree 6, and $\mathrm{Hg}(X) = \mathrm{U}_{F_X}$. If F_X contains no imaginary quadratic subfield, then U_{F_X} is $\mathbb{Q}$-simple, that is, 1 and U_{F_X} are the only subtori of U_{F_X}.

In particular, F_B and F_C are fields. We let $V_A := H^1(A, \mathbb{Q}) = V_B \oplus V_C$, so that $H^*(A, \mathbb{Q}) = \wedge^* V_B \otimes \wedge^* V_C$ by the Künneth formula.

Step 2. In this step, we show that $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p if B and C are isogenous or

$$\mathrm{Hg}(A) = \mathrm{Hg}(B) \times \mathrm{Hg}(C).$$

In particular, (1) holds, and (2) holds if F_B or F_C is $\mathbb{Q}$ or a totally real cubic field.

By the Lefschetz $(1,1)$ theorem (cf. [Moo99, (2.10)]), $\mathrm{Hdg}^1 = \mathcal{D}^1$ for any abelian variety, so the divisor algebra is the subalgebra of the Hodge algebra generated by Hdg^1. Suppose that B and C are isogenous. Then there exists an isogeny $f : A \to B \times B$, and $f^* : H^*(B \times B, \mathbb{Q}) \to H^*(A, \mathbb{Q})$ is an isomorphism of graded $\mathbb{Q}$-algebras and of Hodge structures. Hence f^* maps $\mathrm{Hdg}^p(B \times B)$

onto $\mathrm{Hdg}^p(A)$ and $\mathcal{D}^p(B\times B)$ onto $\mathcal{D}^p(A)$. Since $\mathrm{Hdg}^p(B\times B)=\mathcal{D}^p(B\times B)$ by **Step 1**, we have $\mathrm{Hdg}^p(A)=\mathcal{D}^p(A)$. This proves (1).

Suppose that

$$\mathrm{Hg}(A)=\mathrm{Hg}(B)\times\mathrm{Hg}(C).$$

The group $\mathrm{Hg}(B)\times\mathrm{Hg}(C)$ acts on

$$\wedge^{2p}V_A=\bigoplus_{i+j=2p}\wedge^i V_B\otimes\wedge^j V_C$$

factorwise, and $\eta_X(-1)=-\mathrm{id}_{V_X}$, so $\mathrm{Hg}(X)$ has no non-zero invariants in $\wedge^i V_X$ for odd i. By Lemma 2.3 (2),

$$\mathrm{Hdg}^p(A)=\bigoplus_{i+j=p}\mathrm{pr}_B^*\,\mathrm{Hdg}^i(B)\cup\mathrm{pr}_C^*\,\mathrm{Hdg}^j(C).$$

By **Step 1**, the Hodge algebras of B and C are their divisor algebras, and pullbacks of classes of divisors are classes of divisors. Thus $\mathrm{Hdg}^p(A)\subset\mathcal{D}^p(A)$, and hence $\mathrm{Hdg}^p(A)=\mathcal{D}^p(A)$.

Finally, suppose that B and C are not isogenous and that F_B is $\mathbb{Q}$ or a totally real cubic field. Since B and C are simple of equal dimension, $\mathrm{Hom}(B,C)=0$. By **Step 1** (a),

$$\mathrm{Hg}(A)=\mathrm{Hg}(B)\times\mathrm{Hg}(C),$$

and no imaginary quadratic field embeds into F_B. Hence (2) holds in this case. The case when F_C is $\mathbb{Q}$ or a totally real cubic field is symmetric.

Step 3. In this step, we study the imaginary quadratic subfields of F_B and F_C and the corresponding determinants. From now on, we assume that B and C are not isogenous and that F_B and F_C are as in **Step 1** (b) or **Step 1** (c). Then $\mathrm{Hom}(B,C)=0$ and $\mathrm{Hom}(C,B)=0$. Let $X\in\{B,C\}$.

(i) F_X contains at most one imaginary quadratic subfield: two distinct imaginary quadratic subfields would generate a subfield of degree 4, while $[F_X:\mathbb{Q}]\in\{2,6\}$. In particular, the field K in (3) is unique up to isomorphism. If F_X contains an imaginary quadratic subfield, we denote it by K_X.

(ii) Let $\sigma_X:K_X\hookrightarrow\mathbb{C}$ be an embedding and (r,s) the multiplicities of K_X on $H^{1,0}(X)$. We say that a line in $H^1(X,\mathbb{C})$ is an eigenline of η_X of weight k if $\eta_X(z)$ acts on it by z^k for any $z\in\mathbb{U}$. Since K_X acts by endomorphisms of Hodge structures, and complex conjugation exchanges $H^1(X,\mathbb{C})_{\sigma_X}$ with $H^1(X,\mathbb{C})_{\bar\sigma_X}$ and $H^{1,0}(X)$ with $H^{0,1}(X)$, the subspace $H^1(X,\mathbb{C})_{\sigma_X}$ is the direct sum of r eigenlines of weight 1 and s eigenlines of weight -1. We show that $\{r,s\}=\{1,2\}$. Since $r+s=3$, it suffices to exclude $(r,s)=(3,0)$ and $(r,s)=(0,3)$. If $(r,s)=(3,0)$, then $\eta_X(z)$ acts on $H^1(X,\mathbb{C})_{\sigma_X}$ by z and on $H^1(X,\mathbb{C})_{\bar\sigma_X}$ by $\bar z$, so $\eta_X(\mathbb{U})$ commutes with $\mathrm{End}_{K_X}(V_X)$. The centralizer of $\mathrm{End}_{K_X}(V_X)$ in $\mathrm{GL}(V_X)$ is an algebraic subgroup defined over $\mathbb{Q}$ whose real points contain $\eta_X(\mathbb{U})$, so it contains $\mathrm{Hg}(X)$ (Definition 2.2). By Lemma 2.3 (1), $\mathrm{End}_{K_X}(V_X)\cong M_3(K_X)$ is contained in the image of F_X, which is impossible since F_X is a field. The case $(r,s)=(0,3)$ is symmetric.

(iii) Suppose that F_X contains K_X. Let

$$\chi_X:\mathrm{Hg}(X)\to\mathrm{U}_{K_X}$$

be the K_X-determinant on V_X and $N_X:=\ker\chi_X$. If $F_X=K_X$, then $N_X=\mathrm{SU}_{K_X}(V_X,\psi_X)$ is the derived group of $\mathrm{Hg}(X)$ and $N_{X,\mathbb{C}}\cong\mathrm{SL}_3$. If $[F_X:\mathbb{Q}]=6$, then χ_X is the restriction of Nm_{F_X/K_X} to U_{F_X}, and

$$N_{X,\mathbb{C}}\cong\{(t_1,t_2,t_3)\in\mathbb{G}_m^3\mid t_1t_2t_3=1\}.$$

In both cases, N_X is connected. By (ii), $\sigma_X(\chi_X(\eta_X(z)))=z^{r-s}$ for any $z\in\mathbb{U}$, where $r-s=\pm1$. In particular, $\sigma_X(\chi_X(\eta_X(z)))=z$ if the multiplicities of K_X are $(2,1)$, and χ_X is surjective, since the image of χ_X is a non-trivial connected subgroup of the one-dimensional torus U_{K_X}.

(iv) For a torus T over $\mathbb{Q}$, $X_*(T)$ is the cocharacter group of $T_{\overline{\mathbb{Q}}}$, dual to $X^*(T)$, and $X_*(T)_{\mathbb{Q}}:=X_*(T)\otimes\mathbb{Q}$ (Subsection 8.3), on which $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts. By [Moo99, (1.1)], a surjection

$T \to T'$ of tori induces a surjection $X_*(T)_{\mathbb{Q}} \to X_*(T')_{\mathbb{Q}}$, subtori of T correspond to Galois-stable subspaces of $X_*(T)_{\mathbb{Q}}$, and two tori T and T' are isogenous if and only if $X_*(T)_{\mathbb{Q}} \cong X_*(T')_{\mathbb{Q}}$ as Galois modules. For an imaginary quadratic field L, $X_*(\mathrm{U}_L)_{\mathbb{Q}}$ is the line on which $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts through the quadratic character ϵ_L of L. In particular, if L and L' are imaginary quadratic fields, then U_L and $\mathrm{U}_{L'}$ are isogenous if and only if $L \cong L'$.

Suppose that $[F_X : \mathbb{Q}] = 6$ and F_X contains K_X. Let $R_X := F_X^+$, which is a totally real cubic field, and fix an embedding $\sigma_X : K_X \hookrightarrow \mathbb{C}$. For $\rho \in \mathrm{Hom}(R_X, \mathbb{R})$, let τ_ρ be the complex embedding of F_X with $\tau_\rho|_{R_X} = \rho$ and $\tau_\rho|_{K_X} = \sigma_X$. Then τ_ρ and $\bar{\tau}_\rho$ with $\rho \in \mathrm{Hom}(R_X, \mathbb{R})$ are the six complex embeddings of F_X. Since $\mathrm{U}_{F_X}(\mathbb{C})$ is the group of $(t_\tau)_\tau$ with $t_\tau t_{\bar{\tau}} = 1$, we obtain an isomorphism

$$X_*(\mathrm{U}_{F_X})_{\mathbb{Q}} \cong \mathbb{Q}^{\mathrm{Hom}(R_X,\mathbb{R})},$$

which sends a cocharacter acting on $H^1(X, \mathbb{C})_\tau$ by $z^{n(\tau)}$ to $(n(\tau_\rho))_\rho$, and $g \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on $\mathbb{Q}^{\mathrm{Hom}(R_X,\mathbb{R})}$ by $\epsilon_{K_X}(g)$ times the permutation of coordinates induced by $\rho \to g \circ \rho$. Hence

$$X_*(\mathrm{U}_{F_X})_{\mathbb{Q}} \cong \mathbb{Q} \cdot (1,1,1) \oplus P_X, \quad P_X := \Big\{ x \in \mathbb{Q}^{\mathrm{Hom}(R_X,\mathbb{R})} \;\Big|\; \sum_\rho x_\rho = 0 \Big\},$$

as Galois modules, where $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on $\mathbb{Q} \cdot (1,1,1)$ through ϵ_{K_X}. Let $\widetilde{R}_X \subset \mathbb{R}$ be the normal closure of R_X. Then $\mathrm{Gal}(\widetilde{R}_X/\mathbb{Q})$ acts faithfully and transitively on $\mathrm{Hom}(R_X, \mathbb{R})$, so $\mathrm{Gal}(\widetilde{R}_X/\mathbb{Q})$ is C_3 or S_3. Since $K_X \cap \widetilde{R}_X = \mathbb{Q}$, the group $\mathrm{Gal}(\overline{\mathbb{Q}}/K_X)$ acts on P_X through the standard two-dimensional representation of C_3 or S_3, which is irreducible over $\mathbb{Q}$. For C_3, irreducibility holds since a generator acts on P_X with characteristic polynomial $x^2 + x + 1$. Thus P_X is irreducible.

We conclude the following.

$(*)$ If T is a quotient torus of $\mathrm{Hg}(X)$ with $\dim T = 1$, then F_X contains an imaginary quadratic subfield K_X and $X_*(T)_{\mathbb{Q}} \cong \epsilon_{K_X}$.

If F_X is imaginary quadratic, then T is a quotient of $\mathrm{Hg}(X)/N_X \cong \mathrm{U}_{F_X}$ since N_X is the derived group of $\mathrm{Hg}(X)$. If $[F_X : \mathbb{Q}] = 6$, then $X_*(T)_{\mathbb{Q}}$ is a one-dimensional quotient of $X_*(\mathrm{U}_{F_X})_{\mathbb{Q}}$. If F_X contains no imaginary quadratic subfield, then $X_*(\mathrm{U}_{F_X})_{\mathbb{Q}}$ is irreducible of dimension 3 by **Step 1** (c), which is impossible. Otherwise $X_*(T)_{\mathbb{Q}} \cong \epsilon_{K_X}$ since P_X is irreducible of dimension 2.

Step 4. In this step, we reduce (2) and the description of $\mathrm{Hg}(A)$ in (3) to a statement on common quotients of $\mathrm{Hg}(B)$ and $\mathrm{Hg}(C)$. For $X \in \{B, C\}$, we regard $\mathrm{Hg}(X)$ as a subgroup of $\mathrm{Hg}(B) \times \mathrm{Hg}(C)$ and let N'_X be the identity component of $\mathrm{Hg}(A) \cap \mathrm{Hg}(X)$. By Lemma 2.3 (3) and Goursat's lemma (cf. [MZ99, (3.1)]), N'_X is a normal subgroup of $\mathrm{Hg}(X)$, and for $Q_X := \mathrm{Hg}(X)/N'_X$, the image H of $\mathrm{Hg}(A)$ in $Q_B \times Q_C$ maps onto Q_B and onto Q_C with finite kernels. In particular, Q_B and Q_C are isogenous, and

$$\dim \mathrm{Hg}(A) = \dim \mathrm{Hg}(B) + \dim \mathrm{Hg}(C) - \dim Q_B.$$

Since $\mathrm{Hg}(A)$ is connected,

$$\mathrm{Hg}(A) = \mathrm{Hg}(B) \times \mathrm{Hg}(C)$$

if and only if $\dim Q_B = 0$. By **Step 1** and the assumption at the beginning of **Step 3**, each of F_B and F_C is imaginary quadratic or a CM field of degree 6. We prove the following claim by treating the three possible cases separately.

$(**)$ Q_B is a torus, $\dim Q_B \le 1$, and if $\dim Q_B = 1$, then F_B and F_C contain isomorphic imaginary quadratic fields.

Step 5. In this step, we prove $(**)$ when F_B and F_C are imaginary quadratic. Let $\mathfrak{h} := \mathrm{Lie}\,\mathrm{Hg}(A)$, $\mathfrak{d} := [\mathfrak{h}, \mathfrak{h}]$, and $\mathfrak{s}_X := \mathrm{Lie}\, N_X$ for $X \in \{B, C\}$. By **Step 1** (b) and **Step 3** (iii), $\mathfrak{s}_X$ is the derived algebra of $\mathrm{Lie}\,\mathrm{Hg}(X)$, and $\mathfrak{s}_X \otimes \mathbb{C} \cong \mathfrak{sl}_3$ is simple. By Lemma 2.3 (3), $\mathfrak{d}$ maps onto $\mathfrak{s}_B$ and onto $\mathfrak{s}_C$. By Goursat's lemma (cf. [MZ99, (3.1)]), either $\mathfrak{d} = \mathfrak{s}_B \oplus \mathfrak{s}_C$, or $\mathfrak{d}$ is the graph of an isomorphism $\mathfrak{s}_B \to \mathfrak{s}_C$.

We show that $\mathfrak{d}$ is not a graph. Suppose to the contrary that $\mathfrak{d}$ is the graph of an isomorphism. Let

$$M := \mathrm{Hom}_{\mathfrak{d}}(V_B, V_C) = \{f \in \mathrm{Hom}_{\mathbb{Q}}(V_B, V_C) \mid f \circ x_B = x_C \circ f \text{ for any } (x_B, x_C) \in \mathfrak{d}\}.$$

For $X \in \{B, C\}$, let $V_{X,\sigma_X} := H^1(X, \mathbb{C})_{\sigma_X}$ and $V_{X,\bar{\sigma}_X} := H^1(X, \mathbb{C})_{\bar{\sigma}_X}$. Since ψ_X induces a perfect pairing between the isotropic subspaces V_{X,σ_X} and $V_{X,\bar{\sigma}_X}$, restriction to V_{X,σ_X} induces an isomorphism

$$\mathrm{Hg}(X)_{\mathbb{C}} \cong \mathrm{GL}(V_{X,\sigma_X}),$$

under which $V_{X,\bar{\sigma}_X} \cong V_{X,\sigma_X}^*$. Hence V_{X,σ_X} and $V_{X,\bar{\sigma}_X}$ are non-isomorphic irreducible $\mathfrak{s}_X \otimes \mathbb{C}$-modules of dimension 3. Since the projections

$$\mathfrak{d} \otimes \mathbb{C} \to \mathfrak{s}_X \otimes \mathbb{C}$$

are isomorphisms, V_{C,σ_C} is an irreducible $\mathfrak{d} \otimes \mathbb{C}$-module of dimension 3, and

$$\mathfrak{d} \otimes \mathbb{C} \cong \mathfrak{s}_B \otimes \mathbb{C} \cong \mathfrak{sl}(V_{B,\sigma_B}).$$

Since $\mathrm{SL}(V_{B,\sigma_B})$ is simply connected, the $\mathfrak{sl}(V_{B,\sigma_B})$-module V_{C,σ_C} integrates to an $\mathrm{SL}(V_{B,\sigma_B})$-module, which is irreducible since V_{C,σ_C} is an irreducible $\mathfrak{sl}(V_{B,\sigma_B})$-module. By [Ful97, Section 8.1, Theorem 1] and [Ful97, Section 8.2, Theorem 2 and the paragraph on the special linear group after it], the irreducible $\mathrm{SL}(V_{B,\sigma_B})$-modules of dimension 3 are $E^{(1)} = V_{B,\sigma_B}$ and $E^{(1,1)} = \wedge^2 V_{B,\sigma_B} \cong V_{B,\sigma_B}^*$. Hence V_{C,σ_C} is isomorphic to V_{B,σ_B} or to $V_{B,\sigma_B}^* \cong V_{B,\bar{\sigma}_B}$. Thus $M \otimes \mathbb{C} \neq 0$, and $M \neq 0$.

The group $\mathrm{Hg}(A)$ acts on $\mathrm{Hom}_{\mathbb{Q}}(V_B, V_C)$ by $(g_B, g_C) \cdot f := g_C \circ f \circ g_B^{-1}$, and preserves M since the adjoint action of $\mathrm{Hg}(A)$ preserves $\mathfrak{d}$. Since $\mathrm{Hg}(X)$ commutes with K_X, the group $\mathrm{Hg}(A)_{\mathbb{C}}$ preserves each of the four subspaces $\mathrm{Hom}_{\mathfrak{d}\otimes\mathbb{C}}(Y, Z)$ with $Y \in \{V_{B,\sigma_B}, V_{B,\bar{\sigma}_B}\}$ and $Z \in \{V_{C,\sigma_C}, V_{C,\bar{\sigma}_C}\}$, whose direct sum is $M \otimes \mathbb{C}$. By Schur's lemma, $\dim \mathrm{Hom}_{\mathfrak{d}\otimes\mathbb{C}}(Y, Z) \leq 1$, and any non-zero $f \in \mathrm{Hom}_{\mathfrak{d}\otimes\mathbb{C}}(Y, Z)$ is an isomorphism. For such f, the line $\mathbb{C} \cdot f = \mathrm{Hom}_{\mathfrak{d}\otimes\mathbb{C}}(Y, Z)$ is stable under $\eta_A(\mathbb{U})$, so the circle $\mathbb{U}$ acts on it through a character $z \to z^k$ for some integer k, that is, $\eta_C(z) \circ f \circ \eta_B(z)^{-1} = z^k f$ for any $z \in \mathbb{U}$. Equivalently,

$$f \circ \eta_B(z) = z^{-k} \eta_C(z) \circ f.$$

Thus the non-zero $\mathfrak{d}$-intertwiner f forces the multiset of weights of η_C on Z to coincide with the multiset of weights of η_B on Y shifted by k. By **Step 3** (ii), both multisets contain 1 and -1 and are contained in $\{1, -1\}$. Hence $k = 0$, and $\eta_A(\mathbb{U})$ acts trivially on $M \otimes \mathbb{C}$. The kernel of the action of $\mathrm{Hg}(A)$ on M is an algebraic subgroup defined over $\mathbb{Q}$ whose real points contain $\eta_A(\mathbb{U})$, so $\mathrm{Hg}(A)$ acts trivially on M (Definition 2.2). We regard $\mathrm{Hom}_{\mathbb{Q}}(V_B, V_C)$ as the subspace of $\mathrm{End}_{\mathbb{Q}}(V_A)$ of endomorphisms which vanish on V_C and map V_B into V_C. Then M is contained in the commutant of $\mathrm{Hg}(A)$, which is the image of $\mathrm{End}^0(A)$ by Lemma 2.3 (1). Since $\mathrm{Hom}(B, C) = 0$ and $\mathrm{Hom}(C, B) = 0$, $\mathrm{End}^0(A) = F_B \times F_C$ [Moo99, (2.3)], and the image of $F_B \times F_C$ preserves V_B. Hence $M = 0$, a contradiction.

Therefore $\mathfrak{d} = \mathfrak{s}_B \oplus \mathfrak{s}_C$, so $N_X \subset N_X'$ for $X \in \{B, C\}$. Thus Q_B is a quotient of $\mathrm{Hg}(B)/N_B \cong \mathrm{U}_{F_B}$, so Q_B is a torus with $\dim Q_B \leq 1$. If $\dim Q_B = 1$, then $(*)$ implies that

$$\epsilon_{F_B} \cong X_*(Q_B)_{\mathbb{Q}} \cong X_*(Q_C)_{\mathbb{Q}} \cong \epsilon_{F_C},$$

so $F_B \cong F_C$. This proves $(**)$ in this case.

Step 6. In this step, we prove $(**)$ when one of F_B and F_C is imaginary quadratic and the other is a CM field of degree 6. Since $\dim Q_B = \dim Q_C$, $(**)$ is symmetric in B and C. Possibly exchanging B and C, we may assume that F_B is imaginary quadratic and $[F_C : \mathbb{Q}] = 6$. Since $\mathrm{Hg}(C) = \mathrm{U}_{F_C}$ is a torus, Q_C is a torus, so Q_B is a torus. Hence N_B' contains the derived group N_B of $\mathrm{Hg}(B)$, and Q_B is a quotient of $\mathrm{Hg}(B)/N_B \cong \mathrm{U}_{F_B}$. Thus $\dim Q_B \leq 1$. If $\dim Q_B = 1$, then by $(*)$, F_C contains an imaginary quadratic subfield K_C and

$$\epsilon_{F_B} \cong X_*(Q_B)_{\mathbb{Q}} \cong X_*(Q_C)_{\mathbb{Q}} \cong \epsilon_{K_C},$$

so $F_B \cong K_C$. This proves $(**)$ in this case.

Step 7. In this step, we prove (∗∗) when F_B and F_C are CM fields of degree 6. In this case, $\mathrm{Hg}(X) = \mathrm{U}_{F_X}$ is a torus with $\dim \mathrm{U}_{F_X} = 3$, and Q_X is a quotient torus of U_{F_X} for $X \in \{B, C\}$. If F_X contains no imaginary quadratic subfield for some $X \in \{B, C\}$, then U_{F_X} is $\mathbb{Q}$-simple by **Step 1** (c), so N'_X is trivial or $N'_X = \mathrm{U}_{F_X}$, that is, $\dim Q_B \in \{0, 3\}$. If $\dim Q_B = 1$, then (∗) implies that F_B and F_C contain imaginary quadratic fields K_B and K_C with $\epsilon_{K_B} \cong \epsilon_{K_C}$, so $K_B \cong K_C$. It remains to exclude $\dim Q_B = 3$ and $\dim Q_B = 2$.

Suppose that $\dim Q_B = 3$. Then N'_B and N'_C are trivial, so

$$\dim \mathrm{Hg}(A) = 3 = \dim \mathrm{Hg}(B) = \dim \mathrm{Hg}(C).$$

Since $\mathrm{MT} = \mathbb{G}_m \cdot \mathrm{Hg}$ and $\mathrm{Hg} \subset \mathrm{SL}$, we have $\dim \mathrm{MT} = 1 + \dim \mathrm{Hg}$ (cf. [Moo99, (1.11)]), so

$$\dim \mathrm{MT}(B \times C) = \dim \mathrm{MT}(B) = \dim \mathrm{MT}(C).$$

By [Lom23, Proposition 4.4], B and C are isogenous, a contradiction.

Suppose that $\dim Q_B = 2$. Then F_X contains K_X for $X \in \{B, C\}$, since otherwise $\dim Q_B \in \{0, 3\}$ by the beginning of this step, and we use the notation of **Step 3** (iv). The Galois module $X_*(Q_X)_{\mathbb{Q}}$ is a two-dimensional quotient of $\mathbb{Q} \cdot (1, 1, 1) \oplus P_X$. Since P_X is irreducible of dimension 2, the kernel of the quotient map is $\mathbb{Q} \cdot (1, 1, 1)$, and $X_*(Q_X)_{\mathbb{Q}} \cong P_X$ via the projection along $\mathbb{Q} \cdot (1, 1, 1)$. Since H is a connected subgroup of the torus $Q_B \times Q_C$, it is a torus, and since H maps onto Q_B and onto Q_C with finite kernels, the maps $H \to Q_B$ and $H \to Q_C$ induce Galois-equivariant isomorphisms $X_*(H)_{\mathbb{Q}} \cong X_*(Q_B)_{\mathbb{Q}}$ and $X_*(H)_{\mathbb{Q}} \cong X_*(Q_C)_{\mathbb{Q}}$ by **Step 3** (iv). Let $\phi : P_B \to P_C$ be the composite of the Galois-equivariant isomorphisms

$$P_B \cong X_*(Q_B)_{\mathbb{Q}} \cong X_*(H)_{\mathbb{Q}} \cong X_*(Q_C)_{\mathbb{Q}} \cong P_C.$$

Let μ_X be the cocharacter of $\mathrm{U}_{F_X,\mathbb{C}}$ which acts by z on $H^{1,0}(X)$ and by z^{-1} on $H^{0,1}(X)$, so that $\mu_X|_{\mathbb{U}} = \eta_X$. Under the isomorphism of **Step 3** (iv), μ_X corresponds to the vector $s_X \in \{1, -1\}^{\mathrm{Hom}(R_X, \mathbb{R})}$ with $s_X(\rho) = 1$ if and only if $H^1(X, \mathbb{C})_{\tau_\rho} \subset H^{1,0}(X)$. The number of entries 1 of s_X is the multiplicity of K_X at σ_X, so s_X is not constant by **Step 3** (ii). Since $\mathbb{U}$ is Zariski dense in $\mathbb{G}_m(\mathbb{C})$ and $\eta_A(\mathbb{U}) \subset \mathrm{Hg}(A)(\mathbb{R})$, (μ_B, μ_C) is a cocharacter of $\mathrm{Hg}(A)_{\mathbb{C}}$. Let

$$v_X := s_X - \frac{1}{3}\Big(\sum_\rho s_X(\rho)\Big)(1, 1, 1) \in P_X$$

be the projection of s_X to P_X. The image of (μ_B, μ_C) in $X_*(H)_{\mathbb{Q}}$ maps to the image of μ_X in $X_*(Q_X)_{\mathbb{Q}}$ for $X \in \{B, C\}$, and under $X_*(Q_X)_{\mathbb{Q}} \cong P_X$ the image of μ_X is v_X. Hence $\phi(v_B) = v_C$. Since s_X is a permutation of $\pm(1, 1, -1)$, v_X is a permutation of $\pm(\frac{2}{3}, \frac{2}{3}, -\frac{4}{3})$. In particular, v_X has exactly two equal coordinates.

Let $\widetilde{F}_X \subset \mathbb{C}$ be the normal closure of F_X. Then $\widetilde{F}_X = K_X \widetilde{R}_X$, and restriction induces an isomorphism

$$\mathrm{Gal}(\widetilde{F}_X/\mathbb{Q}) \cong \mathrm{Gal}(K_X/\mathbb{Q}) \times \mathrm{Gal}(\widetilde{R}_X/\mathbb{Q}).$$

The action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on P_X factors through a faithful action of $\mathrm{Gal}(\widetilde{F}_X/\mathbb{Q})$. An element (c^e, g) with $e \in \{0, 1\}$, where c is complex conjugation, acts on P_X by $(-1)^e \pi_X(g)$, where π_X is the standard two-dimensional representation of $\mathrm{Gal}(\widetilde{R}_X/\mathbb{Q})$ on P_X. The representation π_X is faithful, and no element acts by $-\mathrm{id}_{P_X}$, since the transpositions have eigenvalues 1 and -1, and the elements of order 3 have primitive cube roots of unity as eigenvalues. Hence $(-1)^e \pi_X(g) = \mathrm{id}_{P_X}$ implies $e = 0$ and $g = 1$. Since ϕ is Galois-equivariant, the kernels of the Galois actions on P_B and P_C coincide, so $\widetilde{F}_B = \widetilde{F}_C =: \widetilde{F}$. Complex conjugation fixes $\widetilde{R}_X$ and $[\widetilde{F} : \widetilde{R}_X] = 2$, so $\widetilde{R}_X$ is the fixed field of complex conjugation in $\widetilde{F}$, and $\widetilde{R}_B = \widetilde{R}_C =: \widetilde{R}$.

Suppose that $\mathrm{Gal}(\widetilde{R}/\mathbb{Q}) \cong C_3$. Then $\mathrm{Gal}(\widetilde{F}/\mathbb{Q}) \cong C_6$, so $\widetilde{F}$ has a unique quadratic subfield and a unique cubic subfield. Hence $K_B = K_C$, $R_B \cong R_C \cong \widetilde{R}$, and $F_B \cong F_C \cong \widetilde{F}$. For $X \in \{B, C\}$, the CM type

$$\Phi_X := \{\tau \mid H^1(X, \mathbb{C})_\tau \subset H^{1,0}(X)\}$$

of X corresponds to the non-constant vector s_X, after we identify F_X with $\widetilde{F}$. The group $\mathrm{Aut}(\widetilde{F}) \cong C_6$ acts on $\{1,-1\}^3$ by the cyclic permutations of the coordinates and by multiplication by -1, and the six non-constant vectors in $\{1,-1\}^3$ form one orbit under $\mathrm{Aut}(\widetilde{F})$. Thus $\Phi_C = \Phi_B \circ \alpha$ for some $\alpha \in \mathrm{Aut}(\widetilde{F})$. By [MZ99, (4.1)], B and C are isogenous, a contradiction.

Suppose that $\mathrm{Gal}(\widetilde{R}/\mathbb{Q}) \cong S_3$. All cubic subfields of $\widetilde{R}$ are conjugate, so $R_B \cong R_C$. An isomorphism $\alpha_0 : R_B \to R_C$ induces a Galois-equivariant bijection $\mathrm{Hom}(R_C,\mathbb{R}) \to \mathrm{Hom}(R_B,\mathbb{R})$, $\rho \to \rho \circ \alpha_0$. We fix an ordering of $\mathrm{Hom}(R_B,\mathbb{R})$ and regard P_B and P_C as the plane $P \subset \mathbb{Q}^3$ of vectors with coordinate sum 0, on which $g \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts by $\epsilon_{K_B}(g)\pi(g)$ and $\epsilon_{K_C}(g)\pi(g)$, respectively, where π is the permutation action through $\mathrm{Gal}(\widetilde{R}/\mathbb{Q}) \cong S_3$. The argument below does not depend on the choices of σ_B and σ_C: replacing σ_X by $\bar{\sigma}_X$ replaces s_X and v_X by $-s_X$ and $-v_X$. Since $\mathrm{Gal}(\overline{\mathbb{Q}}/K_B)$ acts on P_B by π and maps onto $\mathrm{Gal}(\widetilde{R}/\mathbb{Q}) \cong S_3$, a Galois-equivariant endomorphism of P_B commutes with the three transpositions, so it preserves the three (-1)-eigenlines $\mathbb{Q}(1,-1,0)$, $\mathbb{Q}(0,1,-1)$, and $\mathbb{Q}(1,0,-1)$, which are pairwise distinct lines in the plane P. Hence any Galois-equivariant endomorphism of P_B is a rational scalar.

Let $\mathbb{Q}(\sqrt{f_0})$ be the real quadratic subfield of $\widetilde{R}$, which is the fixed field of A_3, and let K' be the quadratic subfield of $K_B(\sqrt{f_0})$ other than K_B and $\mathbb{Q}(\sqrt{f_0})$. The group $\mathrm{Gal}(\widetilde{F}/\mathbb{Q}) \cong C_2 \times S_3$ has exactly three non-trivial characters with values in $\{\pm 1\}$, namely ϵ_{K_B}, $\epsilon_{\mathbb{Q}(\sqrt{f_0})}$, and

$$\epsilon_{K'} = \epsilon_{K_B}\epsilon_{\mathbb{Q}(\sqrt{f_0})},$$

where $\epsilon_{\mathbb{Q}(\sqrt{f_0})} = \mathrm{sgn} \circ \pi$. Exactly ϵ_{K_B} and $\epsilon_{K'}$ are non-trivial on complex conjugation, so K_B and K' are the imaginary quadratic subfields of $\widetilde{F}$. Since $K_C \subset \widetilde{F}$, we have $K_C = K_B$ or $K_C = K'$.

If $K_C = K_B$, then ϕ is a Galois-equivariant endomorphism of P_B, so $\phi = \lambda \cdot \mathrm{id}_P$ for some $\lambda \in \mathbb{Q}^*$. Since v_B and v_C are permutations of $\pm(\frac{2}{3},\frac{2}{3},-\frac{4}{3})$, the equality $\lambda v_B = v_C$ implies that $\lambda = \pm 1$ and $v_C = \pm v_B$. Since s_X is determined by v_X, we have $s_C = \pm s_B$. We take $\sigma_B = \sigma_C$, and let $\alpha : F_B \to F_C$ be the isomorphism which is the identity on $K_B = K_C$ and α_0 on R_B. If $s_C = s_B$, then $\Phi_C \circ \alpha = \Phi_B$, and if $s_C = -s_B$, then $\Phi_C \circ \alpha$ is the complex conjugate of Φ_B, which is Φ_B composed with complex conjugation of F_B. By [MZ99, (4.1)], B and C are isogenous, a contradiction.

If $K_C = K'$, then

$$\epsilon_{K_C} = \epsilon_{K_B} \cdot (\mathrm{sgn} \circ \pi).$$

Define $J : P \to P$ by

$$J(x_1,x_2,x_3) := (x_2 - x_3, x_3 - x_1, x_1 - x_2).$$

Then $J \circ g = \mathrm{sgn}(g) g \circ J$ for any permutation g of the coordinates, and $J^2 = -3 \cdot \mathrm{id}_P$. Hence J is a Galois-equivariant isomorphism $P_B \to P_C$, and $J^{-1} \circ \phi$ is a Galois-equivariant endomorphism of P_B, so $\phi = \lambda J$ for some $\lambda \in \mathbb{Q}^*$. Since $J(\frac{2}{3},\frac{2}{3},-\frac{4}{3}) = (2,-2,0)$, the vector Jv_B is a permutation of $\pm(2,-2,0)$, so $\lambda J v_B$ has three distinct coordinates. On the other hand, v_C has exactly two equal coordinates. This contradicts $\phi(v_B) = v_C$.

Thus $\dim Q_B \neq 2$, and $(**)$ holds in this case.

Step 8. In this step, we prove (2) and the description of $\mathrm{Hg}(A)$ in (3), and we compute the Hodge classes on A in the situation of (3). By **Steps 5, 6, and 7**, $(**)$ holds. If there exists no imaginary quadratic field which embeds into both F_B and F_C, then $\dim Q_B = 0$, so

$$\mathrm{Hg}(A) = \mathrm{Hg}(B) \times \mathrm{Hg}(C),$$

and (2) follows from **Step 2**. Suppose that K embeds into F_B and F_C. By **Step 3** (ii), there exist embeddings with multiplicities $(2,1)$ on $H^{1,0}(B)$ and $(1,2)$ on $H^{1,0}(C)$. Let

$$G := \{(g_B,g_C) \in \mathrm{Hg}(B) \times \mathrm{Hg}(C) \mid \chi_B(g_B)\chi_C(g_C) = 1\}.$$

By **Step 3** (iii) applied to B, and to C with the conjugate embedding of K, we have $\sigma(\chi_B(\eta_B(z))) = z$ and $\sigma(\chi_C(\eta_C(z))) = \bar{z}$ for any $z \in \mathbb{U}$. Hence $\chi_B(\eta_B(z))\chi_C(\eta_C(z)) = 1$, that is, $\eta_A(\mathbb{U}) \subset G(\mathbb{R})$, and $\mathrm{Hg}(A) \subset G$ by Definition 2.2. Since χ_C is surjective, the projection

$G \to \mathrm{Hg}(B)$ is surjective, and the kernel $\{1\} \times N_C$ of $G \to \mathrm{Hg}(B)$ is connected by **Step 3** (iii). Since $\mathrm{Hg}(B)$ is connected, G is connected. Moreover,

$$\dim G = \dim \mathrm{Hg}(B) + \dim \mathrm{Hg}(C) - 1 \leq \dim \mathrm{Hg}(A)$$

by **Step 4** and $(**)$. Thus $\mathrm{Hg}(A) = G$.

The subgroup $N_B \times N_C$ is normal in G, and $(g_B, g_C) \to \chi_B(g_B)$ induces an isomorphism $G/(N_B \times N_C) \cong \mathrm{U}_K$, since χ_B is surjective by **Step 3** (iii). For $X \in \{B, C\}$, let I_X be the subspace of N_X-invariants in $\wedge^* H^1(X, \mathbb{C})$. By Lemma 2.3 (2), $\bigoplus_p \mathrm{Hdg}^p(A) \otimes \mathbb{C}$ is the subspace of G-invariants in $\wedge^* H^1(B, \mathbb{C}) \otimes \wedge^* H^1(C, \mathbb{C})$, so it is the subspace of U_K-invariants in $I_B \otimes I_C$. By Lemma 8.7 applied to B, and to C with the conjugate embedding of K, we have

$$I_X = \Big(\bigoplus_p \mathrm{Hdg}^p(X) \otimes \mathbb{C} \Big) \oplus \wedge^3 H^1(X, \mathbb{C})_\sigma \oplus \wedge^3 H^1(X, \mathbb{C})_{\bar\sigma},$$

where $\mathrm{Hg}(X)$ acts on $\wedge^3 H^1(X, \mathbb{C})_\sigma$ through $\sigma \circ \chi_X$ and on $\wedge^3 H^1(X, \mathbb{C})_{\bar\sigma}$ through $\bar\sigma \circ \chi_X$. For $(g_B, g_C) \in G$, let $t := \sigma(\chi_B(g_B))$. Since $\bar\sigma(u) = \sigma(u)^{-1}$ for $u \in \mathrm{U}_K$ and $\chi_C(g_C) = \chi_B(g_B)^{-1}$, we have

$$\bar\sigma(\chi_B(g_B)) = t^{-1}, \quad \sigma(\chi_C(g_C)) = t^{-1}, \quad \bar\sigma(\chi_C(g_C)) = t.$$

Since $\mathrm{U}_{K,\mathbb{C}} \cong \mathbb{G}_m$, a tensor product of the summands of I_B and I_C is fixed by G if and only if the product of the corresponding powers of t is 1. Hence the subspace of U_K-invariants in $I_B \otimes I_C$ is

$$\Big(\bigoplus_{i,j} \mathrm{Hdg}^i(B) \otimes \mathrm{Hdg}^j(C) \otimes \mathbb{C} \Big) \oplus \big(\wedge^3 H^1(B, \mathbb{C})_\sigma \otimes \wedge^3 H^1(C, \mathbb{C})_\sigma \big) \oplus \big(\wedge^3 H^1(B, \mathbb{C})_{\bar\sigma} \otimes \wedge^3 H^1(C, \mathbb{C})_{\bar\sigma} \big).$$

Since K acts on A diagonally,

$$H^1(A, \mathbb{C})_\sigma = H^1(B, \mathbb{C})_\sigma \oplus H^1(C, \mathbb{C})_\sigma,$$

so

$$\wedge^3 H^1(B, \mathbb{C})_\sigma \otimes \wedge^3 H^1(C, \mathbb{C})_\sigma = \wedge^6 H^1(A, \mathbb{C})_\sigma,$$

and similarly for $\bar\sigma$. By Definition 2.7, we obtain

$$\bigoplus_p \mathrm{Hdg}^p(A) \otimes \mathbb{C} = \Big(\bigoplus_{i,j} \mathrm{pr}_B^* \mathrm{Hdg}^i(B) \cup \mathrm{pr}_C^* \mathrm{Hdg}^j(C) \Big) \otimes \mathbb{C} \oplus W_K(A) \otimes \mathbb{C}.$$

Both summands on the right are complexifications of $\mathbb{Q}$-subspaces of $H^*(A, \mathbb{Q})$, so

$$\bigoplus_p \mathrm{Hdg}^p(A) = \Big(\bigoplus_{i,j} \mathrm{pr}_B^* \mathrm{Hdg}^i(B) \cup \mathrm{pr}_C^* \mathrm{Hdg}^j(C) \Big) \oplus W_K(A).$$

By **Step 1**, the first summand is contained in the divisor algebra of A. It is a subalgebra which contains $\mathrm{Hdg}^1(A) = \mathcal{D}^1(A)$, since $W_K(A) \subset H^6(A, \mathbb{Q})$. Hence the first summand is the divisor algebra of A, and $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for $p \neq 3$ and $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_K(A)$.

Step 9. In this step, we prove the algebraicity statements. In the situation of (3), $A = B \times C$ is not simple, and the multiplicities of K on $H^{1,0}(A)$ are $(3, 3)$. By Theorem 8.2, any class in $W_K(A)$ is algebraic. This proves (3).

Products of classes of divisors are algebraic, so any class in $\mathcal{D}^p(A)$ is algebraic. By (1), (2), and (3), $\mathrm{Hdg}^p(A)$ is spanned by $\mathcal{D}^p(A)$ and, in the situation of (3) with $p = 3$, by $W_K(A)$. Hence $\mathrm{HC}(A)$ holds. If A' is an abelian sixfold isogenous to A, then $\mathrm{HC}(A')$ holds by Lemma 2.6. □

The next theorem treats the products of a simple abelian threefold with elliptic curves.

Theorem 8.9. *(1) Let B be a simple abelian threefold and T a finite product of elliptic curves. Then $\mathrm{HC}(B \times T)$ holds. More precisely, the Hodge algebra of $B \times T$ is generated by classes of divisors and by the pullbacks of $W_K(B \times E)$ under homomorphisms $\mathrm{id}_B \times f : B \times T \to B \times E$, where E is an elliptic curve with complex multiplication by an imaginary quadratic field K which embeds into $\mathrm{End}^0(B)$, K acts on B through an embedding $K \hookrightarrow \mathrm{End}^0(B)$ and on E through an isomorphism $K \cong \mathrm{End}^0(E)$ such that, for the*

diagonal action, the multiplicities of K on $H^{1,0}(B \times E)$ are $(2,2)$, and $f : T \to E$ is a surjective homomorphism. If no such E exists, then $\mathrm{Hdg}^p(B \times T) = \mathcal{D}^p(B \times T)$ for any p.

(2) Let A be an abelian sixfold which contains a simple abelian threefold. Then $\mathrm{HC}(A)$ holds.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix notation for B, state two facts on Hodge groups of products, and reduce (1) to the case when T is a product of powers of pairwise non-isogenous elliptic curves.

By [MZ99, (2.3)], $F_B := \mathrm{End}^0(B)$ is $\mathbb{Q}$, a totally real cubic field, an imaginary quadratic field, or a CM field of degree 6. Two distinct imaginary quadratic subfields of a field generate a subfield of degree 4, and 4 does not divide $[F_B : \mathbb{Q}]$, so F_B contains at most one imaginary quadratic subfield. Assume that F_B contains an imaginary quadratic field K. If the multiplicities of K on $H^{1,0}(B)$ are $(3,0)$ or $(0,3)$, then $H^1(B,\mathbb{C})_\sigma$ is $H^{1,0}(B)$ or $H^{0,1}(B)$, so any K-linear endomorphism of $H^1(B,\mathbb{Q})$ preserves the Hodge decomposition. Hence $M_3(K) \cong \mathrm{End}_K(H^1(B,\mathbb{Q}))$ is contained in F_B [Moo99, (2.3)], which is impossible since F_B is a field. Thus, possibly composing the embedding $K \hookrightarrow F_B$ with the complex conjugation of K, we may assume that the multiplicities of K on $H^{1,0}(B)$ are $(2,1)$. Let $H_B := \mathrm{Hg}(B)$, let $\chi : H_B \to \mathrm{U}_K$ be the K-determinant on $H^1(B,\mathbb{Q})$, and let $N := \ker \chi$. We also write σ for the character of $\mathrm{U}_{K,\mathbb{C}}$ induced by σ, that is, $[\sigma]$ of Lemma 8.5. Since $\dim H^1(B,\mathbb{C})_\sigma \cap H^{1,0}(B) = 2$ and

$$\dim H^1(B,\mathbb{C})_\sigma \cap H^{0,1}(B) = \dim H^1(B,\mathbb{C})_{\bar\sigma} \cap H^{1,0}(B) = 1,$$

we have

$$\sigma\big(\chi(\eta_B(z))\big) = \det\big(\eta_B(z)|_{H^1(B,\mathbb{C})_\sigma}\big) = z^2\bar z = z \tag{8.1}$$

for any $z \in \mathbb{U}$. In particular, χ is non-trivial, hence χ is surjective onto the one-dimensional torus U_K.

Let E' be an elliptic curve with complex multiplication by K, with the K-action of multiplicities $(0,1)$ on $H^{1,0}(E')$. Then K acts diagonally on $B \times E'$ with multiplicities $(2,2)$, so $(B \times E', K)$ is of Weil type, and $W_K(B \times E') \subset \mathrm{Hdg}^2(B \times E')$ by Lemma 2.10 (2). Hence the subalgebra of $H^*(B \times T, \mathbb{Q})$ generated by the classes of divisors and by the pullbacks in (1) is contained in the Hodge algebra of $B \times T$.

We state two facts. Let $X_1, \dots, X_n$ be abelian varieties and $s_1, \dots, s_n$ positive integers, and let

$$\Delta : \prod_{i=1}^n \mathrm{GL}(H^1(X_i,\mathbb{Q})) \to \mathrm{GL}\Big(\bigoplus_{i=1}^n H^1(X_i,\mathbb{Q})^{\oplus s_i}\Big), \quad (g_1,\dots,g_n) \to (g_1^{\oplus s_1},\dots,g_n^{\oplus s_n}).$$

By Lemma 2.3 (3),

$$\mathrm{Hg}(X_1 \times \cdots \times X_n) \subset \prod_{i=1}^n \mathrm{GL}(H^1(X_i,\mathbb{Q})).$$

Then

$$\mathrm{Hg}(X_1^{s_1} \times \cdots \times X_n^{s_n}) = \Delta\big(\mathrm{Hg}(X_1 \times \cdots \times X_n)\big). \tag{8.2}$$

We have

$$\eta_{X_1^{s_1} \times \cdots \times X_n^{s_n}} = \Delta \circ \eta_{X_1 \times \cdots \times X_n},$$

so the left-hand side of (8.2) is contained in the right-hand side by the minimality in Definition 2.2. Conversely, Δ is a closed embedding defined over $\mathbb{Q}$, and we let $q := \Delta^{-1}$ on the image of Δ, so that $q(g_1^{\oplus s_1},\dots,g_n^{\oplus s_n}) = (g_1,\dots,g_n)$. By the first inclusion, $\mathrm{Hg}(X_1^{s_1} \times \cdots \times X_n^{s_n})$ is contained in the image of Δ, so $q(\mathrm{Hg}(X_1^{s_1} \times \cdots \times X_n^{s_n}))$ is an algebraic subgroup defined over $\mathbb{Q}$ whose real points contain

$$q(\eta_{X_1^{s_1} \times \cdots \times X_n^{s_n}}(\mathbb{U})) = \eta_{X_1 \times \cdots \times X_n}(\mathbb{U}).$$

Hence it contains $\mathrm{Hg}(X_1 \times \cdots \times X_n)$, and

$$\mathrm{Hg}(X_1^{s_1} \times \cdots \times X_n^{s_n}) = \Delta\big(q(\mathrm{Hg}(X_1^{s_1} \times \cdots \times X_n^{s_n}))\big)$$

contains $\Delta(\mathrm{Hg}(X_1 \times \cdots \times X_n))$. The second fact is the following. Let X and Y be abelian varieties such that

$$\mathrm{Hg}(X \times Y) = \mathrm{Hg}(X) \times \mathrm{Hg}(Y).$$

Since the invariants of a product of groups acting on a tensor product of representations are the tensor products of the invariants, Lemma 2.3 (2) and the Künneth formula [Moo99, (2.1)] imply that

$$\mathrm{Hdg}^p(X \times Y) = \sum_{a+b=p} \mathrm{pr}_X^* \mathrm{Hdg}^a(X) \cup \mathrm{pr}_Y^* \mathrm{Hdg}^b(Y) \tag{8.3}$$

for any p.

CM elliptic curves with complex multiplication by one imaginary quadratic field are isogenous [MZ99, (4.1)]. Thus, by [Moo99, (2.3)], T is isogenous to

$$T' := C_1^{s_1} \times \cdots \times C_m^{s_m} \times E^r \times E_1^{r_1} \times \cdots \times E_k^{r_k},$$

where $C_1, \ldots, C_m$ are pairwise non-isogenous elliptic curves without complex multiplication, $r \geq 0$, E is an elliptic curve with complex multiplication by K, E_j is an elliptic curve with complex multiplication by an imaginary quadratic field K_j, $K_1, \ldots, K_k$ are pairwise distinct and distinct from K, and $s_i, r_j \geq 1$. Here $r = 0$ if F_B contains no imaginary quadratic field. When $r \geq 1$, we let K act on E with multiplicities $(0, 1)$ on $H^{1,0}(E)$. An elliptic curve E' is the image of a surjective homomorphism $T \to E'$ if and only if E' is isogenous to a simple factor of T' [Moo99, (2.3)]. Hence an elliptic curve E' as in (1) exists if and only if $r \geq 1$.

Let $h : T \to T'$ be an isogeny. Then

$$(\mathrm{id}_B \times h)^* : H^*(B \times T', \mathbb{Q}) \to H^*(B \times T, \mathbb{Q})$$

is an isomorphism of $\mathbb{Q}$-algebras and of rational Hodge structures, so $(\mathrm{id}_B \times h)^*$ maps the Hodge algebra of $B \times T'$ onto the Hodge algebra of $B \times T$. Moreover, $(\mathrm{id}_B \times h)^*$ maps $\mathcal{D}^p(B \times T')$ into $\mathcal{D}^p(B \times T)$, and

$$(\mathrm{id}_B \times h)^*(\mathrm{id}_B \times f)^* = (\mathrm{id}_B \times (f \circ h))^*,$$

where $f \circ h$ is surjective if f is surjective. Together with Lemma 2.6 and the previous paragraph, (1) for $B \times T'$ implies (1) for $B \times T$. Possibly replacing T with T', we may assume that $T = T'$.

Step 2. In this step, we split off the factors $C_1^{s_1}, \ldots, C_m^{s_m}$.

By [MZ99, Proposition (3.8)] (see also [MZ99, Proposition (2.4)(i),(iii), Lemma (3.4)]), if C is an elliptic curve without complex multiplication and Y is an abelian variety with $\mathrm{Hom}(C, Y) = 0$, then

$$\mathrm{Hg}(C \times Y) = \mathrm{Hg}(C) \times \mathrm{Hg}(Y).$$

Let $Y := B \times E^r \times E_1^{r_1} \times \cdots \times E_k^{r_k}$. For any i, C_i is not isogenous to any simple factor of $C_{i+1} \times \cdots \times C_m \times Y$, so $\mathrm{Hom}(C_i, C_{i+1} \times \cdots \times C_m \times Y) = 0$ [Moo99, (2.3)]. By induction on m, we have

$$\mathrm{Hg}(C_1 \times \cdots \times C_m \times Y) = \mathrm{Hg}(C_1) \times \cdots \times \mathrm{Hg}(C_m) \times \mathrm{Hg}(Y).$$

Applying (8.2) twice, once to $C_1 \times \cdots \times C_m \times Y$ and once to each C_i, we obtain

$$\mathrm{Hg}(B \times T) = \mathrm{Hg}(C_1^{s_1}) \times \cdots \times \mathrm{Hg}(C_m^{s_m}) \times \mathrm{Hg}(Y).$$

Since $\mathrm{Hdg}^p(C_i^{s_i}) = \mathcal{D}^p(C_i^{s_i})$ for any p [MZ99, Theorem (0.1)(iv)], (8.3) applied m times implies that the Hodge algebra of $B \times T$ is generated by the classes of divisors and by the pullbacks of the Hodge algebra of Y under the projection $B \times T \to Y$. For

$$f : E^r \times E_1^{r_1} \times \cdots \times E_k^{r_k} \to E$$

surjective, the pullback of $(\mathrm{id}_B \times f)^* W_K(B \times E)$ under the projection $B \times T \to Y$ is $(\mathrm{id}_B \times f')^* W_K(B \times E)$, where f' is the composition of the projection $T \to E^r \times E_1^{r_1} \times \cdots \times E_k^{r_k}$

with f. Thus (1) for Y implies (1) for $B \times T$. Possibly replacing T with $E^r \times E_1^{r_1} \times \cdots \times E_k^{r_k}$, we may assume that $m = 0$.

Step 3. In this step, we compute $\mathrm{Hg}(B \times E^r)$ when $r \geq 1$.

By [Moo99, (2.6)], $\mathrm{Hg}(E) = \mathrm{U}_K$, where $t \in \mathrm{U}_K$ acts on $H^1(E, \mathbb{C})_\sigma$ by $\sigma(t)$. Since the multiplicities of K on $H^{1,0}(E)$ are $(0, 1)$, we have $H^1(E, \mathbb{C})_\sigma = H^{0,1}(E)$, so $\sigma(\eta_E(z)) = \bar{z}$ for any $z \in \mathbb{U}$. Let

$$\Gamma_1 := \{(g, t) \in H_B \times \mathrm{U}_K \mid \chi(g)t = 1\}.$$

By (8.1), $\sigma(\chi(\eta_B(z))\eta_E(z)) = z\bar{z} = 1$, so $\eta_{B\times E}(\mathbb{U}) \subset \Gamma_1(\mathbb{R})$. Since Γ_1 is the graph of the homomorphism $H_B \to \mathrm{U}_K$, $g \to \chi(g)^{-1}$, Γ_1 is an algebraic subgroup defined over $\mathbb{Q}$, and the projection $\Gamma_1 \to H_B$ is an isomorphism. By the minimality of the Hodge group, $\mathrm{Hg}(B \times E) \subset \Gamma_1$. By Lemma 2.3 (3), $\mathrm{Hg}(B \times E) \to H_B$ is surjective, hence $\mathrm{Hg}(B \times E) = \Gamma_1$. By (8.2),

$$\mathrm{Hg}(B \times E^r) = \{(g, \Delta(t)) \mid g \in H_B,\ t \in \mathrm{U}_K,\ \chi(g)t = 1\},$$

where

$$\Delta : \mathrm{GL}(H^1(E, \mathbb{Q})) \to \mathrm{GL}(H^1(E, \mathbb{Q})^{\oplus r})$$

is the diagonal embedding, and the projection $\mathrm{Hg}(B \times E^r) \to H_B$ is an isomorphism.

We let $H := \mathrm{Hg}(B \times E^r)$ if $r \geq 1$, and $H := H_B$ if $r = 0$. Then H is connected and reductive (Lemma 2.3 (1)), and the projection $H \to H_B$ is an isomorphism.

Step 4. In this step, we show that

$$\mathrm{Hg}(B \times E^r \times E_1^{r_1} \times \cdots \times E_k^{r_k}) = H \times \mathrm{Hg}(E_1^{r_1}) \times \cdots \times \mathrm{Hg}(E_k^{r_k}),$$

and we prove (1) when $r = 0$.

For a connected reductive group G over $\mathbb{Q}$, $X^*(G/G^{\mathrm{der}}) \otimes \mathbb{Q}$ is a representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ through a finite quotient, so any subrepresentation of $X^*(G/G^{\mathrm{der}}) \otimes \mathbb{Q}$ is a direct sum of irreducible constituents. By Lemma 8.5, for two distinct imaginary quadratic fields K' and K'', the one-dimensional representations $X^*(\mathrm{U}_{K'}) \otimes \mathbb{Q}$ and $X^*(\mathrm{U}_{K''}) \otimes \mathbb{Q}$ are not isomorphic, since the kernels of the Galois actions are $\mathrm{Gal}(\overline{\mathbb{Q}}/K')$ and $\mathrm{Gal}(\overline{\mathbb{Q}}/K'')$.

Let ψ be the alternating form on $H^1(B, \mathbb{Q})$ induced by a polarization of B. We list $X^*(H_B/H_B^{\mathrm{der}}) \otimes \mathbb{Q}$ according to F_B.

(1) If F_B is $\mathbb{Q}$ or a totally real cubic field, then H_B is semisimple [Moo99, Proposition (1.24)], so $X^*(H_B/H_B^{\mathrm{der}}) \otimes \mathbb{Q} = 0$.

(2) If $F_B = K$, then $H_B = \mathrm{U}_K(H^1(B, \mathbb{Q}), \psi)$ [MZ99, Theorem (2.7)], $H_B^{\mathrm{der}} = \mathrm{SU}_K(H^1(B, \mathbb{Q}), \psi)$, and $\det_K$ induces an isomorphism $H_B/H_B^{\mathrm{der}} \cong \mathrm{U}_K$. Thus

$$X^*(H_B/H_B^{\mathrm{der}}) \otimes \mathbb{Q} \cong X^*(\mathrm{U}_K) \otimes \mathbb{Q}.$$

(3) If F_B is a CM field of degree 6 which contains no imaginary quadratic subfield, then $H_B = \mathrm{U}_{F_B}$ [MZ99, Theorem (2.7)], and U_{F_B} has no subtorus other than $\{1\}$ and U_{F_B} [MZ99, Proposition (2.4)(ii), Lemma (3.7)]. By [Moo99, (1.1)], $X^*(H_B/H_B^{\mathrm{der}}) \otimes \mathbb{Q}$ is irreducible, and $\dim X^*(H_B/H_B^{\mathrm{der}}) \otimes \mathbb{Q} = 3$.

(4) If F_B is a CM field of degree 6 which contains K, then $H_B = \mathrm{U}_{F_B}$ [MZ99, Theorem (2.7)] and $F_B = KF_B^+$, where F_B^+ is the maximal totally real subfield of F_B. Then

$$\mathrm{Hom}(F_B, \overline{\mathbb{Q}}) = \mathrm{Hom}(K, \overline{\mathbb{Q}}) \times \mathrm{Hom}(F_B^+, \overline{\mathbb{Q}})$$

as Galois sets, and the complex conjugation of F_B acts on the first factor. Let $P_B \subset \mathbb{Q}[\mathrm{Hom}(F_B^+, \overline{\mathbb{Q}})]$ be the subspace of vectors with coordinate sum 0. By Lemma 8.5, we obtain

$$X^*(H_B/H_B^{\mathrm{der}}) \otimes \mathbb{Q} \cong \big(X^*(\mathrm{U}_K) \otimes \mathbb{Q}\big) \otimes \mathbb{Q}[\mathrm{Hom}(F_B^+, \overline{\mathbb{Q}})] = X^*(\mathrm{U}_K) \otimes \mathbb{Q} \oplus \big(\big(X^*(\mathrm{U}_K) \otimes \mathbb{Q}\big) \otimes P_B\big).$$

The Galois group acts transitively on the three elements of $\mathrm{Hom}(F_B^+, \overline{\mathbb{Q}})$, so some element c acts as a 3-cycle. If a line $\mathbb{Q}v \subset P_B$ is Galois stable, then $cv = \pm v$ and $c^3 v = v$, so

$cv = v$, and $v = 0$ since the coordinates of v are equal with sum 0. Thus P_B is irreducible, and $\big(X^*(\mathrm{U}_K) \otimes \mathbb{Q}\big) \otimes P_B$ is irreducible with

$$\dim \big(X^*(\mathrm{U}_K) \otimes \mathbb{Q}\big) \otimes P_B = 2.$$

In all cases, any one-dimensional subrepresentation of $X^*(H_B/H_B^{\mathrm{der}}) \otimes \mathbb{Q}$ is isomorphic to $X^*(\mathrm{U}_K) \otimes \mathbb{Q}$.

By [Moo99, (2.6)], $\mathrm{Hg}(E_j) = \mathrm{U}_{K_j}$ for any j. Let

$$\Gamma := \mathrm{Hg}(B \times E^r \times E_1 \times \cdots \times E_k).$$

By Lemma 2.3 (1), (3), Γ is a connected algebraic subgroup of

$$H \times \mathrm{U}_{K_1} \times \cdots \times \mathrm{U}_{K_k}$$

whose projection to each factor is surjective. Since $H \cong H_B$ (**Step 3**),

$$X^*(H/H^{\mathrm{der}}) \otimes \mathbb{Q} \cong X^*(H_B/H_B^{\mathrm{der}}) \otimes \mathbb{Q}.$$

Since $K_j \neq K$, $X^*(\mathrm{U}_{K_j}) \otimes \mathbb{Q}$ is not isomorphic to a subrepresentation of $X^*(H/H^{\mathrm{der}}) \otimes \mathbb{Q}$ for any j. By Lemma 8.6,

$$\Gamma = H \times \mathrm{U}_{K_1} \times \cdots \times \mathrm{U}_{K_k}.$$

Applying (8.2) to $(B \times E^r) \times E_1^{r_1} \times \cdots \times E_k^{r_k}$ and to each $E_j^{r_j}$, we obtain the equality claimed in this step.

By [MZ99, Theorem (0.1)(iv)], $\mathrm{Hdg}^p(E_j^{r_j}) = \mathcal{D}^p(E_j^{r_j})$ for any p and j. By (8.3), the Hodge algebra of $B \times T$ is generated by the classes of divisors and by the pullbacks of the Hodge algebra of $B \times E^r$ under the projection $B \times T \to B \times E^r$. If $r = 0$, then $\mathrm{Hdg}^p(B) = \mathcal{D}^p(B)$ for any p [MZ99, Theorem (2.7)], so $\mathrm{Hdg}^p(B \times T) = \mathcal{D}^p(B \times T)$ for any p. This proves (1) when $r = 0$, which is the case when no elliptic curve as in (1) exists (**Step 1**). If $r \geq 1$, then as in **Step 2**, it suffices to prove (1) for $B \times E^r$.

Step 5. In this step, we assume that $r \geq 1$, and we show that the Hodge algebra of $B \times E^r$ is generated by the classes of divisors and by the pullbacks $(\mathrm{id}_B \times \mathrm{pr}_i)^* W_K(B \times E)$, where $\mathrm{pr}_i : E^r \to E$ is the i-th projection. This proves (1).

Let $L_\sigma := \wedge^3 H^1(B, \mathbb{C})_\sigma$ and $L_{\bar\sigma} := \wedge^3 H^1(B, \mathbb{C})_{\bar\sigma}$. For $i = 1, \ldots, r$, let e_i and f_i be non-zero vectors in $\mathrm{pr}_i^* H^1(E, \mathbb{C})_\sigma$ and $\mathrm{pr}_i^* H^1(E, \mathbb{C})_{\bar\sigma}$, respectively. Then $e_1, \ldots, e_r, f_1, \ldots, f_r$ form a basis of $H^1(E^r, \mathbb{C})$, and $t \in \mathrm{U}_K$ acts on e_i by $\sigma(t)$ and on f_i by $\sigma(t)^{-1}$. For $I, J \subset \{1, \ldots, r\}$, let $e_I \wedge f_J$ be the wedge product of the e_i with $i \in I$ and the f_j with $j \in J$, in increasing order. Then t acts on $e_I \wedge f_J$ by $\sigma(t)^{|I|-|J|}$. For an integer n, let $M_n \subset \wedge^* H^1(E^r, \mathbb{C})$ be the span of the $e_I \wedge f_J$ with $|I| - |J| = n$.

By **Step 3**, the elements of $\mathrm{Hg}(B \times E^r)$ are $(g, \Delta(\chi(g)^{-1}))$ with $g \in H_B$, and $\mathrm{Hg}(B \times E^r)$ contains $N \times \{1\}$. By Lemma 2.3 (2), the Künneth formula [Moo99, (2.1)], and Lemma 8.7, $\mathrm{Hdg}^*(B \times E^r) \otimes \mathbb{C}$ is the subspace of $\mathrm{Hg}(B \times E^r)$-invariants in

$$\Big(\mathrm{Hdg}^*(B) \otimes \mathbb{C} \oplus L_\sigma \oplus L_{\bar\sigma}\Big) \otimes \wedge^* H^1(E^r, \mathbb{C}).$$

Let $g \in H_B$ and $t := \chi(g)^{-1}$. By Lemma 8.7, $(g, \Delta(t))$ acts on $\mathrm{Hdg}^p(B) \otimes M_n$ by $\sigma(t)^n$, on $L_\sigma \otimes M_n$ by $\sigma(t)^{n-1}$, and on $L_{\bar\sigma} \otimes M_n$ by $\sigma(t)^{n+1}$. Since χ is surjective (**Step 1**), t runs through U_K, and $\sigma : \mathrm{U}_K(\mathbb{C}) \to \mathbb{C}^*$ is an isomorphism. Thus

$$\mathrm{Hdg}^*(B \times E^r) \otimes \mathbb{C} = \Big(\mathrm{Hdg}^*(B) \otimes \mathbb{C} \otimes M_0\Big) \oplus \big(L_\sigma \otimes M_1\big) \oplus \big(L_{\bar\sigma} \otimes M_{-1}\big). \tag{8.4}$$

We describe M_0, M_1, and M_{-1}. Each $e_i \wedge f_j$ is $\mathrm{Hg}(E^r)$-invariant, so

$$e_i \wedge f_j \in \mathrm{Hdg}^1(E^r) \otimes \mathbb{C} = \mathcal{D}^1(E^r) \otimes \mathbb{C}$$

by Lemma 2.3 (2) and the Lefschetz $(1,1)$ theorem [Moo99, (2.10)]. If $|I| = |J| = b$, then

$$e_I \wedge f_J = \pm(e_{i_1} \wedge f_{j_1}) \wedge \cdots \wedge (e_{i_b} \wedge f_{j_b}),$$

where $I = \{i_1, \dots, i_b\}$ and $J = \{j_1, \dots, j_b\}$. If $|I| = |J| + 1$, then $e_I \wedge f_J = \pm e_{i_0} \wedge e_{I'} \wedge f_J$ with $I = I' \sqcup \{i_0\}$ and $|I'| = |J|$. We obtain

$$M_0 = \mathcal{D}^*(E^r) \otimes \mathbb{C}, \quad M_1 = \sum_{i=1}^{r} e_i \wedge M_0, \quad \text{and} \quad M_{-1} = \sum_{i=1}^{r} f_i \wedge M_0.$$

By Lemma 2.3 (2) and [MZ99, Theorem (2.7)], $\mathrm{Hdg}^p(B) = \mathcal{D}^p(B)$ for any p, so the first summand of (8.4) is contained in the complexification of the divisor algebra of $B \times E^r$. Since

$$H^1(B \times E, \mathbb{C})_\sigma = H^1(B, \mathbb{C})_\sigma \oplus H^1(E, \mathbb{C})_\sigma$$

with $\dim H^1(E, \mathbb{C})_\sigma = 1$, the Künneth formula implies that

$$\begin{aligned} L_\sigma \otimes \mathbb{C}e_i \oplus L_{\bar\sigma} \otimes \mathbb{C}f_i &= (\mathrm{id}_B \times \mathrm{pr}_i)^* \big(\wedge^4 H^1(B \times E, \mathbb{C})_\sigma \oplus \wedge^4 H^1(B \times E, \mathbb{C})_{\bar\sigma} \big) \\ &= (\mathrm{id}_B \times \mathrm{pr}_i)^* W_K(B \times E) \otimes \mathbb{C}. \end{aligned}$$

Therefore, the second and the third summands of (8.4) are spanned by the products of the classes in

$$(\mathrm{id}_B \times \mathrm{pr}_i)^* W_K(B \times E) \otimes \mathbb{C}$$

with the pullbacks of $\mathcal{D}^*(E^r) \otimes \mathbb{C}$. Let $\mathcal{A}$ be the subalgebra of $H^*(B \times E^r, \mathbb{Q})$ generated by the classes of divisors and by $(\mathrm{id}_B \times \mathrm{pr}_i)^* W_K(B \times E)$ for $i = 1, \dots, r$. By **Step 1**, $\mathcal{A}$ is contained in the Hodge algebra of $B \times E^r$, and by (8.4), $\mathcal{A} \otimes \mathbb{C}$ contains the complexification of the Hodge algebra of $B \times E^r$. Hence $\mathcal{A}$ is the Hodge algebra of $B \times E^r$.

The fourfold $B \times E$ satisfies $\mathrm{HC}(B \times E)$ (Theorem 2.16), so any class in $W_K(B \times E)$ is algebraic. The pullbacks of algebraic classes under homomorphisms and the cup products of algebraic classes are algebraic, and the classes of divisors are algebraic. Thus any class in $\mathcal{A}$ is algebraic, and $\mathrm{HC}(B \times E^r)$ holds. Together with **Steps 1, 2, and 4**, this proves (1).

Step 6. In this step, we show that $\mathrm{HC}(B \times S \times E')$ holds for any simple abelian surface S and any elliptic curve E'.

By (1), $\mathrm{HC}(B \times E')$ holds. By [MZ99, Theorem (0.1)(iv)], $\mathrm{Hdg}^p(S) = \mathcal{D}^p(S)$ for any p. By (8.3), it suffices to show that

$$\mathrm{Hg}(B \times E' \times S) = \mathrm{Hg}(B \times E') \times \mathrm{Hg}(S).$$

If S is not of CM type, then the equality follows from [MZ99, Proposition (2.4)(i),(iii), Lemma (3.4)], since $\mathrm{Hom}(S, B \times E') = 0$ [Moo99, (2.3)].

Assume that S is of CM type. By [MZ99, (2.2)], $F_S := \mathrm{End}^0(S)$ is a quartic CM field which contains no imaginary quadratic subfield, and $\mathrm{Hg}(S) = \mathrm{U}_{F_S}$. Let L_S be the normal closure of F_S. By Lemma 8.5, $X^*(\mathrm{U}_{F_S}) \otimes \mathbb{Q}$ is an irreducible representation of dimension 2, and the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ in

$$\mathrm{GL}(X^*(\mathrm{U}_{F_S}) \otimes \mathbb{Q})$$

is isomorphic to $\mathrm{Gal}(L_S/\mathbb{Q})$, which has order 4 or 8.

Let $G_1 := \mathrm{Hg}(B \times E')$. By **Steps 2, 3, and 4** applied to $T = E'$, either $G_1 \cong H_B$, or $G_1 = H_B \times \mathrm{Hg}(E')$, where

$$X^*(\mathrm{Hg}(E')/\mathrm{Hg}(E')^{\mathrm{der}}) \otimes \mathbb{Q}$$

is 0 or one-dimensional [Moo99, (2.6)]. By the list in **Step 4**, an irreducible constituent of $X^*(G_1/G_1^{\mathrm{der}}) \otimes \mathbb{Q}$ of dimension 2 exists only in case (4) of the list, and then it is isomorphic to $\big(X^*(\mathrm{U}_K) \otimes \mathbb{Q}\big) \otimes P_B$. Let M' be the normal closure of F_B^+. Since M' is totally real, $K \not\subset M'$, so

$$\mathrm{Gal}(KM'/\mathbb{Q}) \cong \mathrm{Gal}(K/\mathbb{Q}) \times \mathrm{Gal}(M'/\mathbb{Q}),$$

which is of order 6 or 12. The Galois group acts on $\big(X^*(\mathrm{U}_K) \otimes \mathbb{Q}\big) \otimes P_B$ through $\mathrm{Gal}(KM'/\mathbb{Q})$, and

$$(\epsilon, \gamma) \in \mathrm{Gal}(K/\mathbb{Q}) \times \mathrm{Gal}(M'/\mathbb{Q})$$

acts by $\pm\rho(\gamma)$, where ρ is the action on P_B and the sign is -1 if and only if $\epsilon \neq 1$. The representation ρ of $\mathrm{Gal}(M'/\mathbb{Q})$ is faithful, since $\mathrm{Gal}(M'/\mathbb{Q})$ acts faithfully on $\mathrm{Hom}(F_B^+, \overline{\mathbb{Q}})$, and

$-\mathrm{id}_{P_B}$ is not in the image of ρ, since the elements of order 3 act by 3-cycles and the elements of order 2 act by transpositions, whose eigenvalues on P_B are 1 and -1. Thus the image of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ in

$$\mathrm{GL}\big(\big(X^*(\mathrm{U}_K)\otimes\mathbb{Q}\big)\otimes P_B\big)$$

is of order 6 or 12. Isomorphic representations have equal kernels, hence images of equal order, so $X^*(\mathrm{U}_{F_S})\otimes\mathbb{Q}$ is not isomorphic to $\big(X^*(\mathrm{U}_K)\otimes\mathbb{Q}\big)\otimes P_B$. We conclude that no non-zero subrepresentation of $X^*(\mathrm{U}_{F_S})\otimes\mathbb{Q}$ is isomorphic to a subrepresentation of $X^*(G_1/G_1^{\mathrm{der}})\otimes\mathbb{Q}$.

Let $G_2 := \mathrm{Hg}(S)$ and $\Gamma' := \mathrm{Hg}(B\times E'\times S)$. By Lemma 2.3 (1), (3), Γ' is a connected algebraic subgroup of $G_1\times G_2$ whose projections to G_1 and to G_2 are surjective. By the previous paragraph and Lemma 8.6 with $H=G_1$, $k=1$, and $T_1=G_2$, we get $\Gamma'=G_1\times G_2$, as claimed.

Step 7. In this step, we prove (2).

Let $B\subset A$ be a simple abelian threefold. By the Poincaré complete reducibility theorem [Moo99, (1.15), Theorem (2.2)], there exists an isogeny $B\times Y\to A$, where Y is an abelian threefold. Up to isogeny, Y is a simple abelian threefold, or the product of a simple abelian surface and an elliptic curve, or a product of three elliptic curves [Moo99, (2.3)]. In the first case, $\mathrm{HC}(A)$ holds by Theorem 8.8. In the second case and in the third case, $\mathrm{HC}(B\times Y')$ holds for an abelian threefold Y' isogenous to Y by **Step 6** and by (1), respectively. Since $B\times Y'$ is isogenous to A, $\mathrm{HC}(A)$ holds by Lemma 2.6. □

8.5. **A fourfold times a surface.** In this subsection, we treat products $B\times S$ of a simple abelian fourfold and an abelian surface. Sixfolds with a simple factor of dimension three or five are treated in Theorem 8.9 and Proposition 8.4, and sixfolds with a simple factor of dimension one or two that is not of CM type are treated in Subsection 8.6. Apart from such sixfolds, the only remaining obstruction to the Hodge conjecture on nonsimple sixfolds which are not of CM type occurs on the products $B\times S$ such that (B,S) is an exceptional pair in the sense of Definition 8.12.

For an algebraic group H and a character χ of H, written additively, we say that a vector of a representation of H has *weight* χ if H acts on it through χ.

Lemma 8.10. *Let B be a simple abelian fourfold such that $F:=\mathrm{End}^0(B)$ is a CM field of degree 4 and, for an ordering $\phi_1,\phi_2,\bar\phi_1,\bar\phi_2$ of the complex embeddings of F, the multiplicities of F on $H^{1,0}(B)$ at $\phi_1,\phi_2,\bar\phi_1,\bar\phi_2$ are $(2,1,0,1)$. Let $V:=H^1(B,\mathbb{Q})$ and ψ the alternating form on V induced by a polarization of B, and write $V_l:=H^1(B,\mathbb{C})_{\phi_l}$ for $l=1,2$. Then the following hold.*

(1) $\mathrm{Hg}(B)=\mathrm{U}_F(V,\psi)$,

$$\mathrm{Hg}(B)_{\mathbb{C}}\cong\mathrm{GL}(V_1)\times\mathrm{GL}(V_2),$$

the F-determinant $\det_F:\mathrm{Hg}(B)\to\mathrm{U}_F$ is surjective with connected kernel, and $\mathrm{Hdg}^p(B)=\mathcal{D}^p(B)$ for any p.

(2) Let d_l be the character $\det V_l$ of $\mathrm{Hg}(B)_{\mathbb{C}}$. Then $d_1\circ\eta_B$ and $d_2\circ\eta_B$ are z^2 and 1.

(3) Any imaginary quadratic subfield of F has multiplicities $(3,1)$ or $(1,3)$ on $H^{1,0}(B)$.

(4) The subspace $U_B\subset H^2(B,\mathbb{Q})$ with

$$U_B\otimes\mathbb{C}=\bigoplus_{\tau}\wedge^2H^1(B,\mathbb{C})_\tau,$$

the sum running over the four complex embeddings τ of F, is a rational Hodge substructure, and $\mathrm{Hg}(B)$ acts on it with the four weights $\pm d_1,\pm d_2$.

(5) For $l=1,2$, the $\mathrm{SL}(V_l)$-invariants in $\wedge^*(V_l\oplus V_l^*)$ are spanned by 1 in degree 0, by a generator x_l of the line $\wedge^2V_l$, a generator y_l of the line $\wedge^2V_l^*$, and a class $q_l\in V_l\otimes V_l^*$ in degree 2, of weights d_l, $-d_l$, and 0, and by $q_l^2\neq0$ in degree 4; no non-zero invariant has odd degree, and $q_lx_l=q_ly_l=0$.

Proof. Since $\dim_F V = 2$, we have $\dim H^1(B,\mathbb{C})_\tau = 2$ for any complex embedding τ of F. As F preserves the Hodge decomposition,

$$H^1(B,\mathbb{C})_\tau = \big(H^1(B,\mathbb{C})_\tau \cap H^{1,0}(B)\big) \oplus \big(H^1(B,\mathbb{C})_\tau \cap H^{0,1}(B)\big)$$

for any τ. Thus $V_1 \subset H^{1,0}(B)$, and both $V_2 \cap H^{1,0}(B)$ and $V_2 \cap H^{0,1}(B)$ are lines.

We first prove (3). Let $K \subset F$ be an imaginary quadratic subfield and σ a complex embedding of K. The multiplicity of K on $H^{1,0}(B)$ at σ is the sum of the multiplicities of F at both embeddings of F extending σ. If τ extends σ, then $\bar\tau$ extends $\bar\sigma \neq \sigma$. Hence exactly one of $\phi_1, \bar\phi_1$ and exactly one of $\phi_2, \bar\phi_2$ extend σ, and the multiplicity of K at σ is $2+1$ or $0+1$. This proves (3).

(1) The variety B is simple, F is neither $\mathbb{Q}$ nor a quaternion algebra, and by (3) no imaginary quadratic subfield of F has multiplicities $(2,2)$. Thus none of the exceptional cases of [MZ99, Theorem (0.1)] occurs, and [MZ99, Theorem (0.1)(iv)] implies that $\mathrm{Hg}(B)$ is the centralizer of F in the symplectic group of ψ, that is, $\mathrm{Hg}(B) = \mathrm{U}_F(V,\psi)$, and that $\mathrm{Hdg}^p(B) = \mathcal{D}^p(B)$ for any p. Since $\psi(ax,y) = \psi(x,\bar a y)$ for $a \in F$, the form ψ pairs $H^1(B,\mathbb{C})_\tau$ and $H^1(B,\mathbb{C})_{\tau'}$ trivially if $\tau' \neq \bar\tau$, and perfectly if $\tau' = \bar\tau$. Thus ψ induces an isomorphism $H^1(B,\mathbb{C})_{\bar\phi_l} \cong V_l^*$ of representations of $\mathrm{Hg}(B)_\mathbb{C}$, and restriction to V_1 and V_2 induces

$$\mathrm{Hg}(B)_\mathbb{C} \cong \mathrm{GL}(V_1) \times \mathrm{GL}(V_2).$$

Under the isomorphisms

$$\mathrm{Hg}(B)_\mathbb{C} \cong \mathrm{GL}(V_1) \times \mathrm{GL}(V_2)$$

and the isomorphism $\mathrm{U}_{F,\mathbb{C}} \cong \mathbb{G}_m^2$ given by ϕ_1 and ϕ_2, the homomorphism $\det_F$ is $(g_1,g_2) \to (\det g_1, \det g_2)$, which is surjective with kernel $\mathrm{SL}(V_1) \times \mathrm{SL}(V_2)$.

(2) The Hodge circle η_B takes values in $\mathrm{Hg}(B)(\mathbb{R})$, so $\eta_B(z)$ preserves V_1 and V_2. It acts on $V_1 \subset H^{1,0}(B)$ by z, and on V_2 with eigenvalues z and $\bar z$. Hence $d_1(\eta_B(z)) = z^2$ and $d_2(\eta_B(z)) = z\bar z = 1$.

(4) Let $\mathrm{Aut}(\mathbb{C})$ act on $H^*(B,\mathbb{C}) = H^*(B,\mathbb{Q}) \otimes \mathbb{C}$ through the second factor. Since F acts on $H^1(B,\mathbb{Q})$, any $\gamma \in \mathrm{Aut}(\mathbb{C})$ maps $H^1(B,\mathbb{C})_\tau$ onto $H^1(B,\mathbb{C})_{\gamma\circ\tau}$. Thus $\bigoplus_\tau \wedge^2 H^1(B,\mathbb{C})_\tau$ is stable under $\mathrm{Aut}(\mathbb{C})$, so it is $U_B \otimes \mathbb{C}$ for a subspace $U_B \subset H^2(B,\mathbb{Q})$. The group $\mathrm{Hg}(B)$ commutes with F, so it acts on the lines $\wedge^2 V_l$ and $\wedge^2 H^1(B,\mathbb{C})_{\bar\phi_l} \cong \wedge^2 V_l^*$ by d_l and $-d_l$. In particular, U_B is $\mathrm{Hg}(B)$-stable, so U_B is a rational Hodge substructure (cf. [Moo99, (1.12)]).

(5) For $k \geq 0$, we have

$$\wedge^k(V_l \oplus V_l^*) = \bigoplus_{a+b=k} \wedge^a V_l \otimes \wedge^b V_l^*.$$

As $\dim V_l = 2$, the summands for $k = 1$ are V_l and V_l^*, which have no non-zero invariants. For $k = 2$, the summands are the trivial representations $\wedge^2 V_l$ and $\wedge^2 V_l^*$ of $\mathrm{SL}(V_l)$ and $V_l \otimes V_l^* \cong \mathrm{End}(V_l)$, whose invariants are spanned by the class q_l of id_{V_l}. The group $\mathrm{GL}(V_l)$ acts on x_l, y_l, and q_l by d_l, $-d_l$, and 0. The line $\wedge^4(V_l \oplus V_l^*) = \wedge^2 V_l \otimes \wedge^2 V_l^*$ is trivial, so wedge product induces an $\mathrm{SL}(V_l)$-equivariant isomorphism between $\wedge^{4-k}(V_l \oplus V_l^*)$ and the dual of $\wedge^k(V_l \oplus V_l^*)$. Hence there are no non-zero invariants in degree 3, and the invariants in degree 4 form the line $\wedge^4(V_l \oplus V_l^*)$. For a basis e_1, e_2 of V_l with dual basis e_1^*, e_2^*, we have $q_l = e_1 \wedge e_1^* + e_2 \wedge e_2^*$, so $q_l^2 = 2e_1 \wedge e_1^* \wedge e_2 \wedge e_2^* \neq 0$ spans $\wedge^4(V_l \oplus V_l^*)$. Finally, $q_l x_l \in \wedge^3 V_l \otimes V_l^* = 0$ and $q_l y_l \in V_l \otimes \wedge^3 V_l^* = 0$. □

Lemma 8.11. *Let K_1 and K_2 be distinct imaginary quadratic fields, $F := K_1K_2$, and $\phi_1, \phi_2 : F \hookrightarrow \mathbb{C}$ embeddings with $\phi_1|_{K_1} = \phi_2|_{K_1}$ and $\phi_1|_{K_2} = \bar\phi_2|_{K_2}$. Let B be a simple abelian fourfold with $\mathrm{End}^0(B) = F$ and multiplicities $(2,1,0,1)$ at $\phi_1, \phi_2, \bar\phi_1, \bar\phi_2$, let E and E' be elliptic curves, and let $A := B \times E \times E'$.*

1. *If E and E' do not both have complex multiplication by fields in $\{K_1, K_2\}$, then $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p.*
2. *If E and E' both have complex multiplication by K_l for some $l \in \{1,2\}$, then $\mathrm{Hdg}^2(A) = \mathcal{D}^2(A)$ and $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_{K_l}(A)$, where K_l acts on E and E' so that A has multiplicities $(3,3)$. In particular, $\mathrm{HC}(A)$ holds.*

(3) If E has complex multiplication by K_1 and E' by K_2, then there exists no unital embedding of an imaginary quadratic field into $\mathrm{End}^0(A)$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix notation and describe the Galois module $X^*(\mathrm{U}_F)\otimes\mathbb{Q}$.

We use the notation V, ψ, V_l, d_l, x_l, y_l, q_l of Lemma 8.10. Since $\psi(ax,y)=\psi(x,\bar{a}y)$ for $a\in F$, the form ψ pairs $H^1(B,\mathbb{C})_\tau$ and $H^1(B,\mathbb{C})_{\tau'}$ trivially if $\tau'\neq\bar{\tau}$, so, ψ being non-degenerate and invariant under $\mathrm{Hg}(B)=\mathrm{U}_F(V,\psi)$ (Lemma 8.10 (1)), it induces an isomorphism $H^1(B,\mathbb{C})_{\bar{\phi}_l}\cong V_l^*$ of representations of $\mathrm{Hg}(B)_{\mathbb{C}}$, through which we identify $H^1(B,\mathbb{C})_{\bar{\phi}_l}$ with V_l^*. For $l=1,2$, let $\sigma_l:=\phi_1|_{K_l}$, and let $\epsilon_1:=1$ and $\epsilon_2:=-1$. Since $\phi_2|_{K_1}=\sigma_1$ and $\bar{\phi}_2|_{K_2}=\sigma_2$, we have $H^1(B,\mathbb{C})_{\sigma_1}=V_1\oplus V_2$ and

$$H^1(B,\mathbb{C})_{\sigma_2}=V_1\oplus H^1(B,\mathbb{C})_{\bar{\phi}_2}.$$

Thus the lines $\wedge^4H^1(B,\mathbb{C})_{\sigma_1}$ and $\wedge^4H^1(B,\mathbb{C})_{\sigma_2}$ are spanned by x_1x_2 and x_1y_2, and $\mathrm{Hg}(B)_{\mathbb{C}}$ acts on $\wedge^4H^1(B,\mathbb{C})_{\sigma_l}$ by $d_1+\epsilon_l d_2$.

We fix $\overline{\mathbb{Q}}\subset\mathbb{C}$. By Lemma 8.5, $X^*(\mathrm{U}_F)$ is generated by the characters $[\tau]$ given by the complex embeddings τ of F, subject to the relations $[\bar{\tau}]=-[\tau]$, and $\gamma\in\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ sends $[\tau]$ to $[\gamma\circ\tau]$. Through $\det_F$, the character $[\phi_l]$ is d_l. Let $g\in\mathrm{Gal}(F/\mathbb{Q})$ be the non-trivial element fixing K_1. Then $\phi_1\circ g=\phi_2$, since both restrict to σ_1 on K_1 and to $\bar{\sigma}_2$ on K_2. For $\gamma\in\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, write $\gamma\circ\phi_1=\phi_1\circ h$ with $h\in\mathrm{Gal}(F/\mathbb{Q})$. Then $\gamma\circ\phi_2=\phi_1\circ h\circ g=\phi_2\circ h$, and γ acts non-trivially on K_l if and only if h does. Checking the four elements h, we obtain

$$X^*(\mathrm{U}_F)\otimes\mathbb{Q}=\mathbb{Q}(d_1+d_2)\oplus\mathbb{Q}(d_1-d_2),$$

where γ acts on d_1+d_2 (resp. d_1-d_2) by -1 if and only if γ acts non-trivially on K_1 (resp. K_2). Thus the summands are isomorphic to $X^*(\mathrm{U}_{K_1})\otimes\mathbb{Q}$ and $X^*(\mathrm{U}_{K_2})\otimes\mathbb{Q}$, and neither summand is isomorphic to $X^*(\mathrm{U}_K)\otimes\mathbb{Q}$ for an imaginary quadratic field $K\notin\{K_1,K_2\}$.

Step 2. In this step, we fix $l\in\{1,2\}$, an elliptic curve E_0 with complex multiplication by K_l, and $r\in\{1,2\}$, and we compute $\mathrm{Hg}(B\times E_0^r)$ and $\mathrm{Hdg}^*(B\times E_0^r)$.

We view $\mathrm{End}^0(E_0)$ as K_l so that K_l acts on $H^{1,0}(E_0)$ through σ_l. Then $\mathrm{Hg}(E_0)=\mathrm{U}_{K_l}$ [Moo99, (2.6)], and since the Hodge circle of E_0^r is diagonal, $\mathrm{Hg}(E_0^r)$ is U_{K_l} acting diagonally. Let $N_+:=H^{1,0}(E_0^r)$, $N_-:=H^{0,1}(E_0^r)$, and $t:=[\sigma_l]$. Then $\mathrm{U}_{K_l,\mathbb{C}}$ acts on $N_\pm$ by $\pm t$, and $t(\eta_{E_0}(z))=z$. Let $\chi_l:\mathrm{Hg}(B)\to\mathrm{U}_{K_l}$ be the composite of $\det_F$ and the norm $\mathrm{U}_F\to\mathrm{U}_{K_l}$, so $t\circ\chi_l=d_1+\epsilon_l d_2$, and let

$$G_l:=\{(g,u)\in\mathrm{Hg}(B)\times\mathrm{U}_{K_l}\mid\chi_l(g)=u^2\}.$$

We show that $\mathrm{Hg}(B\times E_0^r)=G_l$, with U_{K_l} acting diagonally on $H^1(E_0^r,\mathbb{Q})$. By Lemma 8.10 (2), $t(\chi_l(\eta_B(z)))=z^2=t(\eta_{E_0}(z)^2)$, and t is an isomorphism $\mathrm{U}_{K_l,\mathbb{C}}\to\mathbb{G}_m$, so the Hodge circle of $B\times E_0^r$ takes values in $G_l(\mathbb{R})$. The projection $G_l\to\mathrm{U}_{K_l}$ is surjective, since χ_l is surjective, and the kernel of the projection is $\ker(\chi_l)\times\{1\}$. Over $\mathbb{C}$,

$$\ker(\chi_l)=\{(g_1,g_2)\in\mathrm{GL}(V_1)\times\mathrm{GL}(V_2)\mid\det g_1\cdot(\det g_2)^{\epsilon_l}=1\},$$

which is connected, as the projection $\ker(\chi_l)\to\mathrm{GL}(V_1)$ is surjective with fibers cosets of $\mathrm{SL}(V_2)$. Thus G_l is connected and $\dim G_l=8$. By Definition 2.2, $\mathrm{Hg}(B\times E_0^r)\subset G_l$. Since $\mathrm{Hg}(B\times E_0^r)$ projects onto $\mathrm{Hg}(B)$ (Lemma 2.3 (3)) and $\dim\mathrm{Hg}(B)=8$, we get $\mathrm{Hg}(B\times E_0^r)=G_l$.

The projection $G_l\to\mathrm{Hg}(B)$ is surjective with finite kernel. Thus $X^*(G_l)\otimes\mathbb{Q}$ has basis d_1,d_2, and $t=(d_1+\epsilon_l d_2)/2$ in $X^*(G_l)\otimes\mathbb{Q}$. Moreover, G_l/G_l^{der} is isogenous to

$$\mathrm{Hg}(B)/\mathrm{Hg}(B)^{\mathrm{der}}\cong\mathrm{U}_F,$$

so

$$X^*(G_l/G_l^{\mathrm{der}})\otimes\mathbb{Q}\cong X^*(\mathrm{U}_F)\otimes\mathbb{Q}.$$

By Lemma 2.3 (2), $\mathrm{Hdg}^*(B\times E_0^r)\otimes\mathbb{C}$ is the space of G_l-invariants in

$$\wedge^*(V_1\oplus V_1^*)\otimes\wedge^*(V_2\oplus V_2^*)\otimes\wedge^*(N_+\oplus N_-).$$

The subgroup

$$\mathrm{SL}(V_1) \times \mathrm{SL}(V_2) \times \{1\} \subset G_l$$

acts trivially on $N_\pm$, so by Lemma 8.10 (5), the invariants of

$$\mathrm{SL}(V_1) \times \mathrm{SL}(V_2) \times \{1\}$$

are $I_1 \otimes I_2 \otimes \wedge^*(N_+ \oplus N_-)$, where I_k is spanned by 1, x_k, y_k, q_k, and q_k^2. Let $v_1 \in I_1$, $v_2 \in I_2$, and $c \in \wedge^a N_+ \otimes \wedge^b N_-$, where v_k has weight $w_k d_k$ with $w_k \in \{-1, 0, 1\}$. Then $v_1 v_2 c$ has weight

$$w_1 d_1 + w_2 d_2 + \frac{a-b}{2}(d_1 + \epsilon_l d_2),$$

and $|a-b| \leq r$. The weight of $v_1 v_2 c$ is zero if and only if either $a = b$ and $w_1 = w_2 = 0$, or $a - b = \pm 2$, $w_1 = \mp 1$, and $w_2 = \mp \epsilon_l$.

In the first case, $v_1 v_2$ is a polynomial in q_1, q_2, and c is U_{K_l}-invariant. The classes q_1, q_2 span the weight-zero invariants of degree 2 in $\wedge^* H^1(B, \mathbb{C})$, that is, $\mathrm{Hdg}^1(B) \otimes \mathbb{C}$. The U_{K_l}-invariants in $\wedge^*(N_+ \oplus N_-)$ form

$$\mathrm{Hdg}^*(E_0^r) \otimes \mathbb{C} = \mathcal{D}^*(E_0^r) \otimes \mathbb{C}$$

by Lemma 2.3 (2) and [MZ99, Theorem (0.1)(iv)]. Thus the invariants of the first kind span a subalgebra which is contained in $\mathcal{D}^*(B \times E_0^r) \otimes \mathbb{C}$ and contains all invariants of degree 2, so they span $\mathcal{D}^*(B \times E_0^r) \otimes \mathbb{C}$. Invariants of the second kind exist only for $r = 2$, have degree 6, and span the lines

$$x_1 x_2 \otimes \wedge^2 N_-, \quad y_1 y_2 \otimes \wedge^2 N_+ \quad \text{if } l = 1, \qquad x_1 y_2 \otimes \wedge^2 N_-, \quad y_1 x_2 \otimes \wedge^2 N_+ \quad \text{if } l = 2.$$

Hence $\mathrm{Hdg}^p(B \times E_0) = \mathcal{D}^p(B \times E_0)$ for any p. For $r = 2$, we have

$$\mathrm{Hdg}^p(B \times E_0^2) = \mathcal{D}^p(B \times E_0^2)$$

for $p \neq 3$, and $\mathrm{Hdg}^3(B \times E_0^2) \otimes \mathbb{C}$ is the direct sum of $\mathcal{D}^3(B \times E_0^2) \otimes \mathbb{C}$ and the lines $x_1 x_2 \otimes \wedge^2 N_-$ and $y_1 y_2 \otimes \wedge^2 N_+$ if $l = 1$, or $x_1 y_2 \otimes \wedge^2 N_-$ and $y_1 x_2 \otimes \wedge^2 N_+$ if $l = 2$.

Step 3. In this step, we prove (1).

An isogeny induces an isomorphism of rational cohomology rings which preserves Hodge classes and divisor classes. Thus, possibly replacing E' with E if E and E' are isogenous, we may assume that $A = Y_1 \times \prod_{i=1}^k C_i^{s_i}$, where $s_i \in \{1, 2\}$, the elliptic curves $C_1, \ldots, C_k$ are pairwise non-isogenous, no C_i has complex multiplication by K_1 or K_2, and, by the assumption of (1), either $Y_1 = B$, or $Y_1 = B \times E_0$ for an elliptic curve E_0 with complex multiplication by K_l for some l. By Lemma 8.10 (1) and **Step 2**, $\mathrm{Hdg}^*(Y_1) = \mathcal{D}^*(Y_1)$, and

$$X^*(H/H^{\mathrm{der}}) \otimes \mathbb{Q} \cong X^*(\mathrm{U}_F) \otimes \mathbb{Q}$$

for $H := \mathrm{Hg}(Y_1)$. By **Step 1**, no irreducible constituent of $X^*(H/H^{\mathrm{der}}) \otimes \mathbb{Q}$ is isomorphic to $X^*(\mathrm{U}_K) \otimes \mathbb{Q}$ for an imaginary quadratic field $K \notin \{K_1, K_2\}$.

For any i, the Hodge circle of $C_i^{s_i}$ is diagonal, so $\mathrm{Hg}(C_i^{s_i})$ is $\mathrm{Hg}(C_i)$ acting diagonally, and $\mathrm{Hdg}^*(C_i^{s_i}) = \mathcal{D}^*(C_i^{s_i})$ by [MZ99, Theorem (0.1)(iv)]. If C_i is not of CM type, then $\mathrm{Hom}(C_i, Y_1 \times \prod_{j \neq i} C_j^{s_j}) = 0$, so

$$\mathrm{Hg}(A) = \mathrm{Hg}(C_i^{s_i}) \times \mathrm{Hg}\left(Y_1 \times \prod_{j \neq i} C_j^{s_j}\right)$$

by [MZ99, Proposition (2.4)(i),(iii), Lemma (3.4)]. Repeating the argument for any C_i without complex multiplication, we are left with the curves C_i of CM type. Their fields K_i' are pairwise distinct, since two elliptic curves with complex multiplication by one field are isogenous [MZ99, (4.1)], and $K_i' \notin \{K_1, K_2\}$. Moreover, $\mathrm{Hg}(C_i) = \mathrm{U}_{K_i'}$ [Moo99, (2.6)]. By Lemma 2.3 (1), (3) and Lemma 8.6, the Hodge group of the product of Y_1 and the $C_i^{s_i}$ with C_i of CM type is the product of H and the groups $\mathrm{Hg}(C_i^{s_i})$. Hence

$$\mathrm{Hg}(A) = \mathrm{Hg}(Y_1) \times \prod_{i=1}^k \mathrm{Hg}(C_i^{s_i}).$$

By the Künneth formula and Lemma 2.3 (2), $\mathrm{Hdg}^*(A)$ is the tensor product of $\mathrm{Hdg}^*(Y_1)$ and the $\mathrm{Hdg}^*(C_i^{s_i})$, all of which are generated by divisors. Thus $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p.

Step 4. In this step, we prove (2).

Since E and E' have complex multiplication by K_l, they are isogenous [MZ99, (4.1)]. Let K_l act on E and E' so that K_l acts on $H^{1,0}(E)$ and $H^{1,0}(E')$ through $\bar{\sigma}_l$. As ϕ_1 extends σ_l, the multiplicities of K_l on $H^{1,0}(B)$ are $(3,1)$ by Lemma 8.10 (3), so A has multiplicities $(3,3)$. Any isogeny $E' \to E$ commutes with the K_l-actions on E' and E, so it induces an isomorphism $H^*(A,\mathbb{Q}) \cong H^*(B \times E \times E, \mathbb{Q})$ which preserves Hodge classes, divisor classes, and K_l-actions, hence also Weil planes. Possibly replacing E' with E, we may assume that $E' = E$. We use the notation of **Step 2** for $E_0 := E$ and $r = 2$. For the present K_l-action, $H^1(E^2,\mathbb{C})_{\sigma_l} = N_-$ and $H^1(E^2,\mathbb{C})_{\bar{\sigma}_l} = N_+$. Hence

$$W_{K_l}(A) \otimes \mathbb{C} = \big(\wedge^4 H^1(B,\mathbb{C})_{\sigma_l} \otimes \wedge^2 N_-\big) \oplus \big(\wedge^4 H^1(B,\mathbb{C})_{\bar{\sigma}_l} \otimes \wedge^2 N_+\big).$$

By **Step 1**, the right-hand side is the sum of the lines $x_1x_2 \otimes \wedge^2 N_-$ and $y_1y_2 \otimes \wedge^2 N_+$ for $l = 1$, and of $x_1y_2 \otimes \wedge^2 N_-$ and $y_1x_2 \otimes \wedge^2 N_+$ for $l = 2$. By **Step 2**, $\mathrm{Hdg}^2(A) = \mathcal{D}^2(A)$ and $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_{K_l}(A)$. Since A is not simple, any class in $W_{K_l}(A)$ is algebraic by Theorem 8.2. As divisor classes are algebraic, $\mathrm{HC}^2(A)$ and $\mathrm{HC}^3(A)$ hold, and $\mathrm{HC}(A)$ holds by Lemma 2.6.

Step 5. In this step, we prove (3).

The varieties B, E, and E' are simple, and they are pairwise non-isogenous, since $\mathrm{End}^0(E) = K_1$ and $\mathrm{End}^0(E') = K_2$ are not isomorphic. Hence $\mathrm{End}^0(A) = F \times K_1 \times K_2$ [Moo99, (2.3)]. A unital embedding of an imaginary quadratic field K into $\mathrm{End}^0(A)$ projects to unital homomorphisms $K \to K_1$ and $K \to K_2$, which are isomorphisms. Thus $K_1 \cong K_2$, a contradiction. □

Definition 8.12. Let B be a simple abelian fourfold and S an abelian surface. We say that (B,S) is an *exceptional pair* if the following conditions hold.

(1) $F := \mathrm{End}^0(B)$ is a CM field of degree 4, and for some ordering $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$ of the complex embeddings of F, the multiplicities of F on $H^{1,0}(B)$ at $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$ are $(2,1,0,1)$.

(2) S is either simple of CM type or isogenous to a product of two non-isogenous elliptic curves with complex multiplication.

(3) $U_B \cong T_S$ as rational Hodge structures, where $U_B \subset H^2(B,\mathbb{Q})$ is the Hodge substructure of Lemma 8.10 (4), with

$$U_B \otimes \mathbb{C} = \bigoplus_{\tau} \wedge^2 H^1(B,\mathbb{C})_\tau,$$

and $T_S \subset H^2(S,\mathbb{Q})$ is the orthogonal complement of $NS(S) \otimes \mathbb{Q}$ with respect to the cup product.

If $\mathrm{End}^0(B)$ is a CM field of degree 4, the multiplicity condition in (1) holds for some ordering, so it only fixes the ordering of the complex embeddings used in Lemma 8.10. Indeed, let B be a simple abelian fourfold such that $F := \mathrm{End}^0(B)$ is a CM field of degree 4, let $V := H^1(B,\mathbb{Q})$, and let ψ be the alternating form on V induced by a polarization of B. Since $\dim H^1(B,\mathbb{C})_\tau = 2$ for any complex embedding τ of F and F preserves the Hodge decomposition, the multiplicities of F on $H^{1,0}(B)$ at τ and $\bar{\tau}$ are $(2,0)$, $(1,1)$, or $(0,2)$. If $H^1(B,\mathbb{C})_\tau$ is contained in $H^{1,0}(B)$ or in $H^{0,1}(B)$ for all four embeddings τ, then any F-linear endomorphism of V commutes with the Hodge structure, so $\mathrm{End}^0(B)$ contains a copy of $M_2(F)$, a contradiction. If the four multiplicities are all 1, then $\eta_B(z)$ acts on each $H^1(B,\mathbb{C})_\tau$ with eigenvalues z and $\bar{z}$, so $\det_F \circ \eta_B$ is trivial and $\mathrm{Hg}(B) \subset \mathrm{SU}_F(V,\psi)$. For embeddings ϕ_1, ϕ_2 of F with $\phi_2 \notin \{\phi_1, \bar{\phi}_1\}$ and $V_l := H^1(B,\mathbb{C})_{\phi_l}$, the group $\mathrm{SU}_F(V,\psi)_{\mathbb{C}}$ is $\mathrm{SL}(V_1) \times \mathrm{SL}(V_2)$ acting on $V \otimes \mathbb{C} = V_1 \oplus V_1^* \oplus V_2 \oplus V_2^*$, and $V_l \cong V_l^*$ as representations of $\mathrm{SL}(V_l)$. Hence

$$\dim \mathrm{End}_{\mathrm{Hg}(B)}(V \otimes \mathbb{C}) \geq 8 > \dim F \otimes \mathbb{C} = 4,$$

which contradicts Lemma 2.3 (1). Thus exactly one pair $\{\tau, \bar{\tau}\}$ has multiplicities $(1,1)$, and the multiplicities are $(2,1,0,1)$ for some ordering $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$.

Condition (2) follows from conditions (1) and (3). Indeed, by Lemma 8.10 (4), $\mathrm{Hg}(B)$ acts on U_B through a torus. Hence $\dim T_S = \dim U_B = 4$, and the image of $\mathrm{Hg}(S)$ in $\mathrm{GL}(H^2(S,\mathbb{Q}))$ is a torus. Thus $\dim NS(S) \otimes \mathbb{Q} = 2$, and S is of CM type (Lemma 2.3 (4)) and not isogenous to E^2 for an elliptic curve E. We include condition (2) since Proposition 8.13 assumes conditions (1) and (2) but not condition (3).

An isomorphism $U_B \cong T_S$ of rational Hodge structures induces Hodge classes in $U_B \otimes T_S \subset H^4(B \times S, \mathbb{Q})$ which are not generated by divisors (Proposition 8.14). If (B,S) satisfies conditions (1) and (2) but not condition (3), the Hodge classes on $B \times S$ are generated by divisors (Proposition 8.13).

Proposition 8.13. *Let B be a simple abelian fourfold and S an abelian surface satisfying conditions (1) and (2) of Definition 8.12, let $F := \mathrm{End}^0(B)$, and let d_1, d_2 be the characters $\det H^1(B,\mathbb{C})_{\phi_1}$ and $\det H^1(B,\mathbb{C})_{\phi_2}$ of $\mathrm{Hg}(B)_{\mathbb{C}}$, viewed as characters of $\mathrm{U}_{F,\mathbb{C}}$ through $\det_F$. Then either $\mathrm{Hdg}^p(B \times S) = \mathcal{D}^p(B \times S)$ for any p, or (B,S) is an exceptional pair. If (B,S) is an exceptional pair, then $\mathrm{Hg}(S)$ is a two-dimensional torus, and there exist an isogeny $\varphi : \mathrm{Hg}(S) \to \mathrm{U}_F$ and a basis f_1, f_2 of the character group of $\mathrm{Hg}(S)_{\mathbb{C}}$ such that $\varphi^* d_1 = f_1 + f_2$, $\varphi^* d_2 = \pm(f_1 - f_2)$, and*

$$\mathrm{Hg}(B \times S) = \{(g,t) \in \mathrm{Hg}(B) \times \mathrm{Hg}(S) \mid \det_F(g) = \varphi(t)\};$$

moreover, the following hold.

1. *If F is biquadratic with imaginary quadratic subfields K_1 and K_2, then S is isogenous to $E_1 \times E_2$, where E_l is an elliptic curve with $\mathrm{End}^0(E_l) = K_l$ for $l = 1,2$. Conversely, if F is biquadratic and S is isogenous to such a product $E_1 \times E_2$, then (B,S) is an exceptional pair.*
2. *If F is cyclic over $\mathbb{Q}$, then S is simple and $\mathrm{End}^0(S) \cong F$.*
3. *If F is not Galois over $\mathbb{Q}$, then S is simple, and $\mathrm{End}^0(S)$ is isomorphic to a quartic CM subfield of the normal closure of F which is not isomorphic to F.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we describe $\mathrm{Hg}(S)$ and T_S, and we show that if $\mathrm{Hg}(B \times S) \supset \{1\} \times T'$ for a subtorus $T' \subset \mathrm{Hg}(S)$ without non-zero invariants in T_S, then (B,S) is not an exceptional pair. We also show that $\mathrm{Hdg}^p(B \times S) = \mathcal{D}^p(B \times S)$ for any p if

$$\mathrm{Hg}(B \times S) = \mathrm{Hg}(B) \times \mathrm{Hg}(S).$$

Let $V := H^1(B,\mathbb{Q})$ and ψ the alternating form on V induced by a polarization of B. By condition (2) and Lemma 2.3 (4), $\mathrm{Hg}(S)$ is a torus. If S is simple, then $F_S := \mathrm{End}^0(S)$ is a quartic CM field containing no imaginary quadratic field, and $\mathrm{Hg}(S) = \mathrm{U}_{F_S}$ [MZ99, (2.2)]. In this case, $\dim_{F_S} H^1(S,\mathbb{Q}) = 1$, so $H^1(S,\mathbb{C})$ is the sum of the four lines $H^1(S,\mathbb{C})_\iota$, where ι runs over the complex embeddings of F_S, and $\mathrm{Hg}(S)_{\mathbb{C}}$ acts on $H^1(S,\mathbb{C})_\iota$ by the character $[\iota]$ given by ι. Any such line is contained in $H^{1,0}(S)$ or in $H^{0,1}(S)$, and complex conjugation maps $H^1(S,\mathbb{C})_\iota$ onto $H^1(S,\mathbb{C})_{\bar{\iota}}$. Let Φ_S be the set of embeddings ι with $H^1(S,\mathbb{C})_\iota \subset H^{1,0}(S)$. Then Φ_S contains exactly one of $\iota, \bar{\iota}$ for any ι.

If S is not simple, then S is isogenous to $E_1 \times E_2$, where E_j has complex multiplication by an imaginary quadratic field L_j, and $L_1 \neq L_2$ since E_1 and E_2 are not isogenous [MZ99, (4.1)]. An isogeny $S \to E_1 \times E_2$ induces isomorphisms of rational cohomology rings for S and for $B \times S$ which preserve Hodge structures and divisor classes. Thus it preserves Hodge groups, Hodge classes, divisor algebras, and T_S, and possibly replacing S with $E_1 \times E_2$, we may assume that $S = E_1 \times E_2$. Then $\mathrm{Hg}(E_j) = \mathrm{U}_{L_j}$ [Moo99, (2.6)], and

$$\mathrm{Hg}(S) = \mathrm{U}_{L_1} \times \mathrm{U}_{L_2}$$

by Lemma 8.6 for the trivial group H. Let ρ_j be the embedding of $L_j = \mathrm{End}^0(E_j)$ through which L_j acts on $H^{1,0}(E_j)$. Then $\mathrm{Hg}(S)_{\mathbb{C}}$ acts on $H^{1,0}(E_j)$ and $H^{0,1}(E_j)$ by $[\rho_j]$ and $-[\rho_j]$.

In both cases, $\mathrm{Hg}(S)$ is a two-dimensional torus, and $H^{1,0}(S)$ is the sum of two lines on which $\mathrm{Hg}(S)_{\mathbb{C}}$ acts by characters f_1, f_2 forming a basis of $X^*(\mathrm{Hg}(S))$, namely $[\iota]$ for $\iota \in \Phi_S$, or $[\rho_1]$ and $[\rho_2]$. Moreover, $H^{0,1}(S)$ is the sum of two lines with characters $-f_1$ and $-f_2$, and $f_j(\eta_S(z)) = z$. Hence $H^2(S,\mathbb{C}) = \wedge^2 H^1(S,\mathbb{C})$ is the sum of two lines with character 0 and four lines with the non-zero characters $\pm f_1 \pm f_2$. By Lemma 2.3 (2) and [Moo99, (2.10)], $NS(S)\otimes\mathbb{Q} = \mathrm{Hdg}^1(S)$ is the space of $\mathrm{Hg}(S)$-invariants in $H^2(S,\mathbb{Q})$, so $NS(S)\otimes\mathbb{C}$ is the sum of the lines with character 0. The group $\mathrm{Hg}(S)$ acts trivially on $H^4(S,\mathbb{Q})$ (Lemma 2.3 (2)), so the cup product is $\mathrm{Hg}(S)$-invariant, and it is non-degenerate on $NS(S)\otimes\mathbb{Q}$ by the Hodge index theorem. Thus T_S is an $\mathrm{Hg}(S)$-stable complement of $NS(S)\otimes\mathbb{Q}$, and $T_S\otimes\mathbb{C}$ is the sum of the four lines with characters $\pm f_1 \pm f_2$. In particular, T_S has no non-zero $\mathrm{Hg}(S)$-invariants.

Now assume that $\mathrm{Hg}(B\times S) \supset \{1\}\times T'$ for a subtorus $T' \subset \mathrm{Hg}(S)$ without non-zero invariants in T_S. An isomorphism $U_B \to T_S$ of rational Hodge structures is $\mathrm{Hg}(B\times S)$-equivariant (cf. [Moo99, (1.12)]), so it defines a non-zero $\mathrm{Hg}(B\times S)$-invariant tensor in $U_B^\vee \otimes T_S$. However, $\{1\}\times T'$ acts trivially on $U_B^\vee$, so

$$(U_B^\vee \otimes T_S)^{\{1\}\times T'} = U_B^\vee \otimes T_S^{T'} = 0.$$

Hence (B,S) is not an exceptional pair. If moreover

$$\mathrm{Hg}(B\times S) = \mathrm{Hg}(B)\times\mathrm{Hg}(S),$$

then by the Künneth formula and Lemma 2.3 (2),

$$\mathrm{Hdg}^p(B\times S) = \bigoplus_{i+j=p} \mathrm{Hdg}^i(B)\otimes\mathrm{Hdg}^j(S).$$

By Lemma 8.10 (1), $\mathrm{Hdg}^i(B) = \mathcal{D}^i(B)$. Moreover, $\mathrm{Hdg}^1(S) = \mathcal{D}^1(S)$ [Moo99, (2.10)], and $\mathrm{Hdg}^0(S)$ and $\mathrm{Hdg}^2(S)$ are spanned by 1 and Θ_S^2, where Θ_S is an ample class on S. Thus $\mathrm{Hdg}^p(B\times S) = \mathcal{D}^p(B\times S)$ for any p.

Step 2. In this step, we show that if there exists an isogeny $\varphi : \mathrm{Hg}(S) \to \mathrm{U}_F$ such that $\varphi^* d_1 = f_1 + f_2$ and $\varphi^* d_2 = \pm(f_1 - f_2)$ for the characters f_1, f_2 of **Step 1** in some order, then (B,S) is an exceptional pair and $\mathrm{Hg}(B\times S) = G$, where

$$G := \{(g,t) \in \mathrm{Hg}(B)\times\mathrm{Hg}(S) \mid \det_F(g) = \varphi(t)\}.$$

The group G is defined over $\mathbb{Q}$. The projection $G \to \mathrm{Hg}(S)$ is surjective, since $\det_F$ is surjective, and the kernel $\ker(\det_F)\times\{1\}$ of the projection is connected (Lemma 8.10 (1)). Thus G is connected, and $\dim G = 2+6 = 8$. The characters d_1, d_2 form a basis of $X^*(\mathrm{U}_F)$, and by Lemma 8.10 (2) and **Step 1**,

$$d_1(\det_F(\eta_B(z))) = z^2 = (f_1+f_2)(\eta_S(z)), \qquad d_2(\det_F(\eta_B(z))) = 1 = \big(\pm(f_1-f_2)\big)(\eta_S(z)).$$

Hence the Hodge circle of $B\times S$ takes values in $G(\mathbb{R})$, and $\mathrm{Hg}(B\times S) \subset G$ by Definition 2.2. Since $\mathrm{Hg}(B\times S)$ projects onto $\mathrm{Hg}(B)$ (Lemma 2.3 (3)) and $\dim \mathrm{Hg}(B) = 8$, we get $\mathrm{Hg}(B\times S) = G$.

On G, the character $d_l \circ \det_F$ of the first factor coincides with the character $\varphi^* d_l$ of the second factor. By Lemma 8.10 (4) and **Step 1**, $G_{\mathbb{C}}$ acts on $U_B\otimes\mathbb{C}$ by the characters $\pm(f_1+f_2)$, $\pm(f_1-f_2)$, and on $T_S\otimes\mathbb{C}$ by the characters $\pm f_1 \pm f_2$, each with multiplicity one. Thus $U_B\otimes\mathbb{C} \cong T_S\otimes\mathbb{C}$ as representations of $G_{\mathbb{C}}$. The complexification of the $\mathbb{Q}$-vector space $\mathrm{Hom}_G(U_B, T_S)$ is

$$\mathrm{Hom}_{G_{\mathbb{C}}}(U_B\otimes\mathbb{C}, T_S\otimes\mathbb{C}),$$

which contains an isomorphism. As isomorphisms form a non-empty Zariski open subset and rational points are Zariski dense, there exists a G-equivariant isomorphism $U_B \to T_S$ over $\mathbb{Q}$. Since $G = \mathrm{Hg}(B\times S)$, it is an isomorphism of rational Hodge structures (cf. [Moo99, (1.12)]). Thus (B,S) is an exceptional pair.

Step 3. In this step, we show that if

$$\mathrm{Hg}(B \times S) \neq \mathrm{Hg}(B) \times \mathrm{Hg}(S),$$

then there exist a non-zero $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable subspace $W_1 \subset X^*(\mathrm{U}_F) \otimes \mathbb{Q}$ and an injective $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-equivariant linear map

$$\alpha : W_1 \to X^*(\mathrm{Hg}(S)) \otimes \mathbb{Q}$$

such that

$$\langle w, \det_F \circ \eta_B \rangle = \langle \alpha(w), \eta_S \rangle \quad \text{for any } w \in W_1.$$

Here we fix $\overline{\mathbb{Q}} \subset \mathbb{C}$, for a character χ of $\mathrm{Hg}(S)_{\mathbb{C}}$ we denote by $\langle \chi, \eta_S \rangle$ the integer n with $\chi(\eta_S(z)) = z^n$, for a character χ of $\mathrm{U}_{F,\mathbb{C}}$ we denote by $\langle \chi, \det_F \circ \eta_B \rangle$ the integer n with $\chi(\det_F(\eta_B(z))) = z^n$, and we extend both pairings $\mathbb{Q}$-linearly. By Lemma 8.10 (2), $\langle d_1, \det_F \circ \eta_B \rangle = 2$ and $\langle d_2, \det_F \circ \eta_B \rangle = 0$, and by **Step 1**, $\langle f_j, \eta_S \rangle = 1$.

First, let T'' be a torus over $\mathbb{Q}$ and $H \subset \mathrm{Hg}(B) \times T''$ a connected algebraic subgroup projecting onto $\mathrm{Hg}(B)$. The derived group of H is connected, it projects onto the derived group $\mathrm{SU}_F(V, \psi)$ of $\mathrm{Hg}(B)$ (Lemma 8.10 (1)), and it is contained in $\mathrm{SU}_F(V, \psi) \times \{1\}$. Thus $H^{\mathrm{der}} = \mathrm{SU}_F(V, \psi) \times \{1\}$, so H is the preimage under $\det_F \times \mathrm{id}_{T''}$ of a subtorus of $\mathrm{U}_F \times T''$, which is isomorphic to H/H^{der}.

Now let $H = \mathrm{Hg}(B \times S)$ and $T'' = \mathrm{Hg}(S)$ (Lemma 2.3 (1), (3)). Then $\mathrm{Hg}(B \times S)$ is the preimage of a subtorus $T_0 \subset \mathrm{U}_F \times \mathrm{Hg}(S)$, and $T_0 \neq \mathrm{U}_F \times \mathrm{Hg}(S)$ by assumption. Both projections of T_0 are surjective by Lemma 2.3 (3) and Lemma 8.10 (1). Let W be the kernel of the restriction map

$$\left(X^*(\mathrm{U}_F) \otimes \mathbb{Q}\right) \oplus \left(X^*(\mathrm{Hg}(S)) \otimes \mathbb{Q}\right) \to X^*(T_0) \otimes \mathbb{Q}.$$

By [Moo99, (1.1)], W is a non-zero $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable subspace, and the surjectivity of both projections implies that $W \cap X^*(\mathrm{U}_F) \otimes \mathbb{Q} = 0$ and $W \cap X^*(\mathrm{Hg}(S)) \otimes \mathbb{Q} = 0$. Thus the projection of W to $X^*(\mathrm{U}_F) \otimes \mathbb{Q}$ is injective with non-zero image W_1, and $W = \{(w, -\alpha(w)) \mid w \in W_1\}$ for an injective equivariant linear map

$$\alpha : W_1 \to X^*(\mathrm{Hg}(S)) \otimes \mathbb{Q}.$$

Since $(\det_F(\eta_B(z)), \eta_S(z)) \in T_0(\mathbb{R})$, any element of W is trivial on $(\det_F(\eta_B(z)), \eta_S(z))$, so

$$\langle w, \det_F \circ \eta_B \rangle = \langle \alpha(w), \eta_S \rangle$$

for any $w \in W_1$.

Step 4. In this step, we treat the case where F contains no imaginary quadratic field and S is not simple.

A quartic CM field F is biquadratic if it contains an imaginary quadratic field, since F is then the compositum of that field and the maximal totally real subfield of F, and F is cyclic or not Galois over $\mathbb{Q}$ otherwise (Lemma 8.5).

Assume that F contains no imaginary quadratic field and that $S = E_1 \times E_2$ as in **Step 1**. Then $X^*(\mathrm{Hg}(S)) \otimes \mathbb{Q}$ is the direct sum of the $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable lines spanned by $[\rho_1]$ and $[\rho_2]$. If

$$\mathrm{Hg}(B \times S) \neq \mathrm{Hg}(B) \times \mathrm{Hg}(S),$$

then **Step 3** and the irreducibility of $X^*(\mathrm{U}_F) \otimes \mathbb{Q}$ (Lemma 8.5) imply that $W_1 = X^*(\mathrm{U}_F) \otimes \mathbb{Q}$ and that α is an isomorphism onto $X^*(\mathrm{Hg}(S)) \otimes \mathbb{Q}$, which contradicts the irreducibility of $X^*(\mathrm{U}_F) \otimes \mathbb{Q}$. Thus

$$\mathrm{Hg}(B \times S) = \mathrm{Hg}(B) \times \mathrm{Hg}(S),$$

and by **Step 1**, $\mathrm{Hdg}^p(B \times S) = \mathcal{D}^p(B \times S)$ for any p and (B, S) is not an exceptional pair.

Step 5. In this step, we treat the case where F is biquadratic, and we prove (1).

Let K_1 and K_2 be the imaginary quadratic subfields of F. Since F is Galois over $\mathbb{Q}$, we have $\phi_2 = \phi_1 \circ g$ for some $g \in \mathrm{Gal}(F/\mathbb{Q})$. As $\phi_2 \notin \{\phi_1, \bar{\phi}_1\}$, the element g is neither the identity nor complex conjugation, so the fixed field of g is an imaginary quadratic subfield of F. Possibly renumbering K_1 and K_2, we may assume that it is K_1. Then $\phi_1|_{K_1} = \phi_2|_{K_1}$, and $\phi_1|_{K_2} = \bar{\phi}_2|_{K_2}$

since g acts on K_2 by complex conjugation. Thus B satisfies the assumptions of Lemma 8.11. For $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, write $\gamma \circ \phi_1 = \phi_1 \circ h$ with $h \in \mathrm{Gal}(F/\mathbb{Q})$. Then $\gamma \circ \phi_2 = \phi_2 \circ h$, and checking the four elements h, we obtain

$$X^*(\mathrm{U}_F) \otimes \mathbb{Q} = \mathbb{Q}(d_1 + d_2) \oplus \mathbb{Q}(d_1 - d_2),$$

where γ acts on $d_1 + d_2$ (resp. $d_1 - d_2$) by -1 if and only if γ acts non-trivially on K_1 (resp. K_2). In particular, any non-zero $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable subspace of $X^*(\mathrm{U}_F) \otimes \mathbb{Q}$ contains a stable line.

If S is simple, then F_S contains no imaginary quadratic field, so

$$X^*(\mathrm{Hg}(S)) \otimes \mathbb{Q} = X^*(\mathrm{U}_{F_S}) \otimes \mathbb{Q}$$

contains no stable line by Lemma 8.5. By **Step 3**,

$$\mathrm{Hg}(B \times S) = \mathrm{Hg}(B) \times \mathrm{Hg}(S),$$

and by **Step 1**, $\mathrm{Hdg}^p(B \times S) = \mathcal{D}^p(B \times S)$ for any p and (B, S) is not an exceptional pair.

Assume that $S = E_1 \times E_2$ as in **Step 1**, and that $\{L_1, L_2\} \neq \{K_1, K_2\}$. Then E_1 and E_2 do not both have complex multiplication by fields in $\{K_1, K_2\}$, so $\mathrm{Hdg}^p(B \times S) = \mathcal{D}^p(B \times S)$ for any p by Lemma 8.11 (1). Possibly renumbering E_1 and E_2, we may assume that $L_2 \notin \{K_1, K_2\}$. By **Step 3** applied to $H := \mathrm{Hg}(B \times E_1)$ and $T'' = \mathrm{U}_{L_1}$, the torus H/H^{der} is a subtorus of $\mathrm{U}_F \times \mathrm{U}_{L_1}$. Thus the irreducible constituents of $X^*(H/H^{\mathrm{der}}) \otimes \mathbb{Q}$ are among $X^*(\mathrm{U}_{K_1}) \otimes \mathbb{Q}$, $X^*(\mathrm{U}_{K_2}) \otimes \mathbb{Q}$, and $X^*(\mathrm{U}_{L_1}) \otimes \mathbb{Q}$, none of which is isomorphic to $X^*(\mathrm{U}_{L_2}) \otimes \mathbb{Q}$. By Lemma 2.3 (1), (3) and Lemma 8.6, $\mathrm{Hg}(B \times S) = H \times \mathrm{U}_{L_2}$. Hence

$$\mathrm{Hg}(B \times S) \supset \{1\} \times (\{1\} \times \mathrm{U}_{L_2}),$$

and $\{1\} \times \mathrm{U}_{L_2}$ acts on the four lines of $T_S \otimes \mathbb{C}$ by the non-zero characters $\pm[\rho_2]$. By **Step 1**, (B, S) is not an exceptional pair.

Assume that $S = E_1 \times E_2$ as in **Step 1**, and that $\{L_1, L_2\} = \{K_1, K_2\}$. Possibly renumbering E_1 and E_2, we may assume that $L_j = K_j$ for $j = 1, 2$. Let $f_j := [\rho_j]$, so $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on f_j by -1 if and only if γ acts non-trivially on K_j. The linear map β with $\beta(d_1) = f_1 + f_2$ and $\beta(d_2) = f_1 - f_2$ sends $d_1 + d_2$ to $2f_1$ and $d_1 - d_2$ to $2f_2$, so it is equivariant. It is integral with determinant -2, so by [Moo99, (1.1)], $\beta = \varphi^*$ for an isogeny $\varphi : \mathrm{Hg}(S) \to \mathrm{U}_F$. By **Step 2**, (B, S) is an exceptional pair and $\mathrm{Hg}(B \times S) = G$. Since E_1 and E_2 are arbitrary elliptic curves with $\mathrm{End}^0(E_j) = K_j$, the converse in (1) also holds.

Step 6. In this step, we treat the case where F is cyclic over $\mathbb{Q}$, and we prove (2).

By **Step 4**, we may assume that S is simple. Assume that

$$\mathrm{Hg}(B \times S) \neq \mathrm{Hg}(B) \times \mathrm{Hg}(S).$$

By **Step 3** and the irreducibility in Lemma 8.5, $W_1 = X^*(\mathrm{U}_F) \otimes \mathbb{Q}$ and α is an isomorphism onto $X^*(\mathrm{U}_{F_S}) \otimes \mathbb{Q}$. Thus both Galois modules have the kernel $\mathrm{Gal}(\overline{\mathbb{Q}}/\phi_1(F))$, so by Lemma 8.5, the normal closure of F_S is $\phi_1(F)$, and $F_S \cong F$. This proves (2).

Let $\gamma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ restrict to a generator of $\mathrm{Gal}(\phi_1(F)/\mathbb{Q})$. Then γ^2 restricts to complex conjugation, so γ acts on the embeddings of F and of F_S by 4-cycles

$$\tau \to \gamma \circ \tau \to \bar{\tau} \to \gamma \circ \bar{\tau} \to \tau.$$

Since $\gamma^{-1} \circ \phi_1 = \overline{\gamma \circ \phi_1}$, possibly replacing γ with γ^{-1}, we may assume that $\gamma \circ \phi_1 = \phi_2$. Write $\Phi_S = \{\iota, \iota'\}$. As $\iota' \notin \{\iota, \bar{\iota}\}$, we have $\iota' \in \{\gamma \circ \iota, \gamma^{-1} \circ \iota\}$, and possibly exchanging ι and ι', we may assume that $\iota' = \gamma \circ \iota$. Let $f_1 := [\iota]$ and $f_2 := [\iota']$. Then

$$\gamma(d_1) = d_2, \quad \gamma(d_2) = -d_1, \quad \gamma(f_1) = f_2, \quad \gamma(f_2) = -f_1,$$

and $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on both modules through the powers of γ. Hence $\alpha(d_1) = af_1 + bf_2$ and $\alpha(d_2) = \gamma(\alpha(d_1)) = af_2 - bf_1$ for some $a, b \in \mathbb{Q}$. By **Step 3**, $a + b = 2$ and $a - b = 0$, so $\alpha(d_1) = f_1 + f_2$ and $\alpha(d_2) = -(f_1 - f_2)$. The map α is integral with determinant 2, so by [Moo99, (1.1)], $\alpha = \varphi^*$ for an isogeny $\varphi : \mathrm{Hg}(S) \to \mathrm{U}_F$. By **Step 2**, (B, S) is an exceptional pair and $\mathrm{Hg}(B \times S) = G$.

Step 7. In this step, we treat the case where F is not Galois over $\mathbb{Q}$, and we prove (3).

Let L be the normal closure of $\phi_1(F)$ in $\overline{\mathbb{Q}}$. By Lemma 8.5, $\mathrm{Gal}(L/\mathbb{Q})$ consists of all permutations of $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$ commuting with $\tau \to \bar{\tau}$. Let $r \in \mathrm{Gal}(L/\mathbb{Q})$ act by

$$\phi_1 \to \phi_2 \to \bar{\phi}_1 \to \bar{\phi}_2 \to \phi_1,$$

and let $s \in \mathrm{Gal}(L/\mathbb{Q})$ fix $\phi_1, \bar{\phi}_1$ and exchange $\phi_2, \bar{\phi}_2$. Then r^2 is complex conjugation c, the fixed field of s is $\phi_1(F)$, and sc fixes ϕ_2, so the fixed field of sc is $\phi_2(F)$. Moreover,

$$r(d_1) = d_2, \quad r(d_2) = -d_1, \quad s(d_1) = d_1, \quad s(d_2) = -d_2.$$

By **Step 4**, we may assume that S is simple. Assume that

$$\mathrm{Hg}(B \times S) \neq \mathrm{Hg}(B) \times \mathrm{Hg}(S).$$

As in **Step 6**, α is an isomorphism

$$X^*(\mathrm{U}_F) \otimes \mathbb{Q} \to X^*(\mathrm{U}_{F_S}) \otimes \mathbb{Q},$$

and the normal closure of F_S is L. Thus $\mathrm{Gal}(L/\mathbb{Q})$ acts faithfully on the embeddings of F_S into L. For any embedding ι of F_S, we have $r \circ \iota \notin \{\iota, \bar{\iota}\}$, since $r^2 \circ \iota = \bar{\iota}$, and therefore $\langle r[\iota], \eta_S \rangle = \pm 1$ by **Step 1**.

We claim that $s \circ \iota \notin \{\iota, \bar{\iota}\}$ for any embedding ι of F_S. Otherwise, s fixes or exchanges ι and $\bar{\iota}$, and since $s \notin \{1, c\}$ (s fixes ϕ_1 and c does not), $\mathrm{Gal}(L/\mathbb{Q})$ acts faithfully, and s commutes with complex conjugation, s exchanges or fixes, respectively, the remaining embeddings ι' and $\bar{\iota}'$. Thus the s-invariant line in $X^*(\mathrm{U}_{F_S}) \otimes \mathbb{Q}$ is spanned by $[\iota'']$ for some $\iota'' \in \{\iota, \iota'\}$. As $\alpha(d_1) = \alpha(s(d_1)) = s(\alpha(d_1))$, we have $\alpha(d_1) = a[\iota'']$ for some $a \in \mathbb{Q}$, and $\alpha(d_2) = \alpha(r(d_1)) = ar[\iota'']$. By **Step 3**,

$$0 = \langle d_2, \det_F \circ \eta_B \rangle = \langle \alpha(d_2), \eta_S \rangle = a \langle r[\iota''], \eta_S \rangle = \pm a,$$

which contradicts the injectivity of α. This proves the claim. If $F_S \cong F$, then there exists an embedding ι of F_S with $\iota(F_S) = \phi_1(F)$, and $s \circ \iota = \iota$, which contradicts the claim. Thus $F_S \ncong F$, and F_S is isomorphic to the quartic CM subfield $\iota(F_S) \subset L$. This proves (3).

Fix an embedding ι_1 of F_S, and let $\iota_2 := s \circ \iota_1$. By the claim, $\iota_2 \notin \{\iota_1, \bar{\iota}_1\}$, so the s-invariant line in $X^*(\mathrm{U}_{F_S}) \otimes \mathbb{Q}$ is spanned by $[\iota_1] + [\iota_2]$, and $\alpha(d_1) = a([\iota_1] + [\iota_2])$ for some $a \in \mathbb{Q}$. By **Step 3**,

$$2 = \langle \alpha(d_1), \eta_S \rangle = a\big(\langle [\iota_1], \eta_S \rangle + \langle [\iota_2], \eta_S \rangle\big),$$

so

$$\langle [\iota_1], \eta_S \rangle = \langle [\iota_2], \eta_S \rangle = a = \pm 1.$$

Since $s \circ \bar{\iota}_1 = \bar{\iota}_2$, possibly replacing ι_1, ι_2 with $\bar{\iota}_1, \bar{\iota}_2$, we may assume that $a = 1$. Then $\Phi_S = \{\iota_1, \iota_2\}$, and $f_1 := [\iota_1]$ and $f_2 := [\iota_2]$ are the characters of **Step 1**. As $r \circ \iota_1 \in \{\iota_2, \bar{\iota}_2\}$, we have $r(f_1) = \delta f_2$ with $\delta = \pm 1$, and $r(f_2) = \delta r^2(f_1) = -\delta f_1$. Thus

$$\alpha(d_1) = f_1 + f_2, \qquad \alpha(d_2) = r(\alpha(d_1)) = -\delta(f_1 - f_2).$$

The map α is integral with determinant 2δ, so by [Moo99, (1.1)], $\alpha = \varphi^*$ for an isogeny $\varphi : \mathrm{Hg}(S) \to \mathrm{U}_F$. By **Step 2**, (B, S) is an exceptional pair and $\mathrm{Hg}(B \times S) = G$.

By **Step 4** and Lemma 8.5, F is biquadratic, cyclic over $\mathbb{Q}$, or not Galois over $\mathbb{Q}$. By **Steps 4, 5, 6, and 7**, in any case, either $\mathrm{Hdg}^p(B \times S) = \mathcal{D}^p(B \times S)$ for any p and (B, S) is not an exceptional pair, or there exists φ as in **Step 2**, so that (B, S) is an exceptional pair and $\mathrm{Hg}(B \times S) = G$. This proves the dichotomy and the description of $\mathrm{Hg}(S)$, φ, f_1, f_2, and $\mathrm{Hg}(B \times S)$ for exceptional pairs. If (B, S) is an exceptional pair, then the second alternative holds, and (1), (2), and (3) follow from **Steps 5, 6, and 7**. □

8.6. **The exceptional pairs.** In this subsection, we compute the Hodge classes on $B \times S$ for an exceptional pair (B, S), show that $B \times S$ admits no action of an imaginary quadratic field, and prove Theorem 1.4.

Proposition 8.14. *Let (B, S) be an exceptional pair, $A := B \times S$, and $M := (U_B \otimes T_S) \cap \mathrm{Hdg}^2(A)$, viewed in*

$$H^2(B, \mathbb{Q}) \otimes H^2(S, \mathbb{Q}) \subset H^4(A, \mathbb{Q}).$$

Let Θ_B be an ample class on B, pulled back to A. Then $\dim \mathcal{D}^2(A) = 8$, $\dim M = 4$, $\mathrm{Hdg}^2(A) = \mathcal{D}^2(A) \oplus M$, $\dim \mathcal{D}^3(A) = 10$, multiplication by Θ_B is injective on M, and $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus \Theta_B M$. Moreover, the following are equivalent.

(1) $\mathrm{HC}(A)$ *holds.*
(2) Any class in M is algebraic.
(3) $\mathrm{Hdg}^2(A)$ *contains an algebraic class not in $\mathcal{D}^2(A)$.*

Proof. Let $F := \mathrm{End}^0(B)$, $V := H^1(B, \mathbb{Q})$, and ψ the alternating form on V induced by a polarization of B, and let $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$ be the ordering of the complex embeddings of F in Definition 8.12 (1). For $l = 1, 2$, let $V_l := H^1(B, \mathbb{C})_{\phi_l}$ and $V_l^* := H^1(B, \mathbb{C})_{\bar{\phi}_l}$, which ψ pairs with V_l. Both are two-dimensional, since the multiplicities are $(2, 1, 0, 1)$. Let d_l be the character $\det V_l$ of $\mathrm{Hg}(B)_{\mathbb{C}}$, let x_l and y_l be generators of the lines $\wedge^2 V_l$ and $\wedge^2 V_l^*$, and let q_l be a generator of the line of $\mathrm{SL}(V_l)$-invariants in $V_l \otimes V_l^*$. By Lemma 8.10 (1),

$$H^1(B, \mathbb{C}) = (V_1 \oplus V_1^*) \oplus (V_2 \oplus V_2^*)$$

as representations of

$$\mathrm{Hg}(B)_{\mathbb{C}} \cong \mathrm{GL}(V_1) \times \mathrm{GL}(V_2),$$

and, by Lemma 8.10 (4), $U_B \otimes \mathbb{C}$ is spanned by x_1, y_1, x_2, and y_2. Let $G := \mathrm{Hg}(A)$. By Proposition 8.13, $\mathrm{Hg}(S)$ is a two-dimensional torus, and there exist an isogeny $\varphi : \mathrm{Hg}(S) \to \mathrm{U}_F$ and a basis f_1, f_2 of the character group of $\mathrm{Hg}(S)_{\mathbb{C}}$ such that

$$\lambda_1 := \varphi^* d_1 = f_1 + f_2, \qquad \lambda_2 := \varphi^* d_2 = \pm(f_1 - f_2),$$

and

$$G = \{(g, t) \in \mathrm{Hg}(B) \times \mathrm{Hg}(S) \mid \det_F(g) = \varphi(t)\}.$$

In particular, λ_1 and λ_2 are linearly independent, and $d_l(g) = \lambda_l(t)$ for any $(g, t) \in G_{\mathbb{C}}$ and $l = 1, 2$. We view characters of $\mathrm{Hg}(S)_{\mathbb{C}}$ as characters of $G_{\mathbb{C}}$ through the projection $G_{\mathbb{C}} \to \mathrm{Hg}(S)_{\mathbb{C}}$, so that a vector in a representation of $G_{\mathbb{C}}$ has weight χ (weights as defined at the beginning of Subsection 8.5) if any $(g, t) \in G_{\mathbb{C}}$ acts on it by $\chi(t)$.

Let N be the kernel of the projection $G \to \mathrm{Hg}(S)$. Then $N = \mathrm{SU}_F(V, \psi) \times \{1\}$. Over $\mathbb{C}$, $\det_F$ is the pair (d_1, d_2), so

$$N_{\mathbb{C}} = \mathrm{SL}(V_1) \times \mathrm{SL}(V_2).$$

Let

$$R := \left(\wedge^* H^1(B, \mathbb{C}) \right)^{N_{\mathbb{C}}} = \wedge^*(V_1 \oplus V_1^*)^{\mathrm{SL}(V_1)} \otimes \wedge^*(V_2 \oplus V_2^*)^{\mathrm{SL}(V_2)},$$

and let R^a be the degree a part of R. By Lemma 8.10 (5), the products $u_1 u_2$ with $u_l \in \{1, x_l, y_l, q_l, q_l^2\}$ form a basis of R, $R^a = 0$ for any odd a, and $q_l^3 = 0$. By Lemma 8.10 (5) and the equality $d_l(g) = \lambda_l(t)$, x_l, y_l, and q_l have weights λ_l, $-\lambda_l$, and 0. Since λ_1 and λ_2 are linearly independent, the weight-zero part of R^2 is spanned by q_1, q_2, the weight-zero part of R^4 by $q_1^2, q_1 q_2, q_2^2$, and the weight-zero part of R^6 by $q_1^2 q_2, q_1 q_2^2$. The non-zero weights on R^4 are $\pm\lambda_1$ on $x_1 q_2, y_1 q_2$, $\pm\lambda_2$ on $q_1 x_2, q_1 y_2$, and $\pm\lambda_1 \pm \lambda_2$ on the products $u_1 u_2$ with $u_l \in \{x_l, y_l\}$.

The group $\mathrm{Hg}(S)$ acts trivially on $H^0(S, \mathbb{Q})$, $H^4(S, \mathbb{Q})$, and $NS(S) \otimes \mathbb{Q}$, as they consist of Hodge classes (Lemma 2.3 (2)). By the Hodge index theorem, $H^2(S, \mathbb{Q}) = (NS(S) \otimes \mathbb{Q}) \oplus T_S$. By Definition 8.12 (3), U_B and T_S are isomorphic rational Hodge structures, and any isomorphism $U_B \to T_S$ of rational Hodge structures commutes with G [Moo99, (1.12)]. Thus $T_S \otimes \mathbb{C}$ is spanned by four vectors $t_1^+, t_1^-, t_2^+, t_2^-$, where $t_l^\pm$ has weight $\pm\lambda_l$ (Lemma 8.10 (4)). In particular, $\dim T_S = 4$ and $\dim NS(S) \otimes \mathbb{Q} = 2$.

By Lemma 2.3 (2), $\mathrm{Hdg}^p(A) \otimes \mathbb{C}$ is the space of $G_{\mathbb{C}}$-invariants in

$$H^{2p}(A, \mathbb{C}) = \bigoplus_{a+b=2p} \wedge^a H^1(B, \mathbb{C}) \otimes H^b(S, \mathbb{C}).$$

Since N acts trivially on $H^*(S, \mathbb{C})$, the $G_{\mathbb{C}}$-invariants are the weight-zero vectors in $\bigoplus_{a+b=2p} R^a \otimes H^b(S, \mathbb{C})$, and only even a and b contribute. Since the projection $G \to \mathrm{Hg}(B)$ is surjective (Lemma 2.3 (3)), the weight-zero part of R^a is $\mathrm{Hdg}^{a/2}(B) \otimes \mathbb{C}$, which is $\mathcal{D}^{a/2}(B) \otimes \mathbb{C}$ by Lemma 8.10 (1). In degree two, we obtain

$$\mathrm{Hdg}^1(A) = \mathrm{Hdg}^1(B) \oplus \mathrm{Hdg}^1(S),$$

and $\mathrm{Hdg}^1 = \mathcal{D}^1$ for any abelian variety [Moo99, (2.10)]. Hence

$$\mathcal{D}^p(A) = \sum_{a+b=p} \mathcal{D}^a(B) \cdot \mathcal{D}^b(S),$$

where $\mathcal{D}^a(B) = \mathrm{Hdg}^a(B)$, $\mathcal{D}^1(S) = NS(S) \otimes \mathbb{Q}$, $\mathcal{D}^2(S) = H^4(S, \mathbb{Q})$, and $\mathcal{D}^b(S) = 0$ for $b \geq 3$.

We compute $\mathrm{Hdg}^2(A) \otimes \mathbb{C}$. The weight-zero vectors in $R^4 \otimes H^0(S, \mathbb{C})$ form $\mathcal{D}^2(B) \otimes \mathbb{C}$, of dimension 3, and $R^0 \otimes H^4(S, \mathbb{C}) = \mathcal{D}^2(S) \otimes \mathbb{C}$ is one-dimensional. In $R^2 \otimes H^2(S, \mathbb{C})$, the weight-zero vectors are spanned by $\mathcal{D}^1(B) \otimes \mathcal{D}^1(S) \otimes \mathbb{C}$, of dimension 4, and by the four vectors

$$x_1 t_1^-, \quad y_1 t_1^+, \quad x_2 t_2^-, \quad y_2 t_2^+,$$

where products are cup products of pullbacks. The four vectors form a basis of the weight-zero part of $(U_B \otimes T_S) \otimes \mathbb{C}$, that is, of $M \otimes \mathbb{C}$. Therefore $\dim \mathcal{D}^2(A) = 3 + 4 + 1 = 8$, $\dim M = 4$, and

$$\mathrm{Hdg}^2(A) \otimes \mathbb{C} = (\mathcal{D}^2(A) \otimes \mathbb{C}) \oplus (M \otimes \mathbb{C}).$$

Since $\mathcal{D}^2(A)$ and M are rational subspaces, $\mathrm{Hdg}^2(A) = \mathcal{D}^2(A) \oplus M$.

We compute $\mathrm{Hdg}^3(A) \otimes \mathbb{C}$. The weight-zero vectors in $R^6 \otimes H^0(S, \mathbb{C})$ and in $R^2 \otimes H^4(S, \mathbb{C})$ form $\mathcal{D}^3(B) \otimes \mathbb{C}$ and $\mathcal{D}^1(B) \otimes \mathcal{D}^2(S) \otimes \mathbb{C}$, both of dimension 2. In $R^4 \otimes H^2(S, \mathbb{C})$, the weight-zero vectors are spanned by $\mathcal{D}^2(B) \otimes \mathcal{D}^1(S) \otimes \mathbb{C}$, of dimension 6, and by the four vectors

$$x_1 q_2 t_1^-, \quad y_1 q_2 t_1^+, \quad q_1 x_2 t_2^-, \quad q_1 y_2 t_2^+.$$

The weights $\pm\lambda_1 \pm \lambda_2$ on R^4 do not occur in $H^2(S, \mathbb{C})$, whose weights are 0, $\pm\lambda_1$, and $\pm\lambda_2$. Hence $\dim \mathcal{D}^3(A) = 2 + 6 + 2 = 10$ and $\dim \mathrm{Hdg}^3(A) = 14$.

Since $\Theta_B \in \mathrm{Hdg}^1(B)$, we may write $\Theta_B = a_1 q_1 + a_2 q_2$ with $a_1, a_2 \in \mathbb{C}$. Since $q_1^3 = q_2^3 = 0$,

$$\Theta_B^4 = 6 a_1^2 a_2^2 q_1^2 q_2^2.$$

As Θ_B is ample, $\Theta_B^4 \neq 0$, so $a_1 a_2 \neq 0$. Since $q_l x_l = q_l y_l = 0$, we have

$$\Theta_B x_1 t_1^- = a_2 x_1 q_2 t_1^-, \quad \Theta_B y_1 t_1^+ = a_2 y_1 q_2 t_1^+, \quad \Theta_B x_2 t_2^- = a_1 q_1 x_2 t_2^-, \quad \Theta_B y_2 t_2^+ = a_1 q_1 y_2 t_2^+.$$

Hence multiplication by Θ_B is injective on M, and $\Theta_B M \otimes \mathbb{C}$ is spanned by the four vectors above which are not in $\mathcal{D}^3(A) \otimes \mathbb{C}$. Since $\mathcal{D}^3(A)$ and $\Theta_B M$ are rational subspaces, $\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus \Theta_B M$.

To prove the equivalence, we first construct algebraic correspondences on A. For an abelian variety X, an algebraic class Γ on $X \times X$ acts on $H^*(X, \mathbb{Q})$ by

$$\gamma \to \mathrm{pr}_{2*}(\mathrm{pr}_1^* \gamma \cdot \Gamma),$$

and we call such an action an *algebraic correspondence*. Algebraic correspondences map Hodge classes to Hodge classes and algebraic classes to algebraic classes, and they are closed under sums, compositions, and multiplication by rational numbers. For $c \in \mathrm{End}(X)$, the pullback c^* is an algebraic correspondence. By [Kün93, Section 2], the Künneth projector π_j^X of $H^*(X, \mathbb{Q})$ onto $H^j(X, \mathbb{Q})$ is an algebraic correspondence for any j. If Γ_B and Γ_S are algebraic correspondences on B and S, then $\Gamma_B \times \Gamma_S$ is an algebraic correspondence on A which acts on $H^*(A, \mathbb{Q}) = H^*(B, \mathbb{Q}) \otimes H^*(S, \mathbb{Q})$ by $\Gamma_B \otimes \Gamma_S$.

For any element $c \in F$ and any complex embedding τ of F, c^* acts on $H^1(B,\mathbb{C})_\tau$ by $\tau(c)$. Thus, on

$$H^2(B,\mathbb{C}) = \bigoplus_\tau \wedge^2 H^1(B,\mathbb{C})_\tau \oplus \bigoplus_{\{\tau,\tau'\},\tau\neq\tau'} H^1(B,\mathbb{C})_\tau \otimes H^1(B,\mathbb{C})_{\tau'},$$

c^* acts by $\tau(c)^2$ on $\wedge^2 H^1(B,\mathbb{C})_\tau$ and by $\tau(c)\tau'(c)$ on $H^1(B,\mathbb{C})_\tau \otimes H^1(B,\mathbb{C})_{\tau'}$. The first sum is $U_B \otimes \mathbb{C}$, and we denote by U'_B the rational subspace with $U'_B \otimes \mathbb{C}$ equal to the second sum. After scalar extension to $\mathbb{C}$, the complex embeddings of F are independent coordinates on $F \otimes_{\mathbb{Q}} \mathbb{C}$, so for $\tau' \neq \tau''$ the function $c \to \tau(c)^2 - \tau'(c)\tau''(c)$ is a non-zero polynomial. Since F is Zariski dense in $F \otimes_{\mathbb{Q}} \mathbb{C}$, there exists $\beta \in F$ such that $\tau(\beta)^2 \neq \tau'(\beta)\tau''(\beta)$ for any τ and any $\tau' \neq \tau''$. The inequalities on β are preserved when β is multiplied by a positive integer, so possibly replacing β with a multiple, we may assume that $\beta \in \mathrm{End}(B)$. The polynomials

$$m_1(T) := \prod_\tau \big(T - \tau(\beta)^2\big), \qquad m_2(T) := \prod_{\{\tau,\tau'\},\tau\neq\tau'} \big(T - \tau(\beta)\tau'(\beta)\big)$$

have rational coefficients, since $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ permutes the complex embeddings of F, and they are coprime. Let $Q \in \mathbb{Q}[T]$ be a polynomial with $Q \equiv 1 \bmod m_1$ and $Q \equiv 0 \bmod m_2$, and let $P_U := \pi_2^B \circ Q(\beta^*) \circ \pi_2^B$. Then P_U is an algebraic correspondence on B which acts on $H^2(B,\mathbb{Q})$ as the projection onto U_B along U'_B and kills $H^j(B,\mathbb{Q})$ for $j \neq 2$. Let $P := P_U \times \pi_2^S$. On $H^4(A,\mathbb{Q})$, P is the projection onto $U_B \otimes H^2(S,\mathbb{Q})$ which kills $U'_B \otimes H^2(S,\mathbb{Q})$ and the other Künneth summands. Since $\mathcal{D}^1(B) \otimes \mathbb{C}$ is spanned by q_1, q_2, which belong to $U'_B \otimes \mathbb{C}$, P kills $\mathcal{D}^2(A)$. Moreover, P restricts to the identity on M.

For $c \in F$, let $n \geq 1$ be an integer with $nc \in \mathrm{End}(B)$, and let $c^\# := n^{-2}\pi_2^B \circ (nc)^* \circ \pi_2^B$. Then $c^\#$ is an algebraic correspondence which acts on $\wedge^2 H^1(B,\mathbb{C})_\tau$ by $\tau(c)^2$. Let

$$\kappa_c := \Big(P_U \circ \tfrac{1}{2}\big((c+1)^\# - c^\# - 1^\#\big) \circ P_U\Big) \times \pi_2^S.$$

Then κ_c is an algebraic correspondence on A which preserves $U_B \otimes T_S$ and $\mathrm{Hdg}^2(A)$, hence preserves M. Since $\frac{1}{2}\big((c+1)^2 - c^2 - 1\big) = c$, the correspondence $\frac{1}{2}\big((c+1)^\# - c^\# - 1^\#\big)$ acts on $\wedge^2 H^1(B,\mathbb{C})_\tau$ by $\tau(c)$. On $M \otimes \mathbb{C}$, κ_c acts on $x_l t_l^-$ by $\phi_l(c)$ and on $y_l t_l^+$ by $\bar{\phi}_l(c)$. Thus $c \to \kappa_c|_M$ is a unital homomorphism of $\mathbb{Q}$-algebras $F \to \mathrm{End}_{\mathbb{Q}}(M)$, so M is an F-vector space. Since $\dim_{\mathbb{Q}} M = 4 = [F:\mathbb{Q}]$, M is one-dimensional over F. Hence, for any non-zero $\mu \in M$, $M = \{\kappa_c(\mu) \mid c \in F\}$.

Now we prove the equivalence. If (1) holds, then (3) holds since $M \neq 0$ and $M \cap \mathcal{D}^2(A) = 0$. Assume (3), and let $\alpha \in \mathrm{Hdg}^2(A)$ be an algebraic class not in $\mathcal{D}^2(A)$. Then $P(\alpha)$ is the component of α in M, which is algebraic and non-zero. Thus any class $\kappa_c(P(\alpha))$ is algebraic, so (2) holds. Assume (2). Classes in $\mathcal{D}^p(A)$ are algebraic, and the classes in $\Theta_B M$ are algebraic as products of algebraic classes. Thus $\mathrm{HC}^2(A)$ and $\mathrm{HC}^3(A)$ hold by the decompositions of $\mathrm{Hdg}^2(A)$ and $\mathrm{Hdg}^3(A)$, and $\mathrm{HC}(A)$ holds by Lemma 2.6. □

The following lemma is the reason why Theorem 1.2 has no exceptional case.

Lemma 8.15. *Let (B,S) be an exceptional pair and A an abelian variety isogenous to $B \times S$. Then there exists no unital embedding $K \hookrightarrow \mathrm{End}^0(A)$ of an imaginary quadratic field K.*

Proof. Let $F := \mathrm{End}^0(B)$, and assume that there exists a unital embedding $\iota : K \hookrightarrow \mathrm{End}^0(A)$. An isogeny $f : A \to B \times S$ induces an isomorphism of $\mathbb{Q}$-algebras

$$\mathrm{End}^0(A) \to \mathrm{End}^0(B \times S), \quad u \to f \circ u \circ f^{-1},$$

so we obtain a unital embedding of K into $\mathrm{End}^0(B \times S)$. Since B is simple of dimension 4 and $\dim S = 2$, $\mathrm{Hom}(B,S) = 0$ and $\mathrm{Hom}(S,B) = 0$ [Moo99, (2.3)]. Hence

$$\mathrm{End}^0(B \times S) = F \times \mathrm{End}^0(S).$$

The projection to F is a unital homomorphism $K \to F$, which is injective since K is a field. Let K_F be the image and F^+ the maximal totally real subfield of F, which is a real quadratic field.

Since K_F is imaginary quadratic, $K_F \cap F^+ = \mathbb{Q}$, so the compositum $K_F F^+ \subset F$ has degree 4 and equals F. Since K_F and F^+ are Galois over $\mathbb{Q}$, F is biquadratic.

Let K_1 and K_2 be the imaginary quadratic subfields of F. They are not isomorphic: an imaginary quadratic field has exactly one image in $\mathbb{C}$, so if $K_1 \cong K_2$, then $\tau(K_1) = \tau(K_2)$ for any complex embedding τ of F, and $K_1 = K_2$. Since $\phi_2 \neq \phi_1$ and $\phi_2 \neq \bar{\phi}_1$, the embeddings ϕ_1 and ϕ_2 have equal restrictions to exactly one of K_1 and K_2. Relabelling K_1 and K_2, we may assume that $\phi_1|_{K_1} = \phi_2|_{K_1}$ and $\phi_1|_{K_2} = \bar{\phi}_2|_{K_2}$. By Proposition 8.13 (1), S is isogenous to $E_1 \times E_2$, where E_l is an elliptic curve with $\mathrm{End}^0(E_l) = K_l$ for $l = 1, 2$. Thus A is isogenous to $B \times E_1 \times E_2$, and ι induces a unital embedding of K into $\mathrm{End}^0(B \times E_1 \times E_2)$. This contradicts Lemma 8.11 (3). □

Proof of Theorem 1.4. The proof is long, so we divide it into several steps.

If A is of CM type, then $\mathrm{HC}(A)$ holds by [OAI26a, Theorem 1.1], where the notion of CM type is that of Definition 2.2. Thus we may assume that A is not of CM type.

Step 1. In this step, we show that $\mathrm{HC}(A)$ holds if A has a simple factor of dimension at most 2 which is not of CM type.

By Lemma 2.6, $\mathrm{HC}(A)$ holds if and only if $\mathrm{HC}(A')$ holds for an abelian variety A' isogenous to A, and A is isogenous to a product of simple abelian varieties [Moo99, (2.3)]. Possibly replacing A with such a product, we may assume that $A = \prod_{i=1}^k A_i^{r_i}$, where $A_1, \ldots, A_k$ are simple and pairwise non-isogenous.

We first note two facts. Let X_1 and X_2 be abelian varieties.

(P1) If
$$\mathrm{Hg}(X_1 \times X_2) = \mathrm{Hg}(X_1) \times \mathrm{Hg}(X_2),$$
then any Hodge class on $X_1 \times X_2$ is a sum of products of pullbacks of Hodge classes on X_1 and on X_2. In particular, if moreover $\mathrm{HC}(X_1)$ and $\mathrm{HC}(X_2)$ hold, then $\mathrm{HC}(X_1 \times X_2)$ holds.

(P2) For any integer $r \geq 1$, $\mathrm{Hg}(X_1^r \times X_2)$ is the image of $\mathrm{Hg}(X_1 \times X_2)$ under the diagonal embedding
$$\mathrm{GL}(H^1(X_1, \mathbb{Q})) \times \mathrm{GL}(H^1(X_2, \mathbb{Q})) \to \mathrm{GL}(H^1(X_1, \mathbb{Q})^{\oplus r} \oplus H^1(X_2, \mathbb{Q})).$$

Item (P1) follows from Lemma 2.3 (2), since the invariants of $\mathrm{Hg}(X_1) \times \mathrm{Hg}(X_2)$ in $H^*(X_1, \mathbb{Q}) \otimes H^*(X_2, \mathbb{Q})$ are the tensor products of the invariants of the factors. For (P2), the Hodge circle of $X_1^r \times X_2$ is the image of $\eta_{X_1 \times X_2}$ under the diagonal embedding.

Assume that $C := A_1$ is not of CM type and $\dim C \leq 2$, and let $r := r_1$ and $Y := \prod_{i=2}^k A_i^{r_i}$. If $Y = 0$, then $A = C^r$, and the Hodge algebra of C^r is generated by classes of divisors [MZ99, Theorem (0.1)(iv)], so $\mathrm{HC}(A)$ holds. Assume that $Y \neq 0$. Since no simple factor of Y is isogenous to C, $\mathrm{Hom}(C, Y) = 0$ [Moo99, (2.3)]. By [MZ99, Proposition (2.4), Lemma (3.4)],
$$\mathrm{Hg}(C \times Y) = \mathrm{Hg}(C) \times \mathrm{Hg}(Y).$$
By (P2),
$$\mathrm{Hg}(C^r \times Y) = \mathrm{Hg}(C^r) \times \mathrm{Hg}(Y),$$
where $\mathrm{Hg}(C)$ acts diagonally on $H^1(C, \mathbb{Q})^{\oplus r}$. Since $\dim C^r \leq 5$ and $\dim Y \leq 5$, $\mathrm{HC}(C^r)$ and $\mathrm{HC}(Y)$ hold by Theorem 2.16. Thus $\mathrm{HC}(A)$ holds by (P1).

Step 2. In this step, we treat simple factors of dimension 3 or 5, and we reduce the remaining cases to products $B \times S$ of a simple fourfold B which is not of CM type and an abelian surface S of CM type.

By **Step 1**, we may assume that any A_i with $\dim A_i \leq 2$ is of CM type. If $\dim A_i = 3$ for some i, then $\mathrm{HC}(A)$ holds by Theorem 8.8 if A is a product of two simple threefolds, and by Theorem 8.9 (2) otherwise. If $\dim A_i = 5$ for some i, then $A = A_i \times E$ for an elliptic curve E, and $\mathrm{HC}(A)$ holds by Proposition 8.4.

Assume that $\dim A_i \notin \{3,5\}$ for any i. Since A is not simple, $\dim A_i \neq 6$ for any i. If each A_i were of CM type, with a commutative semisimple subalgebra $L_i \subset \mathrm{End}^0(A_i)$ of dimension $2\dim A_i$, then $\prod_i L_i^{r_i} \subset \mathrm{End}^0(A)$ would show that A is of CM type. Hence some A_i is not of CM type, and $\dim A_i = 4$. Thus $A = B \times S$, where $B := A_i$ is a simple fourfold which is not of CM type and S is a product of abelian varieties of CM type of dimension at most 2. In particular, S is of CM type, and S is isogenous to a simple abelian surface of CM type, to a product of two non-isogenous elliptic curves with complex multiplication, or to E^2 for an elliptic curve E with complex multiplication.

By Lemma 2.3 (3),

$$\mathrm{Hg}(A) \subset \mathrm{Hg}(B) \times \mathrm{Hg}(S)$$

and both projections are surjective. By Lemma 2.3 (4), $\mathrm{Hg}(S)$ is a torus. If

$$\mathrm{Hg}(A) \neq \mathrm{Hg}(B) \times \mathrm{Hg}(S),$$

then Goursat's lemma [MZ99, (3.1)] implies that $\mathrm{Hg}(B)$ and $\mathrm{Hg}(S)$ have a non-trivial common connected quotient Q. As a quotient of $\mathrm{Hg}(S)$, Q is a torus, so Q is a quotient of $\mathrm{Hg}(B)/\mathrm{Hg}(B)^{\mathrm{der}}$. By [Moo99, (1.1)], we obtain the following.

(P3) If no non-zero subrepresentation of $X^*(\mathrm{Hg}(S)) \otimes \mathbb{Q}$ is isomorphic, as a representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, to a subrepresentation of

$$X^*(\mathrm{Hg}(B)/\mathrm{Hg}(B)^{\mathrm{der}}) \otimes \mathbb{Q},$$

then

$$\mathrm{Hg}(A) = \mathrm{Hg}(B) \times \mathrm{Hg}(S),$$

and $\mathrm{HC}(A)$ holds by (P1) and Theorem 2.16.

Step 3. In this step, we show that $\mathrm{HC}(A)$ holds unless $\mathrm{End}^0(B)$ is an imaginary quadratic field, or a quartic CM field with multiplicities $(2,1,0,1)$ on $H^{1,0}(B)$ for an ordering $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$ of the complex embeddings.

Let $D := \mathrm{End}^0(B)$, $V := H^1(B, \mathbb{Q})$, and ψ the alternating form on V induced by a polarization of B. Since B is simple, V is a simple rational Hodge structure [Moo99, Theorem (2.2)]. If B is not of type IV, then $\mathrm{Hg}(B)$ is semisimple [Moo99, Proposition (1.24)], so $\mathrm{Hg}(B)/\mathrm{Hg}(B)^{\mathrm{der}}$ is trivial, and $\mathrm{HC}(A)$ holds by (P3).

Assume that B is of type IV. Let Z_D be the center of D, which is a CM field, and write $[Z_D : \mathbb{Q}] = 2e$, $[D : Z_D] = q_0^2$, and $m := \dim_D V$. Then

$$8 = \dim_{\mathbb{Q}} V = 2eq_0^2 m.$$

Thus either $q_0 = 1$ and $(e,m) \in \{(1,4),(2,2),(4,1)\}$, or $q_0 = 2$ and $e = m = 1$. If $q_0 = 1$ and $e = 4$, then D is a CM field of degree $8 = 2\dim B$, so B is of CM type, which is not the case.

Assume that $q_0 = 2$. Then the multiplicities r and s in $H^{1,0}(B)$ of the simple $D \otimes_{\mathbb{Q}} \mathbb{C}$-modules satisfy $r + s = mq_0 = 2$. If $rs = 0$, we are in case (c) of [Tot15, Theorem 3.1], and if $r = s = 1$, we are in case (e). In both cases, [Tot15, Theorem 3.1] shows that any rational Hodge structure of weight one with an action of D, $m = 1$, and multiplicities (r,s) is a non-trivial direct sum, which contradicts the simplicity of V.

Assume that $q_0 = 1$ and $e = 2$. Then D is a quartic CM field, and the multiplicities are $(2,1,0,1)$ for some ordering $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$, as shown in the paragraph after Definition 8.12.

The remaining case is $q_0 = e = 1$, in which D is an imaginary quadratic field.

Step 4. In this step, we show that $\mathrm{HC}(A)$ holds if $K := \mathrm{End}^0(B)$ is an imaginary quadratic field.

Let (r,s) be the multiplicities of K on $H^{1,0}(B)$ at the complex embeddings $\sigma, \bar{\sigma}$ of K (Definition 2.7), so that $r + s = 4$. If $rs = 0$, then V is a non-trivial direct sum by case (c) of [Tot15, Theorem 3.1], which is impossible. The multiplicities at $\bar{\sigma}$ and σ are (s,r), so possibly replacing σ with $\bar{\sigma}$, we may assume that $(r,s) = (2,2)$ or $(r,s) = (3,1)$.

Assume that $(r,s) = (2,2)$. Then $\det_K \circ \eta_B$ is trivial, since $\eta_B(z)$ acts on $H^1(B,\mathbb{C})_\sigma$ with eigenvalues $z, z, \bar{z}, \bar{z}$. Moreover, $\mathrm{Hg}(B)$ commutes with K (Lemma 2.3 (1)) and preserves ψ, since ψ is induced by the class of an ample divisor, which is invariant by Lemma 2.3 (2). Thus $\mathrm{Hg}(B) \subset \mathrm{SU}_K(V,\psi)$. The center of $\mathrm{Hg}(B)$ commutes with $\mathrm{Hg}(B)$, so it is contained in K by Lemma 2.3 (1). An element $\lambda \in K$ which belongs to $\mathrm{SU}_K(V,\psi)$ satisfies $\lambda^4 = \det_K(\lambda) = 1$. Thus the center of the connected reductive group $\mathrm{Hg}(B)$ is finite, so $\mathrm{Hg}(B)$ is semisimple, and $\mathrm{HC}(A)$ holds by (P3).

Assume that $(r,s) = (3,1)$. By [MZ99, Theorem (0.1)(iv)], $\mathrm{Hg}(B) = \mathrm{U}_K(V,\psi)$. Let $V_\sigma := H^1(B,\mathbb{C})_\sigma$, and let $V_\sigma^* := H^1(B,\mathbb{C})_{\bar\sigma}$, which ψ pairs with V_σ. Then

$$\mathrm{Hg}(B)_{\mathbb{C}} \cong \mathrm{GL}(V_\sigma), \qquad \mathrm{Hg}(B)^{\mathrm{der}} = \mathrm{SU}_K(V,\psi),$$

and $\det_K$ induces an isomorphism

$$\mathrm{Hg}(B)/\mathrm{Hg}(B)^{\mathrm{der}} \cong \mathrm{U}_K\,.$$

Hence

$$X^*(\mathrm{Hg}(B)/\mathrm{Hg}(B)^{\mathrm{der}}) \otimes \mathbb{Q} \cong X^*(\mathrm{U}_K) \otimes \mathbb{Q}$$

is one-dimensional, and $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on it by -1 exactly on the elements which act non-trivially on K.

First assume that S is simple. Then $L := \mathrm{End}^0(S)$ is a quartic CM field without imaginary quadratic subfield, and $\mathrm{Hg}(S) = \mathrm{U}_L$ [MZ99, (2.2)]. By Lemma 8.5, $X^*(\mathrm{U}_L) \otimes \mathbb{Q}$ is an irreducible representation of dimension 2. Hence $\mathrm{HC}(A)$ holds by (P3).

Next assume that S is a product of elliptic curves with complex multiplication. Since elliptic curves with complex multiplication by isomorphic fields are isogenous, after grouping isogenous factors, A is isogenous to $B \times E_K^{r_0} \times \prod_{j=1}^k E_j^{r_j}$, where $r_0 \in \{0,1,2\}$, E_K is an elliptic curve with $\mathrm{End}^0(E_K) \cong K$, and E_j is an elliptic curve with $\mathrm{End}^0(E_j) \cong K_j$ for pairwise non-isomorphic imaginary quadratic fields $K_1, \dots, K_k$, none of which is isomorphic to K. By [Moo99, (2.6)], $\mathrm{Hg}(E_K) = \mathrm{U}_K$ and $\mathrm{Hg}(E_j) = \mathrm{U}_{K_j}$.

Assume that $r_0 \le 1$. Let $H := \mathrm{Hg}(B \times E_K^{r_0})$ and $A' := B \times E_K^{r_0} \times \prod_{j=1}^k E_j$. By (P2), $\mathrm{Hg}(A) = \mathrm{Hg}(A')$ with the diagonal action on repeated factors. By Lemma 2.3 (3),

$$\mathrm{Hg}(A') \subset H \times \prod_j \mathrm{U}_{K_j}$$

is a connected subgroup whose projection to each factor is surjective. The identity component Z^0 of the center of H projects into the center of $\mathrm{Hg}(B)$, which is the torus U_K of scalars, and into $\mathrm{Hg}(E_K) = \mathrm{U}_K$ if $r_0 = 1$. Since H/H^{der} is isogenous to Z^0, any irreducible subrepresentation of $X^*(H/H^{\mathrm{der}}) \otimes \mathbb{Q}$ is isomorphic to $X^*(\mathrm{U}_K) \otimes \mathbb{Q}$, which is not isomorphic to $X^*(\mathrm{U}_{K_j}) \otimes \mathbb{Q}$ for any j, as $K_j \not\cong K$. By Lemma 8.6, $\mathrm{Hg}(A') = H \times \prod_j \mathrm{U}_{K_j}$. By (P2),

$$\mathrm{Hg}(A) = H \times \mathrm{Hg}\Big(\prod_j E_j^{r_j}\Big),$$

and $\mathrm{HC}(A)$ holds by (P1) and Theorem 2.16.

Assume that $r_0 = 2$, that is, A is isogenous to $B \times E_K^2$. Possibly replacing A with $B \times E_K^2$, we may assume that $A = B \times E_K^2$. Let K act on E_K so that the multiplicity of K on $H^{1,0}(E_K)$ at σ is 0, let $\ell := H^1(E_K,\mathbb{C})_\sigma$ and $\bar\ell := H^1(E_K,\mathbb{C})_{\bar\sigma}$, and for $t \in \mathrm{U}_K$ let $\sigma(t) \in \mathbb{C}$ be the scalar by which t acts on ℓ. Let

$$G := \{(g,t) \in \mathrm{U}_K(V,\psi) \times \mathrm{U}_K \mid \det_K(g) t^2 = 1\}.$$

For $z \in \mathbb{U}$, $\det(\eta_B(z)|_{V_\sigma}) = z^3\bar z = z^2$ and $\sigma(\eta_{E_K}(z)) = \bar z$, so $\eta_{B\times E_K}(\mathbb{U}) \subset G(\mathbb{R})$ and $\mathrm{Hg}(B \times E_K) \subset G$. The projection $G \to \mathrm{U}_K$ to the second factor is surjective since $\det_K$ is surjective, with connected kernel $\mathrm{SU}_K(V,\psi)$. Hence G is connected, and $\dim G = \dim \mathrm{U}_K(V,\psi)$. Since

$\mathrm{Hg}(B \times E_K)$ surjects onto $\mathrm{Hg}(B) = \mathrm{U}_K(V, \psi)$, we obtain $\mathrm{Hg}(B \times E_K) = G$ (cf. [MZ99, (5.11)]). By (P2), $\mathrm{Hg}(A) = G$, acting diagonally on $H^1(E_K, \mathbb{Q})^{\oplus 2}$. Over $\mathbb{C}$,

$$G_{\mathbb{C}} = \{(g, t) \in \mathrm{GL}(V_\sigma) \times \mathbb{G}_m \mid \det(g)t^2 = 1\},$$

acting on V_σ by g, on V_σ^* through the dual, on $\ell^{\oplus 2}$ by t, and on $\bar{\ell}^{\oplus 2}$ by t^{-1}, and

$$H^*(A, \mathbb{C}) = \bigoplus_{a,b,c,c'} \wedge^a V_\sigma \otimes \wedge^b V_\sigma^* \otimes \wedge^c(\ell^{\oplus 2}) \otimes \wedge^{c'}(\bar{\ell}^{\oplus 2}).$$

The kernel of the projection $G_{\mathbb{C}} \to \mathbb{G}_m$ is $\mathrm{SL}(V_\sigma)$. The $\mathrm{SL}(V_\sigma)$-invariants in $\wedge^a V_\sigma \otimes \wedge^b V_\sigma^*$ form

$$\mathrm{Hom}_{\mathrm{SL}(V_\sigma)}(\wedge^b V_\sigma, \wedge^a V_\sigma).$$

Since the representations $\wedge^a V_\sigma$ for $0 \le a \le 4$ are irreducible and pairwise non-isomorphic except $\wedge^0 V_\sigma \cong \wedge^4 V_\sigma$, Schur's lemma implies that the $\mathrm{SL}(V_\sigma)$-invariants in $\wedge^a V_\sigma \otimes \wedge^b V_\sigma^*$ are zero unless $a = b$ or $\{a, b\} = \{0, 4\}$, in which case they are one-dimensional. For $a = b$, they are spanned by the identity of $\wedge^a V_\sigma$, which is invariant under $\mathrm{GL}(V_\sigma)$. The group $G_{\mathbb{C}}$ acts on $\wedge^4 V_\sigma$ by $\det(g) = t^{-2}$ and on $\wedge^4 V_\sigma^*$ by t^2, and on

$$\wedge^c(\ell^{\oplus 2}) \otimes \wedge^{c'}(\bar{\ell}^{\oplus 2})$$

by $t^{c-c'}$. Hence the $G_{\mathbb{C}}$-invariants in $H^*(A, \mathbb{C})$ are spanned by the products of the $\mathrm{GL}(V_\sigma)$-invariant classes with $a = b$ and the classes with $c = c'$, together with the lines

$$\wedge^4 V_\sigma \otimes \wedge^2(\ell^{\oplus 2}) = \wedge^6(V_\sigma \oplus \ell^{\oplus 2}), \qquad \wedge^4 V_\sigma^* \otimes \wedge^2(\bar{\ell}^{\oplus 2}) = \wedge^6(V_\sigma^* \oplus \bar{\ell}^{\oplus 2}).$$

The products of the $\mathrm{GL}(V_\sigma)$-invariant classes with $a = b$ and the classes with $c = c'$ are products of pullbacks of Hodge classes on B and on E_K^2, which are algebraic by Theorem 2.16. Let K act diagonally on A. Then $H^1(A, \mathbb{C})_\sigma = V_\sigma \oplus \ell^{\oplus 2}$, the multiplicities of K on $H^{1,0}(A)$ are $(3 + 0 + 0, 1 + 1 + 1) = (3, 3)$, and the lines $\wedge^6(V_\sigma \oplus \ell^{\oplus 2})$ and $\wedge^6(V_\sigma^* \oplus \bar{\ell}^{\oplus 2})$ span $W_K(A) \otimes \mathbb{C}$. Since A is not simple, any class in $W_K(A)$ is algebraic by Theorem 8.2. Thus $\mathrm{HC}(A)$ holds.

Step 5. In this step, we show that $\mathrm{HC}(A)$ holds if $F := \mathrm{End}^0(B)$ is a quartic CM field with multiplicities $(2, 1, 0, 1)$ at $\phi_1, \phi_2, \bar{\phi}_1, \bar{\phi}_2$, unless (B, S) is an exceptional pair.

Here B satisfies condition (1) of Definition 8.12. By **Step 2**, S is isogenous to a simple abelian surface of CM type, to a product of two non-isogenous elliptic curves with complex multiplication, or to E^2 for an elliptic curve E with complex multiplication. In the first two cases, S satisfies condition (2) of Definition 8.12, and by Proposition 8.13, either (B, S) is an exceptional pair, or $\mathrm{Hdg}^p(B \times S) = \mathcal{D}^p(B \times S)$ for any p, in which case $\mathrm{HC}(A)$ holds since $\mathcal{D}^p(B \times S)$ is spanned by products of classes of divisors, and by Lemma 2.6.

Assume that S is isogenous to E^2. Possibly replacing A with $B \times E^2$, we may assume that $A = B \times E \times E$. First assume that F is biquadratic, with imaginary quadratic subfields K_1 and K_2. Since $\phi_2 \neq \phi_1$ and $\phi_2 \neq \bar{\phi}_1$, the embeddings ϕ_1 and ϕ_2 have equal restrictions to exactly one of K_1 and K_2. Relabelling K_1 and K_2, we may assume that $\phi_1|_{K_1} = \phi_2|_{K_1}$ and $\phi_1|_{K_2} = \bar{\phi}_2|_{K_2}$. Then Lemma 8.11 (1) and (2), with $E' = E$, imply $\mathrm{HC}(A)$. Next assume that F is not biquadratic. If a quartic CM field F' contains an imaginary quadratic field K', then F' is the compositum of K' and the maximal totally real subfield of F', hence biquadratic. Thus F has no imaginary quadratic subfield. Since $\mathrm{Hom}(E, B) = 0$ and $\mathrm{End}^0(E)$ does not embed into F, [MZ99, Proposition (3.8)] implies that

$$\mathrm{Hg}(B \times E) = \mathrm{Hg}(B) \times \mathrm{Hg}(E).$$

By (P2),

$$\mathrm{Hg}(A) = \mathrm{Hg}(B) \times \mathrm{Hg}(E^2),$$

and $\mathrm{HC}(A)$ holds by (P1) and Theorem 2.16.

Together with the case of CM type, by **Steps 1–5**, $\mathrm{HC}(A)$ holds unless A is isogenous to $B \times S$ for an exceptional pair (B, S). Since conditions (1)–(3) of Theorem 1.4 are the conditions of Definition 8.12, this proves the theorem. □

9. Proofs of the main theorems

In this section, we prove Theorem 1.2. Theorem 1.3 is proved in Section 6, and Theorem 1.4 is proved in Section 8.

Proof of Theorem 1.2. Let K be $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, let A be an abelian variety of dimension 6, and fix a unital $\mathbb{Q}$-algebra embedding $K \hookrightarrow \mathrm{End}^0(A)$. We regard K as a subfield of $\mathrm{End}^0(A)$. We need to show that $\mathrm{HC}(A)$ holds. We distinguish cases according to $\mathrm{End}^0(A)$.

Case 1. A is of CM type. By [OAI26a, Theorem 1.1], for any abelian variety C of CM type and any integer $p \geq 0$, the cycle class map

$$\mathrm{cl} : CH^p(C) \otimes \mathbb{Q} \to H^{2p}(C, \mathbb{Q}) \cap H^{p,p}(C)$$

is surjective. The notion of CM type in [OAI26a, Theorem 1.1] is that of Definition 2.2. Hence $\mathrm{HC}(A)$ holds.

Case 2. A is not simple and not of CM type. Suppose that A is isogenous to a product $B \times S$ satisfying conditions (1)–(3) of Theorem 1.4. Then (B, S) is an exceptional pair (Definition 8.12), and Lemma 8.15 contradicts the embedding $K \hookrightarrow \mathrm{End}^0(A)$. Hence A is not isogenous to such a product, and $\mathrm{HC}(A)$ holds by Theorem 1.4.

Case 3. A is simple. Then $\mathrm{End}^0(A)$, together with the Rosati involution of a polarization, is of Albert type I, II, III, or IV [Moo99, (1.19)]. We treat the four types separately.

Case 3.1. $\mathrm{End}^0(A)$ is of type I. Then $\mathrm{End}^0(A)$ is a totally real field. A totally real field contains no imaginary quadratic subfield, while $K \subset \mathrm{End}^0(A)$. Thus this case does not occur.

Case 3.2. $\mathrm{End}^0(A)$ is of type II. Since A is simple, $H^1(A, \mathbb{Q})$ is a simple polarizable $\mathbb{Q}$-Hodge structure, and the algebra of Hodge endomorphisms of $H^1(A, \mathbb{Q})$ is the opposite algebra of $\mathrm{End}^0(A)$ [Moo99, Theorem (2.2) and (2.3)]. In particular, the algebra of Hodge endomorphisms of $H^1(A, \mathbb{Q})$ is of type II. Since $\dim A = 6$ is twice the odd prime 3, $\mathrm{HC}(A)$ holds by [Fla17, Corollary 13.5].

Case 3.3. $\mathrm{End}^0(A)$ is of type III. Let $D := \mathrm{End}^0(A)$. By Definition-Theorem 7.6, D is a totally definite quaternion algebra over $\mathbb{Q}$ and the subfield $K \subset D$ has multiplicities $(3, 3)$ on $H^{1,0}(A)$. Hence any class in $W_K(A)$ is algebraic by Theorem 1.3, and $\mathrm{HC}(A)$ holds by Corollary 7.7.

Case 3.4. $\mathrm{End}^0(A)$ is of type IV. By Lemma 2.5, $L := \mathrm{End}^0(A)$ is a CM field of degree 2, 4, 6, or 12 over $\mathbb{Q}$. Let Θ be an ample class. Since $\Theta \in NS(A) \otimes \mathbb{Q}$, the class Θ^p is algebraic for any p. Since $\mathcal{D}^p(A)$ is spanned by products of classes of divisors, any class in $\mathcal{D}^p(A)$ is algebraic.

Suppose that $[L : \mathbb{Q}] = 2$. Since $K \subset L$ and $[K : \mathbb{Q}] = 2$, we have $L = K$. Let $(r, 6 - r)$ be the multiplicities of K on $H^{1,0}(A)$. If $r \neq 3$, then $\mathrm{Hdg}^p(A) = \mathbb{Q}\Theta^p$ for any $0 \leq p \leq 6$ by Proposition 7.1, so $\mathrm{HC}(A)$ holds. If $r = 3$, then $\mathrm{Hdg}^p(A) = \mathbb{Q}\Theta^p$ for $p \neq 3$ and

$$\mathrm{Hdg}^3(A) = \mathbb{Q}\Theta^3 \oplus W_K(A)$$

by Proposition 7.1. Since the multiplicities of K on $H^{1,0}(A)$ are $(3, 3)$, any class in $W_K(A)$ is algebraic by Theorem 1.3. Hence $\mathrm{HC}(A)$ holds.

Suppose that $[L : \mathbb{Q}] = 4$. Then $\mathrm{HC}(A)$ holds by Theorem 7.4.

Suppose that $[L : \mathbb{Q}] = 6$. We first show that L has at most one quadratic subfield. If $E_1 \neq E_2$ are quadratic subfields of L, then $E_1 \cap E_2 = \mathbb{Q}$, so the compositum $E_1E_2 \subset L$ has degree 4 over $\mathbb{Q}$, and by the tower law 4 divides $[L : \mathbb{Q}] = 6$, a contradiction. Thus K is the only quadratic subfield of L. If the multiplicities of K on $H^{1,0}(A)$ are not $(3, 3)$, then L contains no imaginary quadratic subfield whose multiplicities on $H^{1,0}(A)$ are $(3, 3)$, so $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for any p by Proposition 7.2 (1), and $\mathrm{HC}(A)$ holds. If they are $(3, 3)$, then by Proposition 7.2 (2), $\mathrm{Hdg}^p(A) = \mathcal{D}^p(A)$ for $p \neq 3$ and

$$\mathrm{Hdg}^3(A) = \mathcal{D}^3(A) \oplus W_K(A),$$

and any class in $W_K(A)$ is algebraic by Theorem 1.3. Hence $\mathrm{HC}(A)$ holds.

Suppose that $[L:\mathbb{Q}]=12$. Then L is a commutative semisimple $\mathbb{Q}$-subalgebra of $\mathrm{End}^0(A)$ with $\dim_{\mathbb{Q}} L = 12 = 2\dim A$. Thus A is of CM type (Definition 2.2), and this case is covered by Case 1.

Since A is of CM type, or not simple and not of CM type, or simple, A belongs to Case 1, Case 2, or Case 3, and hence $\mathrm{HC}(A)$ holds. □

Appendix A. Explicit monodromy forms

The roots r_j and a_j and the matrix Ω are those of Construction-Theorems 4.4 and 4.5, and $R=[R_0,R_1]$, N, u_k, P_I, Q_I, and the leaf numbering π are as in Definition-Proposition 5.2. This appendix lists the exact data used in the proofs of Proposition 5.4, Proposition 5.6, and Theorem 5.7. All entries are computed by exact arithmetic in K, and any of them can be checked with any computer algebra system.

For $Q\in\mathrm{Herm}_4(K)$ and $1\le j\le 4$ we write $d_j:=Q_{jj}\in\mathbb{Q}$, and for $j<k$ the entry Q_{jk} has two rational coordinates x_{jk},y_{jk}, defined at the beginning of each subsection. We order the sixteen coordinates on $\mathrm{Herm}_4(K)$ as

$$(d_1,d_2,d_3,d_4,x_{12},y_{12},x_{13},y_{13},x_{14},y_{14},x_{23},y_{23},x_{24},y_{24},x_{34},y_{34}).$$

In both subsections n_j denotes the j-th row of N.

The determinants below are computed by Gaussian elimination without scaling of rows. Rows are numbered from 1, always by their current position. At the m-th stage the pivot is the first non-zero entry in column m among rows $m,m+1,\dots$. If the pivot is in row $p\neq m$, then rows m and p are exchanged, and the row exchange (m,p) is listed. Then, for u the pivot, any row below row m with entry c in column m is replaced by itself minus c/u times row m. The determinant equals $(-1)^s$ times the product of the pivots, where s is the number of row exchanges.

A.1. **The Eisenstein forms.** In this subsection $K=\mathbb{Q}(\omega)$, and for $j<k$ the (j,k)-entry of $Q\in\mathrm{Herm}_4(K)$ is $x_{jk}+y_{jk}\omega$. Since $\omega^2=-1-\omega$, we have

$$(a+b\omega)(c+d\omega)=(ac-bd)+(ad+bc-bd)\omega \quad\text{and}\quad \overline{a+b\omega}=(a-b)-b\omega$$

for $a,b,c,d\in\mathbb{Q}$.

Table 1 lists the 10×4 matrix N. Direct multiplication with $\omega^2=-1-\omega$ shows $RN=0$. Since R_0 is invertible and the last four rows of N form I_4, the equation $RN=0$ determines $N=\begin{pmatrix}-R_0^{-1}R_1\\ I_4\end{pmatrix}$, so R_0^{-1} need not be computed.

n_1	$5+4\omega$	$2-2\omega$	$-1-5\omega$	$-4-2\omega$
n_2	$3+3\omega$	$1-\omega$	-3ω	$-2-\omega$
n_3	3	$-1-2\omega$	$-3-3\omega$	-3
n_4	$-5-\omega$	3ω	$2+4\omega$	3
n_5	$1+\omega$	$-\omega$	$-\omega$	-1
n_6	$1+2\omega$	$1-\omega$	$-1-2\omega$	$-2-\omega$
n_7	1	0	0	0
n_8	0	1	0	0
n_9	0	0	1	0
n_{10}	0	0	0	1

Table 1. The rows $n_1,\dots,n_{10}$ of the matrix N for $K=\mathbb{Q}(\omega)$.

As in Proposition 5.4, we write $Q_i:=Q_{\{i\}}$ for $1\le i\le 10$ and $Q_{\le j}:=Q_{\{1,\dots,j\}}$ for $2\le j\le 8$. Since $RP_IN=\sum_{i\in I} r_in_i$ for any set I of leaves, we have

$$Q_i=(r_i^*r_i)n_i^*n_i \quad\text{and}\quad Q_{\le j}=\Big(\sum_{i=1}^{j} r_in_i\Big)^*\Big(\sum_{i=1}^{j} r_in_i\Big),$$

where $r_i^* r_i = 2$ for $i \in \{1,2,3,4,6\}$ and $r_i^* r_i = 6$ for $i \in \{5,7,8,9,10\}$. For example, $RN = 0$ implies $\sum_{i=1}^{8} r_i n_i = -(r_9 n_9 + r_{10} n_{10})$, and the only non-zero entries of $Q_{\leq 8}$ are $d_3 = d_4 = 6$ and $Q_{34} = Q_{43} = 3$.

Let $\mathcal{Q}$ be the 17×16 integer matrix whose rows are the coordinate vectors of

$$Q_1, \ldots, Q_{10}, Q_{\leq 2}, \ldots, Q_{\leq 8},$$

in the order listed. Tables 2 and 3 list columns 1–8 and columns 9–16 of $\mathcal{Q}$, respectively.

	d_1	d_2	d_3	d_4	x_{12}	y_{12}	x_{13}	y_{13}
Q_1	42	24	42	24	−12	−36	−42	−42
Q_2	18	6	18	6	−6	−12	−18	−18
Q_3	18	6	18	18	−6	−12	−18	−18
Q_4	42	18	24	18	−6	−30	−24	−36
Q_5	6	6	6	6	−6	−6	−6	−6
Q_6	6	6	6	6	−6	−6	−6	0
Q_7	6	0	0	0	0	0	0	0
Q_8	0	6	0	0	0	0	0	0
Q_9	0	0	6	0	0	0	0	0
Q_{10}	0	0	0	6	0	0	0	0
$Q_{\leq 2}$	33	18	33	18	−9	−27	−33	−33
$Q_{\leq 3}$	33	18	33	18	−12	−27	−33	−33
$Q_{\leq 4}$	18	9	15	9	−3	−12	−12	−15
$Q_{\leq 5}$	9	6	9	6	0	−3	−6	−6
$Q_{\leq 6}$	6	6	6	6	0	0	0	0
$Q_{\leq 7}$	0	6	6	6	0	0	0	0
$Q_{\leq 8}$	0	0	6	6	0	0	0	0

TABLE 2. Columns 1–8 of the coordinate matrix $\mathcal{Q}$ of the seventeen forms for $K = \mathbb{Q}(\omega)$.

	x_{14}	y_{14}	x_{23}	y_{23}	x_{24}	y_{24}	x_{34}	y_{34}
Q_1	−24	12	12	−24	−24	−24	−12	−36
Q_2	−6	6	6	−6	−6	−6	−6	−12
Q_3	−18	0	6	−6	−6	−12	0	−18
Q_4	−24	6	12	−12	−18	−18	−12	−24
Q_5	0	6	6	0	−6	−6	−6	−6
Q_6	0	6	0	−6	−6	−6	0	−6
Q_7	0	0	0	0	0	0	0	0
Q_8	0	0	0	0	0	0	0	0
Q_9	0	0	0	0	0	0	0	0
Q_{10}	0	0	0	0	0	0	0	0
$Q_{\leq 2}$	−18	9	9	−18	−18	−18	−9	−27
$Q_{\leq 3}$	−21	3	12	−15	−12	−18	−3	−24
$Q_{\leq 4}$	−9	3	6	−6	−6	−6	−3	−9
$Q_{\leq 5}$	−3	0	3	−3	−3	−3	0	−3
$Q_{\leq 6}$	0	0	0	−3	−3	−3	3	0
$Q_{\leq 7}$	0	0	0	−3	−3	−3	3	0
$Q_{\leq 8}$	0	0	0	0	0	0	3	0

TABLE 3. Columns 9–16 of the coordinate matrix $\mathcal{Q}$ of the seventeen forms for $K = \mathbb{Q}(\omega)$.

The coefficient vector of the functional ℓ of Proposition 5.4, in the coordinate order above, is killed by $\mathcal{Q}$:

$$\mathcal{Q}\cdot(0,0,0,0,-3,1,0,0,-1,-2,0,0,-4,4,0,0)^t=0. \tag{A.1}$$

We now compute the 15×15 minor of $\mathcal{Q}/3$ formed by rows 1–13, 15, and 17 of $\mathcal{Q}/3$, namely the rows of

$$Q_1/3,\dots,Q_{10}/3,\ Q_{\le 2}/3,\ Q_{\le 3}/3,\ Q_{\le 4}/3,\ Q_{\le 6}/3,\ Q_{\le 8}/3,$$

and by the fifteen columns other than column 14, which is the column of y_{24}. Moving rows 7–10, the rows of $Q_7/3,\dots,Q_{10}/3$, to the front takes 24 transpositions, so the sign of the determinant does not change. In rows 7–10 of $\mathcal{Q}/3$, the entries in the columns of $d_1,\dots,d_4$ form the block $2I_4$, and all other entries are 0. Thus the minor equals $2^4\det O$, where O is the 11×11 matrix formed by rows $1,2,3,4,5,6,11,12,13,15,17$ of $\mathcal{Q}/3$ and by the columns of

$$x_{12},y_{12},x_{13},y_{13},x_{14},y_{14},x_{23},y_{23},x_{24},x_{34},y_{34},$$

in the order listed. Table 4 lists O.

	x_{12}	y_{12}	x_{13}	y_{13}	x_{14}	y_{14}	x_{23}	y_{23}	x_{24}	x_{34}	y_{34}
$Q_1/3$	−4	−12	−14	−14	−8	4	4	−8	−8	−4	−12
$Q_2/3$	−2	−4	−6	−6	−2	2	2	−2	−2	−2	−4
$Q_3/3$	−2	−4	−6	−6	−6	0	2	−2	−2	0	−6
$Q_4/3$	−2	−10	−8	−12	−8	2	4	−4	−6	−4	−8
$Q_5/3$	−2	−2	−2	−2	0	2	2	0	−2	−2	−2
$Q_6/3$	−2	−2	−2	0	0	2	0	−2	−2	0	−2
$Q_{\le 2}/3$	−3	−9	−11	−11	−6	3	3	−6	−6	−3	−9
$Q_{\le 3}/3$	−4	−9	−11	−11	−7	1	4	−5	−4	−1	−8
$Q_{\le 4}/3$	−1	−4	−4	−5	−3	1	2	−2	−2	−1	−3
$Q_{\le 6}/3$	0	0	0	0	0	0	0	−1	−1	1	0
$Q_{\le 8}/3$	0	0	0	0	0	0	0	0	0	1	0

TABLE 4. The 11×11 matrix O, formed by eleven rows of $\mathcal{Q}/3$ and by the eleven columns other than $d_1,\dots,d_4$ and y_{24}.

Gaussian elimination of the 11×11 matrix O has the five row exchanges

$$(3,4),\ (4,5),\ (6,8),\ (7,8),\ (8,10),$$

and the eleven pivots of O are $-4,2,1,12,-4,-2,-1,-1,1,1/3,-1$, in the order of the stages. Hence

$$\begin{aligned}\det O&=(-1)^5\cdot(-4)\cdot 2\cdot 1\cdot 12\cdot(-4)\cdot(-2)\cdot(-1)\cdot(-1)\cdot 1\cdot\tfrac{1}{3}\cdot(-1)=-256,\\ 2^4\det O&=2^4\cdot(-256)=-4096,\qquad -4096\cdot 3^{15}=-58773123072.\end{aligned} \tag{A.2}$$

Here $2^4\det O=-4096$ is the 15×15 minor of $\mathcal{Q}/3$ described above, and $-4096\cdot 3^{15}=-58773123072$ is the corresponding 15×15 minor of $\mathcal{Q}$.

A.2. **The Gaussian forms.** In this subsection $K=\mathbb{Q}(i)$, and for $j<k$ the (j,k)-entry of $Q\in\mathrm{Herm}_4(K)$ is $x_{jk}+iy_{jk}$. As in Proposition 5.6, we write $L_j:=Q_{\{j\}}$ for $1\le j\le 9$ and $Q_{\le k}:=Q_{\{1,\dots,k\}}$ for $2\le k\le 7$, with the leaves numbered as in Definition-Proposition 5.2.

The matrix $R_0=[a_1,a_3,a_4,a_2,a_5,a_6,a_7,a_8]$ has $\det R_0=-1$. Table 5 lists the rows $n_1,\dots,n_8$ of N, and the last four rows of N form I_4. Direct multiplication shows

$$R_0\begin{pmatrix}n_1\\ \vdots\\ n_8\end{pmatrix}=-[u_1,u_2,u_3,u_4],$$

which is the equation $RN=0$.

n_1	0	0	0	$2i$
n_2	$\frac{1}{2}+\frac{1}{2}i$	$-\frac{1}{2}+\frac{1}{2}i$	$-\frac{1}{2}+\frac{1}{2}i$	$-\frac{1}{2}+\frac{7}{2}i$
n_3	0	$-1+i$	$-1+i$	$-1+5i$
n_4	$-\frac{1}{2}+\frac{1}{2}i$	$-\frac{1}{2}+\frac{1}{2}i$	$-\frac{1}{2}+\frac{1}{2}i$	$-\frac{1}{2}+\frac{5}{2}i$
n_5	0	i	$-1+i$	$-1+4i$
n_6	0	0	$-1+i$	$-1+3i$
n_7	0	0	i	$-1+2i$
n_8	0	0	0	$-1+i$

TABLE 5. The rows $n_1, \dots, n_8$ of the matrix N for $K = \mathbb{Q}(i)$.

For $1 \le j \le 8$ the j-th column of R is $a_{\pi(j)}$, where $(\pi(1), \dots, \pi(9)) = (1, 3, 4, 2, 5, 6, 7, 8, 9)$, and $a_m^* a_m = 2$ for $1 \le m \le 8$, so $L_j = 2n_j^* n_j$. The ninth leaf is attached to the columns $u_1, \dots, u_4$, which are pairwise orthogonal with $u_k^* u_k = 2$, so

$$L_9 = [u_1, u_2, u_3, u_4]^* [u_1, u_2, u_3, u_4] = 2I_4.$$

For $2 \le k \le 7$, the matrix $RP_{\{1,\dots,k\}}N$ is the sum over $1 \le j \le k$ of the j-th column of R times n_j, and $Q_{\le k} = (RP_{\{1,\dots,k\}}N)^*(RP_{\{1,\dots,k\}}N)$. Let $\mathcal{L}$ be the 15×16 integer matrix whose rows are the coordinate vectors of

$$L_1, \dots, L_9, Q_{\le 2}, \dots, Q_{\le 7},$$

in the order listed. Tables 6 and 7 list columns 1–8 and columns 9–16 of $\mathcal{L}$, respectively.

	d_1	d_2	d_3	d_4	x_{12}	y_{12}	x_{13}	y_{13}
L_1	0	0	0	8	0	0	0	0
L_2	1	1	1	25	0	1	0	1
L_3	0	4	4	52	0	0	0	0
L_4	1	1	1	13	1	0	1	0
L_5	0	2	4	34	0	0	0	0
L_6	0	0	4	20	0	0	0	0
L_7	0	0	2	10	0	0	0	0
L_8	0	0	0	4	0	0	0	0
L_9	2	2	2	2	0	0	0	0
$Q_{\le 2}$	1	1	1	19	0	1	0	1
$Q_{\le 3}$	1	3	3	35	0	0	0	0
$Q_{\le 4}$	2	2	2	22	0	0	0	0
$Q_{\le 5}$	2	2	2	14	0	0	0	0
$Q_{\le 6}$	2	2	2	8	0	0	0	0
$Q_{\le 7}$	2	2	2	4	0	0	0	0

TABLE 6. Columns 1–8 of the coordinate matrix $\mathcal{L}$ of the fifteen forms for $K = \mathbb{Q}(i)$.

Let O' be the 13×13 matrix formed by the rows of $\mathcal{L}$ that are the coordinate vectors of

$$L_1, \dots, L_7, L_9, Q_{\le 2}, \dots, Q_{\le 6},$$

in the order listed, and by the columns of

$$d_1, d_2, d_3, d_4, x_{12}, y_{12}, x_{14}, x_{23}, y_{23}, x_{24}, y_{24}, x_{34}, y_{34},$$

in the order listed, which are the thirteen coordinates of ρ in the proof of Theorem 5.7. Gaussian elimination of O' has the nine row exchanges

$$(1,2),\ (2,3),\ (3,5),\ (6,8),\ (7,9),\ (8,10),\ (10,11),\ (11,12),\ (12,13),$$

and the pivots are $1, 4, 2, -12, 2/3, -2, -1, 1, -2, 1, -1, 3, -8$, in the order of the stages. Hence

$$\det O' = (-1)^9 \cdot 1 \cdot 4 \cdot 2 \cdot (-12) \cdot \tfrac{2}{3} \cdot (-2) \cdot (-1) \cdot 1 \cdot (-2) \cdot 1 \cdot (-1) \cdot 3 \cdot (-8) = -6144. \tag{A.3}$$

	x_{14}	y_{14}	x_{23}	y_{23}	x_{24}	y_{24}	x_{34}	y_{34}
L_1	0	0	0	0	0	0	0	0
L_2	3	4	1	0	4	−3	4	−3
L_3	0	0	4	0	12	−8	12	−8
L_4	3	−2	1	0	3	−2	3	−2
L_5	0	0	2	2	8	2	10	−6
L_6	0	0	0	0	0	0	8	−4
L_7	0	0	0	0	0	0	4	2
L_8	0	0	0	0	0	0	0	0
L_9	0	0	0	0	0	0	0	0
$Q_{\le 2}$	2	3	1	0	3	−2	3	−2
$Q_{\le 3}$	0	0	3	0	8	−5	8	−5
$Q_{\le 4}$	0	0	2	0	5	−3	5	−3
$Q_{\le 5}$	0	0	1	1	3	1	4	−2
$Q_{\le 6}$	0	0	0	0	0	0	3	−1
$Q_{\le 7}$	0	0	0	0	0	0	1	1

TABLE 7. Columns 9–16 of the coordinate matrix $\mathcal{L}$ of the fifteen forms for $K = \mathbb{Q}(i)$.

The proof of Theorem 5.7 uses two further pieces of data. The first involves the derivatives $\partial F_0/\partial z_2, \dots, \partial F_0/\partial z_6$ of the function F_0 of (5.12), modulo $\operatorname{Span}_{\mathbb{Q}}(L_1, \dots, L_7, L_9)$, at the point $(z_1, \dots, z_8) = z^0 = (0, 2, 3, 4, 5, 6, 7, 1)$ of the proof of Theorem 5.7, where $z_9 = +\infty$. Their values at z^0 are in the subspace $\mathcal{S}$ of Proposition 5.6, since the coefficients of the logarithms in (5.12) are in $\mathcal{S}$. The relation $L_6 - 2L_7 - 4L_9 - 2Q_{\le 6} + 6Q_{\le 7} = 0$ of Proposition 5.6 implies

$$Q_{\le 7} - Q_{\le 6}/3 \in \operatorname{Span}_{\mathbb{Q}}(L_1, \dots, L_7, L_9).$$

The relation $L_1 - 2L_8 = 0$ implies $L_8 = L_1/2 \in \operatorname{Span}_{\mathbb{Q}}(L_1, \dots, L_7, L_9)$. Since $\dim \mathcal{S} = 13$ and the thirteen matrices $L_1, \dots, L_7, L_9, Q_{\le 2}, \dots, Q_{\le 6}$ are linearly independent by (A.3), they form a basis of $\mathcal{S}$, so the images of $Q_{\le 2}, \dots, Q_{\le 6}$ form a basis of $\mathcal{S}/\operatorname{Span}_{\mathbb{Q}}(L_1, \dots, L_7, L_9)$. By the formulas for $\partial F_0/\partial z_k$ at z^0 and the congruences for the matrices C_{jk} modulo $\operatorname{Span}_{\mathbb{Q}}(L_1, \dots, L_7, L_9)$ in the proof of Theorem 5.7, in particular by

$$Q_{\le 7} - Q_{\le 6}/3 \in \operatorname{Span}_{\mathbb{Q}}(L_1, \dots, L_7, L_9),$$

one obtains

$$\Lambda = \begin{pmatrix} 3 & -2 & 0 & 0 & 0 \\ -2 & 4 & -2 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & -1 & 0 & 3 & -2 \\ 0 & 0 & 0 & -2 & \frac{10}{3} \end{pmatrix}, \qquad \det \Lambda = 3 \cdot \tfrac{8}{3} \cdot \tfrac{5}{4} \cdot \tfrac{12}{5} \cdot \tfrac{5}{3} = 40. \tag{A.4}$$

The matrix Λ of the proof of Theorem 5.7 is (A.4). Gaussian elimination of Λ has no row exchanges, and the pivots are $3, 8/3, 5/4, 12/5, 5/3$, in the order of the stages.

The second piece of data involves the map Π of the proof of Theorem 5.7, restricted to $\operatorname{Herm}_4(K)$, and the matrices $Y_k = i(n_2^* n_k - n_k^* n_2)$, $4 \le k \le 7$, of that proof. Table 8 lists ΠY_k. The value ΠY_k depends only on the entries $(1,2)$, $(1,3)$, and $(1,4)$ of Y_k. For $k = 5, 6, 7$ the first coordinate of n_k is 0, so the first row of Y_k is $i\overline{n_{21}} n_k = \frac{1}{2}(1+i) n_k$, where $n_{21} = \frac{1}{2} + \frac{1}{2}i$ is the first coordinate of n_2. For $k = 4$ the entries $(1,2)$, $(1,3)$, and $(1,4)$ of Y_4 are $-\frac{1}{2} - \frac{1}{2}i$, $-\frac{1}{2} - \frac{1}{2}i$, and $-3 - i$.

k	ΠY_k
4	$(0,0,0)$
5	$(\frac{1}{2},\frac{1}{2},1)$
6	$(1,0,3)$
7	$(\frac{1}{2},-\frac{1}{2},2)$

TABLE 8. The vectors $\Pi Y_k \in \mathbb{Q}^3$ for $4 \leq k \leq 7$.

APPENDIX B. BLOW-UPS OF CYCLIC COVERS ALONG TWO BRANCH DIVISORS

In this appendix, we prove that if a cyclic cover is blown up along the intersection of two branch divisors with exponents a and $\ell - a$, then the parts of the cohomology on which $I = \langle \gamma^a \rangle$ acts by a non-trivial character do not change (Proposition B.5).

Remark B.1. We use the notation of Theorem 3.1. When the quotient map $C \to C/(\mathbb{Z}/\ell)$ is ramified, the proof of [Sch88, Theorem 2.0] regards a finite quotient of C^{2k} as a cyclic cover of the $2k$-th symmetric product $S^{2k}(C/(\mathbb{Z}/\ell))$ [Sch88, p. 12], blows up the intersections of pairs of branch divisors with inverse exponents one after another, and takes normalizations [Sch88, pp. 14–15]. It uses that no primitive character of $\mathbb{Z}/\ell$ occurs in the cokernel of the pullback on cohomology from the cover before each blow-up to the normalized cover after it [Sch88, Lemma 2.8], and the proof of [Sch88, Lemma 2.8] shows that it suffices to exclude the primitive characters from the cohomology of the preimage of the exceptional divisor.

The argument of [Sch88, pp. 18–19] takes a prime number p dividing ℓ such that the quotient cover $C/(p\mathbb{Z}/\ell\mathbb{Z}) \to C/(\mathbb{Z}/\ell)$ of degree p is ramified over a branch point b of the pair defining the blow-up. Since the stabilizer of a point over b is generated by a_b, the quotient cover of degree p is ramified over b if and only if $a_b \notin p\mathbb{Z}/\ell\mathbb{Z}$, so such a prime p can fail to exist exactly when ℓ is not square-free, for instance if $4 \mid \ell$. Indeed, let $\ell = 4$ and let b be a branch point of a pair with $a_b = 2$. Then the stabilizer of a point over b is $2\mathbb{Z}/4\mathbb{Z}$, the unique subgroup of index 2 of $\mathbb{Z}/4$. Thus the quotient cover of degree 2 is unramified over b, and 2 is the only prime factor of 4. Moreover, for such a prime p, the isomorphisms of [Sch88, p. 19], deduced from [Sch88, Sublemma 2.9], show that the invariants of $\mathbb{Z}/\ell$ and of $p\mathbb{Z}/\ell\mathbb{Z}$ in the cohomology of the preimage of the exceptional divisor are equal, which excludes the characters of order p. If ℓ is composite, then $p\mathbb{Z}/\ell\mathbb{Z}$ is non-trivial, so the primitive characters are non-trivial on $p\mathbb{Z}/\ell\mathbb{Z}$ and do not occur in either space of invariants, and the equality does not exclude them. If ℓ is prime, then $p = \ell$ satisfies the condition and $p\mathbb{Z}/\ell\mathbb{Z}$ is trivial.

If $C \to C/(\mathbb{Z}/\ell)$ is unramified, then the proof of [Sch88, Theorem 2.0] involves no blow-up [Sch88, p. 13]. Thus Appendix B is used only when ℓ is composite and $C \to C/(\mathbb{Z}/\ell)$ is ramified. For any $\ell \geq 2$, Proposition B.5 (1) proves the conclusion of [Sch88, Lemma 2.8], and Proposition B.5 (3) proves the non-vanishing of pushforwards that [Sch88, p. 18] deduces from [Sch88, Lemma 2.8]. The last paragraph of Appendix B verifies that the covers of [Sch88, §2] satisfy the conditions of Definition-Lemma B.2. Hence Theorem 3.1, as used in this paper, follows from [Sch88] and Appendix B.

To exclude the faithful characters, we need to control the fibers of the reduced preimage of the exceptional divisor over all points of the center of the blow-up, also over the points where the center intersects other branch divisors. Over such points, the number of components of the fiber, or the degree of a component over the exceptional curve, or both, change (Definition-Lemma B.2 (2)).

Definition-Lemma B.2. Let Y be a smooth algebraic variety, $\ell \geq 2$ an integer, Γ a cyclic group of order ℓ with a generator γ, and ζ a primitive ℓ-th root of unity. Let $\pi : W \to Y$ be a finite morphism of degree ℓ from a normal algebraic variety W, and let Γ act on W by automorphisms over Y such that π induces an isomorphism $W/\Gamma \cong Y$. Assume that the branch divisor D of π is a simple normal crossing divisor, and that any point $y \in Y$ has an open neighborhood U in the analytic topology such that $\pi^{-1}(U)$ is isomorphic over U, compatibly with the action of Γ,

to the normalization of the hypersurface

$$\left\{v^\ell = u\prod_j x_j^{a_j}\right\} \subset U \times \mathbb{C}, \qquad \gamma(v) = \zeta v,$$

where u is a nowhere vanishing holomorphic function on U, the x_j are local equations of the components of D through y, and the a_j are integers with $0 < a_j < \ell$. We call a_j the *exponent* at y of the component $\{x_j = 0\}$. Let D_1 and D_2 be distinct components of D, and let a be an integer with $0 < a < \ell$ such that the exponents of D_1 and D_2 at any point of $Z := D_1 \cap D_2$ are a and $\ell - a$. Let $f : Y' \to Y$ be the blow-up of Y along Z, which is the identity if $Z = \emptyset$, and let $E := f^{-1}(Z)$. Let W' be the normalization of the closure in $W \times_Y Y'$ of the preimage of $Y' \setminus E$, with the action of Γ induced from W, and let $\pi' : W' \to Y'$ and $g : W' \to W$ be the induced morphisms. Let $F := (\pi'^{-1}(E))_{\mathrm{red}}$, $T := (\pi^{-1}(Z))_{\mathrm{red}}$, $q : F \to Z$ the morphism induced by $f \circ \pi'$, and $I \subset \Gamma$ the subgroup generated by γ^a. Then the following hold.

(1) The subgroup I fixes T pointwise.
(2) Let $y \in Z$, let $c_1, \dots, c_r$ be the exponents at y of the components of D through y other than D_1 and D_2, let $N_y \subset \Gamma$ be the subgroup generated by $\gamma^{c_1}, \dots, \gamma^{c_r}$, with $N_y = \{1\}$ if $r = 0$, and let $L_y \subset \Gamma$ be the subgroup generated by γ^a and N_y. Then $F_y := (q^{-1}(y))_{\mathrm{red}}$ is a disjoint union of projective lines $F_{y,1}, \dots, F_{y,e_y}$, the set $\{F_{y,1}, \dots, F_{y,e_y}\}$ with the action of Γ is isomorphic to Γ/L_y, and π' maps each $F_{y,i}$ onto $E_y := f^{-1}(y) \cong \mathbb{P}^1$ with degree $|L_y/N_y|$. In particular, as representations of Γ, $H^j(F_y, \mathbb{C}) \cong \mathbb{C}[\Gamma/L_y]$ for $j = 0, 2$, and $H^j(F_y, \mathbb{C}) = 0$ for $j \neq 0, 2$.
(3) The subgroup I acts trivially on $H^n(F, \mathbb{C})$ for any integer $n \geq 0$.

We prove Definition-Lemma B.2 with explicit local models of W near $\pi^{-1}(y)$ and of W' near $(f \circ \pi')^{-1}(y)$, given by quotients of smooth spaces by finite abelian groups.

Construction B.3. Notation and conditions as in Definition-Lemma B.2. Let $d := \dim Y$ and $\mu_\ell := \{\kappa \in \mathbb{C}^* \mid \kappa^\ell = 1\}$.

(1) Let $y \in Y$, and let $a_1, \dots, a_n$ be the exponents at y of the components of D through y. Since D is a simple normal crossing divisor, after shrinking the neighborhood U of y in Definition-Lemma B.2, U is a polydisc centred at y with coordinates $x_1, \dots, x_d$, where $x_1, \dots, x_n$ are local equations of the components of D through y, and u has an ℓ-th root on U. Let $\widetilde{U}$ be the polydisc with coordinates $(\rho_1, \dots, \rho_n, x_{n+1}, \dots, x_d)$, mapping to U by $x_j = \rho_j^\ell$ for $1 \leq j \leq n$. Let μ_ℓ^n act on $\widetilde{U} \times \mu_\ell$ by

$$(\eta_1, \dots, \eta_n) \cdot (\rho_1, \dots, \rho_n, x_{n+1}, \dots, x_d, \kappa) = \Big(\eta_1\rho_1, \dots, \eta_n\rho_n, x_{n+1}, \dots, x_d, \kappa \prod_{j=1}^n \eta_j^{-a_j}\Big), \tag{B.1}$$

and let $M_U := (\widetilde{U} \times \mu_\ell)/\mu_\ell^n$, with the action of Γ given by

$$\gamma[\rho_1, \dots, \rho_n, x_{n+1}, \dots, x_d, \kappa] = [\rho_1, \dots, \rho_n, x_{n+1}, \dots, x_d, \zeta\kappa].$$

(2) Let $y \in Z$, and let $c_1, \dots, c_r$ be as in Definition-Lemma B.2 (2). In (1), we write the coordinates on U as $(x_1, x_2, z_1, \dots, z_{d-2})$, where $D_1 = \{x_1 = 0\}$, $D_2 = \{x_2 = 0\}$, and the other components of D through y are $\{z_j = 0\}$ for $1 \leq j \leq r$, and we write $\rho_1, \rho_2, w_1, \dots, w_r$ for the root coordinates, so that $x_1 = \rho_1^\ell$, $x_2 = \rho_2^\ell$, and $z_j = w_j^\ell$. Let $U_Z := Z \cap U$, with coordinates $(z_1, \dots, z_{d-2})$, and let $\widetilde{U}_Z$ be the polydisc with coordinates $(w_1, \dots, w_r, z_{r+1}, \dots, z_{d-2})$, mapping to U_Z by $z_j = w_j^\ell$ for $1 \leq j \leq r$. For a point $\omega = (w_1, \dots, w_r, z_{r+1}, \dots, z_{d-2})$ of $\widetilde{U}_Z$ and $\epsilon = (\epsilon_0, \dots, \epsilon_r) \in \mu_\ell^{r+1}$, write

$$\epsilon\omega := (\epsilon_1 w_1, \dots, \epsilon_r w_r, z_{r+1}, \dots, z_{d-2}).$$

Let $V := \mathbb{P}^1_s \times \widetilde{U}_Z \times \mu_\ell$, where s is an inhomogeneous coordinate on $\mathbb{P}^1_s$, let $A := \mu_\ell^{r+1}$ act on V by

$$\epsilon \cdot (s, \omega, \kappa) = \Big(\epsilon_0 s, \epsilon\omega, \kappa\epsilon_0^{-a} \prod_{j=1}^{r} \epsilon_j^{-c_j}\Big), \tag{B.2}$$

and let $Q := V/A$, with the action of Γ given by $\gamma[s, \omega, \kappa] = [s, \omega, \zeta\kappa]$, and with the morphism $h : Q \to \mathbb{P}^1_t \times U_Z$ given by $t = s^\ell$ and $z_j = w_j^\ell$ for $1 \le j \le r$.

Lemma B.4. *Notation and conditions as in Construction B.3. Then the following hold.*

(1) For $y \in Y$, there exists an isomorphism $\pi^{-1}(U) \cong M_U$ over U compatible with the actions of Γ, where $M_U \to U$ is given by $x_j = \rho_j^\ell$ for $1 \le j \le n$.

(2) For $y \in Z$, we have $f^{-1}(U) \cap E = \mathbb{P}^1_t \times U_Z$, where $t = x_1/x_2$, and there exists an isomorphism $F \cap (f \circ \pi')^{-1}(U) \cong Q$ over $\mathbb{P}^1_t \times U_Z$ compatible with the actions of Γ.

(3) Any point ξ of W or of W' has an open neighborhood N such that the pair (N, ξ) is homeomorphic to $(B/H, 0)$, where $B \subset \mathbb{C}^d$ is an open ball centred at 0 and H is a finite group of diagonal matrices whose diagonal entries are roots of unity.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1).

The local equations x_j of the components of D are determined up to units, and a change of the x_j by units is absorbed into u. Thus we may assume that the x_j in Definition-Lemma B.2 are the coordinates of Construction B.3 (1). Replacing v with $u^{-1/\ell}v$, which does not change the relation $\gamma(v) = \zeta v$, we may assume that $u = 1$. Hence $\pi^{-1}(U)$ is isomorphic over U, compatibly with Γ, to the normalization of

$$\Big\{v^\ell = \prod_{j=1}^{n} x_j^{a_j}\Big\} \subset U \times \mathbb{C}, \qquad \gamma(v) = \zeta v. \tag{B.3}$$

The function $v := \kappa \prod_{j=1}^n \rho_j^{a_j}$ on $\widetilde{U} \times \mu_\ell$ is invariant under (B.1) and satisfies $v^\ell = \prod_{j=1}^n x_j^{a_j}$. Thus v defines a morphism from M_U to the hypersurface in (B.3), which is finite, as M_U is finite over U. The space M_U is normal, since normality is local and, on an invariant affine chart, an element of the fraction field of the invariant ring that is integral over the invariant ring is integral over the normal ring of $\widetilde{U} \times \mu_\ell$, hence belongs to it, and is invariant, hence belongs to the invariant ring. If μ_ℓ^n does not act transitively on the labels κ, we apply the normality argument to each orbit of components.

Let $U^\circ := \{x_1 \cdots x_n \ne 0\} \subset U$. Over U°, the group μ_ℓ^n acts freely on $\widetilde{U} \times \mu_\ell$. Two points (ρ, κ) and (ρ', κ') over one point of U° with equal values of v satisfy $\rho'_j = \eta_j \rho_j$ for some $\eta \in \mu_\ell^n$, and then $\kappa' = \kappa \prod_j \eta_j^{-a_j}$, that is, they are related by (B.1). Moreover, varying κ, the function v takes all ℓ roots of (B.3). Hence M_U is isomorphic over U° to the hypersurface in (B.3), which is smooth over U°. Since the preimage of U° is dense in M_U, the uniqueness of normalization implies that M_U is the normalization of the hypersurface in (B.3). The automorphism $\kappa \to \zeta\kappa$ multiplies v by ζ, so it corresponds to γ. This proves (1).

Step 2. In this step we prove (2).

Let $y \in Z$, with the coordinates of Construction B.3 (2). By **Step 1**, $\pi^{-1}(U) \cong M_U$ is the normalization of

$$v^\ell = x_1^a x_2^{\ell-a} \prod_{j=1}^{r} z_j^{c_j}, \tag{B.4}$$

and for

$$(\eta_1, \eta_2, \epsilon_1, \ldots, \epsilon_r) \in \mu_\ell^{r+2},$$

the action (B.1) is

$$(\rho_1,\rho_2,w_1,\dots,w_r,z_{r+1},\dots,z_{d-2},\kappa)\to\Big(\eta_1\rho_1,\eta_2\rho_2,\epsilon_1w_1,\dots,\epsilon_rw_r,z_{r+1},\dots,z_{d-2},\kappa\eta_1^{-a}\eta_2^{a-\ell}\prod_{j=1}^r\epsilon_j^{-c_j}\Big),$$

with $v=\kappa\rho_1^a\rho_2^{\ell-a}\prod_j w_j^{c_j}$.

Write $U=\Delta^2\times U_Z$, where Δ^2 is the bidisc with coordinates (x_1,x_2). Then $f^{-1}(U)=\mathrm{Bl}_0(\Delta^2)\times U_Z$ is covered by the open subsets U_1', with coordinates $(t,x_2,z_1,\dots,z_{d-2})$ and $x_1=tx_2$, and U_2', with coordinates $(t_\infty,x_1,z_1,\dots,z_{d-2})$ and $x_2=t_\infty x_1$. Moreover, $f^{-1}(U)\cap E=\mathbb{P}^1_t\times U_Z$, defined by $x_2=0$ on U_1' and by $x_1=0$ on U_2'. On U_1', the pullback of (B.4) is $v^\ell=t^ax_2^\ell\prod_j z_j^{c_j}$, so the meromorphic function $w:=v/x_2$ is integral and satisfies

$$w^\ell=t^a\prod_{j=1}^r z_j^{c_j}. \tag{B.5}$$

On U_2', the meromorphic function $w_\infty:=v/x_1$ is integral and satisfies

$$w_\infty^\ell=t_\infty^{\ell-a}\prod_{j=1}^r z_j^{c_j},\qquad t_\infty=t^{-1},\qquad w_\infty=w/t. \tag{B.6}$$

Let $V_1:=\{s\neq\infty\}\subset V$ and $V_2:=\{s\neq0\}\subset V$, and let $Q_i:=V_i/A\subset Q$ for $i=1,2$. On V_1 and on V_2, the functions

$$t=s^\ell,\qquad z_j=w_j^\ell,\qquad w=\kappa s^a\prod_{j=1}^r w_j^{c_j} \tag{B.7}$$

and

$$t_\infty=s^{-\ell},\qquad w_\infty=\kappa s^{a-\ell}\prod_{j=1}^r w_j^{c_j} \tag{B.8}$$

respectively are regular and invariant under (B.2), they satisfy (B.5) and (B.6), and $w_\infty=w/t$ on $V_1\cap V_2$. Possibly replacing $\widetilde{U}\times\mu_\ell$ and μ_ℓ^n with V_i and A, the argument of **Step 1** implies that Q_i is normal and finite over the base of (B.5) or (B.6). On the dense open subset of V where $s\notin\{0,\infty\}$ and $w_1\cdots w_r\neq0$, the group A acts freely, and the orbit computation of **Step 1** shows that the fibers of Q_1 map bijectively onto the ℓ roots of (B.5). Thus Q_1 is the normalization of the hypersurface (B.5) over $\mathbb{C}_t\times U_Z$, and Q_2 is the normalization of the hypersurface (B.6) over $\mathbb{C}_{t_\infty}\times U_Z$, glued by $w_\infty=w/t$.

Let N_1 be the preimage of U_1' in $Q_1\times\mathbb{C}_{x_2}$, where $Q_1\times\mathbb{C}_{x_2}$ maps to $\mathbb{C}_t\times\mathbb{C}_{x_2}\times U_Z$. Then N_1 is normal and finite over U_1', and $v=x_2w$ maps N_1 to the pullback of the hypersurface (B.4) to U_1'. Over the dense open subset $U_1'\cap f^{-1}(U^\circ)$, where $U^\circ=\{x_1x_2z_1\cdots z_r\neq0\}$, both N_1 and $M_U\times_U U_1'$ map isomorphically onto the pullback of the hypersurface (B.4), which is smooth over $f^{-1}(U^\circ)$. Let $U_0\subset U$ be an open subset. A holomorphic function ψ on the preimage of U_0 in M_U is integral over the ring of (B.4), so the pullback ψ_1 of ψ to the part of N_1 over $f^{-1}(U_0\cap U^\circ)$, given by the isomorphisms over $f^{-1}(U^\circ)$, is a holomorphic function that is integral over the structure sheaf of N_1. Since N_1 is normal, ψ_1 extends to a holomorphic function on the part of N_1 over $f^{-1}(U_0)$. As M_U is finite over U, the extensions of the functions ψ_1 define a morphism $N_1\to M_U\times_U U_1'$ over U_1', which is finite and an isomorphism over $f^{-1}(U^\circ)$. Hence N_1 is the normalization of the closure in $M_U\times_U U_1'$ of the preimage of $U_1'\setminus E$, that is, $N_1\cong\pi'^{-1}(U_1')$. Replacing x_2, w, and Q_1 with x_1, w_∞, and Q_2, we obtain $N_2\cong\pi'^{-1}(U_2')$, where $v=x_1w_\infty$. The isomorphisms are compatible with Γ, as $\kappa\to\zeta\kappa$ multiplies w, w_∞, and v by ζ, and they are compatible on $\pi'^{-1}(U_1'\cap U_2')$, as they coincide over the dense open subset $f^{-1}(U^\circ)$.

The divisor E is defined by $x_2=0$ on U_1' and by $x_1=0$ on U_2'. Hence $F\cap\pi'^{-1}(U_1')=Q_1\times\{0\}$ and $F\cap\pi'^{-1}(U_2')=Q_2\times\{0\}$, which are reduced since Q_1 and Q_2 are normal. Thus $F\cap(f\circ\pi')^{-1}(U)\cong Q_1\cup Q_2=Q$ over $\mathbb{P}^1_t\times U_Z$, compatibly with Γ. This proves (2).

Step 3. In this step we prove (3).

By **Step 1**, any point of W has a neighborhood isomorphic to an open subset of M_U. The morphism g is an isomorphism over $W \setminus T$, since f is an isomorphism over $Y \setminus Z$ and $W \times_Y (Y' \setminus E) \cong \pi^{-1}(Y \setminus Z)$ is normal. Hence any point of $W' \setminus F$ has a neighborhood isomorphic to an open subset of some M_U. By **Step 2**, any point of F has a neighborhood isomorphic to an open subset of $(V_i \times \mathbb{C})/A$ for $i = 1$ or 2, where A acts trivially on the factor $\mathbb{C}$. Thus any point ξ of W or of W' has a neighborhood isomorphic to an open subset of $\mathcal{V}/G$, where $\mathcal{V}$ is a disjoint union of copies of an open subset of $\mathbb{C}^d$, indexed by μ_ℓ, and G is a finite abelian group acting by permuting the copies and by multiplying each coordinate with a root of unity. For V_2, we use the coordinate s^{-1}, on which ϵ_0 acts by ϵ_0^{-1}.

Let $b \in \mathcal{V}$ be a preimage of ξ, and let $H \subset G$ be the stabilizer of b. If a coordinate of b is non-zero, then any element of H multiplies the coordinate by 1. Thus H acts diagonally in the coordinates of $\mathbb{C}^d$ centred at b. Let B be an open ball centred at b with respect to the coordinates centred at b, small enough that $\sigma(B) \cap B = \emptyset$ for any $\sigma \in G \setminus H$. Then B is H-invariant, and $B/H \to \mathcal{V}/G$ is injective, continuous, and open, so it is a homeomorphism onto an open neighborhood of ξ that maps the image of b to ξ. This proves (3). □

Proof of Definition-Lemma B.2. The proof is long, so we divide it into several steps.

Step 1. In this step we show that the exponent a_j at y depends only on y and on the component $\{x_j = 0\}$.

Let $y \in Y$, and consider two isomorphisms as in Definition-Lemma B.2, one with v, u, and exponents a_j, and one with v', u', and exponents a'_j. Normalization commutes with restriction to open subsets, so we may restrict both isomorphisms to a common open neighborhood U of y. After shrinking U, it is a polydisc centred at y whose coordinates include local equations x_j of the components of D through y. Two local equations of one component differ by a nowhere vanishing holomorphic function ν on U, and replacing one by the other multiplies u' by a power of ν, which is again nowhere vanishing. Thus we may assume that the x_j of the second isomorphism are equal to those of the first. Let $U^\circ := \{\prod_j x_j \neq 0\} \subset U$. On $\pi^{-1}(U^\circ)$, the function v is nowhere zero, as $v^\ell = u \prod_j x_j^{a_j}$, and v'/v is invariant under Γ, as γ multiplies both v and v' by ζ. Since π induces $W/\Gamma \cong Y$, the function v'/v is the pullback of a holomorphic function θ on U°, and

$$\theta^\ell = \frac{u'}{u} \prod_j x_j^{a'_j - a_j}.$$

Along a small circle in U° around $\{x_j = 0\}$ on which only the coordinate x_j varies, the winding number of θ^ℓ is ℓ times the winding number of θ, and the winding number of the right hand side is $a'_j - a_j$, since u'/u has a logarithm on the polydisc U. Thus ℓ divides $a'_j - a_j$, and $0 < a_j, a'_j < \ell$ implies that $a'_j = a_j$.

Step 2. In this step we show that, for any $y \in Z$, the element γ^a is homotopic to the identity on F_y, and we prove (1).

Let $y \in Z$, and let U, U_Z, V, A, Q, and h be as in Construction B.3 (2). By Lemma B.4 (2), $F \cap (f \circ \pi')^{-1}(U) \cong Q$ compatibly with Γ, and q corresponds to the composite of h with the projection to U_Z. For $\lambda \in \mathbb{C}^*$, the map $(s, \omega, \kappa) \to (\lambda s, \omega, \kappa)$ of V commutes with (B.2), is defined at $s = 0$ and at $s = \infty$, and commutes with the projection to $\widetilde{U}_Z$. Hence it induces an automorphism $[s, \omega, \kappa] \to [\lambda s, \omega, \kappa]$ of Q over U_Z. Applying (B.2) with $\epsilon_0 = \zeta$ and $\epsilon_1 = \cdots = \epsilon_r = 1$ to $(s, \omega, \zeta^a \kappa)$, we obtain

$$[\zeta s, \omega, \kappa] = [s, \omega, \zeta^a \kappa] = \gamma^a [s, \omega, \kappa]. \tag{B.9}$$

Let $(\lambda_\tau)_{\tau \in [0,1]}$ be a continuous path in $\mathbb{C}^*$ from $\lambda_0 = 1$ to $\lambda_1 = \zeta$. Then $[s, \omega, \kappa] \to [\lambda_\tau s, \omega, \kappa]$ is a homotopy over U_Z from the identity of Q to γ^a, by (B.9). It preserves the fiber $Q_{y'}$ of Q over any point $y' \in U_Z$, and γ^a is homotopic to the identity on F_y. This holds for any $y \in Z$, also when some components of D other than D_1 and D_2 pass through y.

The group Γ acts on W' over the identity of Y'. For $\lambda \notin \mu_\ell$, the automorphism $[s, \omega, \kappa] \to [\lambda s, \omega, \kappa]$ of Q is over U_Z but not over $\mathbb{P}^1_t \times U_Z$, since it maps t to $\lambda^\ell t$. The homotopy uses only automorphisms of Q over U_Z, which preserve F_y, and only the automorphism for $\tau = 1$, given by $\lambda_1 = \zeta$, has to be the deck transformation γ^a. The $\mathbb{C}^*$-actions on the neighborhoods of different points of Z need not coincide, and we use the homotopy only on a single fiber F_y.

We prove (1). Let $y \in Z$, with the coordinates of Construction B.3 (2). By Lemma B.4 (1), $\pi^{-1}(U) \cong M_U$. Applying the element of μ_ℓ^{r+2} with $\eta_1 = \zeta$ and all other entries 1, we obtain

$$\gamma^a[\rho_1, \rho_2, w_1, \dots, w_r, z_{r+1}, \dots, z_{d-2}, \kappa] = [\zeta\rho_1, \rho_2, w_1, \dots, w_r, z_{r+1}, \dots, z_{d-2}, \kappa].$$

If $\rho_1 = 0$, then the right hand side is $[\rho_1, \rho_2, w_1, \dots, w_r, z_{r+1}, \dots, z_{d-2}, \kappa]$. Thus γ^a fixes any point of $\pi^{-1}(U)$ over $\{x_1 = 0\}$, in particular any point of $T \cap \pi^{-1}(U)$. Since $y \in Z$ is arbitrary, I fixes T pointwise. This proves (1).

Step 3. In this step we prove (2).

Let $y \in Z$, and let U, U_Z, V, A, Q, and h be as in Construction B.3 (2). The coordinates of U are centred at y, so all the coordinates $w_1, \dots, w_r, z_{r+1}, \dots, z_{d-2}$ of $\widetilde{U}_Z$ vanish over y. By Lemma B.4 (2), $q^{-1}(y)$ is the subspace of Q defined by the ideal $J \subset \mathcal{O}_V^A$ generated by $w_1^\ell, \dots, w_r^\ell, z_{r+1}, \dots, z_{d-2}$. Since $\mathcal{O}_V$ is integral over $\mathcal{O}_V^A$, the radical of J is the contraction to $\mathcal{O}_V^A$ of the radical $(w_1, \dots, w_r, z_{r+1}, \dots, z_{d-2})$ of $J\mathcal{O}_V$. As taking invariants under a finite group is exact in characteristic zero,

$$\mathcal{O}_V^A/\sqrt{J} \cong (\mathcal{O}_V/(w_1, \dots, w_r, z_{r+1}, \dots, z_{d-2}))^A,$$

so F_y is the quotient of $V_0 := \mathbb{P}^1_s \times \{0\} \times \mu_\ell$ by A. The nilpotent elements in the structure sheaf of $q^{-1}(y)$ do not affect the topological space or the singular cohomology, so it suffices to describe V_0/A.

We identify μ_ℓ with Γ by $\zeta^k \to \gamma^k$. Then γ acts on the labels by multiplication by ζ, and N_y and L_y correspond to the subgroups of μ_ℓ generated by $\zeta^{c_1}, \dots, \zeta^{c_r}$, and by $\zeta^a, \zeta^{c_1}, \dots, \zeta^{c_r}$, respectively. We first divide V_0 by the elements of A with $\epsilon_0 = 1$. On V_0, they only change the label by $\prod_j \epsilon_j^{-c_j}$, so the quotient is $\mathbb{P}^1_s \times (\Gamma/N_y)$. The remaining factor ϵ_0 acts by $s \to \epsilon_0 s$ and multiplies the label by ϵ_0^{-a}. Thus the label orbits are the cosets of L_y/N_y in Γ/N_y, so the set of label orbits is Γ/L_y, of order $e_y = |\Gamma/L_y| = \gcd(\ell, a, c_1, \dots, c_r)$, and γ acts on it by multiplication.

Fix a label orbit $\mathcal{O}_i \subset \Gamma/N_y$. The image $F_{y,i}$ of $\mathbb{P}^1_s \times \mathcal{O}_i$ in V_0/A is the quotient of $\mathbb{P}^1_s$ by the stabilizer of a label in $\mathcal{O}_i$, which is the group of rotations $s \to \epsilon_0 s$ with $\epsilon_0^a \in N_y$, of order $\ell/|L_y/N_y|$. Thus $F_{y,i} \cong \mathbb{P}^1$. Distinct label orbits are open and closed in Γ/N_y, so the $F_{y,i}$ are pairwise disjoint. Since π' maps $[s, 0, \kappa]$ to $t = s^\ell$ in $E_y = \mathbb{P}^1_t$, the degree of $F_{y,i} \to E_y$ is ℓ divided by the order of the stabilizer, that is, $|L_y/N_y|$.

The group Γ permutes the lines $F_{y,i}$ as Γ/L_y, and the stabilizer L_y of a line acts on it by holomorphic automorphisms, which act trivially on $H^0(\mathbb{P}^1, \mathbb{C})$ and preserve the complex orientation class in $H^2(\mathbb{P}^1, \mathbb{C})$, and $H^j(\mathbb{P}^1, \mathbb{C}) = 0$ for $j \neq 0, 2$. Hence $H^j(F_y, \mathbb{C}) \cong \mathbb{C}[\Gamma/L_y]$ for $j = 0, 2$, and $H^j(F_y, \mathbb{C}) = 0$ for $j \neq 0, 2$. This proves (2).

Step 4. In this step we prove (3).

The group Γ acts on F over the identity of Z, and q is proper, since F is finite over the projective bundle $E \to Z$. By proper base change in the analytic topology [Sta26, Tag 09V6], for any $y \in Z$ and $j \geq 0$ there exists a natural isomorphism

$$(R^j q_* \mathbb{C}_F)_y \cong H^j(F_y, \mathbb{C}). \tag{B.10}$$

Here we use that the cohomology of the constant sheaf on an algebraic variety in the analytic topology is the singular cohomology. By naturality with respect to automorphisms of F over Z, the isomorphism (B.10) is Γ-equivariant. By **Step 2**, any element of I acts as the identity on any stalk in (B.10). An endomorphism of a sheaf is determined by the induced maps on stalks, so I acts as the identity on $R^j q_* \mathbb{C}_F$ for any j.

The Leray spectral sequence [Sta26, Tag 01F2]

$$E_2^{i,j} = H^i(Z, R^j q_* \mathbb{C}_F) \Rightarrow H^{i+j}(F, \mathbb{C}) \tag{B.11}$$

is Γ-equivariant. Thus I acts trivially on any $E_2^{i,j}$, hence on any $E_\infty^{i,j}$, that is, on the graded pieces of the Leray filtration on $H^n(F, \mathbb{C})$. As $E_2^{i,j} = 0$ for $i < 0$ or $j < 0$, the Leray filtration on $H^n(F, \mathbb{C})$ has finite length. A complex representation H of a finite group G is semisimple: for a G-stable subspace $H' \subset H$ and a linear projection p_0 of H onto H', the map $|G|^{-1} \sum_{\sigma \in G} \sigma p_0 \sigma^{-1}$ is a G-equivariant projection onto H', so H' has a G-stable complement. Thus a representation of I with a finite filtration whose graded pieces are trivial is isomorphic to the direct sum of the graded pieces, hence trivial. This proves (3). The argument allows the sheaves $R^j q_* \mathbb{C}_F$ to have arbitrary monodromy and jumps of rank. □

Proposition B.5. *Notation and conditions as in Definition-Lemma B.2. For a complex representation H of Γ and a character $\chi : \Gamma \to \mathbb{C}^*$, let $H_\chi \subset H$ be the subspace on which Γ acts by χ.*

(1) For any character χ of Γ with non-trivial restriction to I and any integer $n \geq 0$, the pullback

$$g^* : H^n(W, \mathbb{C})_\chi \to H^n(W', \mathbb{C})_\chi$$

is an isomorphism. In particular, no faithful character of Γ occurs in the cokernel of $g^* : H^n(W, \mathbb{C}) \to H^n(W', \mathbb{C})$.

(2) Assume that Y is projective of dimension d. Then W and W' are compact rational homology manifolds of real dimension $2d$, oriented by the complex structure, that is, for $M \in \{W, W'\}$ and $x \in M$, $H_k(M, M \setminus \{x\}; \mathbb{Q}) \cong \mathbb{Q}$ for $k = 2d$ and $H_k(M, M \setminus \{x\}; \mathbb{Q}) = 0$ for $k \neq 2d$. Let $g_* : H^n(W', \mathbb{C}) \to H^n(W, \mathbb{C})$ be the pushforward on homology transported by Poincaré duality, and for a subvariety V of W or of W', let $[V]$ be the Poincaré dual of the pushforward of the fundamental class of a resolution of V, extended linearly to algebraic cycles. Then for any χ as in (1), the restriction of g_* to $H^n(W', \mathbb{C})_\chi$ is the inverse of the isomorphism in (1), and $[g_* z] = g_*[z]$ for any algebraic cycle z on W'. In particular, if $[z]$ has a non-zero component in $H^*(W', \mathbb{C})_\chi$, then $[g_* z]$ has a non-zero component in $H^*(W, \mathbb{C})_\chi$.

(3) Assume that Y is projective. Let $W_0 := W$, $Y_0 := Y$, and $\pi_0 := \pi$, and for $1 \leq i \leq k$, let $\pi_i : W_i \to Y_i$ and $g_i : W_i \to W_{i-1}$ be the morphisms π' and g of Definition-Lemma B.2 for $\pi_{i-1} : W_{i-1} \to Y_{i-1}$ and a pair of components of the branch divisor of π_{i-1}, where π_{i-1} satisfies the conditions of Definition-Lemma B.2. Then for any faithful character χ of Γ and any algebraic cycle z on W_k such that $[z]$ has a non-zero component in $H^*(W_k, \mathbb{C})_\chi$, the class $[(g_1 \circ \cdots \circ g_k)_* z]$ has a non-zero component in $H^*(W_0, \mathbb{C})_\chi$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1).

The morphism g is proper, and it is an isomorphism over $W \setminus T$, since f is an isomorphism over $Y \setminus Z$ and $W \times_Y (Y' \setminus E) \cong \pi^{-1}(Y \setminus Z)$ is normal. Since $\pi \circ g = f \circ \pi'$, the preimage $g^{-1}(T)$ is $\pi'^{-1}(E)$, whose underlying topological space is F. Let $i : T \to W$ and $j : F \to W'$ be the closed embeddings, and let $g_F : F \to T$ be the morphism induced by g, so that $g \circ j = i \circ g_F$. We claim that there exists a natural Γ-equivariant distinguished triangle

$$\mathbb{C}_W \to R g_* \mathbb{C}_{W'} \oplus i_* \mathbb{C}_T \to R(g \circ j)_* \mathbb{C}_F \to \mathbb{C}_W[1], \tag{B.12}$$

where the first arrow is the sum of the adjunction maps, and the second arrow is the difference of the restriction maps $R g_* \mathbb{C}_{W'} \to R g_* R j_* \mathbb{C}_F$ and $i_* \mathbb{C}_T \to i_* R g_{F*} \mathbb{C}_F$. We compute the direct images with Godement resolutions, which are functorial, so the restriction maps and the actions of Γ are defined on complexes, and the composites of the first arrow with the restriction maps to $R(g \circ j)_* \mathbb{C}_F$ are equal maps of complexes $\mathbb{C}_W \to R(g \circ j)_* \mathbb{C}_F$. Thus $\mathbb{C}_W$ maps naturally and Γ-equivariantly to the cocone of the second arrow, and it suffices to show that the map from

$\mathbb{C}_W$ to the cocone is a quasi-isomorphism on any stalk. Let $\xi \in W$. If $\xi \notin T$, then the stalks of $i_*\mathbb{C}_T$ and $R(g\circ j)_*\mathbb{C}_F$ at ξ are zero, and $\mathbb{C}_W \to Rg_*\mathbb{C}_{W'}$ is an isomorphism near ξ. If $\xi \in T$, then $g^{-1}(\xi) \subset F$, and by proper base change [Sta26, Tag 09V6], the restriction map

$$(Rg_*\mathbb{C}_{W'})_\xi \to (R(g\circ j)_*\mathbb{C}_F)_\xi$$

is a quasi-isomorphism, both stalks computing $H^*(g^{-1}(\xi),\mathbb{C})$. Let $\mathcal{K}_\xi := (R(g\circ j)_*\mathbb{C}_F)_\xi$, and let $\iota : \mathbb{C} \to \mathcal{K}_\xi$ be the map induced by $(i_*\mathbb{C}_T)_\xi = \mathbb{C}$. Then the stalk of (B.12) at ξ is quasi-isomorphic to

$$\mathbb{C} \to \mathcal{K}_\xi \oplus \mathbb{C} \to \mathcal{K}_\xi \to \mathbb{C}[1], \qquad c \to (\iota(c), c), \qquad (\alpha, c) \to \alpha - \iota(c).$$

The second map is surjective with kernel $\{(\iota(c), c) \mid c \in \mathbb{C}\}$, which is the image of the first map. Thus the triangle is split at ξ, and the claim follows.

Taking cohomology of (B.12) on W, we obtain a Γ-equivariant long exact sequence

(B.13)
$$\cdots \to H^{n-1}(F,\mathbb{C}) \to H^n(W,\mathbb{C}) \xrightarrow{(g^*,i^*)} H^n(W',\mathbb{C})\oplus H^n(T,\mathbb{C}) \to H^n(F,\mathbb{C}) \to H^{n+1}(W,\mathbb{C}) \to \cdots .$$

Let χ be a character of Γ with non-trivial restriction to I. The operator

$$e_\chi := \frac{1}{\ell}\sum_{\sigma\in\Gamma} \chi(\sigma)^{-1}\sigma$$

is a projection of any complex representation H of Γ onto H_χ, and it commutes with Γ-equivariant maps, so applying e_χ to (B.13) yields an exact sequence. By Definition-Lemma B.2 (3) and (1), the group I acts trivially on $H^n(F,\mathbb{C})$ and on $H^n(T,\mathbb{C})$ for any n. If $\alpha \in H_\chi$ and I acts trivially on H, then $\alpha = \gamma^a\alpha = \chi(\gamma^a)\alpha$ with $\chi(\gamma^a) \neq 1$, so $\alpha = 0$. Hence $H^n(F,\mathbb{C})_\chi = H^n(T,\mathbb{C})_\chi = 0$ for any n, and (B.13) implies that

$$g^* : H^n(W,\mathbb{C})_\chi \to H^n(W',\mathbb{C})_\chi$$

is an isomorphism. A faithful character χ is injective on Γ, and $\gamma^a \neq 1$ since $0 < a < \ell$, so χ is non-trivial on I. Since e_χ is exact, the χ-part of the cokernel of g^* is the cokernel of g^* on χ-parts, which is zero. This proves (1).

Step 2. In this step we prove (2).

Let $M \in \{W, W'\}$ and $x \in M$. By Lemma B.4 (3), x has an open neighborhood N such that (N, x) is homeomorphic to $(B/H_x, 0)$, where $B \subset \mathbb{C}^d$ is an open ball centred at 0 and H_x is a finite group of diagonal matrices. Let $\phi : B \to B/H_x$ be the quotient map. Since the fibers of ϕ are finite, proper base change [Sta26, Tag 09V6] implies that $R^k\phi_*\mathbb{Q}_B = 0$ for $k > 0$. Moreover, the operator $|H_x|^{-1}\sum_{\sigma\in H_x}\sigma$ on $\phi_*\mathbb{Q}_B$ is a projection onto the subsheaf of invariants, which is $\mathbb{Q}_{B/H_x}$. Since $\phi^{-1}(0) = \{0\}$, the cohomology of the pair $(B/H_x, (B/H_x)\setminus\{0\})$ with coefficients in $\phi_*\mathbb{Q}_B$ is $H^k(B, B\setminus\{0\};\mathbb{Q})$, and the summand with coefficients in $\mathbb{Q}_{B/H_x}$ is the H_x-invariant part. Together with excision, we obtain

$$H^k(M, M\setminus\{x\};\mathbb{Q}) \cong H^k(B/H_x, (B/H_x)\setminus\{0\};\mathbb{Q}) \cong H^k(B, B\setminus\{0\};\mathbb{Q})^{H_x}.$$

The group $H^k(B, B\setminus\{0\};\mathbb{Q})$ is $\mathbb{Q}$ for $k = 2d$ and zero for $k \neq 2d$, and the complex-linear automorphisms in H_x preserve the real orientation of $\mathbb{C}^d$, hence act trivially on $H^{2d}(B, B\setminus\{0\};\mathbb{Q})$. Thus $H^k(M, M\setminus\{x\};\mathbb{Q}) \cong \mathbb{Q}$ for $k = 2d$ and $H^k(M, M\setminus\{x\};\mathbb{Q}) = 0$ for $k \neq 2d$, and the universal coefficient theorem implies that $H_k(M, M\setminus\{x\};\mathbb{Q}) \cong \mathbb{Q}$ for $k = 2d$ and $H_k(M, M\setminus\{x\};\mathbb{Q}) = 0$ for $k \neq 2d$. The local generators in degree $2d$ are given by the complex orientation. Since Y is projective, Y' is projective, and W and W' are compact, as they are finite over Y and Y'.

Let $[M] \in H_{2d}(M,\mathbb{Q})$ be the fundamental class. By Poincaré duality for compact oriented rational homology manifolds, the cap product with $[M]$ is an isomorphism $H^n(M,\mathbb{C}) \to H_{2d-n}(M,\mathbb{C})$ (see [Bre97, Section V.9]), and g_* is the composite of the cap product with $[W']$, the proper pushforward $H_{2d-n}(W',\mathbb{C}) \to H_{2d-n}(W,\mathbb{C})$, and the inverse of the cap product with $[W]$. Since g is an isomorphism over the dense open subset $W\setminus T$, we have $\deg(g|_{W'_0}) = 1$

for any connected component W'_0 of W', where deg is the topological degree, so $g_*[W'] = [W]$. By the projection formula for the cap product, $g_*(g^*\alpha \cap [W']) = \alpha \cap g_*[W'] = \alpha \cap [W]$, that is,

$$g_* g^* \alpha = \alpha \tag{B.14}$$

for any $\alpha \in H^n(W, \mathbb{C})$. The elements of Γ are holomorphic automorphisms of W and W', so they preserve $[W]$ and $[W']$, and g is Γ-equivariant. Hence g_* is Γ-equivariant and maps $H^n(W', \mathbb{C})_\chi$ to $H^n(W, \mathbb{C})_\chi$. By (1) and (B.14), the restriction of g_* to $H^n(W', \mathbb{C})_\chi$ is the inverse of

$$g^* : H^n(W, \mathbb{C})_\chi \to H^n(W', \mathbb{C})_\chi,$$

and in particular it is injective.

The equality $[g_* z] = g_*[z]$ follows from the definition of $[\cdot]$ and the functoriality of proper pushforward on homology (cf. [Sch88, p. 18]): for a subvariety V' of W', both $g_*[V']$ and $[g_* V']$ are images of classes in $H_{2\dim V'}(g(V'), \mathbb{Q})$, which is zero if $\dim g(V') < \dim V'$ and is one-dimensional if $\dim g(V') = \dim V'$, and $g_*[V']$ and $[g_* V']$ coincide over the dense open subset of $g(V')$ over which $V' \to g(V')$ is a finite covering of smooth varieties. If $[z]$ has a non-zero component in $H^*(W', \mathbb{C})_\chi$, then the χ-component of $[g_* z] = g_*[z]$ is the image of the χ-component of $[z]$ under the injective map g_*, hence non-zero. This proves (2).

Step 3. In this step we prove (3).

For $1 \le i \le k$, let a_i be the integer a of Definition-Lemma B.2 for π_{i-1}, and let $I_i := \langle \gamma^{a_i} \rangle$. The variety Y_i is projective, as it is a blow-up of the projective variety Y_{i-1}. Let χ be a faithful character of Γ. Since $0 < a_i < \ell$, we have $\gamma^{a_i} \neq 1$, so χ is non-trivial on I_i for any i. Applying (2) to $g_k, g_{k-1}, \dots, g_1$ in turn, we obtain that $[(g_i \circ \cdots \circ g_k)_* z]$ has a non-zero component in $H^*(W_{i-1}, \mathbb{C})_\chi$ for $i = k, k-1, \dots, 1$, where

$$(g_i \circ \cdots \circ g_k)_* z = (g_i)_* (g_{i+1} \circ \cdots \circ g_k)_* z$$

by the functoriality of proper pushforward of cycles. For $i = 1$, this is (3). □

In the proof of [Sch88, Theorem 2.0], we use Proposition B.5 as follows. We use the notation of [Sch88, §2], where we write ℓ for the order m of the cyclic group, $2k$ for h, and $C/(\mathbb{Z}/\ell)$ for the quotient curve X. Then $(S^{2k}(C/(\mathbb{Z}/\ell)))_i$ is the blow-up of $(S^{2k}(C/(\mathbb{Z}/\ell)))_{i-1}$ along $S_{2i-1,i-1} \cap S_{2i,i-1}$, and W_i is the normalization of the pullback of W_{i-1} to $(S^{2k}(C/(\mathbb{Z}/\ell)))_i$ [Sch88, pp. 14–15]. By [Sch88, Lemma 2.2] for $i = 1$ and by [Sch88, Lemma 2.5] for $i \ge 2$, the cover $W_{i-1} \to (S^{2k}(C/(\mathbb{Z}/\ell)))_{i-1}$ satisfies the conditions of Definition-Lemma B.2: the branch divisor is the union of the $S_{j,i-1}$, which is a simple normal crossing divisor, the exponents are the α_j, and the exponents of the blown-up pair satisfy $\alpha_{2i-1} + \alpha_{2i} = \ell$. The exceptional divisors of the earlier blow-ups are not branch components, since, for $1 \le i' < i$ and local equations x_1 and x_2 of $S_{2i'-1,i'-1}$ and $S_{2i',i'-1}$, under $x_1 = t x_2$ the monomial $x_1^{\alpha_{2i'-1}} x_2^{\alpha_{2i'}}$ becomes $t^{\alpha_{2i'-1}} x_2^{\ell}$, and x_2^ℓ is absorbed into v. The base is smooth of dimension $2k$, so any irreducible component of the fiber product

$$W_{i-1} \times_{(S^{2k}(C/(\mathbb{Z}/\ell)))_{i-1}} (S^{2k}(C/(\mathbb{Z}/\ell)))_i$$

is of dimension at least $2k$, being locally cut out by $2k$ equations in a variety of dimension $4k$. The part of the fiber product over the exceptional divisor is finite over the exceptional divisor, so the dimension of the part over the exceptional divisor is at most $2k-1$. Thus no irreducible component of the fiber product is contained in the preimage of the exceptional divisor, and W_i is the space W' of Definition-Lemma B.2 for W_{i-1}. Therefore, Proposition B.5 (1) for $\varphi_i : W_i \to W_{i-1}$ yields the conclusion of [Sch88, Lemma 2.8], and Proposition B.5 (3), applied to the r blow-ups, that is, with r in place of the number k of blow-ups in Proposition B.5 (3), implies that for

$$\varphi = \varphi_1 \circ \cdots \circ \varphi_r : W_r \to W_0$$

in [Sch88, p. 18], the class of $\varphi_*(z_\chi)$ has a non-zero component in $H^*(W_0, \mathbb{C})_\chi$ whenever the class of z_χ has a non-zero component in $H^*(W_r, \mathbb{C})_\chi$.

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